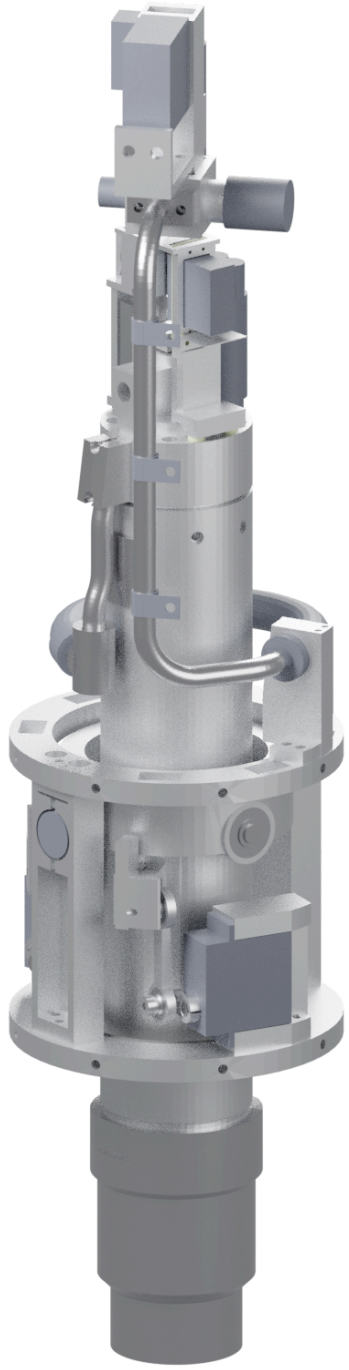
ARGOS H1 · ENGINE PROGRAMME · ARGOS-H1-ENGINE-DATASHEET

Engine technical datasheet

N_2O / printed-ABS hybrid rocket engine · servo-pintle throttle · two-axis gimbal · for a self-landing rocket



CAD v3.4.1 assembly, neutral pose. Visualisation of tessellated display geometry, not a drawing.

DRAFT · rev0-draft

Not physical approval. Nothing in this datasheet is physical approval or acceptance. No value is MEASURED: every value is design, simulation, derivation, assumption, datasheet or requirement data and carries its evidence tag and a trace reference (T###, Appendix A). Missing data are shown as MISSING, TO-DEFINE or PENDING, never estimated by the generator.

RATED THRUST (SL), PC AT THE SOFTWARE CAP

500 N SIM T034

≈ **487** N DERIVED T037 if η_{cf} is at the low end of its band

CHAMBER PRESSURE

2.50 MPa (25.0 bar) SIM T038

THROTTLE FLOOR 5:1

100 N SIM T039

ISP, RATED (SL)

193.1 s SIM T035

CAD MASS, WET (EXCL. TANK)

4.389 kg DERIVED T040

ENVELOPE (CAD BBOX)

139.5 × 139.5 × 593.3 mm DERIVED T044 T045 T041

Configuration	Version	sha256/16
CAD	v3.4.1	GLB 3879a25a4486ee0c
Combustion	v2.4	469727032124329f
Mechanical	v2	4f024ed4ccb4618c
Electronics	v2	da067408c1594814
Space/GNC	v3	3e97c378cfebee95
Requirements	rev 2	fbb9e1205707082f
Airframe ICD	draft 0.2	706455abf4098182
Decision log	176 lines	0c2c6d5539240658
Review register	7 reviews	9736f4974377a7bd
Decision overrides	ovr-3	3666334a809ccf6d

Generated 2026-10-13T13:25:31Z by gen/build_datasheet.py 0.2.0. Quotations: delivery Sønderborg, Denmark, DK-6400. Units SI; pressures in MPa with bar in brackets.

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1 Identification and status

Identification. Text values are quoted from their sources (trace references link to Appendix A).

Item	Value
Document	ARGOS-H1-ENGINE-DATASHEET DESIGN T353
Revision and status	rev0-draft — DRAFT, digital design data only DESIGN T354
Generated	2026-10-13T13:25:31Z DESIGN T355
Engine (requirement ENG-01)	Max thrust 500 N at sea level, with N2O at ~~15–25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition OVR-04. Normal Pc is capped at 25 bar in software. REQ T356
Throttle architecture	Servo-driven variable-area pintle (needle-in-seat) at the head end + separate NC main isolation valve upstream (tank side, off-gimbal) LEAD T357
Propellants and grain (ENG-12)	Propellants: self-pressurised N2O, tank 2.00 kg / 3.17 L conditioned to ~~15–25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition OVR-04, with heater hard-disabled at ARM; N2 supercharge not baselined. 3D-printed ABS grain (PETG backup): straight centred 25 mm port, 53.5 OD × 230 mm, 0.404 kg. Nozzle ε 2.0 with a fine-grain graphite insert. REQ T358
Thrust vector control (ENG-07)	TVC by a two-axis gimbal of the whole engine, pivot at the engine-group mid-burn CG. Range ±7° on the vector norm, 8° conical mechanical stop. Slew ≥ 100°/s at 7° in the worst vector direction at the 7.4 V nominal servo plug (111.2°/s was the v1 load; at the v2 load 99.85°/s, 97.77°/s with the hose at its limit, so ENG-07 is OPEN and marginal until Mechanical's exact kinematics and B-TVC-1; amended 2026-10-13, REV-GNC-1 RG-12). The 6.6 V degraded case (95.3°/s) is accepted on the MC result P 0.982. Linkage ratio n … LEAD T359
Ignition	spark-ignited N2O/propane torch (baseline kept); arc-ABS kept as a research option for bench tests (would remove the propane system) DERIVED T360
CAD configuration	DESIGN MODEL - not released for fabrication; geometric checks are software checks, not physical validation DESIGN T361
Combustion interface status	v2.4 REVISION (digital only, Por revisar). Design point unchanged. No number is MEASURED. The relight claim is gated by CF-4; FA-1 is accepted for design continuation only. DESIGN T362
Electronics interface status	DESIGN + SIL + schematic. Firmware v2.9 runs only in SIL; the ECU v2.3 schematic passes ERC and a netlist cross-check; no board has been built and nothing here is hardware-validated. Values carry tags; open items are listed in open_issues, and the bench plan (06-electronics/docs/bench_plan_v2.md) is where each is closed. DESIGN T363
Review register note	Independent review verdicts for the engine team's digital deliverables. A verdict is not physical approval: no hardware has been tested. DESIGN T364
Decision overrides in force	ovr-3: 9 active, 0 redundant, 0 in error; 4 superseded values withheld (4 interface keys); see §13.1 and §13.2 DERIVED T352

1.1 Separation of evidence

The programme keeps five kinds of evidence apart. This datasheet labels every value with one of them; none of them is a physical test.

Evidence kind	Scope in this datasheet	Source statement
Conceptual geometry	Items defined in text or hand calculation only, not modelled as solids: N2O tank, isolation-valve module mass, head-end bleed path, relief and fill/vent hardware.	v1 concept, not yet in CAD DESIGN T365
CAD solid model	CAD v3.4.1: STEP solids of record, assembly manifest and GLB display mesh. Software checks only.	n: 98; passed: 98 DERIVED T366
Mass budget	CAD volume × density for modelled parts, datasheet or assumed masses for COTS items, explicit exclusions (§4.4).	sum of unit masses x qty (CAD volume x density, or tagged COTS masses); includes the transport-lock items only if they are placed (they are not: group 'lock' is excluded from the instance list); PROVISION lines excluded DERIVED T367
Simulation	Quasi-1D combustion and CEA, plume CFD, 6-DOF Monte Carlo, firmware SIL. Predictions with stated inputs and tags.	M2 v3 (2026-10-12): 6-DOF Monte Carlo (sim6/run6 v2.6, 500 runs per case, same draws) re-baselined on mechanical.json v2 + combustion.json v2.3 (iface_v3c) with flight software v2.6 (tracker, governor with ratchet, pre-MECO throttle-down, relight start rule, NC-closed relight). Lead decisions … SIM T368
Physical validation	None performed. Bench, cold-flow and static-fire tests are listed in §12 and executed by the founders.	Independent review verdicts for the engine team's digital deliverables. A verdict is not physical approval: no hardware has been tested. DESIGN T364

1.2 Evidence tags

SIM model prediction (CEA, quasi-1D, CFD, 6-DOF, SIL)

ASSUMED engineering assumption pending data

TO-VERIFY value or feature to confirm (vendor, CAD or test)

DESIGN design choice or specification value

LEAD lead decision recorded in the CHANGELOG

TO-DEFINE value not yet defined by its owner

SUPERSEDED withdrawn value, not shown (§13.2)

DERIVED calculated from tagged inputs (by a specialist or by this generator)

DATASHEET vendor or catalogue value

LITERATURE published reference

REQ requirement (founder or lead)

MISSING no value in the sources

PENDING analysis not yet delivered

Each value is followed by its tag and a trace reference T###. The trace gives the source file, its version and sha256/16, and the exact key (JSON pointer, markdown row or CHANGELOG line and item). Where a source carries a longer tag text, Appendix A keeps it verbatim.

> **Decisions newer than combustion.json v2.4.** Where an interface file still carries wording that a later lead decision replaced (REV-COMB-2 findings R2-01, R2-02 and R2-05), the stale words are ~~struck through~~ and the decision is inserted with its override id. The override table (§13.1) cites the decision text and checks it on every build; values the Combustion author withdrew are not shown at all (§13.2).

1.3 Recent lead decisions folded in

Last 8 lead decisions in interfaces/CHANGELOG.md (176 lines, sha256/16 0c2c6d5539240658).

Line	Date	Decision (start of text)
161	2026-10-13	On GNC rev1 (lines 157-158; text saved to 01-systems-gnc/staging_v3.1/rev1/REPORT-GNC-rev1.md, lead verified MANIFEST_rev1 2dbe49919289f0ac, rev1_items 799258458f0142a5, FSW record 08c85849a5da950d): (1) FC release predicate ADOPTED (DESIGN): while the drogue is attached, on each new TLM_FAST frame RELEASE iff (d1 & 0x0F) = RUN AND b4 AND [(d2-d3 != 0xFFFF AND >= 300 kPa abs) OR b7], latched; b7 filtered-only (line ... LEAD T369
164	2026-10-13	On Combustion's RG-01 tables and Mechanical's bleed-path line 162: (1) RG-01 design basis = Combustion's minimal set (3): coil re-route + pre-scheduled 2-tap purge on the decided path (equal 0.707 mm restrictors, 0.71 mm drill, each flow-tested for Cd) with 2.5 s of FLOW ending at E0 - 0.05 s + theta <= 2 deg with balanced or axial vent. PTFE lining DROPPED (costs 0.3 s of margin, no P benefit). (2) PURGE command at ... LEAD T370
166	2026-10-13	On REV-ELEC-1: (1) RE-02 light-rule baseline fallback: P0 > +check → keep the measured P0 (never substitute ambient on the high side), log BASE_HIGH and flag the FC; on the pad and the stand this refuses the lift with PC_SENSOR (no commit); at the A1 relight the start proceeds on the measured P0. P0 < -check → P0 := 0 (the -0.75 bar PC_SENSOR path unchanged). Tag LEAD (this line). S13 B extended per RE-02 (2): ... LEAD T371
168	2026-10-13	On REV-GNC-2: (1) Drogue release predicate restated, interim term limited to the invalid case: RELEASE iff RUN AND b4 AND [(Pc valid AND >= 300 kPa abs) OR b7 OR (d2-d3 = 0xFFFF AND t >= t_E2 + 0.10 s)], latched, t_E2 from EVENT 0x120; a late invalidity releases on the first invalid frame. The sentence "the FC never holds the drogue with the engine in RUN beyond E2 + 0.10 s" in line 161 (2) is WITHDRAWN. b7 sets ... LEAD T372
170	2026-10-13	On the Combustion v2.4 handback (line 169): installed 02-combustion/REPORT.md v2.4 (32e366bd78437dc0, from REPORT_v2p4.txt), 05-plume-cfd/REPORT.md v2.2b (140b3c59f8323bba), 02-combustion/REVISION-RESPONSE.md (ee951fb865c1459d); HANDBACK-SUMMARY lines 79-80 replaced with the RC-10 text (1d760e9e197429d8); v2.3 report, v2.3 summary and plume v2.2 archived under superseded/. (1) OI-28 ENG-01 at the cap: the 25 bar ... LEAD T373
172	2026-10-13	On REV-COMB-2: (1) R2-02 ONE stand safing rule after a no-light, effective now: N2 purge >= 60 s at >= 10 g/s, then hold >= 15 min before anyone approaches, with Pc at ambient and head TCs falling, then safing; "wait >= 5 s" is withdrawn everywhere. (2) R2-03 FA-1 load restated as an ENVELOPE: Combustion publishes O/F 4-12 x tf 1-7 ms (1 ms steps) x t_res 0.02/0.06 x pool none/q 2 MW/m2 (head and aft peak, rise ... LEAD T374
173	2026-10-13	On Combustion's FA-1 envelope (02-combustion/data/v2p4a/fa1_envelope_v2p4a.json 64b328150025fa1e; 896 cases, A1 relight, O/F 3-12, tf 1-7 ms; SIM on ASSUMED construct): head peak 83.0 bar abs baseline (vapour once, rounded entrance) / 84.8 sharp / 88.0 upper bracket; aft 0.95-1.00 of head; time above 45 bar <= 3.65 ms (4.45 bracket); above 67.5 bar <= 1.0 ms (1.3); never above 90 bar; impulse <= 726 bar.ms (757); ... LEAD T375
175	2026-10-13	On GNC rev2 (line 174; staging_v3.1/rev2 MANIFEST cb5d2c48e8d8fa27 verified 280/280 by the lead; REPORT-GNC-rev2.md 2edf0360cbc975e5 extracted by the lead; SIM only): (1) A1 apogee target 285 m ADOPTED under line 168 (4) (0 exhausted runs at the 73 g purge; cost 15 m). (2) +0.60 kg bleed-valve mass: no trim below 275 m now; the trigger is the selected valve pair: if it adds > 0.30 kg, GNC re-flies and the lead ... LEAD T376

2 Key performance

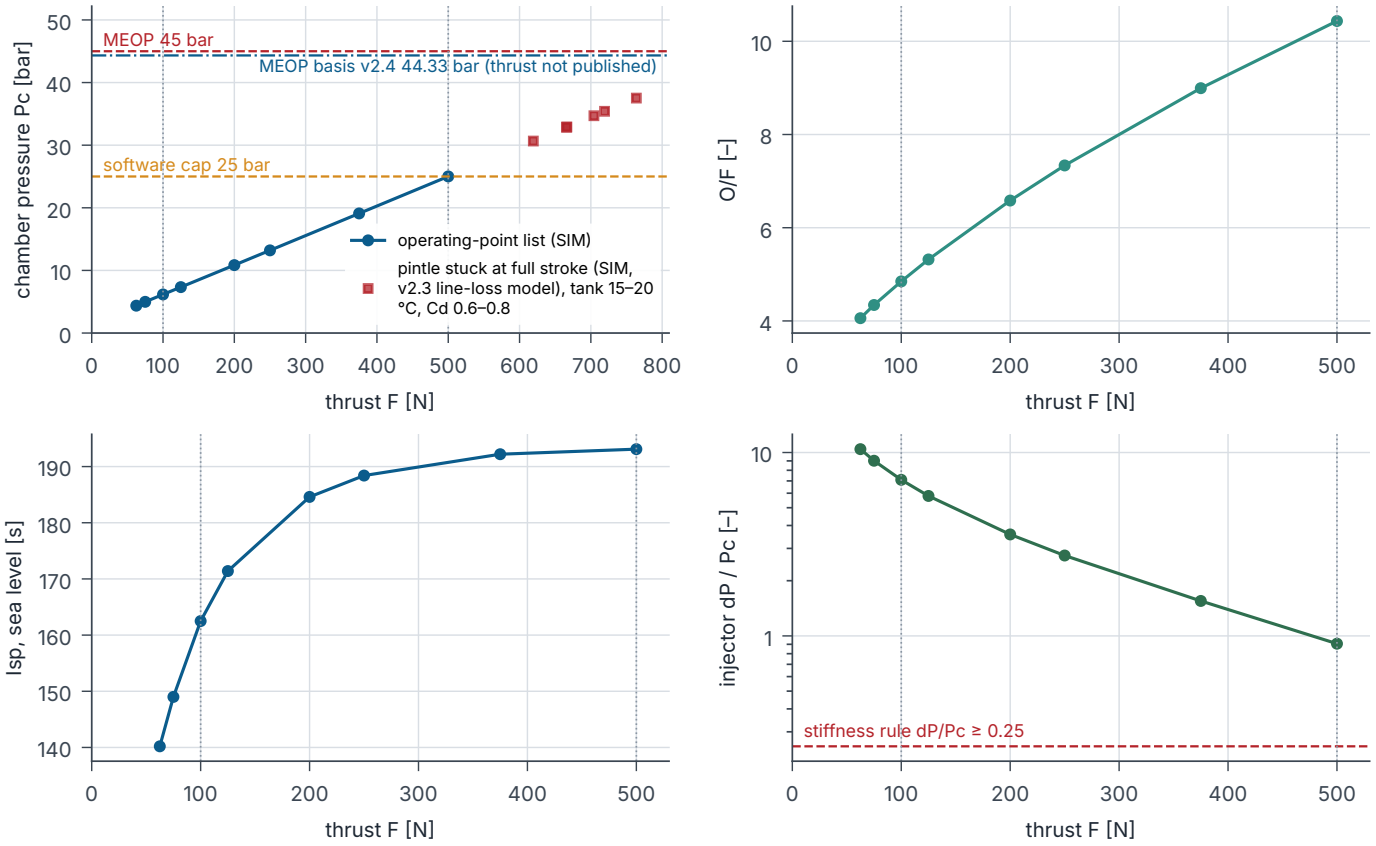
All performance values are model predictions (quasi-1D port model with CEA thermochemistry), with the regression law ASSUMED until the T1–T4 static fires. None is measured. Model basis: 20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975) SIM T377

2.1 Rated point and throttle floor

Operating points from combustion.json /operating_points/throttle_points_20C_initial_port (tank state and port as in the model basis above). Column fractions are the selection keys of the point list.

Quantity	Rated, 100 %	Floor target, 20 % (5:1)	Floor requirement, 25 % (4:1)	Stretch floor, 12.5 % (8:1)
Thrust F (sea level)	**500.0** N SIM T034	**100.0** N SIM T039	**125.0** N SIM T378	**62.5** N SIM T379
Chamber pressure Pc	**2.50** MPa (25.0 bar) SIM T038	**0.62** MPa (6.2 bar) SIM T380	**0.73** MPa (7.3 bar) SIM T381	**0.44** MPa (4.4 bar) SIM T382
Mixture ratio O/F	**10.44** SIM T383	**4.85** SIM T384	**5.32** SIM T385	**4.06** SIM T386
Specific impulse Isp (sea level)	**193.1** s SIM T035	**162.5** s SIM T387	**171.4** s SIM T388	**140.2** s SIM T389
Specific impulse as exhaust velocity	**1894** N·s/kg DERIVED T390	**1594** N·s/kg DERIVED T391	**1681** N·s/kg DERIVED T392	**1375** N·s/kg DERIVED T393
Oxidiser mass flow	**0.2410** kg/s SIM T394	**0.0520** kg/s SIM T395	**0.0626** kg/s SIM T396	**0.0365** kg/s SIM T397
Fuel mass flow	**0.0231** kg/s SIM T398	**0.0107** kg/s SIM T399	**0.0118** kg/s SIM T400	**0.0090** kg/s SIM T401
Total mass flow	**0.2641** kg/s DERIVED T402	**0.0628** kg/s DERIVED T403	**0.0744** kg/s DERIVED T404	**0.0454** kg/s DERIVED T405
Pintle lift	**1.916** mm SIM T406	**0.301** mm SIM T407	**0.367** mm SIM T408	**0.207** mm SIM T409
Injector stiffness dP/Pc	**0.91** SIM T410	**7.10** SIM T411	**5.80** SIM T412	**10.43** SIM T413
Requirement or target thrust	**500.0** N SIM T414	**100.0** N REQ T415	**125.0** N REQ T416	**62.5** N REQ T417
Model status at the point	ENG-01 at the 25 bar cap: see the note below LEAD T036	met (dP/Pc 7.1) SIM T418	met (dP/Pc 5.8) SIM T419	model meets it: dP/Pc 10.4, chug gain margin 38.6 at 10 ms delay, attached nozzle except incipient lip separation … SIM T420

ENG-01 at the software cap. OI-28 ENG-01 at the cap: the 25 bar software Pc cap STAYS (the binding margin is the 30 bar abort, REV-ELEC-1 RE-04); if T1 measures eta_cf at the low end the rated thrust is ~487 N at 25 bar, and the measured k feeds GNC's cap rule; owner may override. LEAD T036



Throttle characteristics over the operating-point list (SIM) and the stuck-pintle full-stroke cases. Dotted lines mark the floor-target and rated points; dashed lines the software cap, the MEOP and the injector-stiffness rule; the dash-dot line is the v2.4 MEOP basis (§2.4). Stuck-pintle rows with a tank above the ARM limit (24 °C, ENG-12 amendment) are not credible ignition states and are not plotted (6 of 12 rows). Data: T034, T035, T038, T039, T378, T379, T380, T381, T382, T383, T384, T385, T386, T387, T388, T389, T410, T411, T412, T413, T421, T422, T423, T424, T425, T426, T427, T428, T429, T430, T431, T432, T433, T434, T435, T436, T437, T438, T439, T440 and 29 more

2.2 Nozzle and chamber

Nozzle and chamber at the rated point (combustion.json).

Quantity	Value
Throat diameter D_t	**13.72** mm DESIGN T470
Exit diameter D_e	**19.41** mm DESIGN T471
Expansion ratio ε	**2.00** DERIVED T472
Throat area A_t	**147.93** mm² DERIVED T473
Exit-to-ambient pressure ratio at rated point	**3.199** SIM T474
Characteristic velocity c* (delivered, η_{c*} 0.92)	**1401** m/s SIM T475
Ratio of specific heats, chamber	**1.1467** SIM T476
Molar mass, chamber	**28.72** kg/kmol SIM T477
Gas-path length, injector face to exit	**338.6** mm DERIVED T478
Free chamber volume, initial	**284.6** cm³ DERIVED T479
Characteristic length L*, initial	**1.924** m DERIVED T480
Chamber gas-side diameter (pre/post chamber)	**53.9** mm DESIGN T481
Case tube OD × wall	**63.50** mm DESIGN T482 × **3.175** mm DESIGN T483
Chamber temperature, ideal (CEA)	**3257** K (2984 °C) SIM T484
Stagnation temperature, delivered (η_{c*}^2)	**2757** K (2484 °C) SIM T485
Throat insert material	isomolded/extruded fine-grain graphite, bulk density >= 1.80-1.85 g/cm3, ash <= 0.1 %; grade TO-QUOTE (Sønderborg DK-6400) DERIVED T486

2.3 Burn, starts and relight

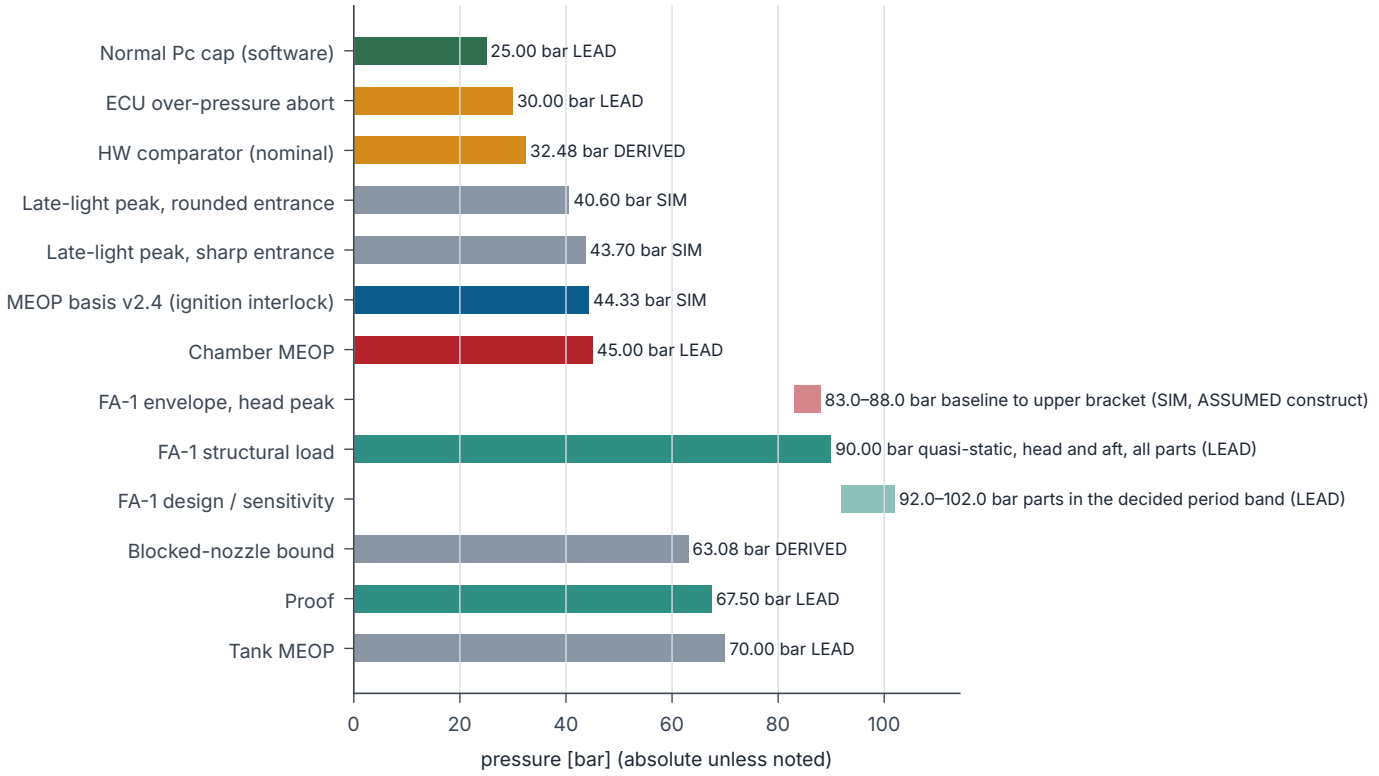
Item	Value
Starts per flight	>= 3; limited by propane (0.055 g/start) and N2O torch flow (0.30 g/start) DERIVED T490

Item	Value
Restart requirement (ENG-04)	Restartable: ≥ 3 starts per flight. Command to 90 % thrust ≤ 0.5 s (target 0.3 s), repeatability ±50 ms; flight software plans relight on 0.81 s (amended 2026-10-13; the 0.6 s rule is withdrawn). DEVIATION recorded 2026-10-13: the ≤ 0.5 s / ±50 ms targets are met only for a liquid-filled segment; vapour-state relights reach 0.81 s to Pc confirm, and the vapour-filled 44 cm³ segment is not yet evaluated (REV-GNC-1 RG-01, RG-12). The ignition lift is held until light is confirmed, and the pintle never opens past the ignition lift before that (amended 2026-10-13: light = Pc rise >= 0.8 bar above the pre-lift baseline [E1-20 ms, E1] on filtered and raw for 10 ms, sanity floor 1.8 bar abs; the 3 … DERIVED T491
Light window W from the lift command: flight first start / flight relight / stand	**0.40** s LEAD T487 / **0.65** s LEAD T488 / **0.35** s LEAD T489
Relight planning value (flight software)	**0.81** s REQ T492
Planning value on an UNPRIMED flag (option c)	**1.41** s REQ T493
A1 burn time, nominal 6-DOF (FSW v2.6 basis)	**7.37** s SIM T494
N2O vented per relight purge, plan maximum	**76** g SIM T495
Grain and throat insert life (ENG-06)	Accumulated burn per flight sized by the duty cycles in gnc.json v3. Grain replaced each flight; web ≥ 1 mm over the whole regression-law envelope (a ×0.6–1.5, n 0.35–0.68). Throat insert: plan on one insert per flight until T1/T4 bore data exist. Model spread is 0.5–4 A1 flights per insert at ΔAt/At 5 %; the literature-anchored case gives 0.46–0.71 (review RC-06). The Pc/thrust loop must tolerate 7–11 % throat-area growth within a flight. Thermal design envelope (amended 2026-10-13): the dispersed duty cycles of gnc.json v3 (index rev d, up to 24.09 s), governed per quantity: A1 p99 … ASSUMED T496

2.4 Pressure limits and physical worst case

Pressure ladder (absolute unless noted). The note column is an editorial pointer, not data.

Limit	Value	Note
Normal chamber pressure cap (software)	**2.50** MPa (25.0 bar) LEAD T466	LEAD decision; ENG-01/ENG-09
ECU over-pressure abort	**3.00** MPa (30.0 bar) LEAD T498	decided channel and persistence: §7.6 (OVR-01)
Hardware comparator trip (Pc transducer, nominal)	**3.248** MPa (32.48 bar) DERIVED T499	absolute; band in electronics.json
Late-light decomposition peak at the pad (key pad_15C): rounded / sharp port entrance	**4.060** MPa (40.6 bar) SIM T500 / **4.370** MPa (43.7 bar) SIM T501	rounded entrance REQUIRED (CHANGELOG 160 (4))
MEOP basis (v2.4) at the ignition interlock, and its margin to MEOP	**4.433** MPa (44.33 bar) SIM T469 at T_{tank} ≤ **24** °C REQ T510; margin **0.67** bar SIM T502	stuck pintle at full stroke, worst corner of the bands; replaces the v2.3 basis (§13.2)
Chamber MEOP	**4.50** MPa (45.0 bar) LEAD T467	kept with the ignition interlock (ENG-09 amendment)
Fuel-assisted relight pulse (FMEA FA-1, key burn_1ms)	**4.89–7.55** MPa (48.9–75.5 bar) SIM T503 single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs OVR-07	MEOP NOT shown → FMEA FA-1 (single-control hazard) SIM T504
FA-1 envelope, head peak: baseline / sharp entrance / upper bracket (v2.4a)	**8.304** MPa (83.0 bar) SIM T529 / **8.476** MPa (84.8 bar) SIM T530 / **8.802** MPa (88.0 bar) SIM T531	SIM on an ASSUMED construct (§2.5)
FA-1 structural load (lead decision): quasi-static at head and aft, all parts	**9.000** MPa (90 bar) LEAD T516	replaces the interim check (CHANGELOG 173)
FA-1 parts with a natural period in the decided band: design / sensitivity pressure	**9.200** MPa (92 bar) LEAD T517 / **10.200** MPa (102 bar) LEAD T518	Mechanical margins reported, not reviewed (§2.5)
Blocked-nozzle bound (chamber at tank vapour pressure)	**6.31** MPa (63.1 bar) DERIVED T505	below yield (FEA PENDING)
Proof pressure (ENG-09) and its minimum margin above the MEOP basis	**6.75** MPa (67.5 bar) LEAD T506; margin ≥ **21.9** bar SIM T507	hydro proof, founders
Tank MEOP	**7.00** MPa (70.0 bar) LEAD T508	tank not yet in CAD
Tank pressure at ARM, saturated: lower end / ignition interlock	**4.500** MPa (45.0 bar) DERIVED T514 / **5.528** MPa (55.28 bar) LEAD T511 less the guard band TO-DEFINE TO-DEFINE T512	interlock per CHANGELOG 172 (4); the combustion.json range still spans the old ARM range (OVR-04)



Pressure ladder. FA-1 is shown as the Combustion v2.4a envelope (SIM on an ASSUMED construct) and the lead structural-load decision (§2.5); the single point in combustion.json v2.4 is annotated (OVR-07) and not plotted. FA-1 is accepted for design continuation only (CHANGELOG 170 (2)). Data: T466, T467, T469, T498, T499, T500, T501, T505, T506, T508, T516, T517, T518, T529, T530, T531

The v2.3 stuck-pintle basis assumed a tank above the ignition interlock, which is no longer a credible ignition state; the thrust at the v2.4 basis is not published, so the structural thrust case stays open (§13.3, C-07).

Physical worst case	Value
Basis	MEOP 45 bar kept with the ignition interlock T_tank <= 24 C at ARM and at ignition; basis 44.33 bar at the interlock (meop_basis_v2p4), margin 0.67 bar **OVR-06** combustion.json still reads: ~~LEAD decision 2026-10-12 (CHANGELOG): basis = pintle full-stroke failure, 30 C tank, Cd 0.8 (data/v2/full_stroke_failure_case_v2.json)~~ LEAD T566
Chamber pressure at the basis (pintle stuck at full stroke)	**4.433** MPa (44.33 bar) SIM T469 at T_{tank} ≤ **24** °C REQ T510
Thrust at the basis	**MISSING** MISSING T563
Full-stroke injector Cd accepted in CF-1/CF-3 (otherwise MEOP is restated)	**0.80** LEAD T565
Lateral thrust at the gimbal range: rated / worst case	**60.9** N DERIVED T562 / **MISSING** MISSING T564
Relight thrust transient (FA-1): interim structural load, confirmed bounding by the envelope	**1.8** kN LEAD T521 × **4** ms LEAD T522
Airframe ICD statement (ICD §3)	876 N at Pc [withheld: superseded value, SUP-04] (pintle stuck at full stroke, ~~30 °C tank~~ ignition interlock T_tank <= 24 C; basis 44.33 bar (meop_basis_v2p4) **OVR-06**, Cd 0.80). Use it as the ultimate axial load case for the thrust rings SIM T567
Pressure-part requirement (ENG-09)	Pressure parts: SF ≥ 1.5 on yield and ≥ 2.0 on ultimate, at MEOP and at temperature. Chamber MEOP 45 bar (~~stuck-pintle case, 30 °C tank~~ ignition interlock T_tank <= 24 C; basis 44.33 bar (meop_basis_v2p4) **OVR-06**), proof 67.5 bar, which covers the blocked-nozzle bound of about 63 bar below yield. Normal Pc ≤ 25 bar; ECU over-pressure abort at about 30 bar. Tank MEOP 7.0 MPa, with a passive relief device (relief valve and/or burst disc, set below proof, T-vent). ~~Trapped-liquid relief ≤ 7.5 MPa.~~ tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check (Cv >= 0.1), disc 9.5 MPa +/- 5 % **OVR-05** Chamber Pc transducer range ≥ 70 bar. LEAD T568
ENG-09 amendment	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. REQ T013

2.5 FA-1 relight-pulse load

The fuel-assisted relight pulse (FMEA FA-1) is published as an envelope over the Combustion v2.4a case grid, computed on a vented three-node construct with ASSUMED inputs: an estimate, not acceptance evidence. The lead set the structural load on it

(CHANGELOG 173). The single point in combustion.json v2.4 is not a load any more and is annotated wherever it is quoted (OVR-07).

Envelope maxima over the case grid. Baseline: fuel vapour counted once; upper bracket: both initial conditions. Absolute pressures. SIM on an ASSUMED construct.

FA-1 envelope (Combustion v2.4a)	Value
Case grid (as published, with the number of cases)	OF_grid: [4.0, 5.0, 6.0, 8.0, 10.0, 12.0]; OF_extension: [3.0, 3.5]; tf_ms: [1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0]; t_res_s: [0.02, 0.06]; pools: none; q2.0; inits: vapour_once: 0.05; as_written: 1.0; entrances: rounded_Cd1.0: 1.0; sharp_Cd0.65: 0.65; n_cases: 896 SIM T552
Head peak: baseline (vapour counted once, rounded entrance) / sharp entrance / upper bracket	**8.304** MPa (83.0 bar) SIM T529 / **8.476** MPa (84.8 bar) SIM T530 / **8.802** MPa (88.0 bar) SIM T531
Aft peak relative to the head peak (range over the cases)	**0.951–0.999** SIM T532
Time above 45 bar at the head (summed): baseline / upper bracket	**3.65** ms SIM T533 / **4.45** ms SIM T534
Time above 67.5 bar at the head (summed): baseline / upper bracket	**1.00** ms SIM T535 / **1.30** ms SIM T536
Time above 90 bar at the head (summed): baseline / upper bracket	**0.00** ms SIM T537 / **0.00** ms SIM T538
Head pressure impulse above the cold-flow pressure: baseline / upper bracket	**726** bar·ms SIM T539 / **757** bar·ms SIM T540 integral of max(P - P0, 0) over 0-25 ms after the light, bar.ms; P0 = cold-flow pressure 1.071 bar SIM T541
Thrust spike, peak (quasi-steady nozzle × η_{cf}): baseline / upper bracket	**1694** N ASSUMED T542 / **1768** N ASSUMED T543
Thrust spike, single-degree-of-freedom equivalent (max over periods), upper bracket; interim rectangular pulse for comparison	**3053** N DERIVED T545; interim pulse **3484–3599** N DERIVED T547
Head equivalent static pressure with the dynamic load factor (max over periods): baseline rounded / upper bracket	**8.898** MPa (89.0 bar) DERIVED T549 / **10.200** MPa (102.0 bar) DERIVED T551
Lead summary of the envelope	On Combustion's FA-1 envelope (02-combustion/data/v2p4a/fa1_envelope_v2p4a.json 64b328150025fa1e; 896 cases, A1 relight, O/F 3-12, tf 1-7 ms; SIM on ASSUMED construct): head peak 83.0 bar abs baseline (vapour once, rounded entrance) / 84.8 sharp / 88.0 upper bracket; aft 0.95-1.00 of head; time above 45 bar <= 3.65 ms (4.45 bracket); above 67.5 bar <= 1.0 ms (1.3); never above 90 bar; impulse <= 726 bar.ms (757); LEAD T523 O/F bounding within the construct (closed-volume max at O/F 3.5); fuel-basis sensitivity +2-3 %; thrust spike <= 1.77 kN, SDOF equivalent <= 3.05 kN (interim 1.8 kN x 4 ms gives 3.48-3.60 kN, so it bounds). LEAD T524 Pressure DLF: <= 1.007 for periods <= 0.1 ms (shell: quasi-static valid); 1.07-1.20 for periods 0.25-1 ms (head equivalent static <= 89.0 baseline rounded, <= 102.0 bracket). LEAD T525
Model tag	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence SIM T554
Not modelled	acoustic/wave peaks along the 230 mm port (lumped nodes; transit ~0.3 ms hot, ~1 ms cold); equilibrium (variable cv, recombination) thermochemistry: kappa fitted at 40 kg/m3 (REV-COMB-2: +3.7 % at 6.6 kg/m3, conservative); frozen-gamma throat outflow (REV-COMB-2: vents ~4 % fast, slightly non-conservative); flame spread, heat loss, combustion roughness; the torch contribution before the light; the fuel-vapour inventory at light (MISSING; first-order bound only) SIM T555
Fuel-vapour inventory at light	MISSING: the fuel-vapour (and aerosol/film) inventory in the warm 32.2 mm port at the moment of light after the 18.67 s A1 coast is not modelled and not measured. Below: first-order bounds only. TO-VERIFY T556 Missing datum (fuel inventory at light) remains TO-VERIFY; LEAD T526 **TO-VERIFY** TO-VERIFY T560
Data file	`02-combustion/data/v2p4a/fa1_envelope_v2p4a.json` 64b328150025fa1e: matches the citation DERIVED T528

Lead decision on the envelope and the margins Mechanical reported against it (quoted as reported, negative values included; SIM, not the assembled FEA of record; §9.2, §13.3 C-14). The proof pressure is in the §2.4 ladder (its absolute or gauge basis is not stated in combustion.json).

FA-1 load decision	Value
FA-1 structural load, quasi-static at head and aft, all parts	**9.000** MPa (90 bar) LEAD T516
Parts with a natural period in the decided band (closures, bolts, J1 lip): design / sensitivity pressure	**0.25–1** ms LEAD T519 T520: **9.200** MPa (92 bar) LEAD T517 / **10.200** MPa (102 bar) LEAD T518

FA-1 load decision	Value
Thrust spike, interim structural load (confirmed bounding)	**1.8** kN LEAD T521 × **4** ms LEAD T522
Mechanical margins at these loads (decision-log result, not independently reviewed)	FA-1 final load rule (lead 172 (2) + 173) closed in S-AX/joints: 90 bar abs quasi-static head+aft every part, 92/102 bar for the 0.25-1 ms parts (modes 1-2: TC-09+nozzle stack 0.33-0.47 ms, TC-02 head on its screws 0.27-0.39 ms; shell/bonnet 0.04-0.08 ms quasi-static), SF 1.0: TC-01 MS_y +0.91/+0.87/+0.69 (ult +1.07/+1.03/+0.83); TC-02 +4.46/+4.34/+3.83; TC-09 316L YIELD -0.40/-0.41/-0.43 (ult +0.80/+0.78/+0.66) → v3.5 TC-09 IN718 +2.81/+2.76/+2.51 (own thermal run); J-FWD 8xM5 A4-70 yield -0.011/-0.033/-0.128, ult +0.60/+0.56/+0.41 (shoulder-screw option +0.86/+0.82/+0.64); seal S13 -0.016 at 102 bar only; NZ-01..04, J-AFT, J-TC03, J-LIP positive. Spike 1.8 kN/4 ms: 608 bearings exceed C0 (MS -0.24 DLF 2.0, -0.21 trapezoid, -0.10 at Combustion SDOF 3.05 kN; C0 AS RECALLED) - brinelling possible, ultimate not demonstrated; TVC linkage 14.8 vs 35.3 N m, pintle link 428 vs 831.6 N, airframe screws MS +3.97 SIM T557 **PENDING** PENDING T559
Physical evidence	IGN-1 with the dynamic Pc channel is the physical evidence LEAD T527
Decision text	DECISION: FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts (covers every baseline peak and the shell); for any closure, bolt, J1-lip or other part with a natural period of 0.25-1 ms Mechanical reports margins also at 92 bar abs (design: baseline rounded with DLF and the +3 % fuel-basis sensitivity) and at 102 bar abs (sensitivity: upper bracket, sharp, DLF); thrust spike 1.8 kN x 4 ms stays the interim load and is confirmed bounding. LEAD T515

3 Throttle and thrust vector control

3.1 Throttle

Item	Value
Throttle requirement (ENG-02)	Throttle floor ≤ 20 % of max (5:1, 100–500 N) with stable combustion. Stretch: 12.5 % (8:1). Throttle by a servo-driven variable-area pintle, with a separate NC isolation valve for shutoff. DERIVED T569
Throttle floor flown by GNC (fraction of rated)	**20** % DERIVED T570
Injector	pintle (needle-in-seat), unchanged from v1 LEAD T571
Pintle seat diameter / stroke	**5.50** mm DESIGN T572 / **4.00** mm DESIGN T573
Discharge coefficient Cd (two-phase N2O)	**0.70** ASSUMED T574
Cd acceptance criterion	Cd_eff (vs Dyer) >= 0.41 measured in N2O cold flow keeps the 15 C ceiling >= 500 N (HEM-like worst case, data/v2/injector_compare_v2.json) DERIVED T575
Pintle drive	MKS HV69 (MKS HV69, MKS no. S0022012 (Lindinger item 9788343)) DATASHEET T576
Pintle servo mass	**21** g DATASHEET T577
Pintle supply rail	PRAIL 7.49 V typ (TPS552882 + LTC4365, see pintle_rail) DESIGN T578
Pintle position resolution	**10** µm DESIGN T579
Throttle response requirement (ENG-03)	Throttle response: t10–90 ≤ max(0.15 s, 0.8·ΔF / 1000 N/s). Ramp (amended 2026-10-13, lead on GNC eng03_rate_v1.1): closing ≥ 2500 N/s in every state; opening ≥ the HV69 corner rate map (06-electronics/results/pintle_rate_map_v1.json, minimum 1188 N/s at 3.5 MPa / 400 N, 2500 N/s below ~40 % thrust); closed-loop t10-90 ≤ 100 ms under load including the drive. 6-DOF loss with the map: −0.07 % (p 0.875), within MC resolution. A flat 2 F_max/s would cost A1 0.992 → 0.978 (REV-GNC-1 RG-05). Pc loop closed and gain-scheduled; plant gain varies ×2.7 down to the floor. DERIVED T580
Start ramp	**2500** N/s LEAD T581
Opening rate, minimum over the map: nominal / corner	**1347** N/s SIM T582 / **1188** N/s SIM T583
Closing rate, minimum over the map (corner)	**2635** N/s SIM T584
Map states opening at ≥ 2500 N/s: nominal / corner	57/95 SIM T585 / 29/95 SIM T586
Chamber filling time constant	**3.4** ms DERIVED T587
Shutdown requirement (ENG-05)	Shutdown: command to < 10 % thrust ≤ 0.10 s. Met with a mandatory pre-MECO throttle-down to ≤ 75 % (a 100 % MECO at 3.5 MPa tank fails at 0.106–0.124 s). No flashback or decomposition in the feed. NC main valve closed during coasts until seat leak-tightness has been measured. DERIVED T588
Actuator speed finding (Combustion)	No combustion basis for 0.08 s. Start transient: Pc overshoot <= 0.34 % for full-stroke times 0.08-0.40 s and starts to 12.5-100 % (chamber fill ~4 ms: the start is quasi-static behind the pintle); max dPc/dt on a start to 100 % falls from 603 bar/s (0.08 s) to 232 bar/s (0.25 s). Hard start: N2O admitted to an unlit chamber before the 0.35 s ignition check 8.7 g at every speed (set by the ignition-lift dwell and … SIM T589

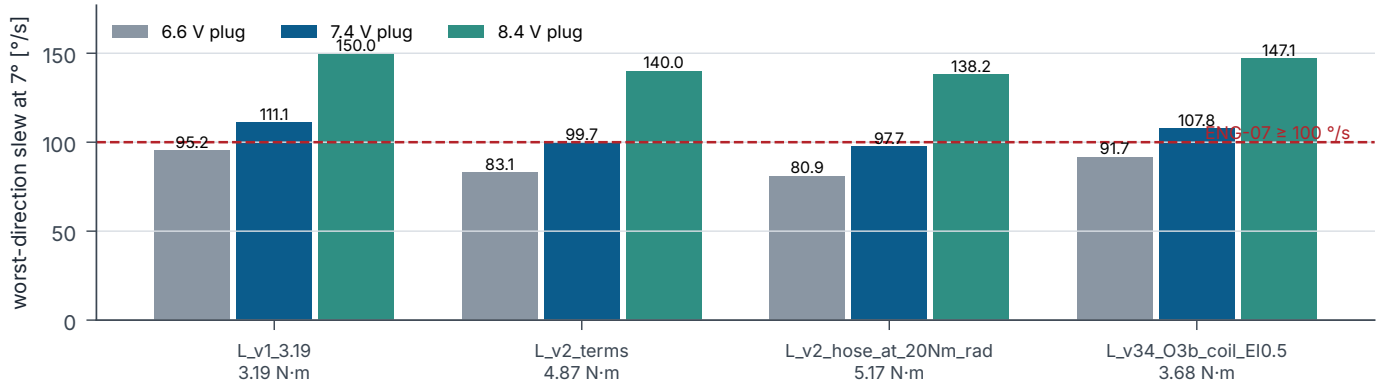
3.2 Thrust vector control

Item	Value
Architecture	two-axis Cardan gimbal of the whole thrust chamber; split engine collar (GM-01/02) clamped on the case at the pivot plane carries the inner pins (+/-Y) in 608-2RS bearings in the ring GM-03; the ring carries the outer pins (+/-X) in 608-2RS bearings in two yoke posts GM-08 on the thrust rings GM-09/GM-10 (fixed module); 8 deg conical PU stop GM-11 DERIVED T590
Gimbal range (vector norm) / mechanical stop	**7.0** ° REQ T561 / **8.0** ° DERIVED T591
Pivot location	at the mid-burn CG of the gimballed group (founder decision FINAL 2026-10-12); re-centred here for the v3.4/feed ESTIMATE items DERIVED T592
Pivot to injector face / to nozzle exit	**105.6** mm DERIVED T593 / **233.0** mm DERIVED T594
Gimbal bearings	4 × 608-2RS (8×22×7); pins 8 h6 17-4PH H1150 TO-VERIFY T595
TVC actuators	Savox SB-2292SG DATASHEET T596
Actuator mass (each)	**0.081** kg DATASHEET T597
Actuator stall torque / no-load speed at 7.4 V	**3.04** N·m DATASHEET T598 / **857** °/s DATASHEET T599
Actuator command rate	**333** Hz DATASHEET T600

Item	Value
Linkage ratio n (servo angle / engine angle)	**4.8** LEAD T601
Engine-angle resolution / backlash	**0.0208 °** DERIVED T602 / **0.0417 °** DERIVED T603
Required slew (ENG-07)	**100** °/s REQ T604
Required gimbal angular acceleration	**34.9** rad/s² REQ T605
Slew status, mechanical.json v2	as built: worst-direction slew at 7 deg 111.2 deg/s at 7.4 V, 118.3 at 7.6 V (>= 100); 95.3 deg/s at the 6.6 V degraded plug voltage, the case GNC validated for n 4.8 (lead FINAL: P(landing) 0.982 at 6.6 V) DERIVED T606
Design engine moment: ENG-10 / Mechanical as built / GNC parts load	Structure (gimbal, mounts, fasteners): SF ≥ 1.5 on yield and ≥ 2.0 on ultimate under design limit loads. Gimbal design loads: engine moment 35.3 N·m (both ... LEAD T607 **37.95** N·m DERIVED T608 / **26.5** N·m REQ T609
Axis stiffness, structure + rods at centre (α / β)	**2442–1608** N·m/rad SIM T610
Stiffness requirement (ENG-11)	Gimbal axis stiffness (structure plus rods) ≥ 3,100 N·m/rad per axis at every pose (hard minimum); goal 5,200 N·m/rad. Alpha/beta spread across poses ≤ ±20 %. Rocking mode as required by ENG-07d. DERIVED T611
Rocking mode, first (range over poses)	**16.0–27.8** Hz SIM T612
Stiffness and rocking-mode status	first rocking mode 16.0-27.8 Hz (excl. J_int) vs the 25 Hz minimum: below it whenever the servo is at the low end of its ASSUMED stiffness; the servo caps the total (servo alone through a rigid structure 16.2-28.7 Hz incl. pose); lead route: ENG-07d closes by control + B-TVC-1 measurement; GNC to state the minimum K / f for a 6 dB flex gain margin, Mechanical stiffens to it (stiffened concept: structure-only ... DERIVED T613

Worst-direction engine slew at 7° (any motion), Mechanical actuator_slew_v2 (SLEW v2). Linear servo torque–speed lines (ASSUMED shape).

Hinge load case	Load	Slew at 6.6V [°/s]	Slew at 7.4V [°/s]	Slew at 8.4V [°/s]
L_v1_3.19	**3.19** N·m DERIVED T614	**95.2** °/s DERIVED T615	**111.1** °/s DERIVED T616	**150.0** °/s DERIVED T617
L_v2_terms	**4.87** N·m DERIVED T618	**83.1** °/s DERIVED T619	**99.7** °/s DERIVED T620	**140.0** °/s DERIVED T621
L_v2_hose_at_20Nm_rad	**5.17** N·m DERIVED T622	**80.9** °/s DERIVED T623	**97.7** °/s DERIVED T624	**138.2** °/s DERIVED T625
L_v34_O3b_coil_EI0.5	**3.68** N·m DERIVED T626	**91.7** °/s DERIVED T627	**107.8** °/s DERIVED T628	**147.1** °/s DERIVED T629
no_load	**0.00** N·m DERIVED T630	**118.3** °/s DERIVED T631	**132.7** °/s DERIVED T632	**168.9** °/s DERIVED T633

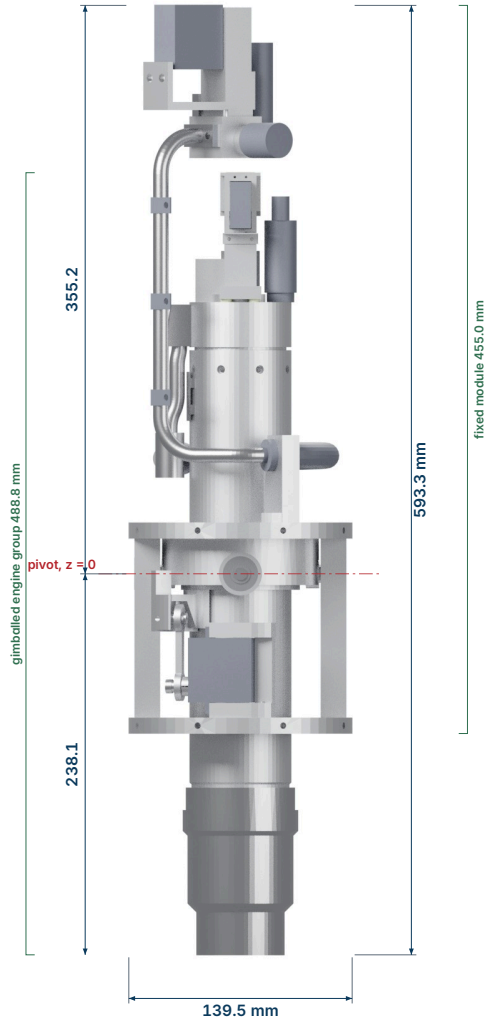


Engine slew at 7° by hinge load case and plug voltage. ENG-07 is OPEN and marginal at the v2 load (REQUIREMENTS amendment); B-TVC-1 measures it.
Data: T615, T616, T617, T619, T620, T621, T623, T624, T625, T627, T628, T629, T631, T632, T633

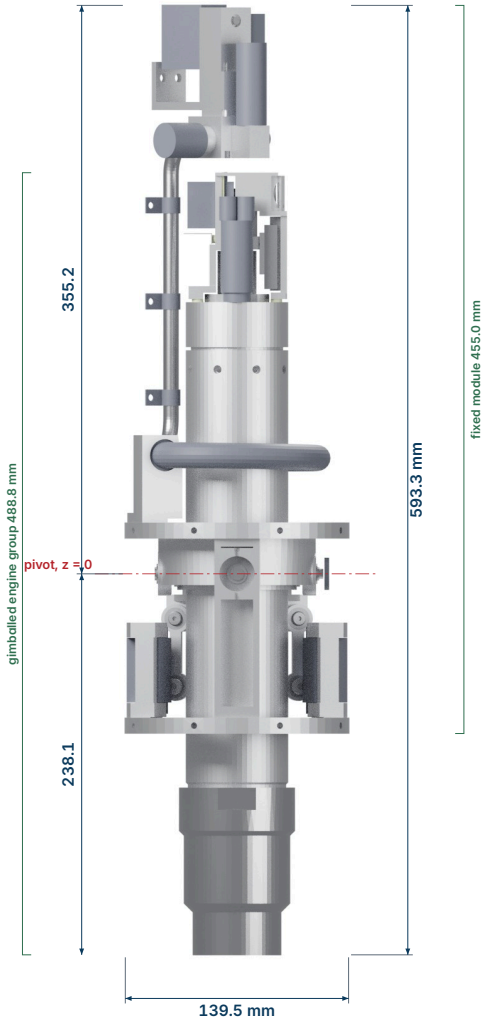
4 Geometry and mass

> **Renders are visualisation only.** They are produced by gen/render_views.sh from the tessellated GLB display mesh of CAD v3.4.1 (GLB sha256/16 3879a25a4486ee0c). They are not drawings and carry no tolerances. Dimensions printed on them are bounding-box extents from the assembly manifest; the STEP solids are the CAD of record.

4.1 Envelope and principal dimensions



Front view (looking along +Y)



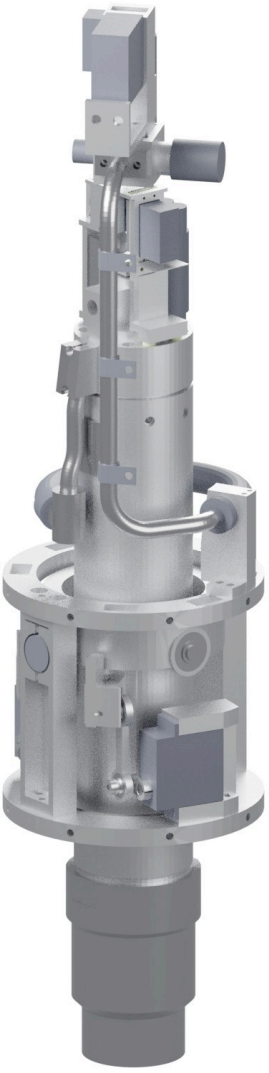
Side view (looking along −X)

Envelope from the union of the 134 instance bounding boxes in the assembly manifest (engine frame, mm, origin at the gimbal pivot, +z forward). An axis-aligned box does not bound the radial extent: the radial sweep check is listed separately.

Dimension	Value
Overall length along the engine axis (z)	**593.3** mm DERIVED T041
Overall width X / Y (axis-aligned bounding box)	**139.50** mm DERIVED T044 / **139.50** mm DERIVED T045
Pivot to forward end / pivot to aft end	**355.2** mm DERIVED T042 / **238.1** mm DERIVED T043
Gimballed engine group: z range / length	**−238.1–250.7** mm DERIVED T047 / **488.8** mm DERIVED T046
Fixed module (thrust frame, valve module, servos): z range / length	**−99.8–355.2** mm DERIVED T049 / **455.0** mm DERIVED T048
Mechanical build record, neutral assembly bounds	[-69.75, -69.75, -238.105, 69.75, 69.75, 355.2] DERIVED T634
Swept radial envelope, worst band (Mechanical checks)	z0_mm: 105.0; z1_mm: 110.0; r_engine_swept_mm: 60.08; by_engine: ENG-FD-03#0; r_ring_swept_mm: 0.0; r_fixed_mm: 71.8; by_fixed: ENG-FD-11#0; r_max_mm: 71.8; clearance_mm: -1.8 TO-VERIFY T635
Airframe bay requirement (ICD §2)	≥ 140 mm over the whole engine zone REQ T636

Dimension	Value
Engine zone (ICD §2)	z = −0.2377 to +0.252 m about the pivot (aft closure at the 8° stop to the top of the pintle stack) DERIVED T637

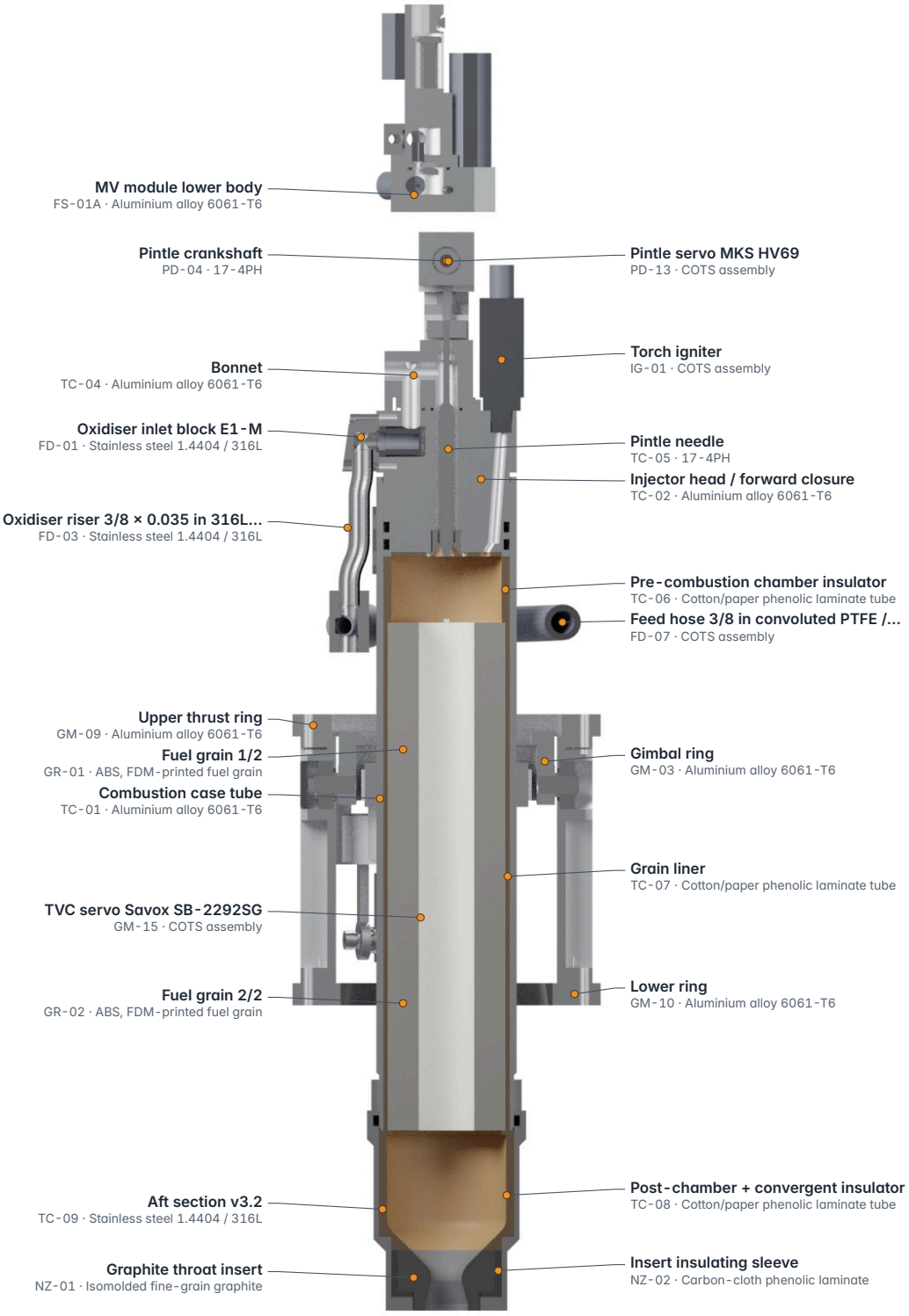
4.2 Assembly views



Isometric view, CAD v3.4.1, neutral pose (render c0baad3b5c2e70a5). Visualisation only.

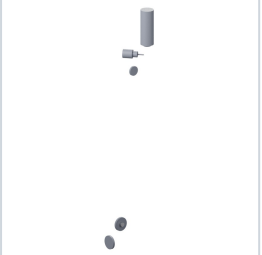


Thrust chamber and pintle stack laid out side by side along the axis order (pintle drive at left, nozzle joint at right), true scale. Axial positions are not preserved; the CAD z range of each group is printed under it. Visualisation only.



Half section in the plane y = 0 (the y < 0 half removed; cut faces darkened). Labels: BOM name, part number and material, placed at the area-weighted centroid of each part's cut face. Visualisation of display geometry, not a drawing. Parts without a cut face in this plane: ENG-FS-10. BOM rows: `T084`, `T184`, `T224`, `T062`, `T066`, `T218`, `T130`, `T154`, `T156`, `T234`, `T170`, `T142`, `T202`, `T166`, `T226`, `T220`, `T228`, `T118`, `T072`, `T230`, `T132`, `T232`, `T172`

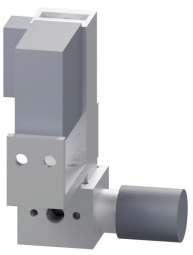
4.3 Part families



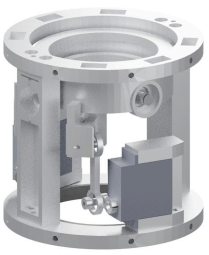
EL · Sensors and encoders
6 DERIVED T254 lines · 0.074 kg DERIVED T255



FD · Oxidiser feed
11 DERIVED T256 lines · 0.268 kg DERIVED T257



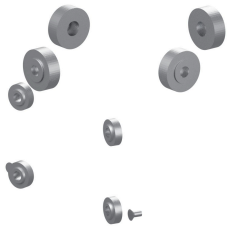
FS · Main valve module (fixed)
15 DERIVED T258 lines · 0.336 kg DERIVED T259



GM · Gimbal and thrust frame
19 DERIVED T260 lines · 1.125 kg DERIVED T261



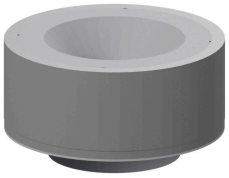
GR · Fuel grain
2 DERIVED T262 lines · 0.424 kg DERIVED T263



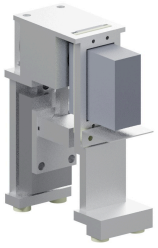
HW · Bearings and hardware
4 DERIVED T264 lines · 0.060 kg DERIVED T265



IG · Igniter
2 DERIVED T266 lines · 0.074 kg DERIVED T267



NZ · Nozzle and joint
4 DERIVED T268 lines · 0.063 kg DERIVED T269



PD · Pintle drive
20 DERIVED T270 lines · 0.175 kg DERIVED T271



TC · Thrust chamber
11 DERIVED T272 lines · 1.731 kg DERIVED T273

Each family is rendered at its assembly position (not exploded). Line counts and masses are sums over the BOM lines of the family (qty × unit mass, DERIVED by this generator; the transport lock is excluded).

4.4 Mass

Mass by subassembly and state, massprops MP v3.4.1. Group "all" is the CAD model total.

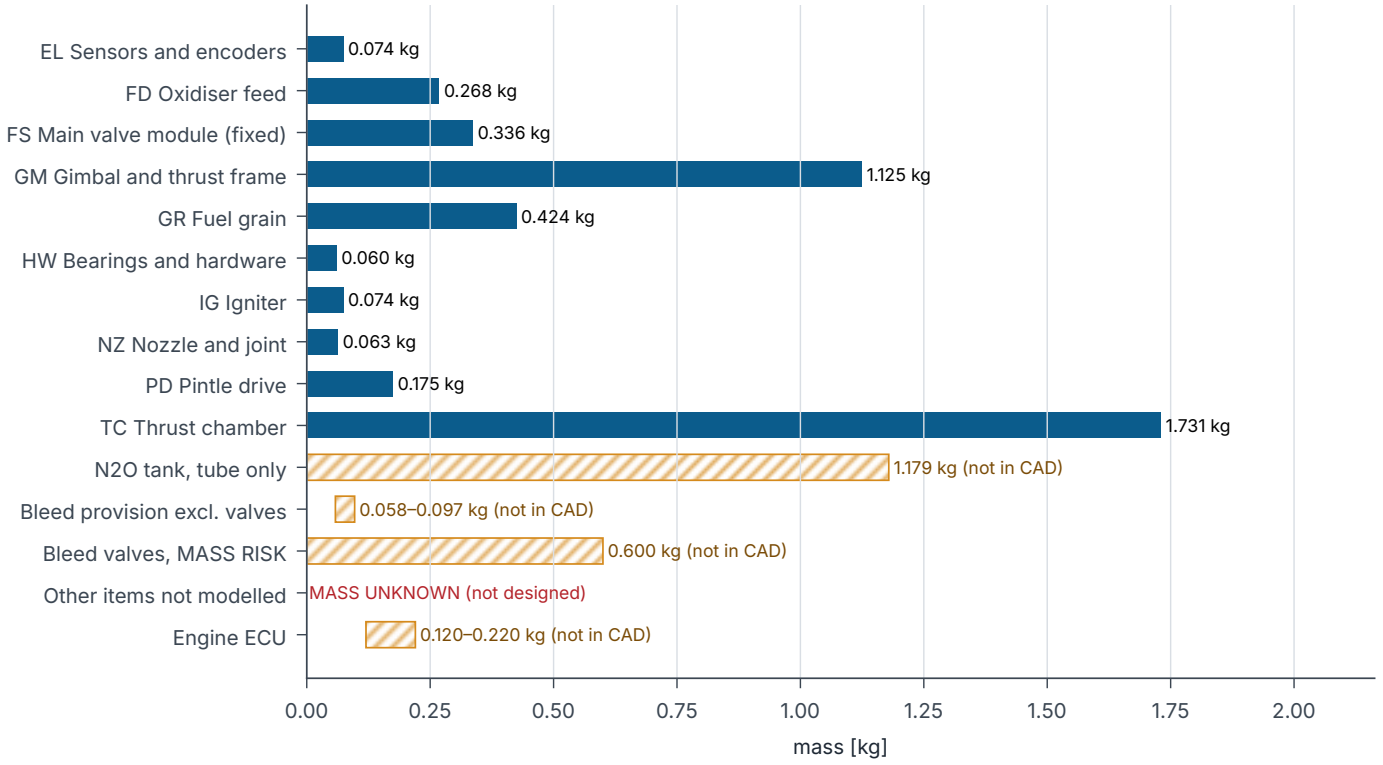
Group	Wet (full grain)	Mid-burn	Burn-out	No grain
gimballed engine	2.7611 kg DERIVED T638	2.6781 kg DERIVED T639	2.5951 kg DERIVED T640	2.3368 kg DERIVED T641
gimbal ring	0.2280 kg DERIVED T642	0.2280 kg DERIVED T643	0.2280 kg DERIVED T644	0.2280 kg DERIVED T645
TVC horns	0.0182 kg DERIVED T646	0.0182 kg DERIVED T647	0.0182 kg DERIVED T648	0.0182 kg DERIVED T649
TVC pushrods	0.0184 kg DERIVED T650	0.0184 kg DERIVED T651	0.0184 kg DERIVED T652	0.0184 kg DERIVED T653
fixed module (thrust frame, MV module, servos)	1.3637 kg DERIVED T654	1.3637 kg DERIVED T655	1.3637 kg DERIVED T656	1.3637 kg DERIVED T657
total	4.3894 kg DERIVED T658	4.3064 kg DERIVED T659	4.2234 kg DERIVED T660	3.9650 kg DERIVED T661

Mass by BOM part family (DERIVED: Σ qty × unit mass over the BOM lines).

BOM family	Lines	Mass
EL · Sensors and encoders	6 DERIVED T254	0.0740 kg DERIVED T255
FD · Oxidiser feed	11 DERIVED T256	0.2680 kg DERIVED T257
FS · Main valve module (fixed)	15 DERIVED T258	0.3364 kg DERIVED T259
GM · Gimbal and thrust frame	19 DERIVED T260	1.1245 kg DERIVED T261
GR · Fuel grain	2 DERIVED T262	0.4244 kg DERIVED T263
HW · Bearings and hardware	4 DERIVED T264	0.0602 kg DERIVED T265
IG · Igniter	2 DERIVED T266	0.0740 kg DERIVED T267
NZ · Nozzle and joint	4 DERIVED T268	0.0634 kg DERIVED T269
PD · Pintle drive	20 DERIVED T270	0.1746 kg DERIVED T271
TC · Thrust chamber	11 DERIVED T272	1.7306 kg DERIVED T273
Sum of BOM lines (excl. transport lock)		4.3301 kg DERIVED T274

Items excluded from the CAD mass. They are not zero: each is carried at its source value or as UNKNOWN.

Not in the CAD mass	Mass	Status / source
N2O tank: tube only (bulkheads, valves, fittings not designed)	**1.179** kg DERIVED T337	6061-T6 tube 101.6 × 3.175 + 2 bolted bulkheads (radial M6, 2 FKM O-rings each) DERIVED T338
Head-end bleed provision, excluding valves (BOM ENG-BL lines)	**0.058–0.097** kg ASSUMED T339	A1 head-end bleed provision (lead 2026-10-13), NOT modelled in CAD v3.4.1; range = ESTIMATE of taps, tube+tee, crossing hose and bulkhead (excl. valves) ASSUMED T340
Bleed valves (MASS RISK, not baseline)	**0.60** kg TO-VERIFY T344	F1 valve mass: 2 × 0.30 kg is held as a MASS RISK, not baseline; Electronics' 12 V NC valve search reports mass with a target <= 0.15 kg per valve; GNC flies +0.60 kg as a sensitivity. LEAD T341
Other items not modelled (mechanical.json v2 list)	**MISSING** MISSING T346	thrust-ring radial screws (16 x M5); isolation valve + actuator (mass UNKNOWN: Electronics/Combustion); tee, filter (<= 15 um, OI-7), relief, fill/vent; hose and lines fixed shares (~0.07 kg ASSUMED); tank MISSING T345
Engine ECU (on the fixed structure, not in CAD)	**0.16** kg ASSUMED T347 (**0.12–0.22** kg T348)	separate engine ECU (Teensy 4.1) on the fixed thrust structure, CAN link to flight computer DERIVED T349
N2O load (propellant)	**2.00** kg LEAD T350	LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill) LEAD T351



CAD mass by BOM family (solid) and items outside the CAD mass (hatched; ranges where the source gives one). Data: T255, T257, T259, T261, T263, T265, T267, T269, T271, T273, T337, T339, T344, T346, T347, T350

4.5 Airframe interface (ICD summary)

Selected rows of 00-requirements/ICD-AIRFRAME.md (ICD draft 0.2). Tag column of the ICD kept as the source tag.

Item	Value (ICD text)
ICD §2. Engine bay	
Engine bay inner diameter	≥ 140 mm over the whole engine zone REQ T636
Engine zone, engine frame	z = −0.2377 to +0.252 m about the pivot (aft closure at the 8° stop to the top of the pintle stack) DERIVED T637
Nozzle exit below the pivot	0.233 m; exit protrudes about 0.03 m below the aft datum DERIVED T662
Gimbal range	±7° on the vector norm; 8° conical mechanical stop REQ T663
Airframe attachment	Thrust rings GM-09 (upper, thrust path) and GM-10 (lower: stop ring, TVC servo base) DERIVED T664
Ring fit	Ring outer radius 69.75 mm in the 140 mm bore; 0.25 mm radial gap DERIVED T665
Ring fasteners	16 × M5 radial screws. Bolt pattern, edge distance and wall bearing check: (v2.1) DERIVED T666

Item	Value (ICD text)
Main-valve module (MV-P, balanced poppet)	Fixed module above the gimbal zone: occupies the bay from +0.252 m to about +0.355 m (engine frame; ESTIMATE). FS-01 axis at r 10 mm, az 219.5°, from z 259.5 mm up; FS-15 fixes it with 2 × M5 radial screws into the wall. Envelope and mass: (v2.1) LEAD T667
Feed across the gimbal	Option A/O3b: coil on the engine case; fixed 3/8 in wall line at az 219.5°, r 64.2 mm, from z 71 to 249.5 mm (engine frame), held by 3 P-clips on the airframe wall at z 110 / 170 / 230 mm. The coil fixed end is being re-routed (retained-vapour fix): it needs an airframe-wall clamp or GM-09 standoff near az 219.5°, z ≈ 71 mm. … REQ T668
ICD §3. Loads	
Rated thrust	500 N (Pc 25 bar, software cap) REQ T669
Over-thrust abort	650 N held for 3 frames on a 20 ms filter. Valid airborne, lit and with no drogue load LEAD T670
Physical worst-case thrust	876 N at Pc [withheld: superseded value, SUP-04] (pintle stuck at full stroke, ~~30 °C tank~~ ignition interlock T_tank <= 24 C; basis 44.33 bar (meop_basis_v2p4) **OVR-06**, Cd 0.80). Use it as the ultimate axial load case for the thrust rings SIM T567
Chamber MEOP / proof	45 bar / 67.5 bar LEAD T671
Lateral thrust at 7°	61 N at 500 N; 107 N at 876 N DERIVED T672
TVC design moment	35.3 N·m engine moment. Rod forces 441 / 552 / 609 N react into the TVC brackets on GM-10 LEAD T673
Peak axial specific force in flight	35.8 m/s² (B1 dispersed max), 33.7 m/s² (B1 p99), 27.7 m/s² (A1 p99) SIM T674
Touchdown, leg contact	50.3 m/s² (B1 p99), 40.3 m/s² (A1 p99). The 530–673 m/s² values in the "disp_max" files are failed-landing impacts, not design loads SIM T675
Axis stiffness, structure plus rods	≥ 3,100 N·m/rad at every pose, goal 5,200. The airframe side of the ring mounts is part of this path LEAD T676
ICD §4. Mass	
Liftoff mass	16.07 kg for A1 (drogue + fins); 15.42 kg for B hops SIM T677
Dry mass / N2O / grain	13.67 kg / 2.00 kg / 0.404 kg DERIVED T678
Engine hardware mass and CG	(v2.1): CAD v3.4 re-issues the gimballed and fixed masses. The main-valve module and tank bulkheads are not yet in CAD MISSING T679

ICD open items (§12).

ICD item	Open item	Owner
ICD-1	Thrust-ring bolt pattern, wall bearing and pull-out at 876 N plus lateral and moment loads TO-DEFINE T680	Mechanical (v2.1) / Lauty DESIGN T681
ICD-2	Main-valve module envelope and mass at the zone top TO-DEFINE T682	Mechanical (v2.1) DESIGN T683
ICD-3	Upper-bay vent area and station; baro-port placement TO-DEFINE T684	Lauty with Mechanical DESIGN T685
ICD-4	Relief and vent port routing (tank relief, 9.5 MPa disc line, fill/vent) TO-DEFINE T686	Mechanical / Lauty DESIGN T687
ICD-5	Tank station and mounting; heater and insulation interface TO-DEFINE T688	Mechanical / Lauty DESIGN T689
ICD-6	Launch-site bay temperature limit TO-DEFINE T690	Founders DESIGN T691
ICD-7	Pad slope specification and leg design TO-DEFINE T692	Lauty DESIGN T693
ICD-8	Drogue opening load, bay volume, ejection method, swivel and release fitting TO-DEFINE T694	Lauty DESIGN T695
ICD-9	Ground plume impingement on the feet TO-DEFINE T696	Combustion DESIGN T697
ICD-10	Airframe first bending mode ≥ 10 Hz on the real tube TO-DEFINE T698	Lauty, then GNC re-check DESIGN T699
ICD-11	Engine hardware mass and CG from CAD v3.4 TO-DEFINE T700	Mechanical (v2.1) DESIGN T701
ICD-12	Wall-line P-clips (3, at z 110 / 170 / 230 mm engine frame) and the coil fixed-end clamp near az 219.5°, z ≈ 71 mm: clamp type, wall fastening, loads (hose end reaction up to 32 N and 1.3 N·m at the 8° stop for EI 0.25) TO-DEFINE T702	Lauty with Mechanical DESIGN T703
ICD-13	Harness routing across the gimbal TO-DEFINE T704	Electronics / Mechanical, TO-DEFINE DESIGN T705
ICD-14	Head-end bleed overboard port (A1 provision): size, station, routing; balanced or axial vent TO-DEFINE T706	Mechanical (v2.1) / Lauty DESIGN T707
ICD-15	Drogue canopy type and oscillation data at the relight condition (≤ 2° design attitude) TO-DEFINE T708	Lauty DESIGN T709

4.6 CAD fit and interference checks

Software geometric checks by Mechanical (CHK v3.4.1). A clean software check is not a physical fit check; the smallest wall gap is reported by part; a part that mounts on the airframe wall (P-clip) touches it by design (see the swept-envelope row above, TO-VERIFY).

Pose	Interferences (not whitelisted)	Unexpected gaps < 3 mm	Smallest unexpected gap	Smallest wall gap
neutral	0 DERIVED T710	41 DERIVED T711	0.500 mm DERIVED T712	0.0 mm DERIVED T713 by ENG-FD-11#0 T714
alpha+	0 DERIVED T715	46 DERIVED T716	0.500 mm DERIVED T717	0.0 mm DERIVED T718 by ENG-FD-11#0 T719
alpha-	0 DERIVED T720	45 DERIVED T721	0.500 mm DERIVED T722	0.0 mm DERIVED T723 by ENG-FD-11#0 T724
beta+	0 DERIVED T725	50 DERIVED T726	0.399 mm DERIVED T727	0.0 mm DERIVED T728 by ENG-FD-11#0 T729
beta-	0 DERIVED T730	50 DERIVED T731	0.345 mm DERIVED T732	0.0 mm DERIVED T733 by ENG-FD-11#0 T734
combined045	0 DERIVED T735	53 DERIVED T736	0.500 mm DERIVED T737	0.0 mm DERIVED T738 by ENG-FD-11#0 T739
combined135	0 DERIVED T740	53 DERIVED T741	0.487 mm DERIVED T742	0.0 mm DERIVED T743 by ENG-FD-11#0 T744
combined225	0 DERIVED T745	53 DERIVED T746	0.487 mm DERIVED T747	0.0 mm DERIVED T748 by ENG-FD-11#0 T749
combined315	0 DERIVED T750	55 DERIVED T751	0.500 mm DERIVED T752	0.0 mm DERIVED T753 by ENG-FD-11#0 T754
stop000	0 DERIVED T755	50 DERIVED T756	0.294 mm DERIVED T757	0.0 mm DERIVED T758 by ENG-FD-11#0 T759
stop045	0 DERIVED T760	53 DERIVED T761	0.421 mm DERIVED T762	0.0 mm DERIVED T763 by ENG-FD-11#0 T764
stop090	0 DERIVED T765	47 DERIVED T766	0.500 mm DERIVED T767	0.0 mm DERIVED T768 by ENG-FD-11#0 T769
stop135	0 DERIVED T770	53 DERIVED T771	0.378 mm DERIVED T772	0.0 mm DERIVED T773 by ENG-FD-11#0 T774
stop180	0 DERIVED T775	50 DERIVED T776	0.238 mm DERIVED T777	0.0 mm DERIVED T778 by ENG-FD-11#0 T779
stop225	0 DERIVED T780	53 DERIVED T781	0.378 mm DERIVED T782	0.0 mm DERIVED T783 by ENG-FD-11#0 T784
stop270	0 DERIVED T785	49 DERIVED T786	0.500 mm DERIVED T787	0.0 mm DERIVED T788 by ENG-FD-11#0 T789
stop315	0 DERIVED T790	55 DERIVED T791	0.421 mm DERIVED T792	0.0 mm DERIVED T793 by ENG-FD-11#0 T794
stopFD07B	0 DERIVED T795	56 DERIVED T796	0.500 mm DERIVED T797	0.0 mm DERIVED T798 by ENG-FD-11#0 T799

5 Parts list

BOM BOM v3.4.1. Price rule: every line TO-QUOTE delivered to Sønderborg DK-6400 unless dated; no zero and no "from" prices.

Item	Value
Delivery address for quotations	Sonderborg, Denmark, DK-6400 REQ T800
Price status	prices not given: every line TO-QUOTE unless dated DESIGN T801
BOM lines / placed instances	95 DERIVED T802 / 129 DERIVED T803
Total cost	TO-QUOTE (not summed) MISSING T276

Columns: part number (with its BOM-row trace), name and supplier as written in the BOM, quantity, material, process, unit mass with its evidence class, and price status. COTS = commercial part. Wetted oxygen-clean parts are marked O_2.

5.1 EL — Sensors and encoders

EL family: 6 lines, 0.0740 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-EL-03 T050	Gimbal encoder, inner axis (beta): AS5048A on a round PCB 20 × 2.8 (REQUEST to Electronics) ams AS5048A on custom 20 mm PCB (Electronics)	1	COTS assembly COTS	COTS/custom PCB	0.0030 ASSUMED T051	TO-QUOTE
ENG-EL-04 T052	Gimbal encoder, outer axis (alpha): AS5048A on a round PCB 20 × 2.8 recessed in the -X post bore (REQUEST ... ams AS5048A on custom 20 mm PCB (Electronics)	1	COTS assembly COTS	COTS/custom PCB	0.0030 ASSUMED T053	TO-QUOTE
ENG-EL-05 T054	Pintle-position encoder AS5048A on a 14 mm round PCB (chip + PCB envelope 14 × 2.4) in the PD-03 cover ams AS5048A on a 14 mm round PCB (Electronics request 2026-10-12)	1	COTS assembly COTS	COTS/custom PCB	0.0020 ASSUMED T055	TO-QUOTE
ENG-EL-06 T056	Diametral NdFeB magnet 6 × 2.5 (AS5048A), bonded on the pin end ams AS5048 evaluation magnet / Magnet-Shop DK	2	NdFeB magnet N35 COTS	COTS	0.0005 DERIVED T057	TO-QUOTE
ENG-EL-11 T058	P_feed transducer J602 TE M3234-000005 class, 1/4-18 NPT male, vertical on the FS-01A boss (Electronics ... O_2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (Electronics)	0.0500 ASSUMED T059	TO-QUOTE
ENG-EL-12 T060	MSS-2 Type T MI thermocouple 1/16 in in a 1/16 x 1/8-27 NPT male connector (Electronics part; tip 9 mm into ... O_2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (Electronics)	0.0150 ASSUMED T061	TO-QUOTE

5.2 FD — Oxidiser feed

FD family: 11 lines, 0.2680 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FD-01 T062	Oxidiser inlet block E1-M (316L): riser socket, 8 mm passages, P_feed tap, face seal to the filter cartridge O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC mill (2 setups) + socket weld of the riser (TIG, back-purged)	0.0354 DERIVED T063	TO-QUOTE
ENG-FD-02 T064	Filter cartridge 15 um absolute, pleated 316L Dutch-twill mesh, area >= 1450 mm2 (custom, in the TC-02 cap) O_2 custom element: Purolator Advanced Filtration / GKN Sinter Metals / Mott (TO-QUO	1	COTS assembly COTS	COTS custom (filter maker)	0.0040 ASSUMED T065	TO-QUOTE
ENG-FD-03 T066	Oxidiser riser 3/8 × 0.035 in 316L seamless, 71 mm developed, 2 S-bends (R >= 24 mm) O_2 316L seamless 3/8 × 0.035 in (ASTM A269): Swagelok SS-T6-S-035-6ME / Sandvik (TO	1	Stainless steel 1.4404 / 316L	tube bender + socket weld into FD-01 (TIG, back-purged) + coil-end ada	0.0137 DERIVED T067	TO-QUOTE
ENG-FD-04 T068	Riser radiation shield: 0.2 mm polished 316 foil + 2 band-clip stand-offs (case OD → riser) in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	cut + roll + spot weld (or 3D-printed PEEK clips, TO-DECIDE by the RC-	0.0056 DERIVED T069	TO-QUOTE
ENG-FD-05 T070	Riser-foot TEE (316L): riser socket (top), hose socket (side, 25.9 deg), MSS-1 1/8-27 NPT (bottom) O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC lathe + mill (2 setups); socket welds of FD-03 and FD-07A (TIG, ba	0.0407 DERIVED T071	TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FD-07 T072	Feed hose 3/8 in convoluted PTFE / 316 braid, rising coil 203 deg, flexible length 180 mm O2 convoluted PTFE 3/8 in, 1× 316 braid, O2-clean, pressure-test certificate: Indun	1	COTS assembly COTS	COTS hose assembly (both ends crimped; FD-07A/B are its end fittings)	0.0379 TO-VERIFY T073	TO-QUOTE
ENG-FD-08 T074	Fixed feed line 3/8 × 0.035 in 316L, 279 mm developed: 3D bend, vertical on the bay wall at az 219.5, R24 … O2 316L seamless 3/8 × 0.035 in (ASTM A269): Swagelok SS-T6-S-035-6ME / Sandvik (TO	1	Stainless steel 1.4404 / 316L	tube bender (R 24 die) + orbital/TIG butt weld to FD-07B + socket weld	0.0538 DERIVED T075	TO-QUOTE
ENG-FD-08F T076	Line flange (316L): socket for FD-08, AS568-012 FKM face seal to FS-01A, 2x M4 O2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC mill	0.0135 DERIVED T077	TO-QUOTE
ENG-FD-09 T078	Fixed hose-end clamp (6061-T6, base + cap, 2x M4 into GM-09 top face, bare aluminium on the sleeve), v3.4.1 … in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0380 DERIVED T079	TO-QUOTE
ENG-FD-11 T080	P-clip 3/8 in, 316 band + silicone cushion, M4 to the airframe wall (insert/rivet nut in the airframe = ICD) Essentra / RS Components DK P-clip 3/8 in SS (TO-QUOTE)	3	COTS assembly COTS	COTS	0.0040 ASSUMED T081	TO-QUOTE
ENG-FD-12F T082	Tank-line flange (316L) on FS-01B: socket for the tank line (route TO-DEFINE with the tank, ICD-5), … O2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC mill	0.0135 DERIVED T083	TO-QUOTE

5.3 FS — Main valve module (fixed)

FS family: 15 lines, 0.3364 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FS-01A T084	MV module lower body (6061-T6): outlet chamber 18 × 12.3 (segment side), line flange face, J602 / MSS-2 / … O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill 3 setups	0.0521 DERIVED T085	TO-QUOTE
ENG-FS-01B T086	MV module upper body (6061-T6): seat web + PCTFE seat counterbore, inlet chamber 16, piston bore 11.9 H8, … O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC lathe + mill	0.0444 DERIVED T087	TO-QUOTE
ENG-FS-02 T088	MV-P poppet: disc 14 × 3 + stem 3 h7, hollow 1.2 (balance hole), M3 into FS-04 O2 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe, stem ground	0.0046 DERIVED T089	TO-QUOTE
ENG-FS-03 T090	MV-P soft seat insert PCTFE: land 11.5-12.5 x 0.3, OD 16 × 2.3 (retention lip TO-DESIGN) O2 in-house / workshop TO-QUOTE	1	PCTFE	lathe from rod	0.0002 DERIVED T091	TO-QUOTE
ENG-FS-04 T092	MV-P balance piston 11.9 f7 + rod 4 h7 (one piece), seal groove, cross hole 1.2 into the balance chamber O2 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe, rod ground	0.0077 DERIVED T093	TO-QUOTE
ENG-FS-06 T094	MV actuator housing (dry side, 6061-T6): spring pocket, cam slot, 2x MR85 bearing bores (servo-side bore … in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0292 DERIVED T095	TO-QUOTE
ENG-FS-07 T096	MV closing spring EN 10270-3 1.4310, d 1.8 / Dm 12 / n 6.6 active, L0 31.3, 85 N closed / 107.6 N open … TO-QUOTE (delivery Sonderborg DK-6400)	1	Stainless spring wire 1.4310 COTS	COTS custom spring (Gutekunst / Lee Spring, TO-QUOTE)	0.0080 ASSUMED T097	TO-QUOTE
ENG-FS-08 T098	MV follower (17-4PH H900, flat hardened face) on the rod top, M3 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe + H900	0.0045 DERIVED T099	TO-QUOTE
ENG-FS-09 T100	MV cam (r 6.0 → 8.8 over 45 deg, 15 deg dwell, width 5; closed 25 deg before the rise) + shaft 5 in 2x … in-house / workshop TO-QUOTE	1	17-4PH	wire EDM + lathe + broach	0.0081 DERIVED T101	TO-QUOTE
ENG-FS-10 T102	MV opener servo Savox SB-2292SG (Electronics, commonality) on the housing +u face, spline coupled to the … TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS	0.0810 DATASHEET T103	TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FS-11 T104	MV latch + release + dashpot + OPEN/CLOSED microswitches: RESERVED ENVELOPE 16 × 12 x 30 (TO-DESIGN with ... TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	TO-DESIGN	0.0250 ASSUMED T105	TO-QUOTE
ENG-FS-12 T106	S1 back-relief check, insert cartridge 5.5 mm class, crack 0.3-0.7 bar (outlet → inlet chamber), ... O2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (TO-SELECT: The Lee Company Chek insert class)	0.0020 ASSUMED T107	TO-QUOTE
ENG-FS-13 T108	Secondary rupture disc 9.5 MPa +/-5 %, reverse-acting, 1/4-18 NPT holder with 1/4 NPT outlet to the ... O2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (TO-SELECT: Fike / BS&B / Oseco mini disc assemblies)	0.0450 ASSUMED T109	TO-QUOTE
ENG-FS-14 T110	Drain port plug 1/8-27 NPT flush socket plug 316 (ground drain/purge; segment low point of the module) O2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS	0.0040 ASSUMED T111	TO-QUOTE
ENG-FS-15 T112	MV module mounting bracket (6061-T6): 2x M4 to FS-01A/B, 2x M5 radial into the airframe wall at az 219.5 ... in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0206 DERIVED T113	TO-QUOTE

5.4 GM — Gimbal and thrust frame

GM family: 19 lines, 1.1245 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-GM-01 T114	Engine collar half +Y (servo 1 lever, 135 deg) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill (pair machined as one ring, then split)	0.0608 DERIVED T115	TO-QUOTE
ENG-GM-02 T116	Engine collar half -Y (servo 2 lever, 225 deg), mirror of GM-01 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill (pair machined as one ring, then split)	0.0620 DERIVED T117	TO-QUOTE
ENG-GM-03 T118	Gimbal ring (Cardan) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill + 4th-axis boring (from 100 mm bar or 16 mm plate)	0.1843 DERIVED T119	TO-QUOTE
ENG-GM-04 T120	Inner pivot pin 8 h6 × 18 (17-4PH H1150) in-house / workshop TO-QUOTE	2	17-4PH	CNC lathe, ground h6	0.0069 DERIVED T121	TO-QUOTE
ENG-GM-05 T122	Outer pivot pin 8 h6 × 18.5 (17-4PH H1150) in-house / workshop TO-QUOTE	2	17-4PH	CNC lathe, ground h6	0.0071 DERIVED T123	TO-QUOTE
ENG-GM-06 T124	Inner-axis spacer sleeve 11/8.1 × 2.85 (304) in-house / workshop TO-QUOTE	2	Stainless steel 1.4301 / AISI 304	lathe from tube	0.0010 DERIVED T125	TO-QUOTE
ENG-GM-07 T126	Outer-axis spacer sleeve 11/8.1 × 1.9 (304) in-house / workshop TO-QUOTE	2	Stainless steel 1.4301 / AISI 304	lathe from tube	0.0006 DERIVED T127	TO-QUOTE
ENG-GM-08 T128	Yoke post (outer-axis bearing housing + frame post), x2 in-house / workshop TO-QUOTE	2	Aluminium alloy 6061-T6	CNC mill from 20 × 30 bar (3 setups)	0.0734 DERIVED T129	TO-QUOTE
ENG-GM-09 T130	Upper thrust ring (engine module to airframe, thrust path) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 12 mm plate	0.1520 DERIVED T131	TO-QUOTE
ENG-GM-10 T132	Lower ring (stop ring, TVC servo base, frame tie) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 12 mm plate	0.1927 DERIVED T133	TO-QUOTE
ENG-GM-11 T134	Conical stop bumper, PU 90A, 8 deg line contact PU 90A tube 100 × 80: Kubo / RS Components DK (TO-QUOTE)	1	Polyurethane bumper 90A	cast or turned from PU 90A tube	0.0092 DERIVED T135	TO-QUOTE
ENG-GM-12 T136	Transport / neutral lock, 2 half rings (REMOVE BEFORE FLIGHT) transport lock, not placed, excluded from mass in-house / workshop TO-QUOTE	1	PETG	FDM print PETG	0.0622 DERIVED T137	TO-QUOTE
ENG-GM-13 T138	TVC servo 1 bracket (C-frame: top/bottom ear blocks + back rail) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 40 × 50 bar (3 setups)	0.0489 DERIVED T139	TO-QUOTE
ENG-GM-14 T140	TVC servo 2 bracket (mirror of GM-13) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 40 × 50 bar (3 setups)	0.0489 DERIVED T141	TO-QUOTE
ENG-GM-15 T142	TVC servo Savox SB-2292SG (electronics.json) Lindinger (AT) item 9738546 per electronics.json v1	2	COTS assembly COTS	COTS	0.0810 DATASHEET T143	119.99 EUR incl. VAT (electronics.json v1, 2026-10-11; DK delivery TO-QUOTE) DK delivery TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-GM-16 T144	TVC horn r 11.1 mm, 25T clamp (7075-T6) in-house / workshop TO-QUOTE	2	Aluminium alloy 7075-T6	CNC mill + wire-EDM 25T spline (or re-machined COTS 25T clamp horn)	0.0022 DERIVED T145	TO-QUOTE
ENG-GM-17 T146	TVC pushrod dog-bone 42.3 mm between eyes (7075-T6), fixed length in-house / workshop TO-QUOTE	2	Aluminium alloy 7075-T6	CNC mill	0.0042 DERIVED T147	TO-QUOTE
ENG-GM-18 T148	Lever pivot stud: flange 7.5 × 1.5 (= washer), 5 h7 shoulder x 7.5, M4 × 4.5, 2.5 hex (17-4PH) in-house / workshop TO-QUOTE	2	17-4PH	CNC lathe	0.0020 DERIVED T149	TO-QUOTE
ENG-GM-19 T150	Joint spacer washer 5.1 × 7.5 × 1.5 (17-4PH H1150), ground faces in-house / workshop TO-QUOTE	6	17-4PH	CNC lathe from bar, faces ground	0.0003 DERIVED T151	TO-QUOTE
ENG-GM-20 T152	Horn-joint shoulder screw: head 8.5 × 4, 5 h7 × 10 shoulder, M4 × 3.5 (17-4PH) in-house / workshop TO-QUOTE	2	17-4PH	CNC lathe	0.0035 DERIVED T153	TO-QUOTE

5.5 GR — Fuel grain

GR family: 2 lines, 0.4244 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-GR-01 T154	Fuel grain 1/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) in-house / workshop TO-QUOTE	1	ABS, FDM-printed fuel grain	FDM print ABS 100 % infill, concentric perimeters	0.2122 DERIVED T155	TO-QUOTE
ENG-GR-02 T156	Fuel grain 2/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) in-house / workshop TO-QUOTE	1	ABS, FDM-printed fuel grain	FDM print ABS 100 % infill, concentric perimeters	0.2122 DERIVED T157	TO-QUOTE

5.6 HW — Bearings and hardware

HW family: 4 lines, 0.0602 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-HW-20 T158	Deep-groove ball bearing 608-2RS (8×22×7) SKF 608-2RSH: RS Components DK / Simply Bearings	4	Carbon/bearing steel COTS	COTS	0.0120 DATASHEET T159	TO-QUOTE
ENG-HW-21 T160	Retaining washer DIN 6799-6 (shaft 7-9 mm), spring steel Bossard DK / RS Components DK	4	Carbon/bearing steel COTS	COTS	0.0003 ASSUMED T161	TO-QUOTE
ENG-HW-22 T162	Spherical plain bearing GE 5 E (steel/steel) - outer ring (pressed in the pushrod eye) SKF / INA / Hepco GE 5 E class: RS Components DK (TO-QUOTE)	4	Carbon/bearing steel COTS	COTS	0.0025 DERIVED T163	TO-QUOTE
ENG-HW-25 T164	Horn axial screw ISO 10642 M3 × 6 (countersunk, 10.9), Loctite 243 Bossard DK / RS Components DK	2	Carbon/bearing steel COTS	COTS	0.0005 ASSUMED T165	TO-QUOTE

5.7 IG — Igniter

IG family: 2 lines, 0.0740 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-IG-01 T166	Torch igniter (N2O/propane, spark) - ENVELOPE 20 × 45 mm + M10×1 nose + 2 inlet fittings (Combustion ... Combustion design: body CNC 316L, micro spark plug (model-engine class, TO-QUOTE	1	COTS assembly COTS	Combustion design (316L body, micro spark plug)	0.0700 ASSUMED T167	TO-QUOTE
ENG-IG-02 T168	Torch inlet check valve, cartridge insert in the torch inlet bore (N2O and propane, x2) O_2 Lee Company Chek cartridge (stainless, 5.5 mm class, cracking <= 0.3 bar, rated	2	COTS assembly COTS	COTS	0.0020 ASSUMED T169	TO-QUOTE

5.8 NZ — Nozzle and joint

NZ family: 4 lines, 0.0634 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-NZ-01 T170	Graphite throat insert (Dt 13.72, De 19.41, OD 44, aft spigot 30) isomolded graphite rod 50 mm (Toyo Tanso IG-110 / SGL R6510 class): Schunk Carbo	1	Isomolded fine-grain graphite	CNC lathe from isomolded graphite rod (dust extraction, sharp carbide)	0.0473 DERIVED T171	TO-QUOTE
ENG-NZ-02 T172	Insert insulating sleeve (carbon-phenolic 50/44, 3 mm) in-house / workshop TO-QUOTE	1	Carbon-cloth phenolic laminate	lathe from carbon-fabric/phenolic tube (tape-wrapped or moulded; suppl	0.0120 DERIVED T173	TO-QUOTE
ENG-NZ-03 T174	Nozzle-joint hot layer J1: carbon-phenolic washer 30.6/49.8 × 2.0 carbon-fabric / phenolic laminate plate >= 2.5 mm (aerospace ablative or industr	1	Carbon-cloth phenolic laminate	water-jet + face-grind from carbon-fabric/phenolic laminate plate (pli	0.0035 DERIVED T175	TO-QUOTE
ENG-NZ-04 T176	Nozzle-joint lip-face seal J1: flexible graphite ring 30.6/49.8 × 0.5, oxidation-inhibited flexible graphite sheet 0.5 mm, oxidation-inhibited grade, >= 98 % C (SGL SIGRAF	1	Flexible graphite sheet	cut from sheet (steel-rule die / water-jet)	0.0007 DERIVED T177	TO-QUOTE

5.9 PD — Pintle drive

PD family: 20 lines, 0.1746 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-PD-01 T178	Pintle drive frame, 2 pieces: A (-Y foot + bridge + leg + servo plate + cap) and B (+Y foot + rail plate + ... in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 6061-T6 block (3 setups per piece) or welded/bolted plat	0.0389 DERIVED T179	TO-QUOTE
ENG-PD-02 T180	Pintle crank housing (bearing seats A 18 and B 19, crank cavity, spring cavity 22) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill + lathe (bores in one setup)	0.0198 DERIVED T181	TO-QUOTE
ENG-PD-03 T182	Crank-housing cover with encoder PCB pocket in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0026 DERIVED T183	TO-QUOTE
ENG-PD-04 T184	Pintle crankshaft (throw r 3.0, pin 3, journals 12 / 10, dog end, spring hub, magnet pocket) in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe + mill (eccentric pin turned in a second setup) + grind jour	0.0125 DERIVED T185	TO-QUOTE
ENG-PD-05 T186	Dog hub on the HV69 6 mm 25T spline (one-sided drive tooth 60 deg) in-house / workshop TO-QUOTE	1	Aluminium alloy 7075-T6	CNC lathe + broach 25T (or re-machined COTS aluminium horn)	0.0006 DERIVED T187	TO-QUOTE
ENG-PD-06 T188	Pintle connecting link 28 mm centres, eyes OD 8 x 3.8 with 4.5 bores, shank 4 × 3 in-house / workshop TO-QUOTE	1	17-4PH	wire EDM from plate + ream	0.0039 DERIVED T189	TO-QUOTE
ENG-PD-07 T190	Pintle carriage: needle clamp arm (M3 stem thread) + link clevis + plate on the MGN09H block in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill, hard anodise	0.0100 DERIVED T191	TO-QUOTE
ENG-PD-08 T192	Miniature linear guide block HIWIN MGN09H (C0 4020 N, M0Y 18.6 N m) HIWIN MGN09H-ZF (preload ZF, end seals) via HIWIN GmbH / Misumi (TO-QUOTE, Sonde	1	Carbon/bearing steel COTS	COTS	0.0300 TO-VERIFY T193	TO-QUOTE
ENG-PD-09 T194	Spiral (clock) return spring, 0.25 N m at closed + 0.12 N m/rad (+/-10 %), envelope 21/8 × 4.5 custom flat spiral spring 1.4310 strip (Gutekunst / Lesjofors DK, TO-QUOTE Sonde	1	Stainless spring wire 1.4310 COTS	COTS custom (spring maker)	0.0055 ASSUMED T195	TO-QUOTE
ENG-PD-10 T196	Radiation shield under the HV69: 0.5 mm polished aluminium L-sheet on 2 G11 stand-offs in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	cut + bend from 0.5 mm 1050A-H14 sheet, polished both faces	0.0007 DERIVED T197	TO-QUOTE
ENG-PD-11 T198	Isolator washer G11 7/3.4 × 2 (frame feet to head top) NEMA G11 sheet 2 mm: Rochling / Isola / RS Components DK (TO-QUOTE, delivery Son	4	Glass-epoxy laminate NEMA LI-1 G11	punched / turned from G11 sheet	0.0001 DERIVED T199	TO-QUOTE
ENG-PD-12 T200	G11 pad 1.5 mm under the HV69 ears (window for the case) NEMA G11 sheet 1.5 mm (TO-QUOTE)	1	Glass-epoxy laminate NEMA LI-1 G11	CNC from G11 sheet	0.0009 DERIVED T201	TO-QUOTE
ENG-PD-13 T202	Pintle servo MKS HV69 (29 × 13 × 28.7, 21 g, 6 mm 25T spline), case below the shaft MKS HV69 (MKS S0022012); Lindinger AT item 9788343	1	COTS assembly COTS	COTS	0.0210 DATASHEET T203	TO-QUOTE ref. Lindinger AT catalogue 62.99 EUR incl. 20 % AT VAT, 2026-10-12, Electr (not a DK quote)
ENG-PD-14 T204	Thin-section ball bearing 12×18×4 (NSK/NMB 6701 class, 'NB701' in pintle_v2p3), C0r 655 N NSK / NMB / SKF 61701 class (TO-QUOTE, Sonderborg DK-6400)	1	Carbon/bearing steel COTS	COTS	0.0031 ASSUMED T205	TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-PD-15 T206	Deep-groove ball bearing 6800-2Z 10×19×5, C0r 840 N SKF 61800-2Z / NSK 6800ZZ (TO-QUOTE, Sonderborg DK-6400)	1	Carbon/bearing steel COTS	COTS	0.0055 ASSUMED T207	TO-QUOTE
ENG-PD-16 T208	Pin 3 g6 × 12 (17-4PH H900) with e-clip grooves (crank pin bush / carriage link pin) in-house / workshop TO-QUOTE	1	17-4PH	turned + ground (or ISO 2338 3 m6 × 12 hardened)	0.0007 DERIVED T209	TO-QUOTE
ENG-PD-17 T210	PTFE-lined steel-backed bush 3 × 4.5 × 3.8 (DU class) GGB DU / SKF PCM class 3 × 4.5 (maker part TO-SELECT, TO-QUOTE)	2	Carbon/bearing steel COTS	COTS	0.0003 ASSUMED T211	TO-QUOTE
ENG-PD-18 T212	Encoder magnet NdFeB N35 diametral 6 × 1 diametral 6 × 1 mm N35 (Supermagnete / Magnetladen, TO-QUOTE)	1	NdFeB magnet N35	COTS	0.0002 DERIVED T213	TO-QUOTE
ENG-PD-19 T214	Miniature rail HIWIN MGNR09R, length 46.1 mm (cut to length) HIWIN MGNR09R cut to 46 mm (TO-QUOTE)	1	Carbon/bearing steel COTS	COTS	0.0175 DATASHEET T215	TO-QUOTE
ENG-PD-20 T216	G11 stand-off 5/2.2 × 2 (shield to servo plate) G11 rod 6 mm (TO-QUOTE)	2	Glass-epoxy laminate NEMA LI-1 G11	turned from G11 rod	0.0001 DERIVED T217	TO-QUOTE

5.10 TC — Thrust chamber

TC family: 11 lines, 1.7306 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-TC-01 T218	Combustion case tube 6061-T6 drawn tube 63.5 × 3.175 (2.5 × 0.125 in): Aluminium-Online.de / Wefa / M	1	Aluminium alloy 6061-T6	cut-stock + CNC lathe (bore fwd seal land, turn aft land and thread) +	0.4625 DERIVED T219	TO-QUOTE
ENG-TC-02 T220	Injector head / forward closure (pintle seat bore, manifold, Pc and torch ports, E1-M filter bore and inlet … O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC lathe + 3-axis mill (2 setups)	0.4415 DERIVED T221	TO-QUOTE
ENG-TC-03 T222	Pintle seat insert (5.5 mm sharp-edged seat, 37.5 deg diffuser) O2 in-house / workshop TO-QUOTE	1	Inconel 718	CNC lathe	0.0120 DERIVED T223	TO-QUOTE
ENG-TC-04 T224	Bonnet: stem seal + guide housing, E1-M inlet boss (8 mm transfer, cross-drilling at the bore top, … O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC lathe + mill	0.0535 DERIVED T225	TO-QUOTE
ENG-TC-05 T226	Pintle needle (15 deg metering cone, 6 mm body, 3 mm stem) O2 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe, stem ground h7 Ra 0.2	0.0166 DERIVED T227	TO-QUOTE
ENG-TC-06 T228	Pre-combustion chamber insulator (phenolic, bore 49.9, wall 3.60 mm, 30 mm long) in-house / workshop TO-QUOTE	1	Cotton/paper phenolic laminate tube	lathe from phenolic tube	0.0244 DERIVED T229	TO-QUOTE
ENG-TC-07 T230	Grain liner (phenolic tube 57.10/53.95) paper- or glass-phenolic tube 2.25 × 0.0625 in class: Aerocon / Kinetic / RS Com	1	Cotton/paper phenolic laminate tube	cut from stock tube, OD sanded to slip fit	0.0853 DERIVED T231	TO-QUOTE
ENG-TC-08 T232	Post-chamber + convergent insulator (phenolic, 3.50 mm over the post-chamber, contour per combustion v2) in-house / workshop TO-QUOTE	1	Cotton/paper phenolic laminate tube	lathe from glass- or silica-phenolic rod/tube	0.0515 DERIVED T233	TO-QUOTE
ENG-TC-09 T234	Aft section v3.2: post-chamber barrel + nozzle holder + retaining lip (SS316L, M63×1.5 socket on the case) in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC lathe (bore from the forward end) + flats	0.5786 DERIVED T235	TO-QUOTE
ENG-TC-10 T236	Injector-face insulator disc, phenolic 2 mm glass- or paper-phenolic sheet 2 mm: RS Components DK / Rochling	1	Cotton/paper phenolic laminate tube	water-jet or lathe from sheet	0.0043 DERIVED T237	TO-QUOTE
ENG-TC-11 T238	Stem guide bushing 5/3 × 4 (PTFE + 25 % glass) O2 PTFE-glass rod 10 mm: RS Components DK / Polyfluor	1	PTFE	lathe from rod	0.0001 DERIVED T239	TO-QUOTE

5.11 Provisions and contingency (not in the CAD model)

PN	Item	Qty	Status	Unit mass [kg]	Price
ENG-BL-01 T240	bleed tap T4 crown take-off fitting with restrictor TO-SELECT	1	provision (not modelled)	MISSING MISSING T241	TO-QUOTE
ENG-BL-02 T242	bleed tap FD-01 radial take-off (plug-tap fitting) TO-SELECT	1	provision (not modelled)	MISSING MISSING T243	TO-QUOTE
ENG-BL-03 T244	bleed line 1/8 in x 0.028 in 316L tube + tee TO-SELECT	1	provision (not modelled)	0.0132 DERIVED T245	TO-QUOTE
ENG-BL-04 T246	bleed crossing hose -2 class PTFE / 316 braid, ~0.2 m TO-SELECT	1	provision (not modelled)	MISSING MISSING T247	TO-QUOTE
ENG-BL-05 T248	bleed valve 2/2 NC solenoid, x2 in series TO-SELECT	2	provision (not modelled)	MISSING MISSING T249	TO-QUOTE
ENG-BL-06 T250	bleed overboard vent: balanced opposed pair or axial outlet TO-SELECT	1	provision (not modelled)	MISSING MISSING T251	TO-QUOTE
ENG-NZ-05 T252	J1b contingency: insulating layer 0.5 mm phlogopite mica between the NZ-03 hot layer and the NZ-04 seal phlogopite mica sheet rated >= 1002 C (J1b FE mica-face peak 901.8 C + 100 K, j1	0	contingency (not fitted)	TO-DEFINE TO-DEFINE T253	TO-QUOTE

Bleed provision mass estimate (excluding the valves): **0.058–0.097** kg ASSUMED T804. Valve mass is a MASS RISK (§4.4).

6 Materials and thermal protection

6.1 Materials

- 6061-T6 (case, head, closures, collar, ring, frame, tank)
- 316L (aft section, flight baseline)
- 17-4PH H1150/H900 (pins, needle, seat, link, crank)
- 7075-T6 (horns, pushrods)
- isomolded graphite (throat)
- paper/glass phenolic (liner, insulators)
- ABS (grain)
- mica (nozzle interlayer)
- PTFE/FKM/PFPE (seals, lubricant)
- flexible graphite (insert gasket)
- polyurethane 90A (stop bumper)
- G10/PEEK (servo-foot washers)
- steel 12.9 zinc-flake / A4-70 / Ti (fasteners)

Source: mechanical.json DERIVED T805

Material usage over the BOM lines (DERIVED by this generator; transport lock excluded).

Material (BOM material_name)	Lines	Mass	Oxygen-clean lines
Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube	21 DERIVED T286	2.1103 kg DERIVED T287	4 DERIVED T288
Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts	8 DERIVED T283	0.7547 kg DERIVED T284	6 DERIVED T285
COTS assembly (mixed materials), mass from datasheet or assumed	17 DERIVED T277	0.5409 kg DERIVED T278	8 DERIVED T279
ABS, FDM-printed fuel grain (100 % infill)	2 DERIVED T310	0.4244 kg DERIVED T311	0 DERIVED T312
Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)	4 DERIVED T331	0.1656 kg DERIVED T332	0 DERIVED T333
Carbon/bearing steel (generic COTS)	9 DERIVED T313	0.1169 kg DERIVED T314	0 DERIVED T315
17-4PH (1.4542) condition H1150, pins	9 DERIVED T289	0.0738 kg DERIVED T290	3 DERIVED T291
Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)	1 DERIVED T316	0.0473 kg DERIVED T317	0 DERIVED T318
17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet	4 DERIVED T298	0.0258 kg DERIVED T299	0 DERIVED T300
Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03	2 DERIVED T319	0.0155 kg DERIVED T320	0 DERIVED T321
Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)	2 DERIVED T295	0.0135 kg DERIVED T296	0 DERIVED T297
Aluminium alloy 7075-T6 (EN AW-7075 T6), bar	3 DERIVED T307	0.0134 kg DERIVED T308	0 DERIVED T309
Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative	1 DERIVED T328	0.0120 kg DERIVED T329	1 DERIVED T330
Polyurethane bumper 90A	1 DERIVED T304	0.0092 kg DERIVED T305	0 DERIVED T306
Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)	2 DERIVED T301	0.0033 kg DERIVED T302	0 DERIVED T303
Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy	3 DERIVED T325	0.0014 kg DERIVED T326	0 DERIVED T327
NdFeB magnet N35 (diametral, Ni-plated)	2 DERIVED T280	0.0012 kg DERIVED T281	0 DERIVED T282
Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face seal NZ-04	1 DERIVED T322	0.0007 kg DERIVED T323	0 DERIVED T324
PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible	1 DERIVED T292	0.0002 kg DERIVED T293	1 DERIVED T294
PTFE	1 DERIVED T334	0.0001 kg DERIVED T335	1 DERIVED T336

6.2 Thermal protection and thermal loads

Item	Value
Grain-zone liner thickness	1.6 mm DESIGN T812
Pre-chamber liner and structural amendments	Structure/thermal: pre-chamber liner TC-06 3.6 mm (D_pre 49.9 mm); case over-temperature abort 448.15 K / 0.2 s while the MV is open (relight refused above 433.15 K); TC-03 seat insert Inconel 718; upper-bay vent >= 0.3 W/K (goal 1.0 W/K) required of the airframe. LEAD T813
Post-chamber insulator	phenolic, thickness set by thermal FEA; 1-D bound needs >= 5 mm and still exceeds the 192 C aft-land allowable (open issue OI-4) DERIVED T814
Peak convective heat flux at the throat (Bartz, 600 K wall)	21.92 MW/m² SIM T806
Peak gas-side heat-transfer coefficient	10259 W/(m²·K) SIM T807

Item	Value
Adiabatic wall temperature at the throat	**2737** K (2464 °C) SIM T808
Incident flux on a cold wall at start (nominal–conservative)	**27.12–31.55** MW/m² SIM T809
Throat insert replacement rule	replace the insert when dA_t/A_t reaches 5 % (measured pin gauge / bore scan after each firing) DERIVED T815
Insert life plan (ENG-06, newest)	Accumulated burn per flight sized by the duty cycles in gnc.json v3. Grain replaced each flight; web ≥ 1 mm over the whole regression-law envelope (a ×0.6–1.5, n 0.35–0.68). Throat insert: plan on one insert per flight until T1/T4 bore data exist. Model spread is 0.5–4 A1 flights per insert at … ASSUMED T496
Nozzle gasket NZ-03 finding	mica/vermiculite 900 C rating vs 1595 C FE: fails by 695 K DERIVED T816
Proposed joint J1	2.0 mm carbon-phenolic flange + 0.5 mm flexible-graphite seal at the lip face DERIVED T817
Joint leak control	N2 pressure-decay leak check at 67.5 bar before and after every firing (criterion TO-DEFINE) DERIVED T818
Case over-temperature abort (MV open) / nozzle-holder limit	**448.15** K (175 °C) LEAD T810 / **523.15** K (250 °C) ASSUMED T811
Thermal FEA status	thermal FEA v1.1 (B1/A1/A2, TS-1 relight, TS-2 soak, servo mount, TC-06 sizing, Al vs 316L retainer) → v2.1 PENDING T819

7 Combustion

> **Reading this section.** combustion.json v2.4 is the source of every value here. Where it still carries wording that a newer lead decision replaced, the old words are ~~struck through~~ and the decision is inserted with its override id (§13.1). The Combustion v2.4a patch is to remove them (CHANGELOG 172 (6)).

7.1 Design point

Item	Value
Fuel	ABS (FDM printed, 100 % infill); PETG backup REQ T820
Oxidiser load and tank volume	**2.00** kg LEAD T350 in **3.17** L LEAD T821
Tank conditioning	Self-pressurised N2O conditioned ~~15–25 C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04** (heater blanket for cold days, removed at liftoff); N2 supercharge NOT baselined DERIVED T822
Model basis	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975) SIM T377
Combustion and nozzle efficiencies η_{c*} / η_{cf}	**0.920** ASSUMED T823 / **0.975** ASSUMED T824
Oxidiser flux at the head, rated, initial port	**491** kg/(m²·s) SIM T825
Regression law (form)	r = a * G_ox^n (oxidiser-flux law, local G_ox along the port) ASSUMED T826
Regression coefficient a / exponent n	**5.770e-05** m/s·(kg/m²s)^-n ASSUMED T827 / **0.50** ASSUMED T828
Regression law note	No verified N2O/ABS law in accessible literature; every result is checked at x0.6 and x1.5 on a. Static fires T1-T2 measure it. ASSUMED T829
T1–T4 measurement specification	02-combustion/T1-T4-REGRESSION-SPEC.md, v2.4 DESIGN T830

7.2 Fuel grain

Item	Value
Segments (BOM)	Fuel grain 1/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) T154 Fuel grain 2/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) T156
Segment length from the CAD manifest (GR-01 / GR-02)	**116.8** mm DERIVED T831 / **115.0** mm DERIVED T832
Grain outer diameter / total length	**53.5** mm DESIGN T834 / **230.0** mm DERIVED T835
Port	single centred straight round port LEAD T836, initial Ø **25.0** mm DESIGN T837
Initial web	**14.25** mm DERIVED T838
Port entrance (decided)	Rounded port entrance r >= 4 mm is now REQUIRED on every grain, flight and stand (decomposition peak 40.6 vs 43.7 bar abs, MEOP 45). The fuel-assisted fast part at relight stays controlled by the torch-first gate alone: FMEA hazard with a single control. LEAD T839
Port entrance in CAD / BOM	not stated in the BOM name or notes of ENG-GR-01/02: rounded entrance TO-VERIFY in CAD TO-VERIFY T833
Density (printed ABS)	**1000** kg/m³ ASSUMED T840
Grain mass: Combustion / CAD	**0.404** kg DERIVED T841 / **0.4244** kg DERIVED T842
Residual web rule	web >= 1 mm at the x1.5 regression bound for every flight duty cycle DERIVED T843
Retention	forward face on the pre-chamber insulator shoulder, aft face on the post-chamber insulator; anti-rotation key (Mechanical) DESIGN T844

7.3 Injector and feed

Item	Value
Feed architecture	tank → NC main isolation valve (off-gimbal) → -6 PTFE-lined braided hose across the gimbal -> <= 15 um filter → pintle injector (on gimbal) DESIGN T845
Injector flow model	Dyer et al. AIAA 2007-5702 (SPI/HEM blend); HEM and SPI bound it (G ratio 0.55-0.68 and 1.34-1.45 at 15 C) LITERATURE T846
Filter	sintered <= 15 um, Cv >= 2.2 DERIVED T847
Isolation valve: minimum Cv / MEOP / closing time	**3.0** DERIVED T848 / **7.00** MPa (70.0 bar) LEAD T849 / **0.15** s DERIVED T850

Item	Value
Relief: trapped liquid, back-relief and tank	tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check MV segment to tank (Cv >= 0.1), secondary disc 9.5 MPa +/- 5 %; the trapped-liquid relief <= 7.5 MPa text is superseded **OVR-05** combustion.json still reads: ~~required (<= 7.5 MPa) between the isolation valve and the pintle~~ DESIGN T851
Coast seat-leak ceiling (v2.4)	**Coast leak ceiling:** 0.0055 g/s (§3.5). DERIVED T852
Hose across the gimbal	-6 accepted: 6.3 m/s at 0.241 kg/s nominal, 6.5-7.1 m/s at 0.26 kg/s (15-25 C); hose friction 8-17 kPa; slow-closure surge (40-150 ms, 0.4-0.8 m line) 14-106 kPa DERIVED T853
Priming rule (first start)	first start of a flight/test (P_feed < 0.5 P_tank): pintle to a bleed lift (0.10 mm) BEFORE the isolation valve moves, so the line is vented through the injector (flow-through, no dead end); isolation valve opening ramp >= 0.3 s DERIVED T854
Pc tap and sense line	head top (Mechanical 2026-10-12) via sense line to the fixed side DESIGN T855 1/8 in OD (1.6 mm ID) SS, length <= 0.3 m, transducer dead volume <= 0.3 cm3 DERIVED T856
Pintle seat and seal review	adopted: rolling-guide carriage, spring 0.25 N m at closed + 0.12 N m/rad (+/-10 %); C2 checked under L1 and L2, C1 under L0; seat 17-4PH, needle+stem H900 / seat H1150, lapped land with blued contact >= 0.24 mm all round, PFPE film on the land; spring-energised PTFE-class rod seal; combustion_check_of_adopted: flight_C2_margin_L1_L2: [1.62, 1.62]; C1_MEOP_assist_L0_Nm: 0.108; crack_open_margin_MEOP_by_mu_seat: ... LEAD T857

7.4 Igniter

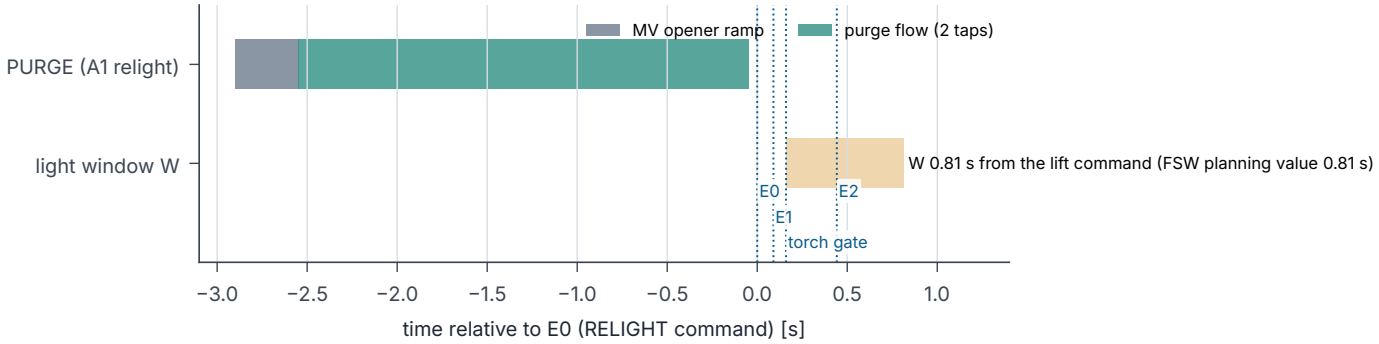
Item	Value
Type	spark-ignited N2O/propane torch, radial into the pre-chamber DESIGN T859
Torch chamber pressure / O/F / throat diameter	**3.0** bar DERIVED T860 / **5.5** DERIVED T861 / **2.0** mm DERIVED T862
Torch flows N2O / propane	**0.503** g/s DERIVED T863 / **0.091** g/s DERIVED T864
Orifice diameters N2O / propane	**0.131** mm DERIVED T865 / **0.254** mm DERIVED T866
Spark: supply / energy / rate	**12** V DERIVED T867 / **0.05** J DERIVED T868 / **50** Hz DERIVED T869
Envelope	radial boss on the head section at x_face 0.015 m, axis inclined 30 deg inward/aft so the plug end stays inside the 120 mm head sweep (answers Mechanical REQUEST 2026-10-12) DERIVED T870
Igniter system mass: Combustion / BOM ENG-IG-01	**0.20–0.30** kg ASSUMED T858 / **0.070** kg ASSUMED T167
Starts per flight	>= 3; limited by propane (0.055 g/start) and N2O torch flow (0.30 g/start) DERIVED T490

7.5 Start, relight and shutdown sequences

Start and relight sequence (combustion.json v2.4 /igniter/sequence; times relative to E0 where stated by the source).

Epoch	t [s]	Action
PURGE (A1 relight only)	**−2.900** s LEAD T872	MV latch and opener ramp (0.35 s) with the pintle seated; V1 and V2 open; both taps flow 2.5 s; close command leads by the measured valve closing time so the flow stops by E0 - 0.05 s; ~~closure confirmed before E1 or BLEED_FAULT to the FC (relight proceeds)~~ E1 = torch confirmed (2oo2) AND (closure confirmed OR deadline <= torch gate); in flight the lift proceeds with BLEED_FAULT (FC LOG-ONLY); stand: abort **OVR-02**. ECU cap = 2.5 s of flow + measured valve times. The relight is never committed before a complete purge: ~~if T_fire would cut the flow below 1.9 s, the FC delays T_fire~~ WITHDRAWN; PROVISIONAL: PURGE commits no later than plain trigger - 1.7 s, otherwise the purge is NOT run and the relight proceeds unpurged **OVR-08** LEAD T873
E0	**0.000** s LEAD T875	START or RELIGHT at T_fire: MV opens with the pintle seated (already open after PURGE); SOL1 (N2O) and SOL2 (propane) open; spark 50 Hz, 50 mJ from E0 + 0.02 s LEAD T876
torch gate	**0.160** s LEAD T878	no E1 by E0 + 0.16 s → TORCH_FAULT (A1); no main N2O admitted LEAD T879
E1	**0.090** s LEAD T881	torch flame confirmed on P_torch AND photodiode (2oo2, FA-1 (a)), 10 ms persistence; pintle commanded to the ignition lift 0.2039 mm (t_lift 29.6 ms; CF-3 lift table keeps <= 37.1 g/s at 20 C, <= 37.4 g/s at the A1 state). From E1 to E2 the torch is supervised on both sensors: loss on BOTH for >= 5 ms closes the pintle and the MV (FA-1 (b), A16) E1 = torch confirmed (2oo2) AND (closure confirmed OR deadline <= torch gate); in flight the lift proceeds with BLEED_FAULT (FC LOG-ONLY); stand: abort **OVR-02** LEAD T882
W	**0.810** s LEAD T884	abort deadline W from the lift command: 0.40 s flight first start, 0.65 s flight relight, 0.35 s stand (A2) LEAD T885
E2	**0.442** s LEAD T887	light: Pc >= 0.8 bar above the pre-lift baseline on filtered AND raw Pc for 10 ms, above 1.8 bar abs; SOL1 off first, SOL2 within 20 ms (torch ends propane-only), spark off, ramp released. LIGHT_SPIKE event logged (replaces A5) LEAD T888

Epoch	t [s]	Action
E3	—	thrust >= 90 % of the command (ENG-04 latency, §7.7) LEAD T891



A1 relight timeline: forced purge before E0, torch, ignition lift at E1, light E2 and the Pc-confirm deadline. E2 is shown at the value listed by the source sequence (not a bound). Data: T872, T875, T878, T881, T884, T887, T890, T892, T893, T894, T895

Phase	Rule
ARM preconditions	~~tank 15–25 C (ENG-12)~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04**; T_tank <= 24 C at ARM and at ignition (liquid-line interlock, §2.8); tank heater hard-disabled at ARM LEAD T896
	pintle seated, encoder healthy; MV-P closed; bleed valves V1 and V2 closed, drivers inhibited LEAD T897
	P_torch auto-zero; ARM refused if P_torch > 1.05 bar abs LEAD T898
	Pc and P_torch transducers 0-70 bar class (ENG-09 >= 70 bar); P_feed, P_tank healthy; Pc reads ambient LEAD T899
	igniter continuity (spark coil, SOL1, SOL2); photodiode healthy (needed for the 2oo2 E1) LEAD T900
	relight only: chamber vented; RELIGHT refused if a case thermocouple reads > 433.15 K LEAD T901
Relight	MV closed during coasts (LEAD); the v2.3 text "the isolation valve may stay open between burns" is withdrawn LEAD T902
	A1 relight: PURGE at E0 - 2.90 s (2.5 s of flow, theta <= 2 deg, balanced or axial vent), then the start sequence from E0 LEAD T903
	settling: liquid at the tank outlet from E0 - 2.90 s - t_settle (§2.4 note); RELIGHT refused if a case thermocouple > 433.15 K LEAD T904
	light window W 0.65 s; E0 → E2 max 0.626 s on the worst purged line state (rg01_line_states_v2p4); 0.81 s plan holds LEAD T905
Shutdown	pre-MECO throttle-down to <= 75 % (ENG-05) LEAD T906
	pintle closes at full drive speed LEAD T907
	MV de-energised on seat confirmation (encoder at seat and Pc < 2 bar abs) or 0.30 s after the command LEAD T908
	trapped liquid relieves to the tank through S1 (§2.7); ground: 30 s hold then safing LEAD T909
Abort if no ignition	pintle closed; torch solenoids closed (SOL1 not after SOL2); MV closed (abort is MV-first) LEAD T910
	no retry into a pooled chamber; ~~ground: wait >= 5 s, N2 purge, safing (§10.2)~~ N2 purge >= 60 s at >= 10 g/s, hold >= 15 min before approach, then safing **OVR-03** LEAD T911
	flight: no retry; vehicle-level response per GNC LEAD T912
	A5 (Pc > 8 bar for 5 ms) is DELETED (LEAD line 160): it tripped on 98 % of the late-light grid; LIGHT_SPIKE records the event instead LEAD T913

FA-1 (fuel-assisted relight pulse). REQUIRED of Electronics; FA-1 accepted for design continuation only; flight acceptance needs IGN-1 (>= 30 lights incl. photodiode false-positive tests) and the Mechanical 75.5 bar[OVR-07] <= 3 ms pulse check
OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs
LEAD T914 FA-1 stays accepted for design continuation only; flight acceptance needs IGN-1 and Mechanical's 75.5 bar[OVR-07] <= 3 ms pulse check.
OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs
LEAD T915
Load restated as an envelope (CHANGELOG 172 (2)). R2-03 FA-1 load restated as an ENVELOPE: Combustion publishes O/F 4-12 x tf 1-7 ms (1 ms steps) x t_res 0.02/0.06 x pool none/q 2 MW/m2 (head and aft peak, rise time, time above 45 and 67.5 bar, impulse). Until it lands Mechanical checks with an INTERIM load: quasi-static 90 bar abs (ENG-09 ultimate design point) at head AND aft, covering shell, closures, bolts and the J1 lip, reporting yield and ultimate margins; parts with periods of 1-5 ms (gimbal, thrust mount, pintle linkage) with a thrust spike of 1.8 kN for 4 ms by DLF/impulse. FMEA consequence: "yield possible; ultimate not demonstrated" until Mechanical reports. LEAD T024
Envelope published and structural load decided (CHANGELOG 173; §2.5). DECISION: FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts (covers every baseline peak and the shell); for any closure, bolt, J1-lip or other part with a natural period of 0.25-1 ms Mechanical reports margins also at 92 bar abs (design: baseline rounded with DLF and the +3 % fuel-basis sensitivity) and at 102 bar abs (sensitivity: upper bracket, sharp, DLF); thrust spike 1.8 kN x 4 ms stays the interim load and is confirmed bounding. LEAD T515

7.6 Abort set

The decided abort set is the newest lead decision for each item (CHANGELOG lines 160, 164, 166, 168, 172 and 175). The Combustion matrix (combustion.json v2.4) and the firmware v2.9 implementation (electronics.json v2) are shown beside it; where they differ, the decision governs and the difference is listed in §13.

ID	Decided (CHANGELOG)	Combustion v2.4 matrix	Firmware v2.9 (SIL only)
A2 No light within the window W	no item of its own; governed by the matrix tag: LEAD lines 160, 164; REPORT v2.4 §7.4 LEAD T916	no E2 within W of the lift command (0.40 / 0.65 / 0.35 s) LEAD T917	**NO_IGNITION** v2.9: no light by the relative rule (rise >= kIgnRise_Pa over P0, floor, kIgnConfirmRelN ticks) within W from the lift command (v2.6-v2.8.3 text: Pc (filtered and raw, kIgnitionConfirmN ticks) not ... DESIGN T918 kIgnRise_Pa = **80000** LEAD T919 kIgnConfirmRelN = **10** LEAD T920 kIgnWindowFirst_s = **0.4** LEAD T921 kIgnWindowRelightLong_s = **0.65** LEAD T922 kIgnWindowFirstStand_s = **0.35** LEAD T923
A4 Chamber over-pressure abort	**L160(2)** 30 bar abort ~~raw~~ FILTERED channel, 5 ms persistence **OVR-01** persistence 3 → 5 ms; comparator stays the fast path for a blocked throat (Electronics confirms the S6 blocked-throat trip still comes first). LEAD T924 **L166(4)** RE-20: the 30 bar firmware abort runs on the FILTERED channel with 5 ms persistence (probe F: comparator stays first; Combustion: filtered light-spike peak <= 21.4 bar); amends the word "raw" in l.160 (2). LEAD T001	Pc > 30 bar ~~raw~~ FILTERED channel, 5 ms persistence **OVR-01** for 5 ms LEAD T925	**PC_OVERPRESSURE** firmware Pc above the over-pressure limit outside IGNITION DESIGN T926 kPcOverPressureAbs_Pa = **3000000** LEAD T927 kPcOverPressureTime_s = **0.003** DESIGN T928
A5 Late-light monitor	**L160(1)** A5 DELETED as an abort, flight and stand; replaced by a LIGHT_SPIKE log event (raw and filtered peak, time above 8 and 30 bar abs, time from light) so IGN-1/T1 measure whether real lights spike. Supersedes item (2) of the Electronics-handback decision (A5 re-spec). OI-A5-LATE closes when Electronics implements this and SIL S6 ... LEAD T929	DELETED → LIGHT_SPIKE log LEAD T930	8.0 bar abs for 5 ms from E1 to the ramp start (unchanged); with the 10 ms light confirmation a fast late light can trip A5 first LEAD T931
A7 Pc plausibility bands, START high side	**L160(3)** A7 START high side in flight must neither switch the Pc loop to open loop nor set telemetered Pc invalid (0xFFFF, which would also block the drogue Pc branch); if the firmware ties them, blank the START high side from E2 to E2 + 0.45 s in flight; on the stand it stays a nuisance abort. Electronics replays Combustion's pool ... LEAD T932 **L166(7)** RE-12 pre-decided: if the SIL pool replay gives a stand A7 trip rate > 1 %, blank the stand START high side from E2 to E2 + 0.45 s as in flight. LEAD T933	Pc loss or implausibility, bands and the flight START high-side rule (Electronics sets values) LEAD T934	HOLD band from E1 (lo P0 - 0.3 bar), START band E2→E3 (lo = A3 threshold on the map, hi 1.3 x max(map, 5.3 bar)), RUN_MAP after E3 (0.7..1.3 x map); 50 ms; IGNITION / stand → PC_SENSOR abort, flight RUN → flag; A7 kept ARMED E1→E3 with ... LEAD T935

ID	Decided (CHANGELOG)	Combustion v2.4 matrix	Firmware v2.9 (SIL only)
A14 Bleed valve fault / closure not confirmed	**L168(2)** FC action on BLEED_FAULT is LOG-ONLY until a leak response is flown and reviewed. LEAD T006 **L168(3)** Bleed closure confirmed from P_mid is a pad go/no-go (launch-commit item) on every flight with the bleed fitted, whatever CF-4 says. LEAD T936	bleed valve fault or ~~closure not confirmed by E0 + 0.05 s~~ E1 = torch confirmed (2oo2) AND (closure confirmed OR deadline <= torch gate); in flight the lift proceeds with BLEED_FAULT (FC LOG-ONLY); stand: abort **OVR-02** → BLEED_FAULT to the FC (FC action LOG-ONLY, line 168); fail-closed -> no purge → no relight on that pass; closure from P_mid is a pad go/no-go (line 168) LEAD T937	**BLEED_FAULT** v2.9 head-end bleed: pad BIT step failed, V1 closure not confirmed by E0 + kHbleedCloseConfirm_s, or (stand) path / leak fault DESIGN T938 kHbleedCloseConfirm_s = **0.15** DESIGN T939 kHbleedLeakTime_s = **0.2** DESIGN T940 kHbleedPathBy_s = **0.1** DESIGN T941
A15 Anomaly during PURGE	no item of its own; governed by the matrix tag: LEAD lines 160, 164; REPORT v2.4 §7.4 LEAD T916	pintle or chamber anomaly during PURGE: inhibit torch, close MV LEAD T942	no matching cause in firmware v2.9 MISSING T943
A16 Torch loss between E1 and E2 (both sensors)	**L164(8)** FA-1: (a) E1 declared only on P_torch AND photodiode (2oo2) and (b) torch supervision E1→E2, where loss on BOTH sensors for >= 5 ms closes the pintle and the MV, are REQUIRED of Electronics (in flight this is an accepted mission loss in exchange for avoiding a > proof pulse). Mechanical adds a <= 3 ms, 75.5 bar[OVR-07] abs head-pulse check of the case and closures (dynamic load factor) to the FEA scope after the current step. FA-1 is accepted as a single-control hazard for DESIGN continuation only; flight acceptance needs IGN-1 (>= 30 lights incl. photodiode false-positive tests) and the Mechanical pulse check. OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs LEAD T944	torch loss on BOTH sensors >= 5 ms between E1 and E2: close pintle and MV LEAD T945	no matching cause in firmware v2.9 MISSING T943
E1 Torch confirmation 2oo2 and closure gate	**L166(3)** RE-05: E1 = torch confirmed (2oo2, FA-1 (a)) AND [closure confirmed OR closure deadline reached]; the deadline is <= the torch gate; at the deadline in flight the lift proceeds, BLEED_FAULT goes to the FC and V2 is commanded closed; on the stand it stays an abort. Bleed drivers get a HARDWARE purge-window enable (FC PURGE_EN ... LEAD T946	—	**TORCH_FAULT** no torch flame (P_torch) by the gate: abort before the pintle step DESIGN T947 kTorchGate_s = **0.16** LEAD T948 kTorchLitAbs_Pa = **200000** DESIGN T949 kTorchConfirmN = **3** DESIGN T950
LR Light-rule baseline fallback	**L166(1)** RE-02 light-rule baseline fallback: P0 > +check → keep the measured P0 (never substitute ambient on the high side), log BASE_HIGH and flag the FC; on the pad and the stand this refuses the lift with PC_SENSOR (no commit); at the A1 relight the start proceeds on the measured P0. P0 < -check → P0 := 0 (the -0.75 bar PC_SENSOR ... LEAD T951	—	P0 = mean of filtered/raw gauge Pc over [E1 - 20 ms, E1]; |P0| > residual + 0.1 bar → ambient (BASE_AMB); light (E2) = filtered AND raw rise over P0 >= 0.8 bar with filtered AND raw Pc >= floor, for 10 consecutive 1 kHz ticks; legacy 3.0 ... LEAD T952
TC Invalid case thermocouple	**L166(6)** RE-08: an invalid case TC is treated as hot, like bay and tank: both case TCs valid required at ARM; one lost → continue on the other; both lost → stand abort, in flight refuse the relight (FC keeps the drogue). LEAD T953	—	**OVER_TEMP** case TC (az 90/270, armed only while the MV is latched or OPEN) or nozzle holder above limit, in a burn or PREPRESS; servo-bracket NTC only flags DESIGN T954 kCaseOverTemp_K = **448.1** LEAD T955 kNozzleHolderOverTemp_K = **523.1** ASSUMED T956 kServoBracketOverTemp_K = **333.1** DERIVED T957 kOverTempTime_s = **0.2** DESIGN T958
RST ECU reset in flight	**L166(8)** RE-28: no in-flight recovery path after an ECU reset; reset → SAFE, MV and bleed closed, the drogue is not released (RUN never reported); HIL proves this; RE-09 power sequencing must remove the E0 brown-out. LEAD T959	—	**WATCHDOG** hardware watchdog reset: reboot lands in SAFE; a lit-session marker becomes a relight lockout DESIGN T960 kHwWatchdog_s = **0.05** DESIGN T961

ID	Decided (CHANGELOG)	Combustion v2.4 matrix	Firmware v2.9 (SIL only)
DRG Drogue release predicate (FC)	**L168(1)** Drogue release predicate restated, interim term limited to the invalid case: RELEASE iff RUN AND b4 AND [(Pc valid AND >= 300 kPa abs) OR b7 ~~OR (d2 - d3 = 0xFFFF AND t >= t_E2 + 0.10 s)~~ REPLACED in design by R2: with Pc invalid, release iff ECU RUN AND accelerometer thrust >= 37.16 N for 3 frames at 100 Hz; otherwise hold the drogue and command ECU ABORT at t_E2 + pcf_tmax (0.8 s, PROPOSAL) **OVR-09**], latched, t_E2 from EVENT 0x120; a late invalidity releases on the first invalid frame. The sentence "the FC never holds the drogue with the engine in RUN beyond E2 + 0.10 s" in line 161 (2) is WITHDRAWN. b7 sets only on a valid filtered channel and never on an invalid one. Release on a dead engine under Pc invalid (sensor fault plus flame-out in [E2, E2 + 0.10 s]) goes to the FMEA; Electronics assesses whether the FA-1 photodiode can serve as a non-Pc flame-out indicator in RUN; GNC amendment (c) flies it. LEAD T962 **L175(5)** Amendment c: the time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc invalid, release iff ECU RUN AND accelerometer thrust >= 37.16 N for 3 frames at 100 Hz; otherwise hold the drogue and command ECU ABORT at t_E2 + pcf_tmax, pcf_tmax 0.8 s (PROPOSAL to be flown; margins 33.0 vs 37.16 N and plateau ... LEAD T031	—	FC (LEAD decision 2026-10-13 CHANGELOG): release at E2 AND [measured Pc >= 3.0 bar abs OR the 0.8 bar rise held continuously to E2 + 0.10 s] DESIGN T963
SAFE Stand safing after a no-light	**L172(1)** R2-02 ONE stand safing rule after a no-light, effective now: N2 purge >= 60 s at >= 10 g/s, then hold >= 15 min before anyone approaches, with Pc at ambient and head TCs falling, then safing; "wait >= 5 s" is withdrawn everywhere. LEAD T008	—	ground procedure, not a firmware cause T964

Abort action. every abort de-energises the NC valve at once and commands the pintle closed in parallel (lead line 56); no in-flight relight retry after a no-light; normal shutdown: pintle first, MV on seat confirmation or the timeout LEAD T965

7.7 Stability

Stability indicators at the initial port (SIM).

Throttle	dP/Pc	Chug gain margin, 5 ms	Chug gain margin, 10 ms	1L acoustic	Helmholtz	Thermal lag
100.0 %	**0.93** SIM T966	**6.5** SIM T967	**4.7** SIM T968	**1460** Hz SIM T969	**1125** Hz SIM T970	**2.17** Hz SIM T971
75.0 %	**1.58** SIM T972	**10.2** SIM T973	**7.5** SIM T974	**1477** Hz SIM T975	**1138** Hz SIM T976	**1.61** Hz SIM T977
50.0 %	**2.78** SIM T978	**17.0** SIM T979	**12.5** SIM T980	**1499** Hz SIM T981	**1155** Hz SIM T982	**1.07** Hz SIM T983
25.0 %	**5.86** SIM T984	**32.1** SIM T985	**23.7** SIM T986	**1532** Hz SIM T987	**1180** Hz SIM T988	**0.56** Hz SIM T989
20.0 %	**7.18** SIM T990	**36.8** SIM T991	**27.2** SIM T992	**1539** Hz SIM T993	**1186** Hz SIM T994	**0.47** Hz SIM T995
12.5 %	**10.54** SIM T996	**52.2** SIM T997	**38.6** SIM T998	**1544** Hz SIM T999	**1190** Hz SIM T1000	**0.33** Hz SIM T1001

injector dP/Pc >= 0.25 everywhere (ceiling rule); chug gain margin from the lumped feed/chamber model; acoustic modes far above the actuator band DERIVED T1002

classical hybrid low-frequency (thermal-lag/boundary-layer) instability 0.3-2 Hz band is identified but its amplitude is NOT predicted; static fire T2/T3 with 1 kHz Pc logging TO-VERIFY T1003

8 Plume CFD

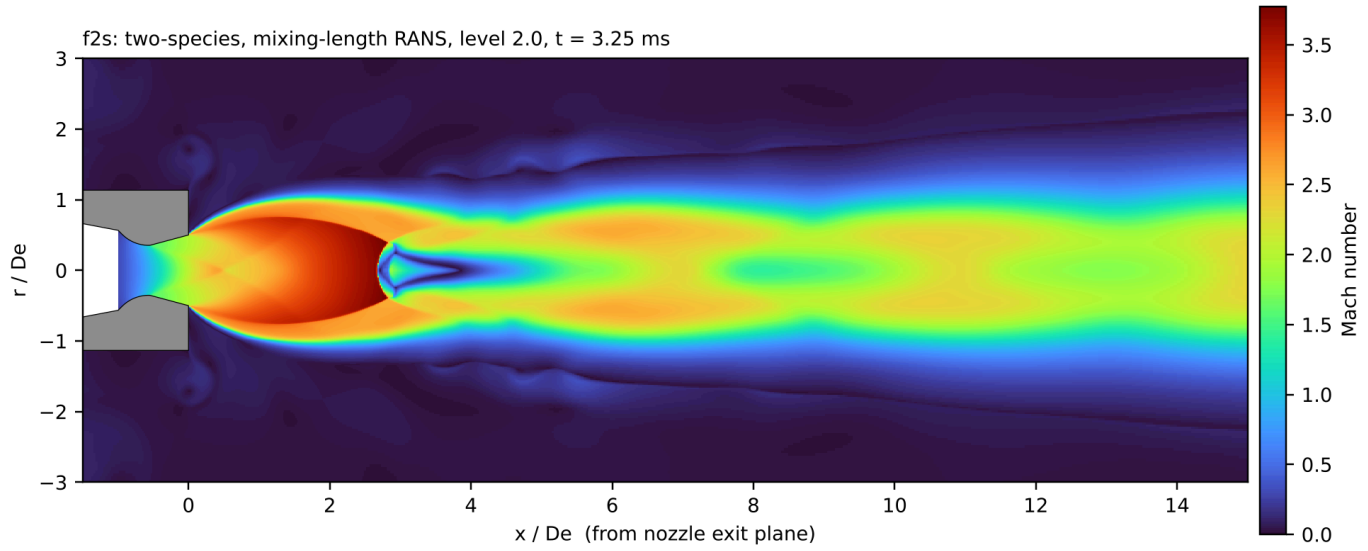
> **Simulation, not measurement.** Version v2.1a for validation and v2.2 for the time-matched afterburn, 2026-10-12. Owner: Combustion and Propulsion. Status: digital simulation only, *Por revisar*. A simulation is a prediction, not a measurement. Text revision v2.2b, 2026-10-13: wording changes for review findings RC-10, RC-11 and RC-19 (`reviews/REVIEW-COMBUSTION.md`, ACCEPT WITH COMMENTS); no case was re-run. SIM T1005

8.1 Key results at the rated point

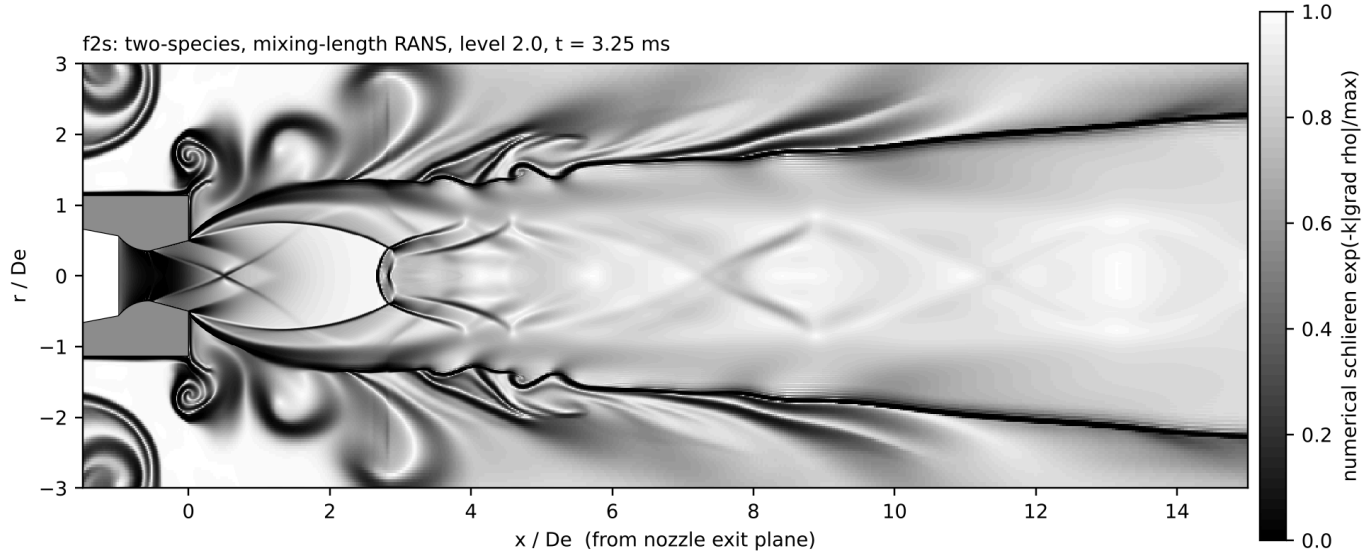
Plume CFD results (PVAL v2.1a, fine grid at the v1 point unless stated; lengths in exit diameters De).

Quantity	Value
Operating point of every grid level	v1 (combustion.json v1): Pc 24.99 bar, O/F 7.76, T0 = eta_c*^2 Tc = 2844.9 K. All grid levels, sensitivities and OpenFOAM runs use it. SIM T1020
Exit Mach number / pe/pa / fully expanded Mach number (1-D)	**2.012** DERIVED T1021 / **3.236** DERIVED T1022 / **2.623** DERIVED T1023
Mach disk position, fine grid (in mm) and Lewis–Carlson correlation	**2.676** De SIM T1006 (**51.9** mm DERIVED T1024) — correlation **2.663** De DERIVED T1007
Mach disk resolvability	grid differences within 2 sigma of the temporal scatter: GCI not meaningful; report the band SIM T1025
Shock-cell spacing, first / mean of three (in mm)	**3.96** De SIM T1008 (**76.8** mm DERIVED T1026) / **4.10** De SIM T1009 (**79.6** mm DERIVED T1027)
Shock-cell spacing, Prandtl–Pack–Tam correlation	**4.82** De DERIVED T1010
Shock-cell spacing grid sensitivity	**6.54** % SIM T1028 — oscillatory: 3 x max-deviation bound SIM T1029
Peak Mach number before the disk	**3.69** SIM T1030
Potential core length, axis Z ≥ 0.95 (in mm)	**14.9** De SIM T1011 (**289** mm DERIVED T1031)
Axis temperature at 10 De / 20 De (frozen, no afterburning)	**2222** K (1949 °C) SIM T1012 / **2058** K (1785 °C) SIM T1013
Thrust, exit integral (fine) vs 1-D with divergence factor; deviation	**500.8** N SIM T1014 vs **505.8** N DERIVED T1015; **−0.98** % SIM T1016
Thrust grid sensitivity	**0.15** % SIM T1032 — monotone without observed convergence (p < 0.5): 3 x max-deviation bound SIM T1033
Mass flow, CFD vs 1-D; deviation	**0.2542** kg/s SIM T1017 vs **0.2558** kg/s DERIVED T1018; **−0.65** % SIM T1019
Starting-jet front at the end of the run	**44.9** De SIM T1034
v2 design point (coarse grid): Mach disk CFD / correlation; thrust deviation	**2.709** De SIM T1035 / **2.666** De DERIVED T1036; **−1.09** % SIM T1037
v2 point remark	same coarse grid and models as own_c_2s_rans (v1 point O/F 7.76); v2 - v1 differences in v2op_vs_v1 carry no grid error but do carry the temporal band of the coarse level SIM T1038

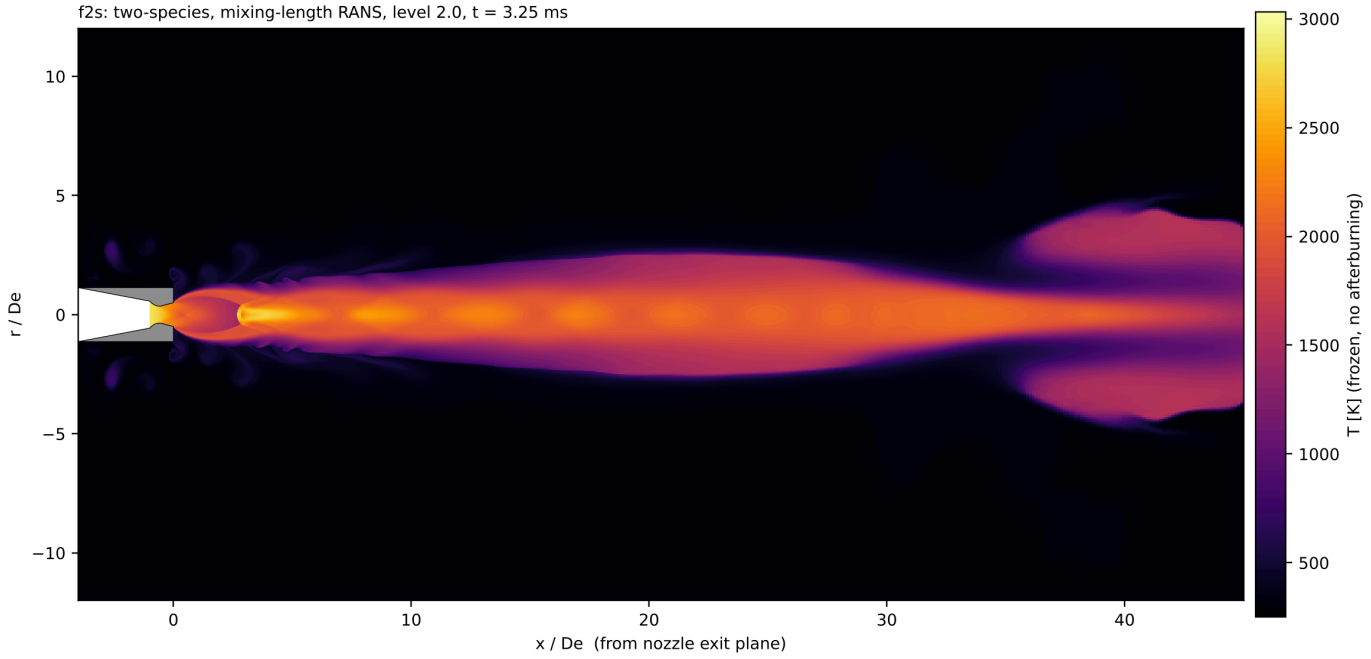
8.2 Fields



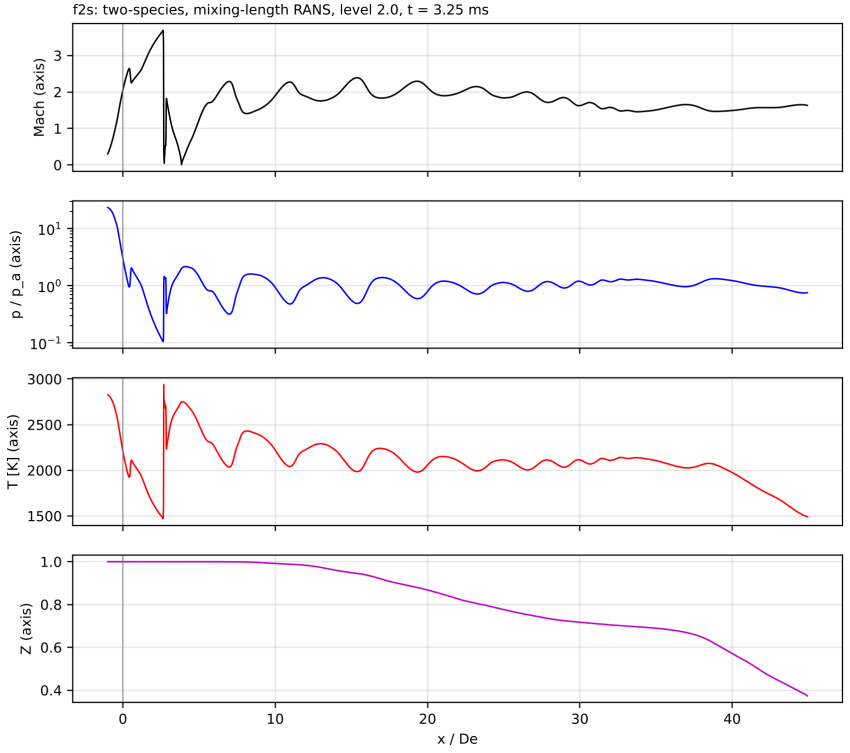
Mach number, near field (SIM, fine grid, v1 point). Source PF-MACH: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_mach_near.png (sha256/16 39cfd050d53b9f68).



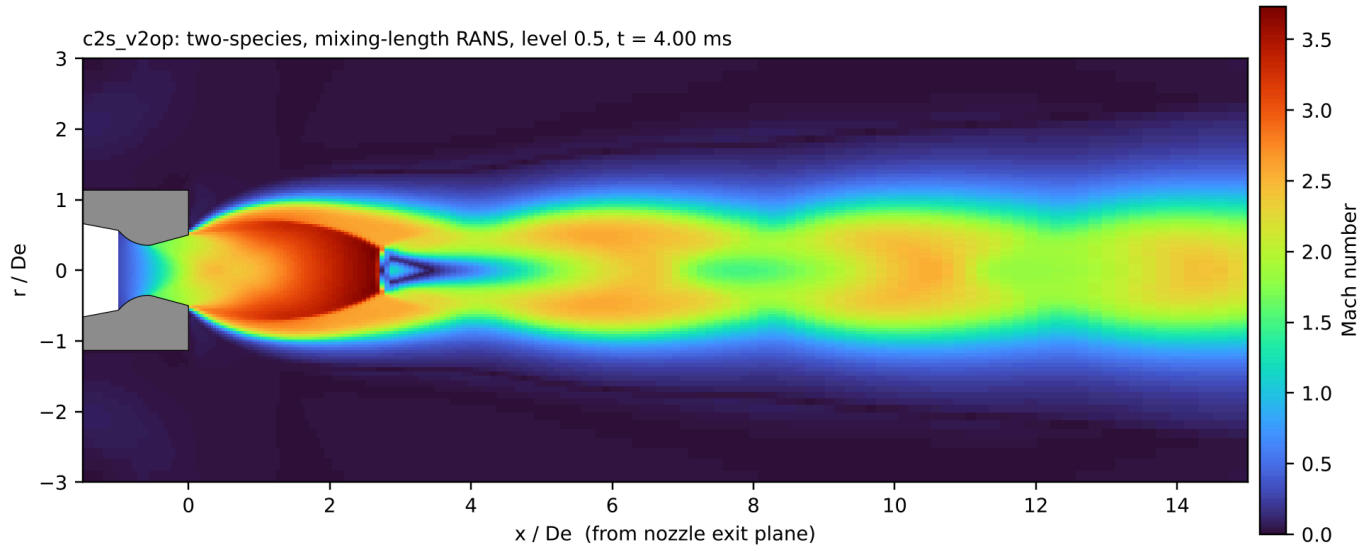
Numerical schlieren $|\nabla\rho|$ (SIM, fine grid, v1 point): Mach disk, reflected shocks and the shock-cell train. Source PF-SCHL: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_schlieren_near.png (sha256/16 5a559717ba7c3a76).



Static temperature, full domain (SIM, frozen chemistry, no afterburning). The far field is not statistically stationary. Source PF-TEMP: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_T_full.png (sha256/16 b7fe094ff216873c).



Centreline profiles (SIM, fine grid). Source PF-AXIS: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_centreline.png (sha256/16 de88abf544299723).

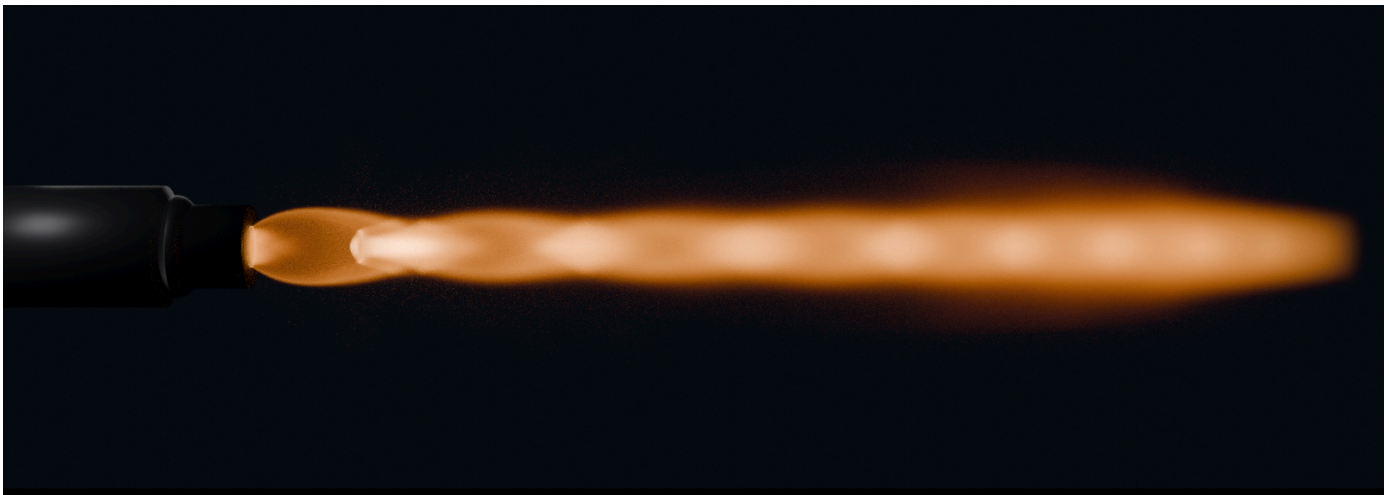


Mach number at the v2 design point (SIM, coarse grid). v2-v1 differences carry the grid error of the coarse level. Source PF-V2MACH: 05-plume-cfd/ownruns/c_2s_100_v2op/fig/c2s_v2op_mach_near.png (sha256/16 034d7f18ff3075f3).

8.3 Afterburning and model limits

Item	Report statement
Afterburning at the v1 point	adds 172-182 K at O/F 7.76 (temporal band 172-178 K; time-matched fine maximum 179.2 K; Richardson extrapolation of the three levels 181.8 K); SIM T1039
Afterburning at the v2 point	adds nothing at the v2 point for an equilibrium exhaust (O/F 10.4, oxidiser-rich; 0.44 K). The stream-B upper bound, which releases the c* deficit by fast chemistry, raises the plume maximum from 2705 K to 3011 K at that point (difference of maxima, +306 K; `ownruns/c_2s_100_v2op/fig/c2s_v2op_afterburn_stats.json`); SIM T1040
Afterburning at the stretch floor	adds up to 1476 K at the fuel-rich 12.5 % floor. SIM T1041
Limit: Soot and radiation	**Soot and radiation.** There is no soot model and no radiative transfer. The synthetic photo uses an ASSUMED soot proxy. SIM T1042
Limit: Finite-rate chemistry	**Finite-rate chemistry and turbulence-chemistry interaction.** Afterburning is a fast-chemistry, mean-Z equilibrium bound. Real afterburning is lower, and possibly zero if the mixture quenches. SIM T1043
Limit: Exhaust gas model	**Exhaust gas model.** The exhaust is a calorically perfect gas with frozen calibrated gamma, which gives core temperature errors up to about -500/+30 K in the strongly expanded region (§4). SIM T1044
Limit: Geometry and setting	**Geometry and setting.** The model is axisymmetric only (no gimbal angle, no 3-D structure) and has no ground plane, so no landing impingement or recirculation is modelled. SIM T1045
Limit: Duration	**Duration.** Simulated time is 3-4 ms. The far field (> 15 De) is not statistically stationary, and the fine level covers only 0.25 ms. SIM T1046
Limit: No acoustics	**No acoustics or plume loads on the vehicle.** SIM T1047

8.4 Illustration



Illustration, not simulation. Blender render of revolved CFD fields with exposure, colour and emitter chosen for appearance (ARTISTIC); it is not a radiation prediction and carries no data. Source PF-ART: 05-plume-cfd/blender/final/argos_plume_artistic_side.png (sha256/16 7c722b7f001b2ae4).

8.5 Review status

Independent reviews of the plume CFD in register order, newest last (reviews/SUPERVISION.json). A review verdict is not physical approval.

Review	Verdict	Comments (or criteria)	Reviewed file hashes vs current files
REV-PLUME-1 T1049	ACCEPT WITH COMMENTS DERIVED T1050	RC-10 thrust GCI R=1.09 diverges: reword; RC-11 correlation claim wording; RC-19 presentation DERIVED T1051	0 unchanged, 1 changed since review (05-plume-cfd/REPORT.md), 0 not found DERIVED T1048
REV-PLUME-2 T1053	ACCEPT DERIVED T1054	R2-12 summary wording (lead file) corrected: HANDBACK-SUMMARY d12c8862bf5524a4 DERIVED T1055	1 unchanged, 0 changed since review (none), 0 not found DERIVED T1052

Item	Value
Report text revision answering the comments	Version v2.1a for validation and v2.2 for the time-matched afterburn, 2026-10-12. Owner: Combustion and Propulsion. Status: digital simulation only, *Por revisar*. A simulation is a prediction, not a measurement. Text revision v2.2b, 2026-10-13: wording changes for review findings RC-10, RC-11 and RC-19 (`reviews/REVIEW-COMBUSTION.md`, ACCEPT WITH COMMENTS); no case was re-run. SIM T1005

9 Structural FEA

> **PENDING.** No structural margin has been published to the hub. The generator looked for `04-fea/**/*margin_table*.csv` under the hub and for `/structural_margins` in mechanical.json ≥ v2.1 (current: v2 DESIGN T1056). The pressure parts and the gimbal structure therefore carry no verified margin in this draft PENDING T1057. Stiffness and modal results from mechanical.json v2 are in §3.2.

9.1 Load cases (inputs)

04-fea/loadcases_v1.json (FEALC v1). The thrust of LC2F is listed in §13.

Case	Title	Chamber pressure	Thrust	SF yield / ultimate
LC1 T1058	Steady max thrust, gimbal neutral DESIGN T1059	**2.50** MPa (25.0 bar) DESIGN T1060	**500.0** N DESIGN T1061	**1.5** REQ T1062 / **2.0** REQ T1063
LC2 T1064	Chamber MEOP + thermal state at the critical time DESIGN T1065	**4.50** MPa (45.0 bar) DESIGN T1066	not applicable	**1.5** REQ T1067 / **2.0** REQ T1068
LC2F T1069	Pintle full-stroke failure thrust (structure) DESIGN T1070	**4.50** MPa (45.0 bar) DESIGN T1071	**907.7** N DESIGN T1072	**1.0** REQ T1073 / **1.5** REQ T1074
LC3 T1075	Max thrust + max gimbal (both axes) + lateral and angular acceleration limit loads DESIGN T1076	**2.50** MPa (25.0 bar) DESIGN T1077	**500.0** N DESIGN T1078	**1.5** REQ T1079 / **2.0** REQ T1080
LC4 T1081	Hydrostatic proof 1.5 x MEOP, room temperature (chamber and tank) DESIGN T1082	**6.75** MPa (67.5 bar) DESIGN T1083	not applicable	**1.0** REQ T1084 / —
BN T1085	Blocked-nozzle bound: chamber at tank pressure (30 C), hot DESIGN T1086	**6.31** MPa (63.1 bar) DESIGN T1087	**0.0** N DESIGN T1088	**1.0** REQ T1089 / —
LC5 T1090	Transient thermal over the flight duty cycle, structural at the critical instants DESIGN T1091	not applicable	not applicable	**1.5** REQ T1092 / **2.0** REQ T1093
LC6 T1094	Landing shock and handling/transport DESIGN T1095	not applicable	not applicable	**1.5** REQ T1096 / **2.0** REQ T1097
MODAL T1098	Gimballed engine on its actuators: first modes vs ENG-11 DESIGN T1099	not applicable	not applicable	— / —

Load cases decided after loadcases_v1 and not yet in a published load-case file.

Load case required by a lead decision	Decision text
LC-TB70 (throat block, assembled FEA)	LC-TB70 (53.4 bar abs, 34 ms) to Mechanical for the assembled FEA. LEAD T1100
Throat-block event in the firmware SIL	id: OI-TB-MEOP; severity: ACCEPTED; issue: S6 throat block Pc 53.4 bar abs > MEOP for 72 ms; owner: Mechanical FEA; closure: LC-TB70 / LC-TB-S in the assembled FEA SIM T1101
FA-1 fuel-assisted relight pulse check	Mechanical adds a <= 3 ms, 75.5 bar[OVR-07] abs head-pulse check of the case and closures (dynamic load factor) to the FEA scope after the current step. OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs LEAD T1102
FA-1 flight acceptance condition	FA-1 stays accepted for design continuation only; flight acceptance needs IGN-1 and Mechanical's 75.5 bar[OVR-07] <= 3 ms pulse check. OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs LEAD T915
FA-1 load envelope and interim check (supersedes the single-point check above)	R2-03 FA-1 load restated as an ENVELOPE: Combustion publishes O/F 4-12 x tf 1-7 ms (1 ms steps) x t_res 0.02/0.06 x pool none/q 2 MW/m2 (head and aft peak, rise time, time above 45 and 67.5 bar, impulse). Until it lands Mechanical checks with an INTERIM load: quasi-static 90 bar abs (ENG-09 ultimate design point) at head AND aft, covering shell, closures, bolts and the J1 lip, reporting yield and ultimate margins; parts with periods of 1-5 ms (gimbal, thrust mount, pintle linkage) with a thrust spike of 1.8 kN for 4 ms by DLF/impulse. FMEA consequence: "yield possible; ultimate not ... LEAD T024
FA-1 structural load decided on the v2.4a envelope (supersedes the interim check; §2.5)	DECISION: FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts (covers every baseline peak and the shell); for any closure, bolt, J1-lip or other part with a natural period of 0.25-1 ms Mechanical reports margins also at 92 bar abs (design: baseline rounded with DLF and the +3 % fuel-basis sensitivity) and at 102 bar abs (sensitivity: upper bracket, sharp, DLF); thrust spike 1.8 kN x 4 ms stays the interim load and is confirmed bounding. LEAD T515

9.2 Margin table

Placeholder: one row per load case until a margin table is published.

Load case	Governing margins

Load case	Governing margins
LC1 T1058	**PENDING** PENDING T1103
LC2 T1064	**PENDING** PENDING T1104
LC2F T1069	**PENDING** PENDING T1105
LC3 T1075	**PENDING** PENDING T1106
LC4 T1081	**PENDING** PENDING T1107
BN T1085	**PENDING** PENDING T1108
LC5 T1090	**PENDING** PENDING T1109
LC6 T1094	**PENDING** PENDING T1110
MODAL T1098	**PENDING** PENDING T1111

Margins reported in the decision log, outside a published margin table (Mechanical result, CHANGELOG 176; FA-1 loads of §2.5). FA-1 final load rule (lead 172 (2) + 173) closed in S-AX/joints: 90 bar abs quasi-static head+aft every part, 92/102 bar for the 0.25-1 ms parts (modes 1-2: TC-09+nozzle stack 0.33-0.47 ms, TC-02 head on its screws 0.27-0.39 ms; shell/bonnet 0.04-0.08 ms quasi-static), SF 1.0: TC-01 MS_y +0.91/+0.87/+0.69 (ult +1.07/+1.03/+0.83); TC-02 +4.46/+4.34/+3.83; TC-09 316L YIELD -0.40/-0.41/-0.43 (ult +0.80/+0.78/+0.66) → v3.5 TC-09 IN718 +2.81/+2.76/+2.51 (own thermal run); J-FWD 8xM5 A4-70 yield -0.011/-0.033/-0.128, ult +0.60/+0.56/+0.41 (shoulder-screw option +0.86/+0.82/+0.64); seal S13 -0.016 at 102 bar only; NZ-01..04, J-AFT, J-TC03, J-LIP positive. Spike 1.8 kN/4 ms: 608 bearings exceed C0 (MS -0.24 DLF 2.0, -0.21 trapezoid, -0.10 at Combustion SDOF 3.05 kN; C0 AS RECALLED) - brinelling possible, ultimate not demonstrated; TVC linkage 14.8 vs 35.3 N m, pintle link 428 vs 831.6 N, airframe screws MS +3.97 SIM T557 Not the assembled FEA of record and not independently reviewed; the CAD changes it proposes are not in CAD v3.4.1. **PENDING** PENDING T559

Schema the generator reads (one row per load case, part and state; the governing row per part is the minimum MS):

Field	Meaning
`case`	load case ID (loadcases file)
`part`	BOM part number
`material`	material code
`state`	critical state or instant
`MS`	margin of safety on yield (combined, at temperature)
`MS_ult`	margin of safety on ultimate
`T_part_max_C`	peak part temperature
`T_limit_C`	usable temperature limit
`SF_y`	yield factor applied
`SF_u`	ultimate factor applied
`criterion`	criterion text
`status`	result statement

A margin table is a simulation result. A positive margin is not a proof test: the hydrostatic proof (LC4) and the founders' pressure tests remain the physical gates for the pressure parts.

10 Electronics and control

Firmware v2.9 and electronics.json v2: REVISE per REV-ELEC-1 (REVISE DERIVED T1112). Blocking findings: RE-01 A5 still an abort; spike source not in SIL; RE-02 BASE_AMB fallback gives false E2 on positive offset; RE-03 b7 not filtered-only; RE-04 S5 up-step overshoot reaches the 30 bar abort; RE-05 bleed logic and hardware vs I.160-161 DERIVED T1113 Lead disposition: Electronics v2.10 publishes to the hub after archiving v2.9 LEAD T1114 Values below are the published v2 interface; where a lead decision changes them, §7.6 and §13 show the decided value.

10.1 Engine control unit and firmware

Item	Value
Architecture	separate engine ECU (Teensy 4.1) on the fixed thrust structure, CAN link to flight computer DERIVED T349
Microcontroller	PJRC Teensy 4.1 (i.MX RT1062 600 MHz, native SDIO microSD, 3 CAN incl. 1 CAN-FD) DESIGN T1115
PCB outline / enclosure reservation	**120.0, 60.0, 1.6** mm DESIGN T1116 / **128, 68, 32** mm DESIGN T1117
ECU mass (range)	**0.16** kg ASSUMED T347 (**0.12–0.22** kg ASSUMED T348)
Schematic	v2.4 (KiCad 7.0.11, generated from hardware/kicad_gen/ecu_design.py; v2.4 = head-end bleed provision drawn) DESIGN T1118
Schematic checks	0 errors, 0 warnings, 9 sheets DERIVED T1119 components 449, nets 277, pins 1256 of 1244 intended, errors 0, warnings 0 DERIVED T1120
Placeholders (not for fabrication)	TI_RPM0026A_PLACEHOLDER (NOT FOR FABRICATION); Pololu_U3V70_WireAttach (wire pads only); Teensy41_Socket (verify vs PJRC drawing) TO-VERIFY T1121
Firmware version / target	v2.9 DESIGN T1122 — PJRC Teensy 4.1 (i.MX RT1062); HAL for the hardware NOT written (core runs in SIL and test_core) TO-VERIFY T1123
Firmware unit tests (v2.9)	287 passed, 0 failed SIM T1124
Interface status	DESIGN + SIL + schematic. Firmware v2.9 runs only in SIL; the ECU v2.3 schematic passes ERC and a netlist cross-check; no board has been built and nothing here is hardware-validated. Values carry tags; open items are listed in open_issues, and the bench plan (06-electronics/docs/bench_plan_v2.md) is where each is closed. DESIGN T363
Constants generated from	src/core/engine_params.h: [["combustion.json", "v2.3", "d9ec54fbf450e243", "current"], ["gnc.json", "v3", "3e97c378cfebee95", "current"], ["pintle_mini_servo_v1.json", null, "62b326482617125a", "current"]]; src/core/mech_params.h: [["mechanical.json", "v2", "4f024ed4ccb4618c", "current"]]; src/core/mv_ladder_table.h: ; src/core/n2o_sat_table.h: [["Table", null, "b11b49519ba2fe30", "not checked ... DESIGN T1125
CAN link	ISO 11898-2 CAN, classic 2.0B 1 Mbit/s, 11-bit IDs; CAN-FD-capable transceivers so FD is a firmware upgrade DESIGN T1126

10.2 Sensors

Item	Value
Pressure transducers (selection)	PC: M3234-000005-070BG; P_TORCH: M3234-000005-070BG; P_FEED: M3234-000005-170BG; P_TANK: M3234-000005-170BG DESIGN T1127
Connectors	Pc J601, P_feed J602 (FS-01 MV-outlet block, Mechanical), P_tank J603, P_torch J901 (pc_transducer_v2 text says J610: wrong, J610 is ENC_A) DESIGN T1128
Hardware over-pressure comparator	31.47 bar g = 32.48 bar abs nominal on the 070BG; band 30.90-34.06 bar abs; firmware abort 30 bar abs DERIVED T1129
Transducer fault window	**0.25** V DESIGN T1130 – **4.75** V DESIGN T1131
Thermocouples	TC1 type K case az 90 (MAX31856 U604); TC2 type K case az 270 (U605); TC3 type K nozzle holder (U606); TC4 type K bore top (U904; CF-3/T1 evidence, flight TO-DEFINE); TC5 type T MSS-1 riser-foot TEE (U905); TC6 type T MSS-2 NC-valve outlet FS-01 (U906) DESIGN T1132
Case over-temperature abort / relight refusal	**448.15** K (175 °C) LEAD T1133 / **433.15** K (160 °C) LEAD T1134
Other temperatures	DS18B20 ×2 tank wall + #3 bay (TVC bracket) on 1-Wire (J608/J614); NTC 10k pintle-servo bracket (J609, flag 60 C) DESIGN T1135
Encoders	AS5048A x3 (gimbal A/B J610/J611, pintle crank J612), SPI1 1 MHz over cable DESIGN T1136
Load cell (stand)	1 kN S-type or compression load cell, 2 mV/V (TO-QUOTE) TO-VERIFY T1137
IMU	FC side (ADIS16470 per the GNC/FC record): filter delay UNKNOWN (over_thrust) TO-VERIFY T1138

10.3 Loop rates, latency and link loss

Item	Value
Pc control loop	**1000** Hz DESIGN T1139
Pc ADC sampling	**4000** Hz DESIGN T1140
TVC command shaping	**1000** Hz DESIGN T1141
Servo PWM	**333** Hz DATASHEET T1142
FC command rate	**500** Hz DESIGN T1143
Telemetry, fast	**500** Hz DESIGN T1144
Telemetry, slow	**50** Hz DESIGN T1145
SD log	**1000** Hz DESIGN T1146
Safety monitor	**1000** Hz DESIGN T1147
Relight-state telemetry	**50** Hz DESIGN T1148
FC thrust estimate	**100** Hz LEAD T1149
Hardware watchdog	**0.050** s DESIGN T1150
Link degraded / lost after	**0.020** s DESIGN T1151 / **0.100** s DESIGN T1152
Latency terms	fc_cmd_to_ecu_tick_max: 0.0012; ecu_to_servo_pwm_max: 0.005; servo_internal_assumed: 0.0015; fc_cmd_to_servo_motion_typ: 0.0045; fc_cmd_to_servo_motion_max: 0.0077; pc_sensor_to_controller: 0.0023; ecu_tlm_to_fc_max: 0.0022 DERIVED T1153

10.4 Light rule and start monitors (firmware v2.9)

Item	Value
Light rule	P0 = mean of filtered/raw gauge Pc over [E1 - 20 ms, E1]; |P0| > residual + 0.1 bar → ambient (BASE_AMB); light (E2) = filtered AND raw rise over P0 >= 0.8 bar with filtered AND raw Pc >= floor, for 10 consecutive 1 kHz ticks; legacy 3.0 bar abs rule WITHDRAWN LEAD T952
Rise over the baseline P0 / confirmation ticks / baseline samples	**80000** Pa LEAD T1154 / **10** LEAD T1155 / **20** LEAD T1156
Floor and baseline check	1.8e5 Pa abs until B-PC-1 shows residual <= 0.3 bar, then 2.0e5 (kBpc1Shown) LEAD T1157 residual + 0.1 bar (0.7 bar now, 0.4 bar after B-PC-1) LEAD T1158
Decided baseline fallback (RE-02)	RE-02 light-rule baseline fallback: P0 > +check → keep the measured P0 (never substitute ambient on the high side), log BASE_HIGH and flag the FC; on the pad and the stand this refuses the lift with PC_SENSOR (no commit); at the A1 relight the start proceeds on the measured P0. P0 < -check → P0 := 0 (the -0.75 bar PC_SENSOR path unchanged). Tag LEAD (this line). S13 B extended per RE-02 (2): both builds, pad and A1, lit and unlit, -0.8..+1.2 bar in 0.05 bar steps; pass = no false E2, every ... LEAD T951
SIL, A1 relight MC: lit runs / light-rule aborts / false lights on unlit runs	**300** SIM T1159 / **0** SIM T1160 / **0** SIM T1161
SIL, E2 after the light: median / maximum	**17** ms SIM T1162 / **18** ms SIM T1163
SIL basis	SIL (ASSUMED/SIM plant); target E2 <= ~15 ms after light NOT met: 16-18 ms = plant rise + 50 Hz filter + 10 ms SIM T1164
A3 start monitor	(Pc - P0)/(Pc_map_lo - P0) < 0.40 for 20 ms in RUN; Pc_map_lo = min(40 ms-lagged map, instantaneous map) at the encoder lift; denominator >= 0.5 bar (the lagged map alone false-tripped on fast throttle-downs: SIL finding) LEAD T1165
A7 plausibility bands	HOLD band from E1 (lo P0 - 0.3 bar), START band E2→E3 (lo = A3 threshold on the map, hi 1.3 x max(map, 5.3 bar)), RUN_MAP after E3 (0.7..1.3 x map); 50 ms; IGNITION / stand → PC_SENSOR abort, flight RUN → flag; A7 kept ARMED E1→E3 with the lit-start band; the rate criterion is not implemented (written reason: no Combustion rate bound yet) LEAD T935
E3 (end of start)	first RUN tick with the ramp done and Pc >= P0 + 0.9 (setpoint - P0), or E2 + 0.6 s DESIGN T1166

10.5 Abort and refusal causes (firmware v2.9)

28 causes in electronics.json /abort_matrix/causes. REVISE per REV-ELEC-1; decided changes in §7.6.

Code	Cause	Trigger	Thresholds
17 T1167	**PC_OVERPRESSURE** T1168	firmware Pc above the over-pressure limit outside IGNITION DESIGN T926	kPcOverPressureAbs_Pa = **3000000** LEAD T927 kPcOverPressureTime_s = **0.003** DESIGN T928
18 T1169	**HW_OVERPRESSURE** T1170	hardware comparator latch (31.47 bar g on the 070BG): ARM bus already off in hardware DESIGN T1171	kHwOvpTime_s = **0.002** DESIGN T1172
19 T1173	**TVC_SUPPLY** T1174	pad only: predicted both-stall TVC plug voltage below the minimum DESIGN T1175	kTvcSupplyMin_V = **6.6** LEAD T1176 kTvcSupplyTime_s = **0.1** DESIGN T1177

Code	Cause	Trigger	Thresholds
20 T1178	**OVERCURRENT** T1179	firmware e-fuse before the first start DESIGN T1180	kEfuse_A = 22 DESIGN T1181 kEfuseTime_s = 1 DESIGN T1182
21 T1183	**MAINT_JAM** T1184	ARM refused: a pintle jam recorded in NV memory and not cleared (bearing A rule) DESIGN T1185	no threshold
22 T1186	**TORCH_FAULT** T1187	no torch flame (P_torch) by the gate: abort before the pintle step DESIGN T947	kTorchGate_s = 0.16 LEAD T948 kTorchLitAbs_Pa = 200000 DESIGN T949 kTorchConfirmN = 3 DESIGN T950
23 T1188	**RELIGHT_LOCKOUT** T1189	ARM/IGNITE/RELIGHT refused after an abort following a lit burn or on a relight, until CLEAR_LOCKOUT (ground) DESIGN T1190	no threshold
24 T1191	**OVER_THRUST** T1192	flight RUN: FC_THRUST above the threshold on consecutive fresh valid frames (with or without a valid Pc; never on the stand) DESIGN T1193	kOverThrust_N = 650 LEAD T1194 kOverThrustFrames = 3 LEAD T1195 kFcThrustRate_Hz = 100 LEAD T1196 kFcThrustStale_s = 0.05 DESIGN T1197
25 T1198	**MODE_MISMATCH** T1199	ARM refused: strap invalid, or strap and load-cell reading disagree DESIGN T1200	no threshold
26 T1201	**BAY_HOT** T1202	ARM refused: bay temperature above the limit or no valid reading DESIGN T1203	kBayArmMax_K = 318.1 LEAD T1204
27 T1205	**TANK_HOT** T1206	ARM refused: tank temperature (max of wall DS18B20 and Tsat(P_tank)) above the limit or none valid DESIGN T1207	kTankIgnMax_K = 297.1 LEAD T1208
28 T1209	**BLEED_FAULT** T1210	v2.9 head-end bleed: pad BIT step failed, V1 closure not confirmed by E0 + kHbleedCloseConfirm_s, or (stand) path / leak fault DESIGN T938	kHbleedCloseConfirm_s = 0.15 DESIGN T939 kHbleedLeakTime_s = 0.2 DESIGN T940 kHbleedPathBy_s = 0.1 DESIGN T941
1 T1211	**CMD** T1212	FC ABORT frame 0x010 or CMD_STATE ABORT DESIGN T1213	no threshold
2 T1214	**HARD_START** T1215	Pc above the over-pressure limit during IGNITION DESIGN T1216	kPcOverPressureAbs_Pa = 3000000 LEAD T1217 kPcOverPressureTime_s = 0.003 DESIGN T1218
3 T1219	**NO_IGNITION** T1220	v2.9: no light by the relative rule (rise >= kIgnRise_Pa over P0, floor, kIgnConfirmRelN ticks) within W from the lift command (v2.6-v2.8.3 text: Pc (filtered and raw, kIgnitionConfirmN ticks) not above the light ... DESIGN T918	kIgnRise_Pa = 80000 LEAD T919 kIgnConfirmRelN = 10 LEAD T920 kIgnWindowFirst_s = 0.4 LEAD T921 kIgnWindowRelightLong_s = 0.65 LEAD T922 kIgnWindowFirstStand_s = 0.35 LEAD T923
4 T1221	**FLAMEOUT** T1222	RUN: gauge Pc below the floor or below a fraction of the expected Pc DESIGN T1223	kFlameoutPc_Pa = 150000 DESIGN T1224 kFlameoutFrac = 0.35 DESIGN T1225 kFlameoutTime_s = 0.02 LEAD T1226
5 T1227	**PC_BAND** T1228	RUN after settling: |Pc - Pc_ref| above both band limits DESIGN T1229	kPcBandFrac = 0.3 DESIGN T1230 kPcBandMin_Pa = 200000 DESIGN T1231 kPcBandTime_s = 0.15 DESIGN T1232 kRunSettle_s = 0.3 DESIGN T1233
6 T1234	**OVER_TEMP** T1235	case TC (az 90/270, armed only while the MV is latched or OPEN) or nozzle holder above limit, in a burn or PREPRESS; servo-bracket NTC only flags DESIGN T954	kCaseOverTemp_K = 448.1 LEAD T955 kNozzleHolderOverTemp_K = 523.1 ASSUMED T956 kServoBracketOverTemp_K = 333.1 DERIVED T957 kOverTempTime_s = 0.2 DESIGN T958
7 T1236	**TANK_OVERPRESSURE** T1237	P_tank above limit in ARMED/PREPRESS (never aborts a burn) DESIGN T1238	kTankOverPressure_Pa = 6500000 DESIGN T1239 kTankOverPressureTime_s = 0.2 DESIGN T1240
8 T1241	**GIMBAL_FAULT** T1242	command-model vs encoder mismatch while burning (allowance grows with the model rate) DESIGN T1243	kGimbalFaultErr_rad = 0.02618 DESIGN T1244 kGimbalFaultTime_s = 0.05 DESIGN T1245 kGimbalFaultRateWin_s = 0.04 DESIGN T1246
9 T1247	**PINTLE_FAULT** T1248	pintle rail lost (LTC4365 trip / regulator failure), jam, encoder invalid at ARM DESIGN T1249	kPrailLost_V = 5.5 DESIGN T1250 kPrailLostTime_s = 0.002 DESIGN T1251 kPintleStallTime_s = 0.1 DESIGN T1252 kPintleStallTimel_s = 0.05 DESIGN T1253
10 T1254	**LINK_LOST** T1255	no valid CMD_FAST: SAFE/ARMED/PREPRESS → SAFE; RUN -> SHUTDOWN by default DESIGN T1256	kLinkDegraded_s = 0.02 DESIGN T1257 kLinkLost_s = 0.1 DESIGN T1258
11 T1259	**PREPRESS_FAIL** T1260	feed not primed (P_feed > fraction of P_tank) in time, bleed lift not confirmed, MV not open DESIGN T1261	kPrepressTimeout_s = 0.6 DESIGN T1262 kFeedPrimedFrac = 0.8 DESIGN T1263
12 T1264	**PC_SENSOR** T1265	Pc transducer output outside the window: IGNITION and stand abort; flight RUN continues open loop DESIGN T1266	kPcSensorMin_V = 0.25 DESIGN T1267 kPcSensorMax_V = 4.75 DERIVED T1268 kPcSensorLossStand_s = 0.005 DESIGN T1269
13 T1270	**BATT_LOW** T1271	VBAT low in pad states before the first start (flag in flight); ARM refused below the rest-voltage rule DESIGN T1272	kBattLow_V = 6.6 DESIGN T1273 kBattLowTime_s = 0.2 DESIGN T1274 kBattArmMin_V = 8.1 LEAD T1275
14 T1276	**MV_FAULT** T1277	MV CLOSED switch missing at ARM; direct relight OPEN switch missing by the deadline: stand abort, flight flag DESIGN T1278	kMvOpenBy_s = 0.95 DESIGN T1279 kMvCloseConfirm_s = 0.5 DESIGN T1280
15 T1281	**ARM_LOST** T1282	arming plug removed or ARM bus not on 50 ms after the request DESIGN T1283	no threshold
16 T1284	**WATCHDOG** T1285	hardware watchdog reset: reboot lands in SAFE; a lit-session marker becomes a relight lockout DESIGN T960	kHwWatchdog_s = 0.05 DESIGN T961

Abort action. every abort de-energises the NC valve at once and commands the pintle closed in parallel (lead line 56); no in-flight relight retry after a no-light; normal shutdown: pintle first, MV on seat confirmation or the timeout LEAD T965

10.6 Power

Item	Value
Battery	LiPo 2S 2.2 Ah (16.3 Wh), usable 13 Wh DESIGN T1286
Logic power, typical / peak	typ: 1.25; peak: 2.6 DERIVED T1287
Energy per flight day	**1.73** Wh DERIVED T1288
Energy per static-fire session	**2.79** Wh DERIVED T1289
Peak current (three servos stalled)	**35.7** A DERIVED T1290
Minimum voltage at the ECU	**6.11** V DERIVED T1291
Fusing	F101: 25 A ATM main; F102: 2 A 1206 logic; F301: 15 A ATM ARM bus; F501: 3 A 1206 12 V boost DESIGN T1292
ARM bus peak vs fuse	TO-CHECK: MV opener stall 9.2 A at 7.4 V (DATASHEET) / 10.4 A at 8.4 V (ASSUMED) + pintle rail input <= 4.33 A + 12 V boost loads vs F301 15 A (Littelfuse 0297015 curve) MISSING T1293

10.7 Head-end bleed provision and fail-safe states

Item	Value
Architecture	2 NC valves in series (V1 at the engine-side tee of the T4 crown + FD-01 radial taps via one -2 line across the gimbal; V1 and V2 on the fixed side), restriction downstream of V2, P_mid transducer between them; ECU drives V1 on pin 15 (U304) and V2 on pin 24 (U307), low-side on +12V_ARM (J509/J510); P_mid comparator on the pin 23 ladder DESIGN T1294
Pad built-in test	PREPRESS steps S0-S4 (valve pattern vs P_mid); any failure → BLEED_FAULT DESIGN T1295
Purge	PURGE state: MV open, pintle seated, V1+V2 open, t_purge 2.5 s; pre-scheduled (Cmd PURGE) or sequence-integrated (RELIGHT); PINTLE_FAULT / encoder / MV OPEN / P_feed fill checks active; V1-first closure confirmed by P_mid, else BLEED_FAULT at E0 + 0.15 s LEAD T1296
Stuck-open bleed during a burn	invisible on P_feed (0.016 bar) and Pc (SIL -0.02..-0.05 bar); seen only by the P_mid leak monitor at +0.21 s: flight flag + EVENT, stand BLEED_FAULT; V2-only stuck open is latent until the next pad BIT; the next purge relight then ends BLEED_FAULT (V1 unconfirmable) SIM T1297
Valve coil	valve: Burkert 6027 HP (stainless, G1/4, PEEK/FKM seat, NC, orifice 1.0 mm: 0-250 bar DC air/water) - DATASHEET via WebFetch summary; coil_42K_DC_W_cold_warm: [16.0, 21.0]; coil_note: column mapping of the coil table ambiguous in the summary: TO-VERIFY on the PDF; coil_voltages_listed: 24 V DC, 230 ... TO-VERIFY T1298
Fail-safe: ECU power loss	MV latch drops → MV-P spring + pressure bias close it (dashpot ~30 ms ESTIMATE); the pintle spring closes the unpowered HV69 (B-PNT-1 back-drive check); igniter dead DESIGN T1299
Fail-safe: MCU hang	hardware watchdog (50 ms) resets MCU and drops ARM bus → same as above; reboot lands in SAFE, never resumes RUN DESIGN T1300
Fail-safe: link loss	see link_loss DESIGN T1301
Fail-safe: Pc sensor fault	RUN: open-loop feed-forward map on P_feed + fault flag; IGNITION: abort DESIGN T1302
Fail-safe: gimbal encoder fault	flag; RUN: continue on servo commands (default) unless mismatch abort criterion met DESIGN T1303
Fail-safe: arming plug removed	ARM bus dead: igniter, pintle and main valve power removed regardless of firmware DESIGN T1304
Fail-safe: pintle rail fail-high	LTC4365 OV disconnect 8.06-8.34 V: servo unpowered, spring closes; PRAIL_OK low drops the MV latch in hardware (U306); firmware PINTLE_FAULT abort DESIGN T1305
Fail-safe: Pc over-pressure	hardware comparator latch 31.47 bar g removes the ARM bus independent of firmware; firmware abort at 30 bar abs DESIGN T1306

10.8 Electronics open issues

electronics.json /open_issues (21 items).

ID	Severity	Issue	Owner
OI-IGN-DET	HIGH (cross-discipline) DESIGN T1308	light-detection margin at the v2.8 ignition lift: S5 dispersed pad-start MC 31/300 NO_IGNITION on a LIT plant since v2.8.1 (v2.7 at 20 % lift: 0). The lit plant Pc at 0.2039 mm sits at the 3.0 bar abs threshold (nominal replay 3.02 bar abs; ... TO-VERIFY T1309	Combustion (threshold and lift) + Electronics (detection logic) DESIGN T1310
OI-RELIGHT-A1	HIGH DESIGN T1312	A1 relight not frozen (RG-01): re-routed segment volume and the GNC re-fly pending; drained-segment fault path to re-run on the published re-route TO-VERIFY T1313	Mechanical / Electronics / GNC DESIGN T1314
OI-GIMBAL-JAM	MEDIUM DESIGN T1316	S6 gimbal_jam_moving_cmd FAIL (small-amplitude jam undetected) TO-VERIFY T1317	Electronics + GNC DESIGN T1318
OI-TB-MEOP	ACCEPTED DESIGN T1320	S6 throat block Pc 53.4 bar abs > MEOP for 72 ms TO-VERIFY T1321	Mechanical FEA DESIGN T1322

ID	Severity	Issue	Owner
OI-HAL	HIGH DESIGN T1324	hardware HAL not written; TLM_SLOW 0x110 undefined; 53 ARM double-promotion warnings TO-VERIFY T1325	Electronics DESIGN T1326
OI-FC	HIGH DESIGN T1328	GNC tracker / FSW not yet in the flight computer; IMU filter delay unknown; RG-16 mass budget TO-VERIFY T1329	FC owner + GNC DESIGN T1330
OI-TIMEBASE	MEDIUM DESIGN T1332	no common time base yet for DAQ, CAN analyser and ECU TO-VERIFY T1333	Electronics DESIGN T1334
OI-PFEED-SPOF	MEDIUM DESIGN T1336	P_feed is a single point for the relight path choice (invalid → UNKNOWN → prep path) and S1 TO-VERIFY T1337	Electronics DESIGN T1338
OI-LATE-FLIP	MEDIUM DESIGN T1340	a coast encoder loss after the last TLM_RELIGHT frame flips the path late (S10 evidence case) TO-VERIFY T1341	GNC planning DESIGN T1342
OI-B1-WIND	MEDIUM DESIGN T1344	B1 launch commit mean wind < 6 m/s until B-PNT-1 shows the drive lag inside the MC tau TO-VERIFY T1345	Electronics bench DESIGN T1346
OI-RAIL	MEDIUM DESIGN T1348	LTC4365 UV ASSUMED; Q803 enhancement; RPM0026A footprint placeholder; ARM-bus peak vs F301 TO-VERIFY T1349	Electronics DESIGN T1350
OI-MV-CAM	LOW DESIGN T1352	MV-P opener cam-limited 61 ms opening (bench B-MV-1) TO-VERIFY T1353	Mechanical + Electronics DESIGN T1354
OI-TRIM	LOW DESIGN T1356	map-trim plateau (Pc loop trim authority +/-40 %) not yet characterised on hardware TO-VERIFY T1357	Electronics + Combustion DESIGN T1358
OI-DOC	LOW DESIGN T1360	pc_transducer_v2 labels P_torch J610 (is J901); protocol.h FC_THRUST valid-window comment wording TO-VERIFY T1361	Electronics DESIGN T1362
OI-A5-LATE	HIGH DESIGN T1364	v2.9 light rule (10 ms) + unchanged A5 (8.0 bar abs, 5 ms): S6 late lights 0.20/0.25 s inside W now abort HARD_START (v2.8.3 continued; plant Pc max 26.9/28.6 bar abs) TO-VERIFY T1365	LEAD / Combustion DESIGN T1366
OI-TEB-FALSE-LIGHT	MEDIUM DESIGN T1368	S9 +1.034 bar post-zero bias: unlit filtered 1.87 bar abs > floor 1.8 → false E2 → RUN → A3 FLAMEOUT +0.14 s; closed by B-PC-1 <= 0.3 bar (floor 2.0) TO-VERIFY T1369	LEAD / Electronics (B-PC-1) DESIGN T1370
OI-E2-LATENCY	LOW DESIGN T1372	E2 - light 16-18 ms in SIL (target ~15 ms): plant rise + 50 Hz filter + 10 ms persistence; IGN-1 measures TO-VERIFY T1373	Combustion / Electronics DESIGN T1374
OI-A3-MARGIN	MEDIUM DESIGN T1376	A3 ratio minimum 0.454 at the A1 relight MC (threshold 0.40) during the start ramp; after E3 0.719 TO-VERIFY T1377	Combustion DESIGN T1378
OI-BLEED-COIL	HIGH DESIGN T1380	Burkert 6027 HP: 24 V DC listed, ECU valve rail +12V_ARM (U3V70F12); 2 × 16 W → +2.7 A for ~2.6 s per purge at 12 V; 12 V coil or a 24 V boost, rail budget and valve mass TO-VERIFY; HP seat PEEK/FKM medium -10..80 C and N2O compatibility TO-VERIFY TO-VERIFY T1381	Electronics / Mechanical DESIGN T1382
OI-BLEED-VOL	MEDIUM DESIGN T1384	unreached vapour with taps (2 deg 4.84 cm3, 10 deg 15.91 cm3) from an UNPUBLISHED Mechanical workdir file TO-VERIFY T1385	Mechanical DESIGN T1386
OI-GNC-V3	MEDIUM DESIGN T1388	gnc.json v3 requirements.ignition no longer carries option_c_unprimed_s / stale_or_missing_flag: FC relight planning source to be re-confirmed TO-VERIFY T1389	GNC DESIGN T1390

11 Mission and landing

11.1 Reference vehicle

gnc.json v3 /reference_vehicle (parametric; the real airframe is owned outside the engine team).

Item	Value
Reference vehicle	RV-1c (GNC reference vehicle on mechanical.json v2, ledger v3d, parametric; real airframe owner: Lauty): CAD v3.3 engine module + fixed TVC module, uniform 146 mm body ASSUMED T1391
Length / body diameter	**2.03** m ASSUMED T1392 / **0.146** m DERIVED T1393
Dry mass	**13.671** kg DERIVED T1394
Fuel grain / N2O load / tank volume	**0.404** kg DESIGN T1395 / **2.00** kg REQ T1396 / **3.167** L REQ T1397
Lift-off mass / landing mass (nominal A1)	**16.075** kg SIM T1398 / **14.767** kg SIM T1399
Lift-off thrust-to-weight	3.156 DERIVED T1400
Landing legs	n: 4; foot_radius_m: 0.4; stroke_m: 0.1; k_per_leg_N_m: 2500.0; c_per_leg_N_s_m: 120.0; deploy: at relight command (A) / fixed (B) ASSUMED T1401
Fins	4 aft fins, span 0.12 m, root/tip chord 0.20/0.10 m, 3 mm GF/PETG, 0.30 kg DERIVED T1402
Tail-first aerodynamics	Tail-first aero (CNa, CP, base/plume interaction) is UNKNOWN to better than ~x2: only CFD (05-plume-cfd), a wind tunnel or flight data close it. The MC draws the whole band. TO-VERIFY T1403

11.2 Mission plan

Item	Value
A1 role and profile	BASELINE flight REQ T1409: A-D: TVC ascent, pre-MECO throttle-down, nose-first coast with aft fins, nose drogue at apogee, NC-closed relight, throttled TVC landing REQ T1410
A1 apogee target (newest lead decision)	**285** m LEAD T1406 A1 apogee target 285 m ADOPTED under line 168 (4) (0 exhausted runs at the 73 g purge; cost 15 m). LEAD T1405
A1 apogee target in gnc.json v3 (planning value, superseded; §13.3 C-08)	~~300 m~~ REQ ~~T1404~~
A1 apogee trim rule	A1 tail draws: the re-fly flies the purge at the ECU cap (<= 73.0 g) plus the +0.60 kg valve-mass sensitivity and sizes the A1 apogee trim that removes new exhaustions; a trim to >= 275 m is ADOPTED without further decision (300 m was a lead default), a larger trim comes back to the lead. LEAD T1407
Launch-commit condition added (newest lead decision)	Pc VALID is now a launch-commit condition. LEAD T1408
A1 nominal: apogee / relight height / burn time / N2O burned / reserve	**303.5** m SIM T1411 / **64.7** m SIM T1412 / **7.40** s SIM T1413 / **1.177** kg SIM T1414 / **0.398** SIM T1415
B1 role	development hop (continuous burn, 20 % floor) REQ T1416
B1 hop cap	**62** m REQ T1417 — 62 m PROVISIONAL (lead 2026-10-13: the GNC 62.9 m rounded down) until the T1-T4 regression data; 85 m (v3 planning value) kept for sizing (b1_regression_band.lead_cap) LEAD T1418
B1 sizing height	**85** m REQ T1419
B1 cap rule	Cap rule ADOPTED as the method to set the B1 cap after T1-T4; 62 m stays provisional and holds only if the corner k >= 1.00 (k 0.97 → 60.0 m). LEAD T1420
B2	development hop (unchanged) REQ T1421, **50** m REQ T1422
Order of first flights	30-100 m (charter): B2 then B1, before A1 REQ T1423
Launch commit (wind)	Launch commit PROVISIONAL for A1 and B1: W10 < 6 m/s AND sigma_u(3 m) <= 1.0 m/s, 10-min statistics before ARM, anemometer at 3 m; supersedes the interim 'B1 mean wind < 6 m/s'; ACEMA site data and B-PNT-1 TO-DEFINE. LEAD T1424
Mission statement (REQUIREMENTS)	Mission B1: hop cap 62 m PROVISIONAL (was 85 m) until T1-T4 regression data; B1 flies only after the T1-T4 regression law, with a burn limit at t(web = 1 mm) using 1.1 x measured a. 85 m dispersed cycles remain the thermal and erosion sizing cases. LEAD T1425

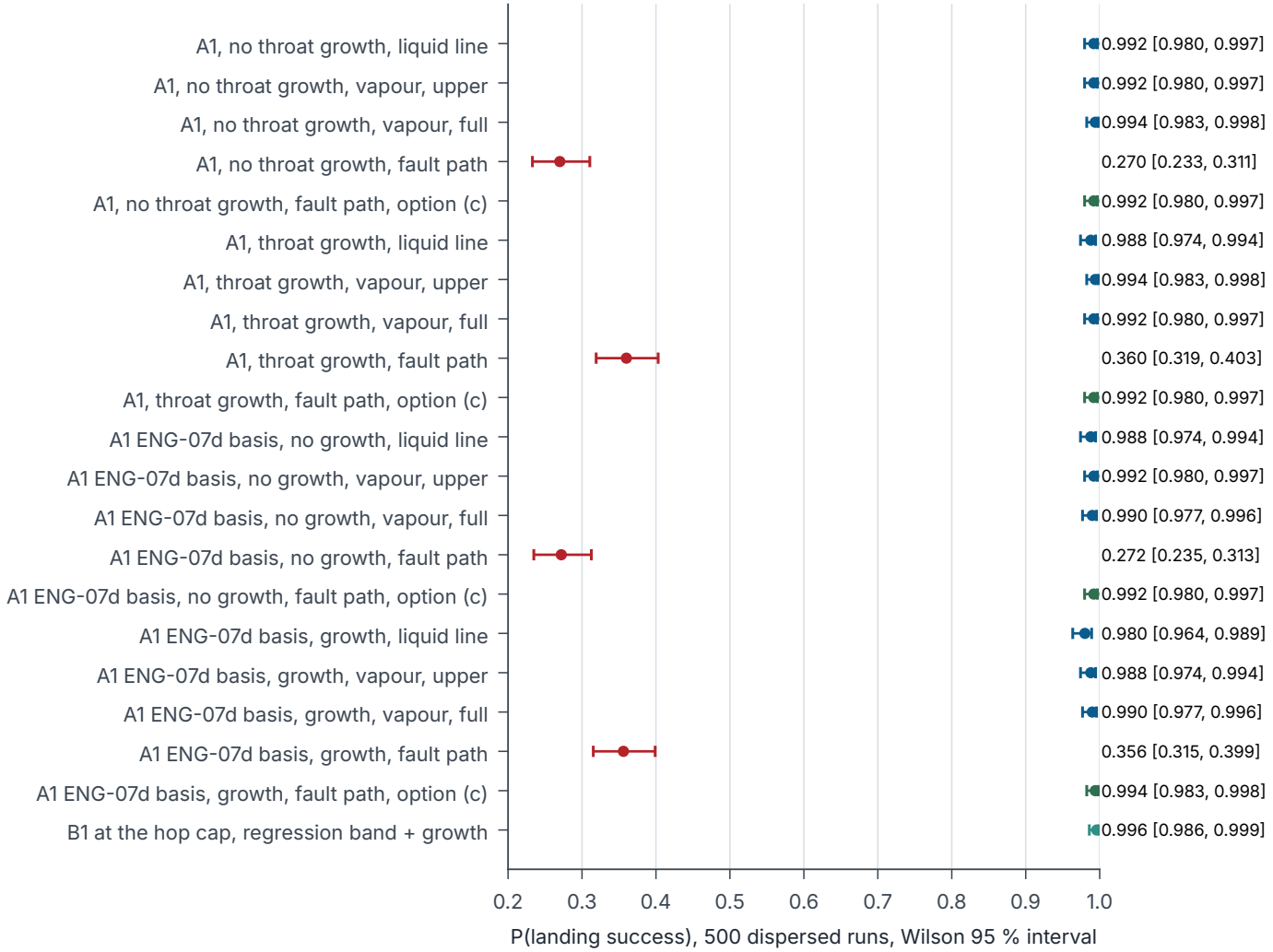
11.3 Landing Monte Carlo

P(landing success), A1 relight at the planning value 0.81 s (T1468); option (c) plans 1.41 s (T493) on an UNPRIMED flag. Wilson 95 % intervals DERIVED by the generator.

Basis	Liquid line	Vapour, upper	Vapour, full	Fault path (plan value)	Fault path, option (c)

Basis	Liquid line	Vapour, upper	Vapour, full	Fault path (plan value)	Fault path, option (c)
A1, no throat growth	**0.992** SIM T1428 [0.980–0.997 T1429]	**0.992** SIM T1430 [0.980–0.997 T1431]	**0.994** SIM T1432 [0.983–0.998 T1433]	**0.270** SIM T1434 [0.233–0.311 T1435]	**0.992** SIM T1436 [0.980–0.997 T1437]
A1, throat growth	**0.988** SIM T1438 [0.974–0.995 T1439]	**0.994** SIM T1440 [0.983–0.998 T1441]	**0.992** SIM T1442 [0.980–0.997 T1443]	**0.360** SIM T1444 [0.319–0.403 T1445]	**0.992** SIM T1446 [0.980–0.997 T1447]
A1 ENG-07d basis, no growth	**0.988** SIM T1448 [0.974–0.995 T1449]	**0.992** SIM T1450 [0.980–0.997 T1451]	**0.990** SIM T1452 [0.977–0.996 T1453]	**0.272** SIM T1454 [0.235–0.313 T1455]	**0.992** SIM T1456 [0.980–0.997 T1457]
A1 ENG-07d basis, growth	**0.980** SIM T1458 [0.964–0.989 T1459]	**0.988** SIM T1460 [0.974–0.995 T1461]	**0.990** SIM T1462 [0.977–0.996 T1463]	**0.356** SIM T1464 [0.315–0.399 T1465]	**0.994** SIM T1466 [0.983–0.998 T1467]

Case	Value
B1 at the provisional hop cap: P with regression band and throat growth [interval as published]	**0.996** SIM T1469 [**0.986** T1470, **0.999** T1471]
B1 finding	B1 at the lead's 62 m cap: the nominal flight closes the +25 % rule at the worst band corner (a0.6|n0.68) with g 0.11 (reserve 25.8 %) and at every other corner (corner caps 62.9 m to >= 85 m, the search bound); 500 runs with band + growth: P 0.996 [0.986, 0.999] vs growth only 0.996 (2 lost / 2 gained vs growth only at 62 m, McNemar p = 1), 0 runs exhaust the liquid, usable liquid p01 0.188 kg (growth only 0.261 … SIM T1472



Landing success by basis and line state at the relight (6-DOF Monte Carlo, SIM). Red: drained-segment fault path at the planning value; green: the same fault path with option (c). Data: T1427, T1469

Item	Value
Monte Carlo basis	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN Electronics v2.8 sequence + REQ lead decisions 2026-10-13 SIM T1426
Planning status	planning value 0.81 s ADOPTED by the lead (2026-10-13, the GNC rule; the 0.6 s + sequence rule withdrawn); option (c) (1.41 s on an ECU UNPRIMED flag) ADOPTED PROVISIONALLY and verified paired here; the stale / missing-flag plan is the GNC decision the lead asked for (option_c_fault_aware_plan.stale_or_missing_flag) LEAD T1473

Item	Value
Not physical acceptance	simulation and analysis only; no item here is a bench or flight validation SIM T1474
RG-01 (relight evidence)	CONDICIONADO DERIVED T1475: RG-01 relight on the current design basis not re-flown DERIVED T1476
RG-01 re-fly	GNC RG-01 re-fly started on these tables LEAD T1477

RG-01 OPEN. The A1 relight probabilities above come from the v2.8 relight study on the earlier segment basis. The re-fly on the decided purge path and the re-routed coil (CHANGELOG 164) had not been published when this draft was generated; the values are planning evidence, not a relight claim. CF-4 is the physical gate.

12 Verification status

> **Nothing in this datasheet is physical approval.** Independent review verdicts for the engine team's digital deliverables. A verdict is not physical approval: no hardware has been tested. DESIGN T364 No value here is a measurement. Review verdicts, matching file hashes, clean CAD checks, passing unit tests and this document's own trace check are digital evidence only.

12.1 Review register

reviews/SUPERVISION.json (9736f4974377a7bd), 7 entries in register order. A verdict covers the hashed files only.

Review	Deliverable	Verdict	What it is not; file state
REV-COMB-1 T1478 2026-10-12/13	02-combustion (REPORT.md v2.3, combustion.json v2.3, HANDBACK-SUMMARY.md) DERIVED T1481	**REVISE** DERIVED T1480 Blocking: RC-01 ignition/abort logic; RC-02 hard-start bound DERIVED T1482	Not physical approval; not a review of any file changed after the listed hashes; the reviewed files have since changed, so the verdict covers an earlier state; not acceptance of the open findings. 0 of 3 reviewed files unchanged; changed since review: 02-combustion/REPORT.md, interfaces/combustion.json, 02-combustion/HANDBACK-SUMMARY.md T1483
REV-PLUME-1 T1049 2026-10-12/13	05-plume-cfd (REPORT.md) DERIVED T1485	**ACCEPT WITH COMMENTS** DERIVED T1050	Not physical approval; not a review of any file changed after the listed hashes; the reviewed files have since changed, so the verdict covers an earlier state. 0 of 1 reviewed files unchanged; changed since review: 05-plume-cfd/REPORT.md T1486
REV-GNC-1 T1487 2026-10-13	01-systems-gnc v3 (REPORT-v3.md, gnc.json v3) with addendum on v3.1 staging (FC_THRUST) DERIVED T1490	**REVISE** DERIVED T1489 Blocking: RG-01 relight evidence on superseded 21.4 cm3 segment; flag blind to vapour DERIVED T1491	Not physical approval; not a review of any file changed after the listed hashes; not acceptance of the open findings. 5 of 5 reviewed files unchanged T1492
REV-ELEC-1 T1493 2026-10-13	06-electronics v2.9 (firmware v2.9, SIL sil_v2p9, KiCad ECU v2.4, BOM v2.1, bench_plan_v2.md, REPORT-v2.9-HANDBACK.md, interfaces/electronics.json v2) DERIVED T1495	**REVISE** DERIVED T1112 Blocking: RE-01 A5 still an abort; spike source not in SIL; RE-02 BASE_AMB fallback gives false E2 on positive offset; RE-03 b7 not filtered-only; RE-04 S5 up-step overshoot reaches the 30 bar abort; RE-05 bleed logic and hardware vs I.160-161 DERIVED T1113	Not physical approval; not a review of any file changed after the listed hashes; not acceptance of the open findings. 10 of 10 reviewed files unchanged T1496 Lead disposition: CHANGELOG line 166 LEAD T1497
REV-GNC-2 T1498 2026-10-13	01-systems-gnc/staging_v3.1/rev1 (REPORT-GNC-rev1.md, rev1_items.json, FSW record v1.4, sim6 v2.8-dev MWFPRS) responding to REV-GNC-1 DERIVED T1500	**CONDICIONADO** DERIVED T1475 Blocking: RG-01 relight on the current design basis not re-flown DERIVED T1476	Not physical approval; not a review of any file changed after the listed hashes; not acceptance of the open findings. 6 of 6 reviewed files unchanged T1501 Lead disposition: CHANGELOG line 168 LEAD T1502
REV-COMB-2 T1503 2026-10-13	02-combustion v2.4 (REPORT.md, REVISION-RESPONSE.md, CF-4-SPEC.md, T1-T4-REGRESSION-SPEC.md, combustion.json v2.4) DERIVED T1506	**REVISE (targeted)** DERIVED T1505	Not physical approval; not a review of any file changed after the listed hashes. 5 of 5 reviewed files unchanged T1507 Lead disposition: CHANGELOG line 172 LEAD T1508
REV-PLUME-2 T1053 2026-10-13	05-plume-cfd v2.2b (REPORT.md) DERIVED T1510	**ACCEPT** DERIVED T1054	Not physical approval; not a review of any file changed after the listed hashes. 1 of 1 reviewed files unchanged T1511 Lead disposition: CHANGELOG line 172 LEAD T1512

12.2 Open items and blockers

Item	Status	Evidence (quoted)
RG-01 A1 relight evidence on the current design basis (rev2 re-fly staged, SIM; independent review REV-GNC-3 pending)	**BLOCKING**	RG-01 relight on the current design basis not re-flown DERIVED T1476 GNC RG-01 re-fly started on these tables LEAD T1477 REV-GNC-1 reviewer resumed for the rev1 response; RG-01 stays BLOCKING until the re-fly on Combustion's re-routed-coil tables and CF-4 LEAD T1514 GNC rev2 (response to REV-GNC-2 and lead lines 164-172) STAGED in 01-systems-gnc/staging_v3.1/rev2 (280 files; rev1 and interfaces/gnc.json untouched). RG-01 re-fly on ONE flight FSW (FSW-R2) with the decided purge sequence (sim6 v2.9-dev amendment G: PURGE ... SIM T1515 Independent review REV-GNC-3 of rev2 and amendments G, K, V launched (max). gnc.json v3.1 is published only after that review LEAD T1516

Item	Status	Evidence (quoted)
FA-1 Fuel-assisted relight pulse above proof at its upper bound (single control)	**DESIGN CONTINUATION ONLY; margins reported, not reviewed**	fuel-assisted fast part at an A1 relight, 48.9-75.5 bar[OVR-07] at 1 ms OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs DERIVED T1517 FA-1 stays accepted for design continuation only; flight acceptance needs IGN-1 and Mechanical's 75.5 bar[OVR-07] <= 3 ms pulse check. OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs LEAD T915 R2-03 FA-1 load restated as an ENVELOPE: Combustion publishes O/F 4-12 x tf 1-7 ms (1 ms steps) x t_res 0.02/0.06 x pool none/q 2 MW/m2 (head and aft peak, rise time, time above 45 and 67.5 bar, impulse). Until it lands Mechanical checks with an INTERIM load: ... LEAD T024 DECISION: FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts (covers every baseline peak and the shell); for any closure, bolt, J1-lip or other part with a natural period of 0.25-1 ms Mechanical reports margins also at 92 bar abs ... LEAD T515 FA-1 final load rule (lead 172 (2) + 173) closed in S-AX/joints: 90 bar abs quasi-static head+aft every part, 92/102 bar for the 0.25-1 ms parts (modes 1-2: TC-09+nozzle stack 0.33-0.47 ms, TC-02 head on its screws 0.27-0.39 ms; shell/bonnet 0.04-0.08 ms ... SIM T557 PENDING PENDING T559
CF-4 Relight-purge cold flow (gate for the A1 relight claim)	**NOT RUN (founders)**	A1 relight lights within W 0.65 s with P_abort <= 1e-3, venting <= 76 g, theta <= 2 deg LEAD T1518 CF-4 spec ACCEPTED as the gate for the relight claim (founders execute); Combustion publishes it as a hub document at handback. LEAD T1519
IGN-1 Torch reliability, photodiode false positives, light spike on hardware	**NOT RUN (founders)**	**Record at 1 kHz:** raw Pc and filtered Pc; P0 (the log v10 `ign_ref_hPa`); the rise (`ign_rise_hPa`); the `det` flags; `tlm_flags`; the plant light time from a photodiode or the torch TC. **Evidence wanted:** Lit starts: rise above P0 against time. SIL: E2 ... DESIGN T1520 A5 DELETED as an abort, flight and stand; replaced by a LIGHT_SPIKE log event (raw and filtered peak, time above 8 and 30 bar abs, time from light) so IGN-1/T1 measure whether real lights spike. LEAD T1521
B-PC-1 Pc chain residual and drift; sets the light-rule floor	**NOT RUN (founders)**	post-zero residual: sets `kBpc1Shown` (floor 1.8 → 2.0 bar abs, baseline check 0.7 → 0.4 bar) DESIGN T1522 B-PC-1 measures zero TC and 20 s drift (Mechanical supplies the transducer temperature MECO→RELIGHT); LEAD T1523 RE-06 HAL is an Electronics open item, owner Electronics, required before B-PC-1. LEAD T1524
BLEED-MASS Head-end bleed valve mass (MASS RISK, not baseline)	**TO-VERIFY**	F1 valve mass: 2 × 0.30 kg is held as a MASS RISK, not baseline; Electronics' 12 V NC valve search reports mass with a target <= 0.15 kg per valve; GNC flies +0.60 kg as a sensitivity. LEAD T341 Burkert 6027 HP: 24 V DC listed, ECU valve rail +12V_ARM (U3V70F12); 2 × 16 W → +2.7 A for ~2.6 s per purge at 12 V; 12 V coil or a 24 V boost, rail budget and valve mass TO-VERIFY; HP seat PEEK/FKM medium -10..80 C and N2O compatibility TO-VERIFY TO-VERIFY T1381 +0.60 kg bleed-valve mass: no trim below 275 m now; the trigger is the selected valve pair: if it adds > 0.30 kg, GNC re-flies and the lead chooses between a trim below 275 m and a propellant/landing-reserve change. LEAD T1525
ENG-11 Gimbal axis stiffness and rocking mode	**NOT MET (structure only)**	first rocking mode 16.0-27.8 Hz (excl. J_int) vs the 25 Hz minimum: below it whenever the servo is at the low end of its ASSUMED stiffness; the servo caps the total (servo alone through a rigid structure 16.2-28.7 Hz incl. pose); lead route: ENG-07d closes by ... DERIVED T613
ENG-07 Gimbal slew at the v2 hinge load	**MARGINAL**	as built: worst-direction slew at 7 deg 111.2 deg/s at 7.4 V, 118.3 at 7.6 V (>= 100); 95.3 deg/s at the 6.6 V degraded plug voltage, the case GNC validated for n 4.8 (lead FINAL: P(landing) 0.982 at 6.6 V) DERIVED T606
FEA Structural margins of the pressure parts and gimbal	**PENDING**	PENDING PENDING T1526
ENVELOPE Swept radial envelope at the airframe wall	**TO-VERIFY**	z0_mm: 105.0; z1_mm: 110.0; r_engine_swept_mm: 60.08; by_engine: ENG-FD-03#0; r_ring_swept_mm: 0.0; r_fixed_mm: 71.8; by_fixed: ENG-FD-11#0; r_max_mm: 71.8; clearance_mm: -1.8 TO-VERIFY T635
GRAIN-R4 Rounded port entrance r ≥ 4 mm shown in CAD/BOM	**TO-VERIFY**	not stated in the BOM name or notes of ENG-GR-01/02: rounded entrance TO-VERIFY in CAD TO-VERIFY T833

Item	Status	Evidence (quoted)
REV-COMB-2 Independent re-review of combustion v2.4	**REVISE (targeted)**	R2-01 line 166 (3)/(4) not implemented; R2-02 two stand safing rules; R2-03 FA-1 pulse single point at ASSUMED O/F; R2-04 T1-T4 law test cannot resolve; default gate less safe DERIVED T1513 CHANGELOG line 172 LEAD T1508 R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion v2.4a text/interface patch (relief_v2p4 block added); the ENG-09 "trapped-liquid relief <= 7.5 MPa" text is superseded by the tank relief 6.9 MPa / MAWP 8.42 MPa / … LEAD T017
COMB-v2.4a Combustion interface patch for the stale values (override table §13.1)	**IN PROGRESS**	R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion v2.4a text/interface patch (relief_v2p4 block added); the ENG-09 "trapped-liquid relief <= 7.5 MPa" text is superseded by the tank relief 6.9 MPa / MAWP 8.42 MPa / … LEAD T017 ovr-3: 9 active, 0 redundant, 0 in error; 4 superseded values withheld (4 interface keys); see §13.1 and §13.2 DERIVED T352
INTERLOCK-GB Guard band of the tank interlock (channel uncertainty)	**TO-DEFINE**	R2-07 interlock guard band: T_tank and P_tank thresholds sit below 24 C / 55.28 bar abs by the total channel uncertainty (Electronics states it); CF-1/CF-3 acceptance includes full-stroke Cd <= 0.80 and line loss >= model at the failure flow, otherwise MEOP … LEAD T014
ELEC-v2.10 Electronics v2.10 answering REV-ELEC-1	**IN PROGRESS**	RE-01 A5 still an abort; spike source not in SIL; RE-02 BASE_AMB fallback gives false E2 on positive offset; RE-03 b7 not filtered-only; RE-04 S5 up-step overshoot reaches the 30 bar abort; RE-05 bleed logic and hardware vs I.160-161 DERIVED T1113 Electronics v2.10 publishes to the hub after archiving v2.9 LEAD T1114

12.3 Requirement verification methods

00-requirements/REQUIREMENTS.md §2 and its amendments (REQ rev 2). No requirement has been verified physically.

ID	Requirement (rev 2 text, start)	Amendments after rev 2 (sentences naming the ID)	Verification method
ENG-01	Max thrust 500 N at sea level, with N2O at ~~15–25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04**. Normal Pc is capped at 25 bar in software. REQ T356	ENG-01 at the cap (CHANGELOG 170): the 25 bar software Pc cap stays; if T1 measures the low end of the eta_cf band (0.95), rated thrust is ~487 N at 25 bar (DERIVED) and the measured thrust coefficient k feeds the GNC cap rule. REQ T1543	Analysis, then static fire REQ T1542
ENG-02	Throttle floor ≤ 20 % of max (5:1, 100–500 N) with stable combustion. Stretch: 12.5 % (8:1). Throttle by a servo-driven variable-area pintle, with a separate NC isolation valve for shutoff. DERIVED T569	none	Analysis, cold flow, static fire T3 with 1 kHz Pc REQ T1544
ENG-03	Throttle response: t10–90 ≤ max(0.15 s, 0.8·ΔF / 1000 N/s). Ramp (amended 2026-10-13, lead on GNC eng03_rate_v1.1): closing ≥ 2500 N/s in every state; opening ≥ the HV69 corner rate map … DERIVED T580	none	SIL, B-PNT-1, static fire REQ T1545
ENG-04	Restartable: ≥ 3 starts per flight. Command to 90 % thrust ≤ 0.5 s (target 0.3 s), repeatability ±50 ms; flight software plans relight on 0.81 s (amended 2026-10-13; the 0.6 s rule is withdrawn). DEVIATION recorded 2026-10-13: … DERIVED T491	ENG-04 / relight: deterministic relight (RELIGHT at planned T_fire; REQ T1547	Analysis, static fire REQ T1546
ENG-05	Shutdown: command to < 10 % thrust ≤ 0.10 s. Met with a mandatory pre-MECO throttle-down to ≤ 75 % (a 100 % MECO at 3.5 MPa tank fails at 0.106–0.124 s). No flashback or decomposition in the feed. NC main valve closed during … DERIVED T588	ENG-05 / hard start (stand): unignited-inventory allowable set by the two-volume relief analysis; stand rule R1-R4; stand relief per the 2026-10-13 decision, fallback (b) ADOPTED: R1 = R2 = 0.35 s windows, R3 deleted, rounded grain port entrance r >= 4 mm, … REQ T1549	Analysis, B-PNT-1 closing time at 25 bar, static fire REQ T1548
ENG-06	Accumulated burn per flight sized by the duty cycles in gnc.json v3. Grain replaced each flight; web ≥ 1 mm over the whole regression-law envelope (a ×0.6–1.5, n 0.35–0.68). Throat insert: plan on one insert per flight until … ASSUMED T496	none	Analysis, static fire bore measurement REQ T1550
ENG-07	TVC by a two-axis gimbal of the whole engine, pivot at the engine-group mid-burn CG. Range ±7° on the vector norm, 8° conical mechanical stop. Slew ≥ 100°/s at 7° in the worst vector direction at the 7.4 V nominal servo plug … LEAD T359	none	SIL, B-TVC-1 REQ T1551

ID	Requirement (rev 2 text, start)	Amendments after rev 2 (sentences naming the ID)	Verification method
ENG-07b	No actuator saturation episode longer than 0.1 s during the landing burn. Logged in flight as SAT_LONG. DERIVED T1552	none	MC, flight logs REQ T1553
ENG-07c	Reflected servo rotor inertia J_int ≤ 1e-3 kg m² at the output (provisional), unless a saturation-aware controller meets ENG-07b up to 2.7e-3. Estimated value for the SB-2292SG: 1.6e-3 (0.76–2.7e-3). DERIVED T1554	none	B-TVC-1 REQ T1555
ENG-07d	Flex gain margin ≥ 6 dB at the gimbal rocking mode. ECU notch required, with separate alpha and beta centre frequencies set from B-TVC-1. DERIVED T1556	none	CSI analysis, B-TVC-1 REQ T1557
ENG-08	Gimbal sweep fits the airframe ICD: bay ID ≥ 140 mm, zone [−0.238, +0.252] m. Feed across the gimbal (amended 2026-10-13; Option A/O3b supersedes the 0.40 m radial line): coil on the case, fixed 3/8 wall line at az 219.5° clamped ... REQ T1558	none	CAD, B-FD-1 REQ T1559
ENG-09	Pressure parts: SF ≥ 1.5 on yield and ≥ 2.0 on ultimate, at MEOP and at temperature. Chamber MEOP 45 bar (~~stuck-pintle case, 30 °C tank~~ ignition interlock T_tank <= 24 C; basis 44.33 bar (meop_basis_v2p4) **OVR-06**), proof 67.5 bar, which covers the blocked-nozzle bound of about 63 bar below yield. Normal Pc ≤ 25 bar; ECU over-pressure abort at about 30 bar. Tank MEOP 7.0 MPa, with a passive relief device (relief valve and/or burst disc, set below proof, T-vent). ~~Trapped-liquid relief ≤ 7.5 MPa.~~ tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check (Cv >= 0.1), disc 9.5 MPa +/- 5 % **OVR-05** Chamber Pc transducer range ≥ 70 bar. LEAD T568	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. REQ T1561 ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check Cv >= 0.1, secondary disc 9.5 MPa +/- 5 %. REQ T1562	FEA, hydro proof REQ T1560
ENG-10	Structure (gimbal, mounts, fasteners): SF ≥ 1.5 on yield and ≥ 2.0 on ultimate under design limit loads. Gimbal design loads: engine moment 35.3 N·m (both servos at 8.4 V stall); rod force 441 N at centre, 552 N at 7°, 609 N for ... LEAD T607	none	FEA, hand calculations REQ T1563
ENG-11	Gimbal axis stiffness (structure plus rods) ≥ 3,100 N·m/rad per axis at every pose (hard minimum); goal 5,200 N·m/rad. Alpha/beta spread across poses ≤ ±20 %. Rocking mode as required by ENG-07d. DERIVED T611	none	Modal and stiffness FEA, B-TVC-1 REQ T1564
ENG-12	Propellants: self-pressurised N2O, tank 2.00 kg / 3.17 L conditioned to ~~15–25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04**, with heater hard-disabled at ARM; N2 supercharge not baselined. 3D-printed ABS grain (PETG backup): straight centred 25 mm port, 53.5 OD × 230 mm, 0.404 kg. Nozzle ε 2.0 with a fine-grain graphite insert. REQ T358	ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM. REQ T1566	Trade studies, static fire REQ T1565
ENG-13	Engine controller (ECU) works with a Teensy-class flight computer. It provides closed-loop thrust and gimbal control, telemetry, arming interlocks and logging. ARM requires predicted both-servo-stall plug voltage ≥ 6.6 V and ... DERIVED T1567	none	Bench, SIL, HIL REQ T1568
ENG-14	Cost: cheapest viable. Every BOM line has a source and price, or is marked TO-QUOTE with a named vendor (delivery: Sønderborg, DK-6400). No zero or "from" prices. REQ T1569	none	BOM review REQ T1570
ENG-15	Manufacturability: printed parts on a hobby FDM printer; metal parts from stock tube and bar by online CNC or a simple lathe; a PETG fit-check version of every part. REQ T1571	none	CAD review REQ T1572
ENG-16	Safety: N2O-compatible, oxygen-clean wetted paths (PFPE lubricant only); passive relief plus remote vent; arming plug; remote firing; written hazard list. Cold flows only into a back-pressured catch volume or with N2 at ≤ 3.5 ... REQ T1573	none	Review, bench REQ T1574

12.4 Founder test list

Physical tests executed by the founders. None has been run. Cost: TO-QUOTE, delivered to Sønderborg, DK-6400.

Kind	Test	Gate or measurement	Needs / source
Bench step 1	B-PWR-1, B-PWR-2 DESIGN T1575	rails, e-fuse, ARM interlock DESIGN T1576	bare ECU, bench supply DESIGN T1577
Bench step 2	B-ECU-1 DESIGN T1578	I/O, CAN, watchdog, log v10 DESIGN T1579	step 1 DESIGN T1580
Bench step 3	B-PC-1 DESIGN T1581	post-zero residual: sets `kBpc1Shown` (floor 1.8 → 2.0 bar abs, baseline check 0.7 → 0.4 bar) DESIGN T1522	step 2, N2 0-70 bar g, reference gauge DESIGN T1582
Bench step 4	B-MV-1, B-PNT-1, B-TVC-1 DESIGN T1583	actuator timing, ENG-03, MV switch ladder DESIGN T1584	step 2 DESIGN T1585
Bench step 5	B-BLEED-1 (new) DESIGN T1586	head-end bleed: pad BIT, P_mid comparator, V1 closure confirmation time DESIGN T1587	steps 2-4, valves + tee DESIGN T1588
Bench step 6	B-IGN-1, B-FC-1 (+ drogue telemetry) DESIGN T1589	spark EMI on Pc, FC link, TLM_FAST d1 bits DESIGN T1590	step 2, FC DESIGN T1591
Bench step 7	HIL-OT-1, HIL-TR-1 DESIGN T1592	over-thrust and transient replay DESIGN T1593	steps 2, 6 DESIGN T1594
Bench step 8	LEAK-1, CF-3, CF-4 DESIGN T1595	feed leak, ignition-lift table, segment void DESIGN T1596	Mechanical / Combustion rigs DESIGN T1597
Bench step 9	IGN-1, T1-T4 DESIGN T1598	relative light rule on hardware, start latency, F_acc DESIGN T1599	hot-fire stand DESIGN T1600
Early purchase (REQ §4)	**B-PNT-1**	Commanded travel ≥ 103°; unpowered spring closing at the MEOP-equivalent load; guide μ ≤ 0.20; break-out after a dwell; closing time at 25 bar; HV69 torque-speed under load (0.42-0.55 N·m at 7.335 and 7.49 V) against … REQ T1602	1× MKS HV69 (EUR 62.99 at Lindinger, catalogue price incl. AT VAT; TO-QUOTE to DK-6400) REQ T1603
Early purchase (REQ §4)	**B-TVC-1**	Measure servo output stiffness K_s at 6.6 and 7.4 V, damping, J_int and the rotor/structure split; stall-hold heating. If K_s < ~100 N·m/rad, switch to the actuator contingency (maxon ECX 22 M class). Reduced landing … REQ T1605	2× SB-2292SG (TO-QUOTE) with the n 4.8 linkage on a 0.0545 kg·m² inertia dummy REQ T1606
Early purchase (REQ §4)	**B-PWR-1/2**	Source and branch resistance against the 15 / 20 mΩ assumptions. REQ T1608	Flight battery pack and harness REQ T1609
Early purchase (REQ §4)	**B-FD-1**	Hinge stiffness while pressurised ≤ 20 N·m/rad per axis. REQ T1611	Gimbal hose REQ T1612
CF-4 pass criterion	**CF-4**	P1 liquid at the pintle inlet <= 0.293 s after reaching the lift LEAD T1613	02-combustion/CF-4-SPEC.md 1.0
CF-4 pass criterion	**CF-4**	P2 >= 0.35 s continuous liquid or inside the void budget LEAD T1614	02-combustion/CF-4-SPEC.md 1.0
CF-4 pass criterion	**CF-4**	P3 cap clears FD-01 crown <= 2.3 s after flow start LEAD T1615	02-combustion/CF-4-SPEC.md 1.0
CF-4 pass criterion	**CF-4**	P4 each tap >= 80 % of model flow LEAD T1616	02-combustion/CF-4-SPEC.md 1.0
CF-4 pass criterion	**CF-4**	P5 <= 76 g vented (IS-1, IS-2) LEAD T1617	02-combustion/CF-4-SPEC.md 1.0
CF-4 pass criterion	**CF-4**	P6 ~~closure by E0 − 0.05 s~~ E1 = torch confirmed (2oo2) AND (closure confirmed OR deadline <= torch gate); in flight the lift proceeds with BLEED_FAULT (FC LOG-ONLY); stand: abort **OVR-02** and BLEED_FAULT on a held-open valve LEAD T1618	02-combustion/CF-4-SPEC.md 1.0
CF-4 pass criterion	**CF-4**	P7 no unfiltered path to the seat LEAD T1619	02-combustion/CF-4-SPEC.md 1.0
T1–T4 measurement	**T1–T4**	Grain mass before and after each firing (balance resolution <= 0.1 g), dried to constant mass. DESIGN T1620	02-combustion/T1-T4-REGRESSION-SPEC.md v2.4
T1–T4 measurement	**T1–T4**	Port diameter at six stations, x/L = 0.10, 0.30, 0.50, 0.70, 0.90 and 0.97 from the head end, on two orthogonal diameters (bore gauge, resolution <= 0.02 mm), or a 3-D scan of the bore with the same stations extracted. x/L 0.97 is where the total-flux law … DESIGN T1621	02-combustion/T1-T4-REGRESSION-SPEC.md v2.4

Kind	Test	Gate or measurement	Needs / source
T1–T4 measurement	**T1–T4**	Time histories: Pc at 1 kHz, oxidiser mass flow from the tank load cell and P_feed, thrust, burn time from the Pc rise and fall (E2 to the tail-off at 10 % of the run Pc). **DESIGN** T1622	02-combustion/T1-T4-REGRESSION-SPEC.md v2.4
T1–T4 measurement	**T1–T4**	At least two oxidiser fluxes: T1 at 100 % and the T2 throttle steps, so that n is fitted rather than assumed. **DESIGN** T1623	02-combustion/T1-T4-REGRESSION-SPEC.md v2.4

13 Decision overrides and open contradictions

13.1 Decision overrides

Table gen/decision_overrides.json (ovr-3, sha256/16 3666334a809ccf6d). Each decision is quoted from its source; the generator checks that the cited text contains the key phrases and every number of the decision, and applies a target only while the stale wording is still in it. An entry with no active target is reported REDUNDANT and can be dropped.

Override	Decision shown	Cited sources (quoted)	Targets: key, mode, state, displays
OVR-01 30 bar over-pressure abort: channel and persistence basis: REV-COMB-2 R2-01	30 bar firmware abort on the FILTERED channel with 5 ms persistence (amends 'raw' in CHANGELOG 160 (2)) LEAD T004 **ACTIVE** T1640	CHANGELOG.md L166(4) RE-20: the 30 bar firmware abort runs on the FILTERED channel with 5 ms persistence (probe F: comparator stays first; Combustion: filtered light-spike peak <= 21.4 bar); amends the word "raw" in l.160 (2). LEAD T001 REQUIREMENTS.md §Amendments after rev 2 bullet "Electronics review decisions (CHANGELOG 166)" [sentence containing "filtered channel, 5 ms"] Electronics review decisions (CHANGELOG 166): light-rule baseline never substituted by ambient on the high side (pad/stand refuse with PC_SENSOR, A1 relight proceeds on measured P0); nominal-profile plant Pc <= 28 bar abs at p99.9 with zero PC_OVERPRESSURE in S5 (30 bar abort unchanged, filtered channel, 5 ms); REQ T003	combustion.json /abort_matrix_v2p4/v/A4: annotate, ACTIVE, shown 1× CHANGELOG.md L160(2): annotate, ACTIVE, shown 1× T1639
OVR-02 E1 gate and bleed closure deadline basis: REV-COMB-2 R2-01	E1 = torch confirmed (2oo2) AND (closure confirmed OR closure deadline reached), deadline <= the torch gate; at the deadline in flight the lift proceeds with BLEED_FAULT to the FC (FC action LOG-ONLY); on the stand it stays an abort LEAD T007 **ACTIVE** T1642	CHANGELOG.md L166(3)[sentence containing "E1 = torch confirmed"] RE-05: E1 = torch confirmed (2oo2, FA-1 (a)) AND [closure confirmed OR closure deadline reached]; the deadline is <= the torch gate; at the deadline in flight the lift proceeds, BLEED_FAULT goes to the FC and V2 is commanded closed; on the stand it stays an abort. LEAD T005 CHANGELOG.md L168(2) FC action on BLEED_FAULT is LOG-ONLY until a leak response is flown and reviewed. LEAD T006	combustion.json /igniter/sequence/v/start/0/action: annotate, ACTIVE, shown 1× combustion.json /igniter/sequence/v/start/3/action: append, ACTIVE, shown 1× combustion.json /abort_matrix_v2p4/v/A14: annotate, ACTIVE, shown 1× combustion.json /cf4_gate/v/pass_criteria/5: annotate, ACTIVE, shown 1× T1641
OVR-03 Stand safing after a no-light basis: REV-COMB-2 R2-02	N2 purge >= 60 s at >= 10 g/s, then hold >= 15 min before anyone approaches (Pc at ambient, head TCs falling), then safing; 'wait >= 5 s' is withdrawn LEAD T011 **ACTIVE** T1644	CHANGELOG.md L172(1) R2-02 ONE stand safing rule after a no-light, effective now: N2 purge >= 60 s at >= 10 g/s, then hold >= 15 min before anyone approaches, with Pc at ambient and head TCs falling, then safing; "wait >= 5 s" is withdrawn everywhere. LEAD T008 REQUIREMENTS.md §Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)" [sentence containing "stand safing"] Combustion re-review decisions (CHANGELOG 172): stand safing after a no-light = N2 purge >= 60 s at >= 10 g/s, hold >= 15 min before approach (Pc ambient, head TCs falling), then safing; "wait >= 5 s" withdrawn. REQ T010	combustion.json /igniter/sequence/v/abort_if_no_ignition/1: annotate, ACTIVE, shown 1× T1643
OVR-04 Tank temperature range at ARM and ignition interlock basis: REV-COMB-2 R2-05, R2-07	tank temperature range at ARM 15-24 C (was 15-25 C); T_tank <= 24 C at ARM and at ignition; thresholds sit below 24 C / 55.28 bar abs by the channel uncertainty LEAD T015 **ACTIVE** T1646	REQUIREMENTS.md §Amendments after rev 2 bullet "ENG-12:" ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM. REQ T012 REQUIREMENTS.md §Amendments after rev 2 bullet "ENG-09:" ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. REQ T013 CHANGELOG.md L172(4) R2-07 interlock guard band: T_tank and P_tank thresholds sit below 24 C / 55.28 bar abs by the total channel uncertainty (Electronics states it); CF-1/CF-3 acceptance includes full-stroke Cd <= 0.80 and line loss >= model at the failure flow, otherwise MEOP is restated. LEAD T014	combustion.json /decisions/tank_pressurisation: annotate, ACTIVE, shown 1× combustion.json /igniter/sequence/v/arm_preconditions/0: annotate, ACTIVE, shown 1× REQUIREMENTS.md §2. Top-level row "ENG-12" col 1: annotate, ACTIVE, shown 2× REQUIREMENTS.md §2. Top-level row "ENG-01" col 1: annotate, ACTIVE, shown 2× T1645

Override	Decision shown	Cited sources (quoted)	Targets: key, mode, state, displays
OVR-05 Relief set points (feed and tank) basis: REV-COMB-2 R2-05; CHANGELOG 172 (6)	tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check MV segment to tank (Cv >= 0.1), secondary disc 9.5 MPa +/- 5 %; the trapped-liquid relief <= 7.5 MPa text is superseded LEAD T019 **ACTIVE** T1648	REQUIREMENTS.md §Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)" [sentence containing "trapped-liquid relief"] ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check Cv >= 0.1, secondary disc 9.5 MPa +/- 5 %. REQ T016 CHANGELOG.md L172(6) R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion v2.4a text/interface patch (relief_v2p4 block added); the ENG-09 "trapped-liquid relief <= 7.5 MPa" text is superseded by the tank relief 6.9 MPa / MAWP 8.42 MPa / back-relief check / 9.5 MPa disc. LEAD T017 REQUIREMENTS.md §Amendments after rev 2 bullet "Relief:" Relief: back-relief check MV segment → tank (Cv >= 0.1); secondary rupture disc 9.5 MPa +/- 5 % on FS-01, discharge overboard. REQ T018	combustion.json /valves_and_actuators/isolation_valve/trapped_liquid_relief: replace, ACTIVE, shown 1× combustion.json /valves_and_actuators/priming_rule/v/2: annotate, ACTIVE, shown 0× REQUIREMENTS.md §2. Top-level row "ENG-09" col 1: annotate, ACTIVE, shown 2× T1647
OVR-06 MEOP basis: ignition interlock instead of the v2.3 tank temperature basis: REV-COMB-2 R2-05 (RC-08 closed); CHANGELOG 172 (4)	MEOP 45 bar kept with the ignition interlock T_tank <= 24 C at ARM and at ignition; basis 44.33 bar at the interlock (meop_basis_v2p4), margin 0.67 bar LEAD T021 **ACTIVE** T1650	REQUIREMENTS.md §Amendments after rev 2 bullet "ENG-09:" ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. REQ T013 combustion.json /meop_basis_v2p4/v {"basis_bar": 44.33, "margin_to_MEOP_bar": 0.67, "proof_margin_min_bar": 21.9, "supersedes": "v2.3 basis [withheld: superseded value, SUP-01] bar (RC-08)"} SIM T020	combustion.json /decisions/chamber_MEOP_Pa/tag: replace, ACTIVE, shown 1× REQUIREMENTS.md §2. Top-level row "ENG-09" col 1: annotate, ACTIVE, shown 2× ICD-AIRFRAME.md §3. Loads row "Physical worst-case thrust" col 1: annotate, ACTIVE, shown 2× T1649
OVR-07 FA-1 relight pulse load basis: REV-COMB-2 R2-03; lead decision on the Combustion v2.4a envelope	75.5 bar is a single point at ASSUMED O/F 8. The FA-1 load is the published envelope (head peak 83.0 bar abs baseline, 88.0 upper bracket) and the FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts LEAD T026 **ACTIVE** T1652	SUP /reviews/5/major/2 R2-03 FA-1 pulse single point at ASSUMED O/F DERIVED T022 RV2 line 441 **What was specified.** Line 164 (8) asks Mechanical for a "≤ 3 ms, 75.5 bar[OVR-07] abs head-pulse check". These are the grid maxima at O/F 8 with tf ∈ {1, 5, 10} ms. At O/F 8 they reproduce within my bracket (71.2–78.6 bar, 2.5–3.5 ms). OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs DERIVED T023 CHANGELOG.md L172(2) R2-03 FA-1 load restated as an ENVELOPE: Combustion publishes O/F 4-12 x tf 1-7 ms (1 ms steps) x t_res 0.02/0.06 x pool none/q 2 MW/m2 (head and aft peak, rise time, time above 45 and 67.5 bar, impulse). Until it lands Mechanical checks with an INTERIM load: quasi-static 90 bar abs (ENG-09 ultimate design point) at ... LEAD T024 CHANGELOG.md L173 On Combustion's FA-1 envelope (02-combustion/data/v2p4a/fa1_envelope_v2p4a.json 64b328150025fa1e; 896 cases, A1 relight, O/F 3-12, tf 1-7 ms; SIM on ASSUMED construct): head peak 83.0 bar abs baseline (vapour once, rounded entrance) / 84.8 sharp / 88.0 upper bracket; aft 0.95-1.00 of head; time above 45 bar <= 3.65 ms ... LEAD T025	annotation of every displayed text quoting the single-point value: ACTIVE, shown 9× T1651
OVR-08 Relight purge commit: the flight computer no longer delays the fire time for the purge basis: GNC rev2, H-G3 refuted	Line 164 "the FC delays T_fire instead" is WITHDRAWN. PROVISIONAL rule: PURGE commits no later than plain trigger - 1.7 s; with no commit by then the purge is NOT run and the relight proceeds unpurged LEAD T030 **ACTIVE** T1654	CHANGELOG.md L175(3) Line 164 "the FC delays T_fire instead" is WITHDRAWN (commit at the plain trigger gives P 0; H-G3 refuted). PROVISIONAL rule: PURGE commits no later than plain trigger - 1.7 s (T_fire delay <= 0.6 s costs no measurable P); with no commit by then the purge is NOT run and the relight proceeds unpurged (PURGE NOT RUN LO, ... LEAD T027 REQUIREMENTS.md §Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)"[sentence containing "PURGE commits"] PURGE commits no later than the plain relight trigger - 1.7 s; otherwise no purge (T_fire is not delayed for the purge beyond 0.6 s). REQ T029	combustion.json /igniter/sequence/v/start/0/action: annotate, ACTIVE, shown 1× T1653

Override	Decision shown	Cited sources (quoted)	Targets: key, mode, state, displays

OVR-09
Drogue release with Pc invalid: accelerometer-confirmed thrust instead of the time-only term
basis: GNC rev2 amendment c
	The time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc invalid, release iff ECU RUN AND accelerometer thrust >= 37.16 N for 3 frames at 100 Hz; otherwise hold the drogue and command ECU ABORT at t_E2 + pcf_tmax, pcf_tmax 0.8 s (PROPOSAL to be flown) LEAD T033
	ACTIVE T1656
		`CHANGELOG.md L175(5)`
		Amendment c: the time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc invalid, release iff ECU RUN AND accelerometer thrust >= 37.16 N for 3 frames at 100 Hz; otherwise hold the drogue and command ECU ABORT at t_E2 + pcf_tmax, pcf_tmax 0.8 s (PROPOSAL to be flown; margins 33.0 vs 37.16 N ... LEAD T031
		`REQUIREMENTS.md §Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)"[sentence containing "accelerometer-confirmed thrust"]`
		With Pc invalid after E2 the drogue is released only on accelerometer-confirmed thrust (>= 37.16 N, 3 frames at 100 Hz) with the ECU in RUN; otherwise the drogue is held and the ECU aborted at E2 + 0.8 s (proposal, to be flown). REQ T032
			CHANGELOG.md L168(1): annotate, ACTIVE, shown 1× T1655

13.2 Superseded values withheld

Values the Combustion author withdrew in REPORT v2.4 §13 that are still live in combustion.json v2.4 (REV-COMB-2 R2-05). They are not shown in this datasheet; masked text reads "[withheld: superseded value, SUP-xx]".

ID	Shown instead	Keys withheld	Text masks	In the author's withdrawal list
SUP-01 v2.3 MEOP basis (stuck pintle, tank above the ignition interlock)	v2.4 basis /meop_basis_v2p4 (OVR-06) basis: REV-COMB-2 R2-05; combustion.json /superseded_v2p4	`combustion.json /operating_points/pressure_limits/full_stroke_failure_S230_max_bar: WITHHELD; combustion.json /operating_points/pressure_limits/full_stroke_failure_S230/v/11/Pc_bar: WITHHELD` T1657	1 displayed field masked T1658	yes T1659
SUP-02 v2.3 throat-insert life in A1 flights	one insert per flight (ENG-06 as amended) basis: REV-COMB-2 R2-05; combustion.json /superseded_v2p4	`combustion.json /thermal_and_loads/flights_per_insert_A1 : WITHHELD` T1660	0 displayed fields masked T1661	yes T1662
SUP-03 v2.3 coast seat-leak ceiling	coast leak ceiling from the Combustion report v2.4 (§7.3 of this datasheet) basis: REV-COMB-2 R2-05; combustion.json /superseded_v2p4	`combustion.json /valves_and_actuators/pintle_actuator/seat_and_seal_review_v2p3/v/b_seat/leak_tightness: WITHHELD` T1663	0 displayed fields masked T1664	yes T1665
SUP-04 ICD-AIRFRAME quotes the v2.3 MEOP-basis pressure rounded	v2.4 basis (OVR-06); worst-case thrust at that basis not published basis: same physical case as SUP-01 (ICD tag: SIM, Combustion v2.3)	`no key listed` T1666	3 displayed fields masked T1667	yes T1668

13.3 Open contradictions

Each entry cites every side. The column "shown in this datasheet" names the value used elsewhere in the document (the newest decided one). Entries are generated only when the current sources actually disagree.

ID	Sides (quoted)	Shown in this datasheet; next step
C-02 Fuel grain mass: Combustion vs CAD density	**Combustion v2.4:** 0.404 DERIVED T841 **CAD v3.4.1 mass properties:** 0.42436 DERIVED T842 **BOM GR-01 notes:** port per combustion grain v2 S230 (lead relay; combustion.json v1 was a helical port); replaced every flight (ENG-06); weigh each grain (density ASSUMED 1000 kg/m3); total mass 0.404 kg per Combustion DESIGN T1624 **ENG-12:** Propellants: self-pressurised N2O, tank 2.00 kg / 3.17 L conditioned to ~~15–25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04**, with heater hard-disabled at ARM; N2 supercharge not baselined. 3D-printed ABS grain (PETG backup): straight centred 25 mm port, 53.5 OD × 230 mm, 0.404 kg. Nozzle ε 2.0 with a fine-grain graphite insert. REQ T358	Combustion value (ENG-12). CAD material density differs from the Combustion and BOM-note density; weigh every grain (ENG-12). T1669

ID	Sides (quoted)	Shown in this datasheet; next step
C-03 30 bar over-pressure abort: persistence and channel	**firmware v2.9:** 0.003 DESIGN T1625 **CHANGELOG 160 (2):** 30 bar abort ~~raw~~ FILTERED channel, 5 ms persistence OVR-01 persistence 3 → 5 ms; comparator stays the fast path for a blocked throat (Electronics confirms the S6 blocked-throat trip still comes first). LEAD T924 **CHANGELOG 166 (4):** RE-20: the 30 bar firmware abort runs on the FILTERED channel with 5 ms persistence (probe F: comparator stays first; Combustion: filtered light-spike peak <= 21.4 bar); amends the word "raw" in l.160 (2). LEAD T001 **combustion.json v2.4 A4:** Pc > 30 bar ~~raw~~ FILTERED channel, 5 ms persistence OVR-01 for 5 ms LEAD T925	CHANGELOG 166 (4): filtered channel, 5 ms. Electronics v2.10 to implement (REV-ELEC-1). T1670
C-04 A5 late-light monitor: deleted by decision, still an abort in firmware v2.9	**CHANGELOG 160 (1):** A5 DELETED as an abort, flight and stand; replaced by a LIGHT_SPIKE log event (raw and filtered peak, time above 8 and 30 bar abs, time from light) so IGN-1/T1 measure whether real lights spike. Supersedes item (2) of the Electronics-handback decision (A5 ... LEAD T929 **firmware v2.9:** 8.0 bar abs for 5 ms from E1 to the ramp start (unchanged); with the 10 ms light confirmation a fast late light can trip A5 first LEAD T931	deleted (LIGHT_SPIKE log event). REV-ELEC-1 RE-01 blocking; Electronics v2.10. T1671
C-05 Relight option (c) planning value: present in gnc.json v3 or not	**electronics.json v2:** NOT in gnc.json v3 requirements.ignition (key removed; FC rule source to be re-confirmed by GNC) DESIGN T1627 **gnc.json v3:** 1.41 REQ T493 **CHANGELOG 161 (9):** OI-GNC-V3: lead verified gnc.json v3 (3e97c378cfebee95) carries requirements.ignition.relight_planning_value_s. {option_c_unprimed_s 1.41, stale_or_missing_flag}; the Electronics generator looks one level too high; Electronics fixes the lookup and closes ... LEAD T1628	present (lead verified, CHANGELOG 161 (9)). Electronics fixes the lookup (OI-GNC-V3). T1672
C-06 Head-end bleed: "fallback, no design" vs decided purge path	**electronics.json /relight_interface:** FALLBACK only (lead line 126); no design (coordinator) DESIGN T1629 **electronics.json /head_end_bleed_provision:** 2 NC valves in series (V1 at the engine-side tee of the T4 crown + FD-01 radial taps via one -2 line across the gimbal; V1 and V2 on the fixed side), restriction downstream of V2, P_mid transducer between them; ECU drives V1 on pin 15 (U304) and V2 on pin 24 ... DESIGN T1294 **CHANGELOG 164 (1):** RG-01 design basis = Combustion's minimal set (3): coil re-route + pre-scheduled 2-tap purge on the decided path (equal 0.707 mm restrictors, 0.71 mm drill, each flow-tested for Cd) with 2.5 s of FLOW ending at E0 - 0.05 s + theta <= 2 deg with balanced or ... LEAD T1630	CHANGELOG 164 (1): decided purge path. stale text in electronics.json v2; Electronics v2.10. T1673
C-07 Structural thrust case: LC2F and ICD bases vs the decided ignition interlock	**loadcases_v1 LC2F thrust:** 907.7 DESIGN T1072 **LC2F source:** Combustion data/v2/full_stroke_failure_case_v2.json max F_N 907.7 N (v1 helix, 30 C, Cd 0.8) DESIGN T1631 **ICD §3:** 876 N at Pc [withheld: superseded value, SUP-04] (pintle stuck at full stroke, ~~30 °C tank~~ ignition interlock T_tank <= 24 C; basis 44.33 bar (meop_basis_v2p4) OVR-06, Cd 0.80). Use it as the ultimate axial load case for the thrust rings SIM T567 **ENG-09 amendment:** ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. REQ T013 **combustion.json v2.4 MEOP basis:** 44.33 SIM T469 **thrust at the v2.4 basis:** MISSING MISSING T563	v2.4 MEOP basis (pressure); the thrust at that basis is MISSING. Combustion to publish the thrust at the MEOP basis; Mechanical to restate LC2F and the ICD row. Until then no thrust value is shown as bounding. T1674
C-08 A1 apogee: planning target vs the target adopted after the purge sizing	**gnc.json v3:** 300.0 REQ T1404 **REQUIREMENTS §1 mission:** 4. Coast nose-first, stabilised by 4 aft fins, to a **300 m** apogee. REQ T1632 **CHANGELOG 168 (4) trim rule:** A1 tail draws: the re-fly flies the purge at the ECU cap (<= 73.0 g) plus the +0.60 kg valve-mass sensitivity and sizes the A1 apogee trim that removes new exhaustions; a trim to >= 275 m is ADOPTED without further decision (300 m was a lead default), a ... LEAD T1407 **CHANGELOG 175 (1):** A1 apogee target 285 m ADOPTED under line 168 (4) (0 exhausted runs at the 73 g purge; cost 15 m). LEAD T1405	CHANGELOG 175 (1). gnc.json v3.1 (published after REV-GNC-3) and the REQUIREMENTS mission text to restate the target. T1675
C-09 Igniter mass: Combustion system estimate vs BOM line	**combustion.json v2.4:** [0.2, 0.3] ASSUMED T858 **BOM ENG-IG-01:** 0.07 ASSUMED T167 **BOM ENG-IG-01 row:** ENG-IG-01: Torch igniter (N2O/propane, spark) - ENVELOPE 20 × 45 mm + M DESIGN T166	both shown (scopes differ). BOM line covers body and plug; solenoids and propane supply to be placed in the BOM or the exclusions. T1676

ID	Sides (quoted)	Shown in this datasheet; next step
C-10 Mechanical interface basis vs CAD of this datasheet	**mechanical.json v2:** M2. Mass properties from the CAD v3.3 solids (engine + gimbal, CG pivot iterated in CAD) with the v3.4 pintle stack (HV69 in-line) and the feed/line engine shares added as tagged ESTIMATE items and the pivot re-centred on the mid-burn CG; gimbal axis ... DESIGN T1633 **CAD build:** v3.4.1 DESIGN T1634 **Mechanical open item:** CAD v3.4: HV69 in-line pintle stack with rolling guide, feed parts (hose/elbow/lines), stiffened levers/ring → mass properties and K re-issued (v2.1); v3.4 sweep must clear r 67 mm at z 215-260 PENDING T1635	CAD v3.4.1 for geometry, BOM and mass; mechanical.json v2 for stiffness and loads until v2.1. mechanical.json v2.1 pending. T1677
C-11 Firmware constants generated from an earlier combustion interface	**electronics.json generated headers:** src/core/engine_params.h: [["combustion.json", "v2.3", "d9ec54fbf450e243", "current"], ["gnc.json", "v3", "3e97c378cfebee95", "current"], ["pintle_mini_servo_v1.json", null, "62b326482617125a", "current"]]; src/core/mech_params.h: [["mechanical.json", "v2", ... DESIGN T1125 **current combustion.json:** v2.4 DESIGN T1638	combustion.json v2.4. regenerate engine_params.h in Electronics v2.10. T1678
C-12 combustion.json v2.4 wording behind the lead decisions	**combustion.json version:** v2.4 DESIGN T1638 **REV-COMB-2 majors:** R2-01 line 166 (3)/(4) not implemented; R2-02 two stand safing rules; R2-03 FA-1 pulse single point at ASSUMED O/F; R2-04 T1-T4 law test cannot resolve; default gate less safe DERIVED T1513 **CHANGELOG 172 (6):** R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion v2.4a text/interface patch (relief_v2p4 block added); the ENG-09 "trapped-liquid relief <= 7.5 MPa" text is superseded by the tank relief 6.9 MPa / MAWP 8.42 MPa / ... LEAD T017	the decisions, via overrides OVR-01, OVR-02, OVR-03, OVR-04, OVR-05, OVR-06, OVR-07, OVR-08, OVR-09 (§13.1). Combustion v2.4a patch; drop each override when the generator reports it REDUNDANT. T1679
C-13 CAD mass: BOM mass_sum_kg vs the sum of its lines	**BOM mass_sum_kg:** 4.3923 DERIVED T1636 **BOM mass_sum_note:** sum of unit masses x qty (CAD volume x density, or tagged COTS masses); includes the transport-lock items only if they are placed (they are not: group 'lock' is excluded from the instance list); PROVISION lines excluded DERIVED T367 **generator sum, transport lock excluded:** 4.33007 DERIVED T274 **transport-lock lines:** 0.06225 DERIVED T275	generator sum with the transport lock excluded, as the BOM note states. the difference equals the transport-lock lines: mass_sum_kg includes them although the note excludes them. Mechanical to restate mass_sum_kg or the note. T1680
C-14 FA-1 load vs the parts of CAD v3.4.1: margins reported by Mechanical	**FA-1 load decision:** DECISION: FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts (covers every baseline peak and the shell); for any closure, bolt, J1-lip or other part with a natural period of 0.25-1 ms Mechanical reports margins also at 92 bar abs (design: baseline rounded with DLF and the +3 % fuel-basis sensitivity) and at 102 bar abs (sensitivity: upper bracket, sharp, DLF); thrust ... LEAD T515 **Mechanical result, CHANGELOG 176 (sentences on yield, joints and bearings, quoted in full):** TC-09 316L YIELD -0.40/-0.41/-0.43 (ult +0.80/+0.78/+0.66) → v3.5 TC-09 IN718 +2.81/+2.76/+2.51 (own thermal run); J-FWD 8xM5 A4-70 yield -0.011/-0.033/-0.128, ult +0.60/+0.56/+0.41 (shoulder-screw option +0.86/+0.82/+0.64); seal S13 -0.016 at 102 bar only; Spike 1.8 kN/4 ms: 608 bearings exceed C0 (MS -0.24 DLF 2.0, -0.21 trapezoid, -0.10 at Combustion SDOF 3.05 kN; C0 AS RECALLED) - brinelling possible, ultimate not demonstrated; SIM T558 **BOM ENG-TC-09 material (CAD v3.4.1):** Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts DESIGN T1637	CAD v3.4.1 as built (no lead decision on the result yet). lead decision on the Mechanical result; CAD v3.5 with the proposed part changes and its independent review; the datasheet is regenerated on that CAD. T1681

14 Unknowns register

Every value the sources mark MISSING, TO-DEFINE, PENDING or TO-VERIFY, plus every key the generator looked for and did not find. They are listed so that none is mistaken for a design value.

MISSING 19 · PENDING 18 · TO-DEFINE 18 · TO-VERIFY 40

Trace	Class	Source key	Value or text
T346	MISSING	—: mass of the items listed as not modelled in mechanical.json v2 (CAD v3.3 basis)	MISSING
T563	MISSING	—: Physical worst-case thrust at the v2.4 MEOP basis: combustion.json v2.4 publishes the basis pressure only	MISSING
T564	MISSING	—: Lateral thrust at the gimbal range for the physical worst case (needs the worst-case thrust)	MISSING
T241	MISSING	BOM: /provision_lines/0/unit_mass_kg	MISSING
T243	MISSING	BOM: /provision_lines/1/unit_mass_kg	MISSING
T247	MISSING	BOM: /provision_lines/3/unit_mass_kg	MISSING
T249	MISSING	BOM: /provision_lines/4/unit_mass_kg	MISSING
T251	MISSING	BOM: /provision_lines/5/unit_mass_kg	MISSING
T276	MISSING	BOM: /currency	TO-QUOTE (not summed)
T943	MISSING	ELE: /abort_matrix/causes (generator search)	no matching cause in firmware v2.9
T1293	MISSING	ELE: /power_budget/arm_bus_peak_vs_F301	TO-CHECK: MV opener stall 9.2 A at 7.4 V (DATASHEET) / 10.4 A at 8.4 V (ASSUMED) + pintle rail input <= 4.33 A + 12 V boost loads vs F301 15 A (Littelfuse 0297015 curve)
T679	MISSING	ICD: §4. Mass row "Engine hardware mass and CG" col 1	(v2.1): CAD v3.4 re-issues the gimballed and fixed masses. The main-valve module and tank bulkheads are not yet in CAD
T345	MISSING	MEC: /fixed_module/not_modelled	['thrust-ring radial screws (16 x M5)', 'isolation valve + actuator (mass UNKNOWN: Electronics/Combustion)', 'tee, filter (<= 15 um, OI-7), relief, fill/vent', 'hose and lines fixed shares (~0.07 kg ASSUMED)', 'tank']
T1527	MISSING	generator: open-item status RG-01	BLOCKING
T1528	MISSING	generator: open-item status FA-1	DESIGN CONTINUATION ONLY; margins reported, not reviewed
T1529	MISSING	generator: open-item status CF-4	NOT RUN (founders)
T1530	MISSING	generator: open-item status IGN-1	NOT RUN (founders)
T1531	MISSING	generator: open-item status B-PC-1	NOT RUN (founders)
T1533	MISSING	generator: open-item status ENG-11	NOT MET (structure only)
T559	PENDING	—: Independent review and lead decision on the Mechanical FA-1 margins (CHANGELOG 176) and on the CAD changes they propose	PENDING
T1057	PENDING	—: no published margin table found by the generator	PENDING
T1103	PENDING	—: FEA margins for load case LC1	PENDING
T1104	PENDING	—: FEA margins for load case LC2	PENDING
T1105	PENDING	—: FEA margins for load case LC2F	PENDING
T1106	PENDING	—: FEA margins for load case LC3	PENDING
T1107	PENDING	—: FEA margins for load case LC4	PENDING
T1108	PENDING	—: FEA margins for load case BN	PENDING
T1109	PENDING	—: FEA margins for load case LC5	PENDING
T1110	PENDING	—: FEA margins for load case LC6	PENDING
T1111	PENDING	—: FEA margins for load case MODAL	PENDING
T1526	PENDING	—: Structural FEA margins (see §9)	PENDING
T819	PENDING	MEC: /open_items/3	thermal FEA v1.1 (B1/A1/A2, TS-1 relight, TS-2 soak, servo mount, TC-06 sizing, Al vs 316L retainer) → v2.1
T1635	PENDING	MEC: /open_items/0	CAD v3.4: HV69 in-line pintle stack with rolling guide, feed parts (hose/elbow/lines), stiffened levers/ring → mass properties and K re-issued (v2.1); v3.4 sweep must clear r 67 mm at z 215-260
T1535	PENDING	generator: open-item status FEA	PENDING
T1538	PENDING	generator: open-item status REV-COMB-2	REVISE (targeted)
T1539	PENDING	generator: open-item status COMB-v2.4a	IN PROGRESS
T1541	PENDING	generator: open-item status ELEC-v2.10	IN PROGRESS
T512	TO-DEFINE	—: Tank interlock guard band = total channel uncertainty (Electronics states it; CHANGELOG 172 (4))	TO-DEFINE
T253	TO-DEFINE	BOM: /contingency_lines/0/unit_mass_kg	TO-DEFINE
T680	TO-DEFINE	ICD: §12 row "ICD-1" col 1	Thrust-ring bolt pattern, wall bearing and pull-out at 876 N plus lateral and moment loads
T682	TO-DEFINE	ICD: §12 row "ICD-2" col 1	Main-valve module envelope and mass at the zone top
T684	TO-DEFINE	ICD: §12 row "ICD-3" col 1	Upper-bay vent area and station; baro-port placement
T686	TO-DEFINE	ICD: §12 row "ICD-4" col 1	Relief and vent port routing (tank relief, 9.5 MPa disc line, fill/vent)
T688	TO-DEFINE	ICD: §12 row "ICD-5" col 1	Tank station and mounting; heater and insulation interface
T690	TO-DEFINE	ICD: §12 row "ICD-6" col 1	Launch-site bay temperature limit
T692	TO-DEFINE	ICD: §12 row "ICD-7" col 1	Pad slope specification and leg design
T694	TO-DEFINE	ICD: §12 row "ICD-8" col 1	Drogue opening load, bay volume, ejection method, swivel and release fitting

Trace	Class	Source key	Value or text
T696	TO-DEFINE	ICD: §12 row "ICD-9" col 1	Ground plume impingement on the feet
T698	TO-DEFINE	ICD: §12 row "ICD-10" col 1	Airframe first bending mode ≥ 10 Hz on the real tube
T700	TO-DEFINE	ICD: §12 row "ICD-11" col 1	Engine hardware mass and CG from CAD v3.4
T702	TO-DEFINE	ICD: §12 row "ICD-12" col 1	Wall-line P-clips (3, at z 110 / 170 / 230 mm engine frame) and the coil fixed-end clamp near az 219.5°, z ≈ 71 mm: clamp type, wall fastening, loads (hose end reaction up to 32 N and 1.3 N·m at the 8° stop for EI 0.25)
T704	TO-DEFINE	ICD: §12 row "ICD-13" col 1	Harness routing across the gimbal
T706	TO-DEFINE	ICD: §12 row "ICD-14" col 1	Head-end bleed overboard port (A1 provision): size, station, routing; balanced or axial vent
T708	TO-DEFINE	ICD: §12 row "ICD-15" col 1	Drogue canopy type and oscillation data at the relight condition (≤ 2° design attitude)
T1540	TO-DEFINE	generator: open-item status INTERLOCK-GB	TO-DEFINE
T560	TO-VERIFY	—: FA-1 fuel-vapour inventory at light (data file: first-order bound only; lead decision: remains TO-VERIFY)	TO-VERIFY
T073	TO-VERIFY	BOM: /lines/11/unit_mass_kg	0.0379
T193	TO-VERIFY	BOM: /lines/71/unit_mass_kg	0.03
T833	TO-VERIFY	BOM: /lines/*[ENG-GR-01, ENG-GR-02]/name+notes (generator text search)	not stated in the BOM name or notes of ENG-GR-01/02: rounded entrance TO-VERIFY in CAD
T635	TO-VERIFY	CHK: /swept_worst_band	{'z0_mm': 105.0, 'z1_mm': 110.0, 'r_engine_swept_mm': 60.08, 'by_engine': 'ENG-FD-03#0', 'r_ring_swept_mm': 0.0, 'r_fixed_mm': 71.8, 'by_fixed': 'ENG-FD-11#0', 'r_max_mm': 71.8, 'clearance_mm': -1.8}
T1003	TO-VERIFY	CMB: /stability/hybrid_LF_note	classical hybrid low-frequency (thermal-lag/boundary-layer) instability 0.3-2 Hz band is identified but its amplitude is NOT predicted; static fire T2/T3 with 1 kHz Pc logging
T1121	TO-VERIFY	ELE: /ecu/schematic/placeholders	['TI_RPM0026A_PLACEHOLDER (NOT FOR FABRICATION)', 'Pololu_U3V70_WireAttach (wire pads only)', 'Teensy41_Socket (verify vs PJRC drawing)']
T1123	TO-VERIFY	ELE: /firmware/target	PJRC Teensy 4.1 (i.MX RT1062); HAL for the hardware NOT written (core runs in SIL and test_core)
T1137	TO-VERIFY	ELE: /sensors/load_cell/model	1 kN S-type or compression load cell, 2 mV/V (TO-QUOTE)
T1138	TO-VERIFY	ELE: /sensors/imu	FC side (ADIS16470 per the GNC/FC record): filter delay UNKNOWN (over_thrust)
T1298	TO-VERIFY	ELE: /head_end_bleed_provision/coil	{'valve': 'Burkert 6027 HP (stainless, G1/4, PEEK/FKM seat, NC, orifice 1.0 mm: 0-250 bar DC air/water) - DATASHEET via WebFetch summary', 'coil_42K_DC_W_cold_warm': [16.0, 21.0], 'coil_note': 'column mapping of the coil table ambiguous in the summary: TO-VERI...
T1309	TO-VERIFY	ELE: /open_issues/0/issue	light-detection margin at the v2.8 ignition lift: S5 dispersed pad-start MC 31/300 NO_IGNITION on a LIT plant since v2.8.1 (v2.7 at 20 % lift: 0). The lit plant Pc at 0.2039 mm sits at the 3.0 bar abs threshold (nominal replay 3.02 bar abs; low-Cd/low-efficien...
T1313	TO-VERIFY	ELE: /open_issues/1/issue	A1 relight not frozen (RG-01): re-routed segment volume and the GNC re-fly pending; drained-segment fault path to re-run on the published re-route
T1317	TO-VERIFY	ELE: /open_issues/2/issue	S6 gimbal_jam_moving_cmd FAIL (small-amplitude jam undetected)
T1321	TO-VERIFY	ELE: /open_issues/3/issue	S6 throat block Pc 53.4 bar abs > MEOP for 72 ms
T1325	TO-VERIFY	ELE: /open_issues/4/issue	hardware HAL not written; TLM_SLOW 0x110 undefined; 53 ARM double-promotion warnings
T1329	TO-VERIFY	ELE: /open_issues/5/issue	GNC tracker / FSW not yet in the flight computer; IMU filter delay unknown; RG-16 mass budget
T1333	TO-VERIFY	ELE: /open_issues/6/issue	no common time base yet for DAQ, CAN analyser and ECU
T1337	TO-VERIFY	ELE: /open_issues/7/issue	P_feed is a single point for the relight path choice (invalid → UNKNOWN → prep path) and S1
T1341	TO-VERIFY	ELE: /open_issues/8/issue	a coast encoder loss after the last TLM_RELIGHT frame flips the path late (S10 evidence case)
T1345	TO-VERIFY	ELE: /open_issues/9/issue	B1 launch commit mean wind < 6 m/s until B-PNT-1 shows the drive lag inside the MC tau
T1349	TO-VERIFY	ELE: /open_issues/10/issue	LTC4365 UV ASSUMED; Q803 enhancement; RPM0026A footprint placeholder; ARM-bus peak vs F301
T1353	TO-VERIFY	ELE: /open_issues/11/issue	MV-P opener cam-limited 61 ms opening (bench B-MV-1)
T1357	TO-VERIFY	ELE: /open_issues/12/issue	map-trim plateau (Pc loop trim authority +/-40 %) not yet characterised on hardware
T1361	TO-VERIFY	ELE: /open_issues/13/issue	pc_transducer_v2 labels P_torch J610 (is J901); protocol.h FC_THRUST valid-window comment wording
T1365	TO-VERIFY	ELE: /open_issues/14/issue	v2.9 light rule (10 ms) + unchanged A5 (8.0 bar abs, 5 ms): S6 late lights 0.20/0.25 s inside W now abort HARD_START (v2.8.3 continued; plant Pc max 26.9/28.6 bar abs)
T1369	TO-VERIFY	ELE: /open_issues/15/issue	S9 +1.034 bar post-zero bias: unlit filtered 1.87 bar abs > floor 1.8 → false E2 → RUN → A3 FLAMEOUT +0.14 s; closed by B-PC-1 <= 0.3 bar (floor 2.0)
T1373	TO-VERIFY	ELE: /open_issues/16/issue	E2 - light 16-18 ms in SIL (target ~15 ms): plant rise + 50 Hz filter + 10 ms persistence; IGN-1 measures
T1377	TO-VERIFY	ELE: /open_issues/17/issue	A3 ratio minimum 0.454 at the A1 relight MC (threshold 0.40) during the start ramp; after E3 0.719
T1381	TO-VERIFY	ELE: /open_issues/18/issue	Burkert 6027 HP: 24 V DC listed, ECU valve rail +12V_ARM (U3V70F12); 2 × 16 W → +2.7 A for ~2.6 s per purge at 12 V; 12 V coil or a 24 V boost, rail budget and valve mass TO-VERIFY; HP seat PEEK/FKM medium -10..80 C and N2O compatibility TO-VERIFY
T1385	TO-VERIFY	ELE: /open_issues/19/issue	unreached vapour with taps (2 deg 4.84 cm3, 10 deg 15.91 cm3) from an UNPUBLISHED Mechanical workdir file
T1389	TO-VERIFY	ELE: /open_issues/20/issue	gnc.json v3 requirements.ignition no longer carries option_c_unprimed_s / stale_or_missing_flag: FC relight planning source to be re-confirmed

Trace	Class	Source key	Value or text
T556	TO-VERIFY	FA1: /fuel_vapour_inventory/status	MISSING: the fuel-vapour (and aerosol/film) inventory in the warm 32.2 mm port at the moment of light after the 18.67 s A1 coast is not modelled and not measured. Below: first-order bounds only.
T1403	TO-VERIFY	GNC: /reference_vehicle/aero/unknowns	Tail-first aero (CNa, CP, base/plume interaction) is UNKNOWN to better than ~x2: only CFD (05-plume-cfd), a wind tunnel or flight data close it. The MC draws the whole band.
T595	TO-VERIFY	MEC: /gimbal/bearings	4 × 608-2RS (8×22×7); pins 8 h6 17-4PH H1150
T344	TO-VERIFY	generator:	0.6
T1532	TO-VERIFY	generator: open-item status BLEED-MASS	TO-VERIFY
T1534	TO-VERIFY	generator: open-item status ENG-07	MARGINAL
T1536	TO-VERIFY	generator: open-item status ENVELOPE	TO-VERIFY
T1537	TO-VERIFY	generator: open-item status GRAIN-R4	TO-VERIFY

15 Revision history and sources

15.1 Revision history

Revision	Generated (UTC)	CAD	Generator	Note
rev0-draft	2026-10-13T13:25:31Z	v3.4.1	build_datasheet.py 0.2.0	first draft generated from the hub state; FEA, RG-01 re-fly, Electronics v2.10 and the Combustion v2.4a interface patch pending (FA-1 envelope data folded in)

15.2 Sources

ID	Role	Path	Version	sha256/16	Modified (UTC)	Reads
CMB	Combustion interface (design point, grain, injector, igniter, sequences)	hub: interfaces/combustion.json	v2.4	469727032124329f	2026-10-13T11:22:00Z	582
MEC	Mechanical interface (gimbal, TVC, envelope, materials)	hub: interfaces/mechanical.json	v2	4f024ed4ccb4618c	2026-10-12T23:04:37Z	30
ELE	Electronics interface (ECU, sensors, firmware, abort causes)	hub: interfaces/electronics.json	v2	da067408c1594814	2026-10-13T09:35:16Z	381
GNC	Space/GNC interface (reference vehicle, mission, Monte Carlo)	hub: interfaces/gnc.json	v3	3e97c378cfebee95	2026-10-13T03:24:09Z	90
BASE	Engine baseline v0 (seed values, superseded)	hub: interfaces/ENGINE-BASELINE-v0.json	v0	508faa652ccbd539	2026-10-12T17:22:30Z	0
CL	Decision log (line-numbered)	hub: interfaces/CHANGELOG.md	176 lines	0c2c6d5539240658	2026-10-13T13:16:07Z	0
REQ	Engine requirements	hub: 00-requirements/REQUIREMENTS.md	rev 2	fbb9e1205707082f	2026-10-13T13:09:55Z	0
ICD	Engine-airframe ICD	hub: 00-requirements/ICD-AIRFRAME.md	draft 0.2	706455abf4098182	2026-10-13T11:13:06Z	0
SUP	Independent review register	hub: reviews/SUPERVISION.json	7 reviews	9736f4974377a7bd	2026-10-13T12:24:24Z	63
RV2	Independent Combustion re-review REV-COMB-2 (text)	hub: reviews/REVIEW-COMBUSTION-R2.md	—	e6bf189fb271e652	2026-10-13T12:23:43Z	0
OVR	Decision override table (datasheet generator input; drop entries when combustion.json v2.4a lands)	hub: 07-datasheet/gen/decision_overrides.json	ovr-3	3666334a809ccf6d	2026-10-13T13:17:59Z	0
FA1	Combustion v2.4a FA-1 relight-pulse load envelope (SIM on an ASSUMED construct)	hub: 02-combustion/data/v2p4a/fa1_envelope_v2p4a.json	v2.4a	64b328150025fa1e	2026-10-13T12:49:52Z	46
RCMB	Combustion report	hub: 02-combustion/REPORT.md	2.4	32e366bd78437dc0	2026-10-13T11:24:48Z	0
CF4	CF-4 relight-purge cold-flow specification	hub: 02-combustion/CF-4-SPEC.md	1.0	c9e1efbb18177155	2026-10-13T11:11:28Z	0
T14	T1-T4 regression measurement specification	hub: 02-combustion/T1-T4-REGRESSION-SPEC.md	v2.4	25a207af8fe61f21	2026-10-13T11:22:15Z	0
RPL	Plume CFD report	hub: 05-plume-cfd/REPORT.md	v2.2b	140b3c59f8323bba	2026-10-13T11:24:48Z	0
PVAL	Plume CFD validation results	hub: 05-plume-cfd/validation/results/validation_v2.json	v2.1a	cbfbce89c41689f3	2026-10-12T21:31:13Z	28
FEALC	FEA load cases (inputs)	hub: 04-fea/loadcases_v1.json	v1	cbafa22eb69ef039	2026-10-12T18:38:02Z	55
SLEW	TVC slew calculation v2 (Mechanical, hub copy)	hub: 03-cad/calcs/actuator_slew_v2.json	v2	7f69321838404705	2026-10-13T05:09:00Z	42
BENCH	Electronics bench and test plan	hub: 06-electronics/docs/bench_plan_v2.md	v2	ece8ad4a1465886d	2026-10-13T09:34:09Z	0
RELE	Electronics v2.9 handback report	hub: 06-electronics/REPORT-v2.9-HANDBACK.md	v2.9	a7f047b1858f1293	2026-10-13T09:37:36Z	0
RGNC	Space/GNC report v3	hub: 01-systems-gnc/REPORT-v3.md	v3	2d79920b8cc1537d	2026-10-13T03:27:36Z	0
BOM	Bill of materials (CAD v3.4.1)	CAD: BOM_v3.4.1.json	v3.4.1	d028143d22b09f60	2026-10-13T09:59:04Z	13
MP	Mass properties (CAD v3.4.1)	CAD: massprops_v3.4.1.json	v3.4.1	4fbb9d192af6e38e	2026-10-13T09:59:04Z	27
MAN	Assembly manifest, instance bounds (CAD v3.4.1)	CAD: assembly/assembly_manifest_v3.4.1.json	v3.4.1	4fd02f57b93a8aae	2026-10-13T09:58:51Z	5
GLB	Assembly GLB, visualisation geometry (CAD v3.4.1)	CAD: assembly/ENGINE_v3.4.1_neutral.glb	v3.4.1	3879a25a4486ee0c	2026-10-13T09:59:03Z	0
PROP	CAD build properties and results	CAD: properties.json	v3.4.1	8e30376fc16dafa4	2026-10-13T11:20:38Z	4

ID	Role	Path	Version	sha256/16	Modified (UTC)	Reads
CHK	CAD software geometric checks (CAD v3.4.1)	CAD: checks/checks_summary_v3.4.1.json	v3.4.1	2dc1c50043abcf1d	2026-10-13T11:20:38Z	93
RMAN	Render manifest (this generator)	hub: 07-datasheet/build/renders/v3.4.1/render_manifest.json	v3.4.1	0933a1acf0ccf049	2026-10-13T12:06:32Z	0
PF-MACH	Plume figure	hub: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_mach_near.png	figure	39cfd050d53b9f68	2026-10-12T21:31:30Z	0
PF-SCHL	Plume figure	hub: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_schlieren_near.png	figure	5a559717ba7c3a76	2026-10-12T21:31:35Z	0
PF-TEMP	Plume figure	hub: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_T_full.png	figure	b7fe094ff216873c	2026-10-12T21:31:32Z	0
PF-AXIS	Plume figure	hub: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_centreline.png	figure	de88abf544299723	2026-10-12T21:31:36Z	0
PF-V2MACH	Plume figure	hub: 05-plume-cfd/ownruns/c_2s_100_v2op/fig/c2s_v2op_mach_near.png	figure	034d7f18ff3075f3	2026-10-12T21:29:26Z	0
PF-ART	Plume figure	hub: 05-plume-cfd/blender/final/argos_plume_artistic_side.png	figure	7c722b7f001b2ae4	2026-10-12T20:43:12Z	0
R-ISO	Render metadata: iso (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/iso_meta.json	—	2121fc233088f3e3	2026-10-13T11:55:44Z	0
R-FRONT	Render metadata: front (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/front_meta.json	—	7fc0263918c59104	2026-10-13T11:58:05Z	0
R-SIDE	Render metadata: side (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/side_meta.json	—	4cbd647ca19e6719	2026-10-13T11:59:44Z	0
R-SECTION	Render metadata: section (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/section_meta.json	—	3498e82e92acfae3	2026-10-13T12:01:15Z	0
R-LAYOUT	Render metadata: layout (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/layout_meta.json	—	7b8e8e58356b21aa	2026-10-13T12:02:36Z	0
R-FAM-EL	Render metadata: family_EL (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_EL_meta.json	—	d6ff80bf2412728b	2026-10-13T12:02:41Z	0
R-FAM-FD	Render metadata: family_FD (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_FD_meta.json	—	512c4823bf89f0b7	2026-10-13T12:02:55Z	0
R-FAM-FS	Render metadata: family_FS (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_FS_meta.json	—	268f64b4d50bdc1a	2026-10-13T12:03:48Z	0
R-FAM-GM	Render metadata: family_GM (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_GM_meta.json	—	2007d3d795d007c4	2026-10-13T12:04:56Z	0
R-FAM-GR	Render metadata: family_GR (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_GR_meta.json	—	940506e5736ec69a	2026-10-13T12:05:15Z	0
R-FAM-HW	Render metadata: family_HW (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_HW_meta.json	—	dd0b06dc474067ee	2026-10-13T12:05:31Z	0
R-FAM-IG	Render metadata: family_IG (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_IG_meta.json	—	19b2c9fc816aa7d8	2026-10-13T12:05:43Z	0
R-FAM-NZ	Render metadata: family_NZ (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_NZ_meta.json	—	12c92ce3e5094a4c	2026-10-13T12:06:00Z	0
R-FAM-PD	Render metadata: family_PD (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_PD_meta.json	—	a41ca9d350d97609	2026-10-13T12:06:24Z	0
R-FAM-TC	Render metadata: family_TC (camera, anchors)	hub: 07-datasheet/build/renders/v3.4.1/family_TC_meta.json	—	46238777d2a21e12	2026-10-13T12:06:31Z	0

Hub root: `/home/claude/work/shared-fallback/engine`. CAD root: `/home/claude/work/mechanical/engine/03-cad`. The same table, with absolute paths, is written to out/sources.json.

Appendix A Value trace

Every traced value used in this document: trace id, value as displayed, evidence class, source and version, key (JSON pointer, markdown row, CHANGELOG line and item, or generator formula) and the source tag text. Identical source keys share one id.

ID	Value	Class	Source	Key	Source tag / formula
T001	RE-20: the 30 bar firmware abort runs on the FILTERED channel with 5 ms persistence (pro...	LEAD	CL 176 lines	L166(4)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T002	Electronics review decisions (CHANGELOG 166): light-rule baseline never substituted by a...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Electronics review decisions (CHANGELOG 166)"	REQ bullet
T003	Electronics review decisions (CHANGELOG 166): light-rule baseline never substituted by a...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Electronics review decisions (CHANGELOG 166)"[sentence containing "filtered channel, 5 ms"]	REQ bullet
T004	30 bar firmware abort on the FILTERED channel with 5 ms persistence (amends 'raw' in CHA...	LEAD	OVR ovr-3	/overrides/0/value	override OVR-01 (ovr-3); numbers and key phrases checked against the cited sources
T005	RE-05: E1 = torch confirmed (2oo2, FA-1 (a)) AND [closure confirmed OR closure deadline ...	LEAD	CL 176 lines	L166(3)[sentence containing "E1 = torch confirmed"]	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T006	FC action on BLEED_FAULT is LOG-ONLY until a leak response is flown and reviewed.	LEAD	CL 176 lines	L168(2)	lead (decision) 2026-10-13 (line sha 3ad186c5fa9490ff)
T007	E1 = torch confirmed (2oo2) AND (closure confirmed OR closure deadline reached), deadlin...	LEAD	OVR ovr-3	/overrides/1/value	override OVR-02 (ovr-3); numbers and key phrases checked against the cited sources
T008	R2-02 ONE stand safing rule after a no-light, effective now: N2 purge >= 60 s at >= 10 g...	LEAD	CL 176 lines	L172(1)	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T009	Combustion re-review decisions (CHANGELOG 172): stand safing after a no-light = N2 purge...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)"	REQ bullet
T010	Combustion re-review decisions (CHANGELOG 172): stand safing after a no-light = N2 purge...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)"[sentence containing "stand safing"]	REQ bullet
T011	N2 purge >= 60 s at >= 10 g/s, then hold >= 15 min before anyone approaches (Pc at ambie...	LEAD	OVR ovr-3	/overrides/2/value	override OVR-03 (ovr-3); numbers and key phrases checked against the cited sources
T012	ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM.	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-12:"	REQ bullet
T013	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition.	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-09:"	REQ bullet
T014	R2-07 interlock guard band: T_tank and P_tank thresholds sit below 24 C / 55.28 bar abs ...	LEAD	CL 176 lines	L172(4)	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T015	tank temperature range at ARM 15-24 C (was 15-25 C); T_tank <= 24 C at ARM and at igniti...	LEAD	OVR ovr-3	/overrides/3/value	override OVR-04 (ovr-3); numbers and key phrases checked against the cited sources
T016	ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa, pro...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)"[sentence containing "trapped-liquid relief"]	REQ bullet
T017	R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion...	LEAD	CL 176 lines	L172(6)	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T018	Relief: back-relief check MV segment → tank (Cv >= 0.1); secondary rupture disc 9.5 MPa...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Relief:"	REQ bullet
T019	tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check MV segment t...	LEAD	OVR ovr-3	/overrides/4/value	override OVR-05 (ovr-3); numbers and key phrases checked against the cited sources
T020	{"basis_bar": 44.33, "margin_to_MEOP_bar": 0.67, "proof_margin_min_bar": 21.9, "supersed...	SIM	CMB v2.4	/meop_basis_v2p4/v	SIM meop_basis_v2p4.json
T021	MEOP 45 bar kept with the ignition interlock T_tank <= 24 C at ARM and at ignition; basi...	LEAD	OVR ovr-3	/overrides/5/value	override OVR-06 (ovr-3); numbers and key phrases checked against the cited sources
T022	R2-03 FA-1 pulse single point at ASSUMED O/F	DERIVED	SUP 7 reviews	/reviews/5/major/2	(class set by generator: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T023	**What was specified.** Line 164 (8) asks Mechanical for a "≤ 3 ms, 75.5 bar abs head-pu... [OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs]	DERIVED	RV2	line 441	RV2 line 441 (independent review text)
T024	R2-03 FA-1 load restated as an ENVELOPE: Combustion publishes O/F 4-12 x tf 1-7 ms (1 ms...	LEAD	CL 176 lines	L172(2)	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T025	On Combustion's FA-1 envelope (02-combustion/data/v2p4a/fa1_envelope_v2p4a.json 64b32815...	LEAD	CL 176 lines	L173	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T026	75.5 bar is a single point at ASSUMED O/F 8. The FA-1 load is the published envelope (he... [OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs]	LEAD	OVR ovr-3	/overrides/6/value	override OVR-07 (ovr-3); numbers and key phrases checked against the cited sources
T027	Line 164 "the FC delays T_fire instead" is WITHDRAWN (commit at the plain trigger gives ...	LEAD	CL 176 lines	L175(3)	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T028	GNC rev2 decisions (CHANGELOG 175): A1 apogee target 285 m. PURGE commits no later than ...	REQ	REQ rev 2	§Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)"	REQ bullet
T029	PURGE commits no later than the plain relight trigger - 1.7 s; otherwise no purge (T_fir...	REQ	REQ rev 2	§Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)" [sentence containing "PURGE commits"]	REQ bullet
T030	Line 164 "the FC delays T_fire instead" is WITHDRAWN. PROVISIONAL rule: PURGE commits no...	LEAD	OVR ovr-3	/overrides/7/value	override OVR-08 (ovr-3); numbers and key phrases checked against the cited sources
T031	Amendment c: the time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R...	LEAD	CL 176 lines	L175(5)	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T032	With Pc invalid after E2 the drogue is released only on accelerometer-confirmed thrust (...	REQ	REQ rev 2	§Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)" [sentence containing "accelerometer-confirmed thrust"]	REQ bullet
T033	The time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc in...	LEAD	OVR ovr-3	/overrides/8/value	override OVR-09 (ovr-3); numbers and key phrases checked against the cited sources
T034	500 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T035	193.1 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T036	OI-28 ENG-01 at the cap: the 25 bar software Pc cap STAYS (the binding margin is the 30 ...	LEAD	CL 176 lines	L170(1)	lead (decision) 2026-10-13 (line sha 0e5d7d63a5e8dd26)
T037	487 N	DERIVED	CL 176 lines	L170(1) [regex ~(\d+) N at]	lead (decision) 2026-10-13 (line sha 0e5d7d63a5e8dd26) [rated thrust if T1 measures eta_cf at the low end of its band (CHANGELOG 170 (1))]
T038	2501007 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T039	100 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T040	4.389 kg	DERIVED	MP v3.4.1	/states/wet/all/m_kg	(class set by generator: DERIVED) [CAD model total, wet (full grain)]
T041	593.3 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: overall z extent over assembly-manifest instance bounds
T042	355.2 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z of the forward end (pivot at z 0) over assembly-manifest instance bounds

ID	Value	Class	Source	Key	Source tag / formula
T043	238.1 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: distance pivot to aft end over assembly-manifest instance bounds
T044	139.5 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: x extent over assembly-manifest instance bounds
T045	139.5 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: y extent over assembly-manifest instance bounds
T046	488.8 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent over assembly-manifest instance bounds (group = engine)
T047	−238.1–250.7 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z range over assembly-manifest instance bounds (group = engine)
T048	455 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent over assembly-manifest instance bounds (group = fixed)
T049	−99.8–355.2 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z range over assembly-manifest instance bounds (group = fixed)
T050	ENG-EL-03: Gimbal encoder, inner axis (beta): AS5048A on a round PCB 20	DESIGN	BOM v3.4.1	/lines/0	BOM row (CAD parts/ENG-EL-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T051	0.003 kg	ASSUMED	BOM v3.4.1	/lines/0/unit_mass_kg	COTS mass (ASSUMED 2-5 g incl. holder); CG and inertia of a uniform bounding box
T052	ENG-EL-04: Gimbal encoder, outer axis (alpha): AS5048A on a round PCB 2	DESIGN	BOM v3.4.1	/lines/1	BOM row (CAD parts/ENG-EL-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T053	0.003 kg	ASSUMED	BOM v3.4.1	/lines/1/unit_mass_kg	COTS mass (ASSUMED 2-5 g); CG and inertia of a uniform bounding box
T054	ENG-EL-05: Pintle-position encoder AS5048A on a 14 mm round PCB (chip +	DESIGN	BOM v3.4.1	/lines/2	BOM row (CAD parts/ENG-EL-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T055	0.002 kg	ASSUMED	BOM v3.4.1	/lines/2/unit_mass_kg	COTS mass (ASSUMED 1-3 g); CG and inertia of a uniform bounding box
T056	ENG-EL-06: Diametral NdFeB magnet 6 × 2.5 (AS5048A), bonded on the pin	DESIGN	BOM v3.4.1	/lines/3	BOM row (CAD parts/ENG-EL-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T057	0.0005 kg	DERIVED	BOM v3.4.1	/lines/3/unit_mass_kg	COTS mass (DERIVED 6 × 2.5 mm x 7.5 g/cm3); CG and inertia of a uniform bounding box
T058	ENG-EL-11: P_feed transducer J602 TE M3234-000005 class, 1/4-18 NPT mal	DESIGN	BOM v3.4.1	/lines/4	BOM row (CAD parts/ENG-EL-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T059	0.05 kg	ASSUMED	BOM v3.4.1	/lines/4/unit_mass_kg	COTS mass (ASSUMED 50 g (electronics.json)); CG and inertia of a uniform bounding box
T060	ENG-EL-12: MSS-2 Type T MI thermocouple 1/16 in in a 1/16 x 1/8-27 NPT	DESIGN	BOM v3.4.1	/lines/5	BOM row (CAD parts/ENG-EL-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T061	0.015 kg	ASSUMED	BOM v3.4.1	/lines/5/unit_mass_kg	COTS mass (ESTIMATE 15 g incl. connector); CG and inertia of a uniform bounding box
T062	ENG-FD-01: Oxidiser inlet block E1-M (316L): riser socket, 8 mm passage	DESIGN	BOM v3.4.1	/lines/6	BOM row (CAD parts/ENG-FD-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T063	0.03544 kg	DERIVED	BOM v3.4.1	/lines/6/unit_mass_kg	CAD volume x rho(SS316L)
T064	ENG-FD-02: Filter cartridge 15 um absolute, pleated 316L Dutch-twill me	DESIGN	BOM v3.4.1	/lines/7	BOM row (CAD parts/ENG-FD-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T065	0.004 kg	ASSUMED	BOM v3.4.1	/lines/7/unit_mass_kg	COTS mass (ESTIMATE 3-6 g (envelope 13.9 x 19, 316L)); CG and inertia of a uniform bounding box
T066	ENG-FD-03: Oxidiser riser 3/8 x 0.035 in 316L seamless, 71 mm developed	DESIGN	BOM v3.4.1	/lines/8	BOM row (CAD parts/ENG-FD-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T067	0.0137 kg	DERIVED	BOM v3.4.1	/lines/8/unit_mass_kg	CAD volume x rho(SS316L)
T068	ENG-FD-04: Riser radiation shield: 0.2 mm polished 316 foil + 2 band-cl	DESIGN	BOM v3.4.1	/lines/9	BOM row (CAD parts/ENG-FD-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T069	0.00555 kg	DERIVED	BOM v3.4.1	/lines/9/unit_mass_kg	CAD volume x rho(SS316L)

ID	Value	Class	Source	Key	Source tag / formula
T070	ENG-FD-05: Riser-foot TEE (316L): riser socket (top), hose socket (side	DESIGN	BOM v3.4.1	/lines/10	BOM row (CAD parts/ENG-FD-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T071	0.04066 kg	DERIVED	BOM v3.4.1	/lines/10/unit_mass_kg	CAD volume x rho(SS316L)
T072	ENG-FD-07: Feed hose 3/8 in convoluted PTFE / 316 braid, rising coil 20	DESIGN	BOM v3.4.1	/lines/11	BOM row (CAD parts/ENG-FD-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T073	0.0379 kg	TO-VERIFY	BOM v3.4.1	/lines/11/unit_mass_kg	COTS mass (ESTIMATE 0.21 kg/m x 0.180 m (datasheet TO-VERIFY)); CG and inertia of a uniform bounding box
T074	ENG-FD-08: Fixed feed line 3/8 x 0.035 in 316L, 279 mm developed: 3D be	DESIGN	BOM v3.4.1	/lines/12	BOM row (CAD parts/ENG-FD-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T075	0.05379 kg	DERIVED	BOM v3.4.1	/lines/12/unit_mass_kg	CAD volume x rho(SS316L)
T076	ENG-FD-08F: Line flange (316L): socket for FD-08, AS568-012 FKM face sea	DESIGN	BOM v3.4.1	/lines/13	BOM row (CAD parts/ENG-FD-08F.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T077	0.01346 kg	DERIVED	BOM v3.4.1	/lines/13/unit_mass_kg	CAD volume x rho(SS316L)
T078	ENG-FD-09: Fixed hose-end clamp (6061-T6, base + cap, 2x M4 into GM-09	DESIGN	BOM v3.4.1	/lines/14	BOM row (CAD parts/ENG-FD-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T079	0.03803 kg	DERIVED	BOM v3.4.1	/lines/14/unit_mass_kg	CAD volume x rho(AL6061-T6)
T080	ENG-FD-11: P-clip 3/8 in, 316 band + silicone cushion, M4 to the airfra	DESIGN	BOM v3.4.1	/lines/15	BOM row (CAD parts/ENG-FD-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T081	0.004 kg	ASSUMED	BOM v3.4.1	/lines/15/unit_mass_kg	COTS mass (ESTIMATE 3-5 g); CG and inertia of a uniform bounding box
T082	ENG-FD-12F: Tank-line flange (316L) on FS-01B: socket for the tank line	DESIGN	BOM v3.4.1	/lines/16	BOM row (CAD parts/ENG-FD-12F.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T083	0.0135 kg	DERIVED	BOM v3.4.1	/lines/16/unit_mass_kg	CAD volume x rho(SS316L)
T084	ENG-FS-01A: MV module lower body (6061-T6): outlet chamber 18 × 12.3 (se	DESIGN	BOM v3.4.1	/lines/17	BOM row (CAD parts/ENG-FS-01A.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T085	0.0521 kg	DERIVED	BOM v3.4.1	/lines/17/unit_mass_kg	CAD volume x rho(AL6061-T6)
T086	ENG-FS-01B: MV module upper body (6061-T6): seat web + PCTFE seat counte	DESIGN	BOM v3.4.1	/lines/18	BOM row (CAD parts/ENG-FS-01B.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T087	0.04435 kg	DERIVED	BOM v3.4.1	/lines/18/unit_mass_kg	CAD volume x rho(AL6061-T6)
T088	ENG-FS-02: MV-P poppet: disc 14 × 3 + stem 3 h7, hollow 1.2 (balance ho	DESIGN	BOM v3.4.1	/lines/19	BOM row (CAD parts/ENG-FS-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T089	0.00464 kg	DERIVED	BOM v3.4.1	/lines/19/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T090	ENG-FS-03: MV-P soft seat insert PCTFE: land 11.5-12.5 x 0.3, OD 16 × 2	DESIGN	BOM v3.4.1	/lines/20	BOM row (CAD parts/ENG-FS-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T091	0.00023 kg	DERIVED	BOM v3.4.1	/lines/20/unit_mass_kg	CAD volume x rho(PCTFE)
T092	ENG-FS-04: MV-P balance piston 11.9 f7 + rod 4 h7 (one piece), seal gro	DESIGN	BOM v3.4.1	/lines/21	BOM row (CAD parts/ENG-FS-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T093	0.00772 kg	DERIVED	BOM v3.4.1	/lines/21/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T094	ENG-FS-06: MV actuator housing (dry side, 6061-T6): spring pocket, cam	DESIGN	BOM v3.4.1	/lines/22	BOM row (CAD parts/ENG-FS-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T095	0.02918 kg	DERIVED	BOM v3.4.1	/lines/22/unit_mass_kg	CAD volume x rho(AL6061-T6)
T096	ENG-FS-07: MV closing spring EN 10270-3 1.4310, d 1.8 / Dm 12 / n 6.6 a	DESIGN	BOM v3.4.1	/lines/23	BOM row (CAD parts/ENG-FS-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T097	0.008 kg	ASSUMED	BOM v3.4.1	/lines/23/unit_mass_kg	COTS mass (ESTIMATE wire volume 8 g); CG and inertia of a uniform bounding box
T098	ENG-FS-08: MV follower (17-4PH H900, flat hardened face) on the rod top	DESIGN	BOM v3.4.1	/lines/24	BOM row (CAD parts/ENG-FS-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T099	0.00448 kg	DERIVED	BOM v3.4.1	/lines/24/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)

ID	Value	Class	Source	Key	Source tag / formula
T100	ENG-FS-09: MV cam (r 6.0 → 8.8 over 45 deg, 15 deg dwell, width 5; clo	DESIGN	BOM v3.4.1	/lines/25	BOM row (CAD parts/ENG-FS-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T101	0.00813 kg	DERIVED	BOM v3.4.1	/lines/25/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)
T102	ENG-FS-10: MV opener servo Savox SB-2292SG (Electronics, commonality) o	DESIGN	BOM v3.4.1	/lines/26	BOM row (CAD parts/ENG-FS-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T103	0.081 kg	DATASHEET	BOM v3.4.1	/lines/26/unit_mass_kg	COTS mass (DATASHEET 81 g (electronics.json)); CG and inertia of a uniform bounding box
T104	ENG-FS-11: MV latch + release + dashpot + OPEN/CLOSED microswitches: RE	DESIGN	BOM v3.4.1	/lines/27	BOM row (CAD parts/ENG-FS-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T105	0.025 kg	ASSUMED	BOM v3.4.1	/lines/27/unit_mass_kg	COTS mass (ESTIMATE reserve 25 g (no design yet)); CG and inertia of a uniform bounding box
T106	ENG-FS-12: S1 back-relief check, insert cartridge 5.5 mm class, crack 0	DESIGN	BOM v3.4.1	/lines/28	BOM row (CAD parts/ENG-FS-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T107	0.002 kg	ASSUMED	BOM v3.4.1	/lines/28/unit_mass_kg	COTS mass (ESTIMATE 1-3 g); CG and inertia of a uniform bounding box
T108	ENG-FS-13: Secondary rupture disc 9.5 MPa +/-5 %, reverse-acting, 1/4-1	DESIGN	BOM v3.4.1	/lines/29	BOM row (CAD parts/ENG-FS-13.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T109	0.045 kg	ASSUMED	BOM v3.4.1	/lines/29/unit_mass_kg	COTS mass (ESTIMATE 35-60 g holder + disc); CG and inertia of a uniform bounding box
T110	ENG-FS-14: Drain port plug 1/8-27 NPT flush socket plug 316 (ground dra	DESIGN	BOM v3.4.1	/lines/30	BOM row (CAD parts/ENG-FS-14.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T111	0.004 kg	ASSUMED	BOM v3.4.1	/lines/30/unit_mass_kg	COTS mass (ESTIMATE 4 g); CG and inertia of a uniform bounding box
T112	ENG-FS-15: MV module mounting bracket (6061-T6): 2x M4 to FS-01A/B, 2x	DESIGN	BOM v3.4.1	/lines/31	BOM row (CAD parts/ENG-FS-15.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T113	0.0206 kg	DERIVED	BOM v3.4.1	/lines/31/unit_mass_kg	CAD volume x rho(AL6061-T6)
T114	ENG-GM-01: Engine collar half +Y (servo 1 lever, 135 deg)	DESIGN	BOM v3.4.1	/lines/32	BOM row (CAD parts/ENG-GM-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T115	0.06076 kg	DERIVED	BOM v3.4.1	/lines/32/unit_mass_kg	CAD volume x rho(AL6061-T6)
T116	ENG-GM-02: Engine collar half -Y (servo 2 lever, 225 deg), mirror of GM	DESIGN	BOM v3.4.1	/lines/33	BOM row (CAD parts/ENG-GM-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T117	0.06199 kg	DERIVED	BOM v3.4.1	/lines/33/unit_mass_kg	CAD volume x rho(AL6061-T6)
T118	ENG-GM-03: Gimbal ring (Cardan)	DESIGN	BOM v3.4.1	/lines/34	BOM row (CAD parts/ENG-GM-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T119	0.1843 kg	DERIVED	BOM v3.4.1	/lines/34/unit_mass_kg	CAD volume x rho(AL6061-T6)
T120	ENG-GM-04: Inner pivot pin 8 h6 x 18 (17-4PH H1150)	DESIGN	BOM v3.4.1	/lines/35	BOM row (CAD parts/ENG-GM-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T121	0.00693 kg	DERIVED	BOM v3.4.1	/lines/35/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T122	ENG-GM-05: Outer pivot pin 8 h6 x 18.5 (17-4PH H1150)	DESIGN	BOM v3.4.1	/lines/36	BOM row (CAD parts/ENG-GM-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T123	0.00713 kg	DERIVED	BOM v3.4.1	/lines/36/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T124	ENG-GM-06: Inner-axis spacer sleeve 11/8.1 × 2.85 (304)	DESIGN	BOM v3.4.1	/lines/37	BOM row (CAD parts/ENG-GM-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T125	0.001 kg	DERIVED	BOM v3.4.1	/lines/37/unit_mass_kg	CAD volume x rho(SS304)
T126	ENG-GM-07: Outer-axis spacer sleeve 11/8.1 × 1.9 (304)	DESIGN	BOM v3.4.1	/lines/38	BOM row (CAD parts/ENG-GM-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T127	0.00065 kg	DERIVED	BOM v3.4.1	/lines/38/unit_mass_kg	CAD volume x rho(SS304)

ID	Value	Class	Source	Key	Source tag / formula
T128	ENG-GM-08: Yoke post (outer-axis bearing housing + frame post), x2	DESIGN	BOM v3.4.1	/lines/39	BOM row (CAD parts/ENG-GM-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T129	0.07338 kg	DERIVED	BOM v3.4.1	/lines/39/unit_mass_kg	CAD volume x rho(AL6061-T6)
T130	ENG-GM-09: Upper thrust ring (engine module to airframe, thrust path)	DESIGN	BOM v3.4.1	/lines/40	BOM row (CAD parts/ENG-GM-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T131	0.152 kg	DERIVED	BOM v3.4.1	/lines/40/unit_mass_kg	CAD volume x rho(AL6061-T6)
T132	ENG-GM-10: Lower ring (stop ring, TVC servo base, frame tie)	DESIGN	BOM v3.4.1	/lines/41	BOM row (CAD parts/ENG-GM-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T133	0.1927 kg	DERIVED	BOM v3.4.1	/lines/41/unit_mass_kg	CAD volume x rho(AL6061-T6)
T134	ENG-GM-11: Conical stop bumper, PU 90A, 8 deg line contact	DESIGN	BOM v3.4.1	/lines/42	BOM row (CAD parts/ENG-GM-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T135	0.0092 kg	DERIVED	BOM v3.4.1	/lines/42/unit_mass_kg	CAD volume x rho(ELASTOMER-PU)
T136	ENG-GM-12: Transport / neutral lock, 2 half rings (REMOVE BEFORE FLIGHT	DESIGN	BOM v3.4.1	/lines/43	BOM row (CAD parts/ENG-GM-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T137	0.06225 kg	DERIVED	BOM v3.4.1	/lines/43/unit_mass_kg	CAD volume x rho(PETG)
T138	ENG-GM-13: TVC servo 1 bracket (C-frame: top/bottom ear blocks + back r	DESIGN	BOM v3.4.1	/lines/44	BOM row (CAD parts/ENG-GM-13.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T139	0.04893 kg	DERIVED	BOM v3.4.1	/lines/44/unit_mass_kg	CAD volume x rho(AL6061-T6)
T140	ENG-GM-14: TVC servo 2 bracket (mirror of GM-13)	DESIGN	BOM v3.4.1	/lines/45	BOM row (CAD parts/ENG-GM-14.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T141	0.04893 kg	DERIVED	BOM v3.4.1	/lines/45/unit_mass_kg	CAD volume x rho(AL6061-T6)
T142	ENG-GM-15: TVC servo Savox SB-2292SG (electronics.json)	DESIGN	BOM v3.4.1	/lines/46	BOM row (CAD parts/ENG-GM-15.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T143	0.081 kg	DATASHEET	BOM v3.4.1	/lines/46/unit_mass_kg	COTS mass (INTERFACE electronics.json v2 (actuators.tvc_A.mass_kg)); CG and inertia of a uniform bounding box
T144	ENG-GM-16: TVC horn r 11.1 mm, 25T clamp (7075-T6)	DESIGN	BOM v3.4.1	/lines/47	BOM row (CAD parts/ENG-GM-16.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T145	0.00218 kg	DERIVED	BOM v3.4.1	/lines/47/unit_mass_kg	CAD volume x rho(AL7075-T6)
T146	ENG-GM-17: TVC pushrod dog-bone 42.3 mm between eyes (7075-T6), fixed l	DESIGN	BOM v3.4.1	/lines/48	BOM row (CAD parts/ENG-GM-17.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T147	0.00421 kg	DERIVED	BOM v3.4.1	/lines/48/unit_mass_kg	CAD volume x rho(AL7075-T6)
T148	ENG-GM-18: Lever pivot stud: flange 7.5 × 1.5 (= washer), 5 h7 shoulder	DESIGN	BOM v3.4.1	/lines/49	BOM row (CAD parts/ENG-GM-18.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T149	0.002 kg	DERIVED	BOM v3.4.1	/lines/49/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T150	ENG-GM-19: Joint spacer washer 5.1 × 7.5 × 1.5 (17-4PH H1150), ground f	DESIGN	BOM v3.4.1	/lines/50	BOM row (CAD parts/ENG-GM-19.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T151	0.00028 kg	DERIVED	BOM v3.4.1	/lines/50/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T152	ENG-GM-20: Horn-joint shoulder screw: head 8.5 × 4, 5 h7 × 10 shoulder,	DESIGN	BOM v3.4.1	/lines/51	BOM row (CAD parts/ENG-GM-20.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T153	0.00352 kg	DERIVED	BOM v3.4.1	/lines/51/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T154	ENG-GR-01: Fuel grain 1/2: ABS 53.5 OD x 115.0, straight centred port 2	DESIGN	BOM v3.4.1	/lines/52	BOM row (CAD parts/ENG-GR-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T155	0.2122 kg	DERIVED	BOM v3.4.1	/lines/52/unit_mass_kg	CAD volume x rho(ABS)
T156	ENG-GR-02: Fuel grain 2/2: ABS 53.5 OD x 115.0, straight centred port 2	DESIGN	BOM v3.4.1	/lines/53	BOM row (CAD parts/ENG-GR-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T157	0.2122 kg	DERIVED	BOM v3.4.1	/lines/53/unit_mass_kg	CAD volume x rho(ABS)

ID	Value	Class	Source	Key	Source tag / formula
T158	ENG-HW-20: Deep-groove ball bearing 608-2RS (8×22×7)	DESIGN	BOM v3.4.1	/lines/54	BOM row (CAD parts/ENG-HW-20.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T159	0.012 kg	DATASHEET	BOM v3.4.1	/lines/54/unit_mass_kg	COTS mass (DATASHEET SKF 608-2RSH 0.012 kg (verify)); CG and inertia of a uniform bounding box
T160	ENG-HW-21: Retaining washer DIN 6799-6 (shaft 7-9 mm), spring steel	DESIGN	BOM v3.4.1	/lines/55	BOM row (CAD parts/ENG-HW-21.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T161	0.0003 kg	ASSUMED	BOM v3.4.1	/lines/55/unit_mass_kg	COTS mass (ASSUMED 0.2-0.4 g); CG and inertia of a uniform bounding box
T162	ENG-HW-22: Spherical plain bearing GE 5 E (steel/steel) - outer ring (p	DESIGN	BOM v3.4.1	/lines/56	BOM row (CAD parts/ENG-HW-22.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T163	0.0025 kg	DERIVED	BOM v3.4.1	/lines/56/unit_mass_kg	CAD volume x rho(STEEL)
T164	ENG-HW-25: Horn axial screw ISO 10642 M3 × 6 (countersunk, 10.9), Locti	DESIGN	BOM v3.4.1	/lines/57	BOM row (CAD parts/ENG-HW-25.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T165	0.0005 kg	ASSUMED	BOM v3.4.1	/lines/57/unit_mass_kg	COTS mass (ASSUMED 0.4-0.6 g); CG and inertia of a uniform bounding box
T166	ENG-IG-01: Torch igniter (N2O/propane, spark) - ENVELOPE 20 × 45 mm + M	DESIGN	BOM v3.4.1	/lines/58	BOM row (CAD parts/ENG-IG-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T167	0.07 kg	ASSUMED	BOM v3.4.1	/lines/58/unit_mass_kg	COTS mass (ASSUMED body 50 g + plug 10 g (ignition_v2.json) + 2 fittings 10 g); CG and inertia of a uniform bounding box
T168	ENG-IG-02: Torch inlet check valve, cartridge insert in the torch inlet	DESIGN	BOM v3.4.1	/lines/59	BOM row (CAD parts/ENG-IG-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T169	0.002 kg	ASSUMED	BOM v3.4.1	/lines/59/unit_mass_kg	COTS mass (ASSUMED 1-3 g cartridge); CG and inertia of a uniform bounding box
T170	ENG-NZ-01: Graphite throat insert (Dt 13.72, De 19.41, OD 44, aft spigo	DESIGN	BOM v3.4.1	/lines/60	BOM row (CAD parts/ENG-NZ-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T171	0.04726 kg	DERIVED	BOM v3.4.1	/lines/60/unit_mass_kg	CAD volume x rho(GRAPHITE-ISO)
T172	ENG-NZ-02: Insert insulating sleeve (carbon-phenolic 50/44, 3 mm)	DESIGN	BOM v3.4.1	/lines/61	BOM row (CAD parts/ENG-NZ-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T173	0.01196 kg	DERIVED	BOM v3.4.1	/lines/61/unit_mass_kg	CAD volume x rho(CARBON-PHENOLIC)
T174	ENG-NZ-03: Nozzle-joint hot layer J1: carbon-phenolic washer 30.6/49.8	DESIGN	BOM v3.4.1	/lines/62	BOM row (CAD parts/ENG-NZ-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T175	0.00352 kg	DERIVED	BOM v3.4.1	/lines/62/unit_mass_kg	CAD volume x rho(CARBON-PHENOLIC)
T176	ENG-NZ-04: Nozzle-joint lip-face seal J1: flexible graphite ring 30.6/4	DESIGN	BOM v3.4.1	/lines/63	BOM row (CAD parts/ENG-NZ-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T177	0.00068 kg	DERIVED	BOM v3.4.1	/lines/63/unit_mass_kg	CAD volume x rho(FLEX-GRAPHITE-OI)
T178	ENG-PD-01: Pintle drive frame, 2 pieces: A (-Y foot + bridge + leg + se	DESIGN	BOM v3.4.1	/lines/64	BOM row (CAD parts/ENG-PD-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T179	0.03886 kg	DERIVED	BOM v3.4.1	/lines/64/unit_mass_kg	CAD volume x rho(AL6061-T6)
T180	ENG-PD-02: Pintle crank housing (bearing seats A 18 and B 19, crank cav	DESIGN	BOM v3.4.1	/lines/65	BOM row (CAD parts/ENG-PD-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T181	0.01983 kg	DERIVED	BOM v3.4.1	/lines/65/unit_mass_kg	CAD volume x rho(AL6061-T6)
T182	ENG-PD-03: Crank-housing cover with encoder PCB pocket	DESIGN	BOM v3.4.1	/lines/66	BOM row (CAD parts/ENG-PD-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T183	0.00261 kg	DERIVED	BOM v3.4.1	/lines/66/unit_mass_kg	CAD volume x rho(AL6061-T6)
T184	ENG-PD-04: Pintle crankshaft (throw r 3.0, pin 3, journals 12 / 10, dog	DESIGN	BOM v3.4.1	/lines/67	BOM row (CAD parts/ENG-PD-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T185	0.01249 kg	DERIVED	BOM v3.4.1	/lines/67/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)

ID	Value	Class	Source	Key	Source tag / formula
T186	ENG-PD-05: Dog hub on the HV69 6 mm 25T spline (one-sided drive tooth 6	DESIGN	BOM v3.4.1	/lines/68	BOM row (CAD parts/ENG-PD-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T187	0.00061 kg	DERIVED	BOM v3.4.1	/lines/68/unit_mass_kg	CAD volume x rho(AL7075-T6)
T188	ENG-PD-06: Pintle connecting link 28 mm centres, eyes OD 8 x 3.8 with 4	DESIGN	BOM v3.4.1	/lines/69	BOM row (CAD parts/ENG-PD-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T189	0.00394 kg	DERIVED	BOM v3.4.1	/lines/69/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T190	ENG-PD-07: Pintle carriage: needle clamp arm (M3 stem thread) + link cl	DESIGN	BOM v3.4.1	/lines/70	BOM row (CAD parts/ENG-PD-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T191	0.01003 kg	DERIVED	BOM v3.4.1	/lines/70/unit_mass_kg	CAD volume x rho(AL6061-T6)
T192	ENG-PD-08: Miniature linear guide block HIWIN MGN09H (C0 4020 N, M0Y 18	DESIGN	BOM v3.4.1	/lines/71	BOM row (CAD parts/ENG-PD-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T193	0.03 kg	TO-VERIFY	BOM v3.4.1	/lines/71/unit_mass_kg	COTS mass (DATASHEET HIWIN MG catalogue 2023 (copy) 0.03 kg incl. rail share? - block only TO-VERIFY); CG and inertia of a uniform bounding box
T194	ENG-PD-09: Spiral (clock) return spring, 0.25 N m at closed + 0.12 N m/	DESIGN	BOM v3.4.1	/lines/72	BOM row (CAD parts/ENG-PD-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T195	0.0055 kg	ASSUMED	BOM v3.4.1	/lines/72/unit_mass_kg	COTS mass (ESTIMATE 5-6 g (strip 4.5 x ~0.71, ~206 mm active, 1.4310 +C1900 class: calcs/pd09_spring_fit_v1.json; v3.4 erratum: the earlier 4.5 × 0.25 strip note gave 9.7 GPa at 0.455 N m)); CG and in
T196	ENG-PD-10: Radiation shield under the HV69: 0.5 mm polished aluminium L	DESIGN	BOM v3.4.1	/lines/73	BOM row (CAD parts/ENG-PD-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T197	0.0007 kg	DERIVED	BOM v3.4.1	/lines/73/unit_mass_kg	CAD volume x rho(AL6061-T6)
T198	ENG-PD-11: Isolator washer G11 7/3.4 × 2 (frame feet to head top)	DESIGN	BOM v3.4.1	/lines/74	BOM row (CAD parts/ENG-PD-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T199	0.00011 kg	DERIVED	BOM v3.4.1	/lines/74/unit_mass_kg	CAD volume x rho(G11)
T200	ENG-PD-12: G11 pad 1.5 mm under the HV69 ears (window for the case)	DESIGN	BOM v3.4.1	/lines/75	BOM row (CAD parts/ENG-PD-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T201	0.00086 kg	DERIVED	BOM v3.4.1	/lines/75/unit_mass_kg	CAD volume x rho(G11)
T202	ENG-PD-13: Pintle servo MKS HV69 (29 × 13 × 28.7, 21 g, 6 mm 25T spline	DESIGN	BOM v3.4.1	/lines/76	BOM row (CAD parts/ENG-PD-13.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T203	0.021 kg	DATASHEET	BOM v3.4.1	/lines/76/unit_mass_kg	COTS mass (DATASHEET 21 g (Electronics pintle_mini_servo_v1, retailer copies of MKS data)); CG and inertia of a uniform bounding box
T204	ENG-PD-14: Thin-section ball bearing 12×18×4 (NSK/NMB 6701 class, 'NB70	DESIGN	BOM v3.4.1	/lines/77	BOM row (CAD parts/ENG-PD-14.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T205	0.0031 kg	ASSUMED	BOM v3.4.1	/lines/77/unit_mass_kg	COTS mass (ASSUMED catalogue class 3 g (6701)); CG and inertia of a uniform bounding box
T206	ENG-PD-15: Deep-groove ball bearing 6800-2Z 10×19×5, C0r 840 N	DESIGN	BOM v3.4.1	/lines/78	BOM row (CAD parts/ENG-PD-15.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T207	0.0055 kg	ASSUMED	BOM v3.4.1	/lines/78/unit_mass_kg	COTS mass (ASSUMED catalogue class 5.5 g (61800)); CG and inertia of a uniform bounding box
T208	ENG-PD-16: Pin 3 g6 × 12 (17-4PH H900) with e-clip grooves (crank pin b	DESIGN	BOM v3.4.1	/lines/79	BOM row (CAD parts/ENG-PD-16.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T209	0.00066 kg	DERIVED	BOM v3.4.1	/lines/79/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)
T210	ENG-PD-17: PTFE-lined steel-backed bush 3 × 4.5 × 3.8 (DU class)	DESIGN	BOM v3.4.1	/lines/80	BOM row (CAD parts/ENG-PD-17.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T211	0.0003 kg	ASSUMED	BOM v3.4.1	/lines/80/unit_mass_kg	COTS mass (ESTIMATE 0.3 g); CG and inertia of a uniform bounding box

ID	Value	Class	Source	Key	Source tag / formula
T212	ENG-PD-18: Encoder magnet NdFeB N35 diametral 6 × 1	DESIGN	BOM v3.4.1	/lines/81	BOM row (CAD parts/ENG-PD-18.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T213	0.00021 kg	DERIVED	BOM v3.4.1	/lines/81/unit_mass_kg	CAD volume x rho(NDFEB)
T214	ENG-PD-19: Miniature rail HIWIN MGNR09R, length 46.1 mm (cut to length)	DESIGN	BOM v3.4.1	/lines/82	BOM row (CAD parts/ENG-PD-19.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T215	0.0175 kg	DATASHEET	BOM v3.4.1	/lines/82/unit_mass_kg	COTS mass (DATASHEET 0.38 kg/m (HIWIN MG catalogue copy)); CG and inertia of a uniform bounding box
T216	ENG-PD-20: G11 stand-off 5/2.2 × 2 (shield to servo plate)	DESIGN	BOM v3.4.1	/lines/83	BOM row (CAD parts/ENG-PD-20.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T217	6e-05 kg	DERIVED	BOM v3.4.1	/lines/83/unit_mass_kg	CAD volume x rho(G11)
T218	ENG-TC-01: Combustion case tube	DESIGN	BOM v3.4.1	/lines/84	BOM row (CAD parts/ENG-TC-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T219	0.4625 kg	DERIVED	BOM v3.4.1	/lines/84/unit_mass_kg	CAD volume x rho(AL6061-T6)
T220	ENG-TC-02: Injector head / forward closure (pintle seat bore, manifold,	DESIGN	BOM v3.4.1	/lines/85	BOM row (CAD parts/ENG-TC-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T221	0.4415 kg	DERIVED	BOM v3.4.1	/lines/85/unit_mass_kg	CAD volume x rho(AL6061-T6)
T222	ENG-TC-03: Pintle seat insert (5.5 mm sharp-edged seat, 37.5 deg diffus	DESIGN	BOM v3.4.1	/lines/86	BOM row (CAD parts/ENG-TC-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T223	0.01205 kg	DERIVED	BOM v3.4.1	/lines/86/unit_mass_kg	CAD volume x rho(IN718)
T224	ENG-TC-04: Bonnet: stem seal + guide housing, E1-M inlet boss (8 mm tra	DESIGN	BOM v3.4.1	/lines/87	BOM row (CAD parts/ENG-TC-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T225	0.05354 kg	DERIVED	BOM v3.4.1	/lines/87/unit_mass_kg	CAD volume x rho(AL6061-T6)
T226	ENG-TC-05: Pintle needle (15 deg metering cone, 6 mm body, 3 mm stem)	DESIGN	BOM v3.4.1	/lines/88	BOM row (CAD parts/ENG-TC-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T227	0.01663 kg	DERIVED	BOM v3.4.1	/lines/88/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T228	ENG-TC-06: Pre-combustion chamber insulator (phenolic, bore 49.9, wall	DESIGN	BOM v3.4.1	/lines/89	BOM row (CAD parts/ENG-TC-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T229	0.02445 kg	DERIVED	BOM v3.4.1	/lines/89/unit_mass_kg	CAD volume x rho(PHENOLIC)
T230	ENG-TC-07: Grain liner (phenolic tube 57.10/53.95)	DESIGN	BOM v3.4.1	/lines/90	BOM row (CAD parts/ENG-TC-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T231	0.08531 kg	DERIVED	BOM v3.4.1	/lines/90/unit_mass_kg	CAD volume x rho(PHENOLIC)
T232	ENG-TC-08: Post-chamber + convergent insulator (phenolic, 3.50 mm over	DESIGN	BOM v3.4.1	/lines/91	BOM row (CAD parts/ENG-TC-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T233	0.0515 kg	DERIVED	BOM v3.4.1	/lines/91/unit_mass_kg	CAD volume x rho(PHENOLIC)
T234	ENG-TC-09: Aft section v3.2: post-chamber barrel + nozzle holder + reta	DESIGN	BOM v3.4.1	/lines/92	BOM row (CAD parts/ENG-TC-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T235	0.5786 kg	DERIVED	BOM v3.4.1	/lines/92/unit_mass_kg	CAD volume x rho(SS316L)
T236	ENG-TC-10: Injector-face insulator disc, phenolic 2 mm	DESIGN	BOM v3.4.1	/lines/93	BOM row (CAD parts/ENG-TC-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T237	0.00429 kg	DERIVED	BOM v3.4.1	/lines/93/unit_mass_kg	CAD volume x rho(PHENOLIC)
T238	ENG-TC-11: Stem guide bushing 5/3 × 4 (PTFE + 25 % glass)	DESIGN	BOM v3.4.1	/lines/94	BOM row (CAD parts/ENG-TC-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T239	0.00011 kg	DERIVED	BOM v3.4.1	/lines/94/unit_mass_kg	CAD volume x rho(PTFE)
T240	ENG-BL-01: bleed tap T4 crown take-off fitting with restrictor	DESIGN	BOM v3.4.1	/provision_lines/0	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T241	MISSING	MISSING	BOM v3.4.1	/provision_lines/0/unit_mass_kg	ESTIMATE 10-15 g (no datasheet)
T242	ENG-BL-02: bleed tap FD-01 radial take-off (plug-tap fitting)	DESIGN	BOM v3.4.1	/provision_lines/1	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]

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T243	MISSING	MISSING	BOM v3.4.1	/provision_lines/1/unit_mass_kg	ESTIMATE 10-15 g (no datasheet)
T244	ENG-BL-03: bleed line 1/8 in x 0.028 in 316L tube + tee	DESIGN	BOM v3.4.1	/provision_lines/2	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T245	0.0132 kg	DERIVED	BOM v3.4.1	/provision_lines/2/unit_mass_kg	CALC tube only: 5.504 mm2 × 8000 kg/m3 x 0.3 m (route length ESTIMATE); tee ESTIMATE +8 g
T246	ENG-BL-04: bleed crossing hose -2 class PTFE / 316 braid, ~0.2 m	DESIGN	BOM v3.4.1	/provision_lines/3	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T247	MISSING	MISSING	BOM v3.4.1	/provision_lines/3/unit_mass_kg	ESTIMATE 15-25 g (no datasheet)
T248	ENG-BL-05: bleed valve 2/2 NC solenoid, x2 in series	DESIGN	BOM v3.4.1	/provision_lines/4	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T249	MISSING	MISSING	BOM v3.4.1	/provision_lines/4/unit_mass_kg	TO-VERIFY (datasheet)
T250	ENG-BL-06: bleed overboard vent: balanced opposed pair or axial outlet	DESIGN	BOM v3.4.1	/provision_lines/5	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T251	MISSING	MISSING	BOM v3.4.1	/provision_lines/5/unit_mass_kg	ESTIMATE 10-20 g (no datasheet)
T252	ENG-NZ-05: J1b contingency: insulating layer 0.5 mm phlogopite mica bet	DESIGN	BOM v3.4.1	/contingency_lines/0	BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T253	TO-DEFINE	TO-DEFINE	BOM v3.4.1	/contingency_lines/0/unit_mass_kg	not fitted
T254	6	DERIVED	generator	formula: count of BOM lines with part-number family EL	DERIVED by build_datasheet.py: count of BOM lines with part-number family EL ← T050, T052, T054, T056, T058, T060
T255	0.074 kg	DERIVED	generator	formula: Σ qty × unit mass over the EL lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the EL lines ← T051, T053, T055, T057, T059, T061
T256	11	DERIVED	generator	formula: count of BOM lines with part-number family FD	DERIVED by build_datasheet.py: count of BOM lines with part-number family FD ← T062, T064, T066, T068, T070, T072, T074, T076 ...
T257	0.268 kg	DERIVED	generator	formula: Σ qty × unit mass over the FD lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the FD lines ← T063, T065, T067, T069, T071, T073, T075, T077 ...
T258	15	DERIVED	generator	formula: count of BOM lines with part-number family FS	DERIVED by build_datasheet.py: count of BOM lines with part-number family FS ← T084, T086, T088, T090, T092, T094, T096, T098 ...
T259	0.3364 kg	DERIVED	generator	formula: Σ qty × unit mass over the FS lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the FS lines ← T085, T087, T089, T091, T093, T095, T097, T099 ...
T260	19	DERIVED	generator	formula: count of BOM lines with part-number family GM	DERIVED by build_datasheet.py: count of BOM lines with part-number family GM ← T114, T116, T118, T120, T122, T124, T126, T128 ...
T261	1.125 kg	DERIVED	generator	formula: Σ qty × unit mass over the GM lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the GM lines ← T115, T117, T119, T121, T123, T125, T127, T129 ...
T262	2	DERIVED	generator	formula: count of BOM lines with part-number family GR	DERIVED by build_datasheet.py: count of BOM lines with part-number family GR ← T154, T156
T263	0.4244 kg	DERIVED	generator	formula: Σ qty × unit mass over the GR lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the GR lines ← T155, T157
T264	4	DERIVED	generator	formula: count of BOM lines with part-number family HW	DERIVED by build_datasheet.py: count of BOM lines with part-number family HW ← T158, T160, T162, T164
T265	0.0602 kg	DERIVED	generator	formula: Σ qty × unit mass over the HW lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the HW lines ← T159, T161, T163, T165
T266	2	DERIVED	generator	formula: count of BOM lines with part-number family IG	DERIVED by build_datasheet.py: count of BOM lines with part-number family IG ← T166, T168
T267	0.074 kg	DERIVED	generator	formula: Σ qty × unit mass over the IG lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the IG lines ← T167, T169
T268	4	DERIVED	generator	formula: count of BOM lines with part-number family NZ	DERIVED by build_datasheet.py: count of BOM lines with part-number family NZ ← T170, T172, T174, T176
T269	0.06342 kg	DERIVED	generator	formula: Σ qty × unit mass over the NZ lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the NZ lines ← T171, T173, T175, T177
T270	20	DERIVED	generator	formula: count of BOM lines with part-number family PD	DERIVED by build_datasheet.py: count of BOM lines with part-number family PD ← T178, T180, T182, T184, T186, T188, T190, T192 ...
T271	0.1746 kg	DERIVED	generator	formula: Σ qty × unit mass over the PD lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the PD lines ← T179, T181, T183, T185, T187, T189, T191, T193 ...
T272	11	DERIVED	generator	formula: count of BOM lines with part-number family TC	DERIVED by build_datasheet.py: count of BOM lines with part-number family TC ← T218, T220, T222, T224, T226, T228, T230, T232 ...
T273	1.731 kg	DERIVED	generator	formula: Σ qty × unit mass over the TC lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the TC lines ← T219, T221, T223, T225, T227, T229, T231, T233 ...

ID	Value	Class	Source	Key	Source tag / formula
T274	4.33 kg	DERIVED	generator	formula: Σ qty × unit mass over all BOM lines (transport lock excluded)	DERIVED by build_datasheet.py: Σ qty × unit mass over all BOM lines (transport lock excluded) ← T255, T257, T259, T261, T263, T265, T267, T269 …
T275	0.06225 kg	DERIVED	generator	formula: Σ qty × unit mass over the transport-lock lines (group lock)	DERIVED by build_datasheet.py: Σ qty × unit mass over the transport-lock lines (group lock) ← T137
T276	TO-QUOTE (not summed)	MISSING	BOM v3.4.1	/currency	BOM /currency: prices not given, every line TO-QUOTE unless dated [total not summed: no line carries a DK-6400 quotation]
T277	17	DERIVED	generator	formula: count of BOM lines with material_name "COTS assembly (mixed materials), mass from datasheet or assumed"	DERIVED by build_datasheet.py: count of BOM lines with material_name "COTS assembly (mixed materials), mass from datasheet or assumed" ← T050, T052, T054, T058, T060, T064, T072, T080 …
T278	0.5409 kg	DERIVED	generator	formula: Σ qty × unit mass, material "COTS assembly (mixed materials), mass from datasheet or assumed"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "COTS assembly (mixed materials), mass from datasheet or assumed" ← T051, T053, T055, T059, T061, T065, T073, T081 …
T279	8	DERIVED	generator	formula: count of oxygen-clean lines, material "COTS assembly (mixed materials), mass from datasheet or assumed"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "COTS assembly (mixed materials), mass from datasheet or assumed" ← T050, T052, T054, T058, T060, T064, T072, T080 …
T280	2	DERIVED	generator	formula: count of BOM lines with material_name "NdFeB magnet N35 (diametral, Ni-plated)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "NdFeB magnet N35 (diametral, Ni-plated)" ← T056, T212
T281	0.00121 kg	DERIVED	generator	formula: Σ qty × unit mass, material "NdFeB magnet N35 (diametral, Ni-plated)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "NdFeB magnet N35 (diametral, Ni-plated)" ← T057, T213
T282	0	DERIVED	generator	formula: count of oxygen-clean lines, material "NdFeB magnet N35 (diametral, Ni-plated)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "NdFeB magnet N35 (diametral, Ni-plated)" ← T056, T212
T283	8	DERIVED	generator	formula: count of BOM lines with material_name "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts" ← T062, T066, T068, T070, T074, T076, T082, T234
T284	0.7547 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts" ← T063, T067, T069, T071, T075, T077, T083, T235
T285	6	DERIVED	generator	formula: count of oxygen-clean lines, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts" ← T062, T066, T068, T070, T074, T076, T082, T234
T286	21	DERIVED	generator	formula: count of BOM lines with material_name "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube" ← T078, T084, T086, T094, T112, T114, T116, T118 …
T287	2.11 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube" ← T079, T085, T087, T095, T113, T115, T117, T119 …
T288	4	DERIVED	generator	formula: count of oxygen-clean lines, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube" ← T078, T084, T086, T094, T112, T114, T116, T118 …
T289	9	DERIVED	generator	formula: count of BOM lines with material_name "17-4PH (1.4542) condition H1150, pins"	DERIVED by build_datasheet.py: count of BOM lines with material_name "17-4PH (1.4542) condition H1150, pins" ← T088, T092, T120, T122, T148, T150, T152, T188 …
T290	0.07377 kg	DERIVED	generator	formula: Σ qty × unit mass, material "17-4PH (1.4542) condition H1150, pins"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "17-4PH (1.4542) condition H1150, pins" ← T089, T093, T121, T123, T149, T151, T153, T189 …
T291	3	DERIVED	generator	formula: count of oxygen-clean lines, material "17-4PH (1.4542) condition H1150, pins"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "17-4PH (1.4542) condition H1150, pins" ← T088, T092, T120, T122, T148, T150, T152, T188 …
T292	1	DERIVED	generator	formula: count of BOM lines with material_name "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible"	DERIVED by build_datasheet.py: count of BOM lines with material_name "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible" ← T090
T293	0.00023 kg	DERIVED	generator	formula: Σ qty × unit mass, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible" ← T091

ID	Value	Class	Source	Key	Source tag / formula
T294	1	DERIVED	generator	formula: count of oxygen-clean lines, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible" ← T090
T295	2	DERIVED	generator	formula: count of BOM lines with material_name "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)" ← T096, T194
T296	0.0135 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)" ← T097, T195
T297	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)" ← T096, T194
T298	4	DERIVED	generator	formula: count of BOM lines with material_name "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet"	DERIVED by build_datasheet.py: count of BOM lines with material_name "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet" ← T098, T100, T184, T208
T299	0.02576 kg	DERIVED	generator	formula: Σ qty × unit mass, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet" ← T099, T101, T185, T209
T300	0	DERIVED	generator	formula: count of oxygen-clean lines, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet" ← T098, T100, T184, T208
T301	2	DERIVED	generator	formula: count of BOM lines with material_name "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)" ← T124, T126
T302	0.0033 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)" ← T125, T127
T303	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)" ← T124, T126
T304	1	DERIVED	generator	formula: count of BOM lines with material_name "Polyurethane bumper 90A"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Polyurethane bumper 90A" ← T134
T305	0.0092 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Polyurethane bumper 90A"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Polyurethane bumper 90A" ← T135
T306	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Polyurethane bumper 90A"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Polyurethane bumper 90A" ← T134
T307	3	DERIVED	generator	formula: count of BOM lines with material_name "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar" ← T144, T146, T186
T308	0.01339 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar" ← T145, T147, T187
T309	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar" ← T144, T146, T186
T310	2	DERIVED	generator	formula: count of BOM lines with material_name "ABS, FDM-printed fuel grain (100 % infill)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "ABS, FDM-printed fuel grain (100 % infill)" ← T154, T156
T311	0.4244 kg	DERIVED	generator	formula: Σ qty × unit mass, material "ABS, FDM-printed fuel grain (100 % infill)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "ABS, FDM-printed fuel grain (100 % infill)" ← T155, T157
T312	0	DERIVED	generator	formula: count of oxygen-clean lines, material "ABS, FDM-printed fuel grain (100 % infill)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "ABS, FDM-printed fuel grain (100 % infill)" ← T154, T156
T313	9	DERIVED	generator	formula: count of BOM lines with material_name "Carbon/bearing steel (generic COTS)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Carbon/bearing steel (generic COTS)" ← T158, T160, T162, T164, T192, T204, T206, T210 …
T314	0.1169 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Carbon/bearing steel (generic COTS)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Carbon/bearing steel (generic COTS)" ← T159, T161, T163, T165, T193, T205, T207, T211 …

ID	Value	Class	Source	Key	Source tag / formula
T315	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Carbon/bearing steel (generic COTS)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Carbon/bearing steel (generic COTS)" ← T158, T160, T162, T164, T192, T204, T206, T210 ...
T316	1	DERIVED	generator	formula: count of BOM lines with material_name "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)" ← T170
T317	0.04726 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)" ← T171
T318	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)" ← T170
T319	2	DERIVED	generator	formula: count of BOM lines with material_name "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-0	DERIVED by build_datasheet.py: count of BOM lines with material_name "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03" ← T172, T174
T320	0.01548 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03" ← T173, T175
T321	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-0	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03" ← T172, T174
T322	1	DERIVED	generator	formula: count of BOM lines with material_name "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 m	DERIVED by build_datasheet.py: count of BOM lines with material_name "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face sea
T323	0.00068 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face seal NZ-04" ←
T324	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 m	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face sea
T325	3	DERIVED	generator	formula: count of BOM lines with material_name "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy" ← T198, T200, T216
T326	0.00142 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy" ← T199, T201, T217
T327	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy" ← T198, T200, T216
T328	1	DERIVED	generator	formula: count of BOM lines with material_name "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative" ← T222
T329	0.01205 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative" ← T223
T330	1	DERIVED	generator	formula: count of oxygen-clean lines, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative" ← T222
T331	4	DERIVED	generator	formula: count of BOM lines with material_name "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)" ← T228, T230, T232, T236

ID	Value	Class	Source	Key	Source tag / formula
T332	0.1656 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)" ← T229, T231, T233, T237
T333	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)" ← T228, T230, T232, T236
T334	1	DERIVED	generator	formula: count of BOM lines with material_name "PTFE"	DERIVED by build_datasheet.py: count of BOM lines with material_name "PTFE" ← T238
T335	0.00011 kg	DERIVED	generator	formula: Σ qty × unit mass, material "PTFE"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "PTFE" ← T239
T336	1	DERIVED	generator	formula: count of oxygen-clean lines, material "PTFE"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "PTFE" ← T238
T337	1.179 kg	DERIVED	MEC v2	/fixed_module/tank/tube_mass_kg	DERIVED
T338	6061-T6 tube 101.6 × 3.175 + 2 bolted bulkheads (radial M6, 2 FKM O-rings each)	DERIVED	MEC v2	/fixed_module/tank/concept	DERIVED
T339	0.058–0.097 kg	ASSUMED	BOM v3.4.1	/provision_mass_estimate_kg/estimate_range_kg	(class set by generator: ASSUMED)
T340	A1 head-end bleed provision (lead 2026-10-13), NOT modelled in CAD v3.4.1; range = ESTIM...	ASSUMED	BOM v3.4.1	/provision_mass_estimate_kg/note	(class set by generator: ASSUMED)
T341	F1 valve mass: 2 × 0.30 kg is held as a MASS RISK, not baseline; Electronics' 12 V NC va...	LEAD	CL 176 lines	L164(7)	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T342	2	LEAD	CL 176 lines	L164(7) [regex (\d+) x ([\d.]+) kg]	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T343	0.3 kg	LEAD	CL 176 lines	L164(7) [regex (\d+) x ([\d.]+) kg]	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T344	0.6 kg	TO-VERIFY	generator	formula: n valves × mass per valve (MASS RISK, not baseline)	DERIVED by build_datasheet.py: n valves × mass per valve (MASS RISK, not baseline) ← T342, T343
T345	thrust-ring radial screws (16 x M5); isolation valve + actuator (mass UNKNOWN: Electroni...	MISSING	MEC v2	/fixed_module/not_modelled	OPEN
T346	MISSING	MISSING	—		[mass of the items listed as not modelled in mechanical.json v2 (CAD v3.3 basis)]
T347	0.16 kg	ASSUMED	ELE v2	/ecu/mass_kg	ASSUMED (PCB+Teensy+connectors 0.11, enclosure 0.05)
T348	0.12–0.22 kg	ASSUMED	ELE v2	/ecu/mass_kg/range	(class set by generator: ASSUMED)
T349	separate engine ECU (Teensy 4.1) on the fixed thrust structure, CAN link to flight compu...	DERIVED	ELE v2	/ecu/architecture	DERIVED trade (REPORT Architecture)
T350	2.002 kg	LEAD	CMB v2.4	/decisions/N2O_load_and_tank/v/m_N2O_kg	LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill)
T351	LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill)	LEAD	CMB v2.4	/decisions/N2O_load_and_tank/tag	(class set by generator: LEAD)
T352	ovr-3: 9 active, 0 redundant, 0 in error; 4 superseded values withheld (4 interface keys...	DERIVED	generator	formula: count of override entries by status on this build	DERIVED by build_datasheet.py: count of override entries by status on this build ← T004, T007, T011, T015, T019, T021, T026, T030 ...
T353	ARGOS-H1-ENGINE-DATASHEET	DESIGN	generator	DOC_ID	generator configuration (command line / run time)
T354	rev0-draft — DRAFT, digital design data only	DESIGN	generator	--rev	generator configuration (command line / run time)
T355	2026-10-13T13:25:31Z	DESIGN	generator	run time (UTC)	generator configuration (command line / run time)
T356	Max thrust 500 N at sea level, with N2O at 15–25 °C. Normal Pc is capped at 25 bar in so...	REQ	REQ rev 2	§2. Top-level row "ENG-01" col 1	REQ
T357	Servo-driven variable-area pintle (needle-in-seat) at the head end + separate NC main is...	LEAD	CMB v2.4	/decisions/throttle_architecture	LEAD decision 2026-10-11; v2 trade (data/v2/throttle_trade_v2.json) finds no showstopper
T358	Propellants: self-pressurised N2O, tank 2.00 kg / 3.17 L conditioned to 15–25 °C, with h...	REQ	REQ rev 2	§2. Top-level row "ENG-12" col 1	REQ / LEAD
T359	TVC by a two-axis gimbal of the whole engine, pivot at the engine-group mid-burn CG. Ran...	LEAD	REQ rev 2	§2. Top-level row "ENG-07" col 1	LEAD (GNC MC: 80°/s worse, p 0.022)
T360	spark-ignited N2O/propane torch (baseline kept); arc-ABS kept as a research option for b...	DERIVED	CMB v2.4	/decisions/igniter	DERIVED weighted trade data/v2/ignition_v2.json (torch 0.80, arc-ABS 0.675, pyro 0.64, catalytic 0.525)
T361	DESIGN MODEL - not released for fabrication; geometric checks are software checks, not p...	DESIGN	PROP v3.4.1	/status	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T362	v2.4 REVISION (digital only, Por revisar). Design point unchanged. No number is MEASURED...	DESIGN	CMB v2.4	/status	(class set by generator: DESIGN)
T363	DESIGN + SIL + schematic. Firmware v2.9 runs only in SIL; the ECU v2.3 schematic passes ...	DESIGN	ELE v2	/status	(class set by generator: DESIGN)
T364	Independent review verdicts for the engine team's digital deliverables. A verdict is not...	DESIGN	SUP 7 reviews	/note	(class set by generator: DESIGN)
T365	v1 concept, not yet in CAD	DESIGN	MEC v2	/fixed_module/tank/concept/src	(class set by generator: DESIGN)
T366	n: 98; passed: 98	DERIVED	PROP v3.4.1	/results/parts_summary	(class set by generator: DERIVED)
T367	sum of unit masses x qty (CAD volume x density, or tagged COTS masses); includes the tra...	DERIVED	BOM v3.4.1	/mass_sum_note	(class set by generator: DERIVED)
T368	M2 v3 (2026-10-12): 6-DOF Monte Carlo (sim6/run6 v2.6, 500 runs per case, same draws) re...	SIM	GNC v3	/status	(class set by generator: SIM)
T369	On GNC rev1 (lines 157-158; text saved to 01-systems-gnc/staging_v3.1/rev1/REPORT-GNC-re...	LEAD	CL 176 lines	L161	lead (decision) 2026-10-13 (line sha 175f8cd442f74c80)
T370	On Combustion's RG-01 tables and Mechanical's bleed-path line 162: (1) RG-01 design basi...	LEAD	CL 176 lines	L164	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T371	On REV-ELEC-1: (1) RE-02 light-rule baseline fallback: P0 > +check → keep the measured ...	LEAD	CL 176 lines	L166	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T372	On REV-GNC-2: (1) Drogue release predicate restated, interim term limited to the invalid...	LEAD	CL 176 lines	L168	lead (decision) 2026-10-13 (line sha 3ad186c5fa9490ff)
T373	On the Combustion v2.4 handback (line 169): installed 02-combustion/REPORT.md v2.4 (32e3...	LEAD	CL 176 lines	L170	lead (decision) 2026-10-13 (line sha 0e5d7d63a5e8dd26)
T374	On REV-COMB-2: (1) R2-02 ONE stand safing rule after a no-light, effective now: N2 purge...	LEAD	CL 176 lines	L172	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T375	On Combustion's FA-1 envelope (02-combustion/data/v2p4a/fa1_envelope_v2p4a.json 64b32815...	LEAD	CL 176 lines	L173	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T376	On GNC rev2 (line 174; staging_v3.1/rev2 MANIFEST cb5d2c48e8d8fa27 verified 280/280 by t...	LEAD	CL 176 lines	L175	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T377	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 m...	SIM	CMB v2.4	/operating_points/basis	(class set by generator: SIM)
T378	125 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T379	62.5 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T380	616171 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T381	733390 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T382	436985 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T383	10.44	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,

ID	Value	Class	Source	Key	Source tag / formula
T384	4.848	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T385	5.319	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T386	4.059	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T387	162.5 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T388	171.4 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T389	140.2 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T390	1894 N·s/kg	DERIVED	generator	formula: Isp × g0, g0 = 9.80665 m/s² (defined constant)	DERIVED by build_datasheet.py: Isp × g0, g0 = 9.80665 m/s² (defined constant) ← T035
T391	1594 N·s/kg	DERIVED	generator	formula: Isp × g0, g0 = 9.80665 m/s² (defined constant)	DERIVED by build_datasheet.py: Isp × g0, g0 = 9.80665 m/s² (defined constant) ← T387
T392	1681 N·s/kg	DERIVED	generator	formula: Isp × g0, g0 = 9.80665 m/s² (defined constant)	DERIVED by build_datasheet.py: Isp × g0, g0 = 9.80665 m/s² (defined constant) ← T388
T393	1375 N·s/kg	DERIVED	generator	formula: Isp × g0, g0 = 9.80665 m/s² (defined constant)	DERIVED by build_datasheet.py: Isp × g0, g0 = 9.80665 m/s² (defined constant) ← T389
T394	0.241 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T395	0.05202 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T396	0.06261 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T397	0.03646 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T398	0.02309 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T399	0.01073 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T400	0.01177 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T401	0.00898 kg/s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T402	0.2641 kg/s	DERIVED	generator	formula: ṁ_ox + ṁ_fuel	DERIVED by build_datasheet.py: ṁ_ox + ṁ_fuel ← T394, T398
T403	0.06275 kg/s	DERIVED	generator	formula: ṁ_ox + ṁ_fuel	DERIVED by build_datasheet.py: ṁ_ox + ṁ_fuel ← T395, T399

ID	Value	Class	Source	Key	Source tag / formula
T404	0.07438 kg/s	DERIVED	generator	formula: ṁ_ox + ṁ_fuel	DERIVED by build_datasheet.py: ṁ_ox + ṁ_fuel ← T396, T400
T405	0.04544 kg/s	DERIVED	generator	formula: ṁ_ox + ṁ_fuel	DERIVED by build_datasheet.py: ṁ_ox + ṁ_fuel ← T397, T401
T406	1.916 mm	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T407	0.301 mm	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T408	0.367 mm	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T409	0.207 mm	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T410	0.907	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/0/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T411	7.097	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/5/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T412	5.796	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/4/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T413	10.43	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/7/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T414	500 N	SIM	CMB v2.4	/operating_points/max_thrust/F_N	SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975)
T415	100 N	REQ	CMB v2.4	/operating_points/floor_target_20pct/F_N	REQ
T416	125 N	REQ	CMB v2.4	/operating_points/floor_requirement_25pct/F_N	REQ
T417	62.5 N	REQ	CMB v2.4	/operating_points/floor_stretch_12p5pct/F_N	REQ stretch
T418	met (dP/Pc 7.1)	SIM	CMB v2.4	/operating_points/floor_target_20pct/status	(class set by generator: SIM)
T419	met (dP/Pc 5.8)	SIM	CMB v2.4	/operating_points/floor_requirement_25pct/status	(class set by generator: SIM)
T420	model meets it: dP/Pc 10.4, chug gain margin 38.6 at 10 ms delay, attached nozzle except...	SIM	CMB v2.4	/operating_points/floor_stretch_12p5pct/status	(class set by generator: SIM)
T421	24 °C	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-12:" [regex at ARM 15-(\d+) C]	REQ bullet
T422	619.3 N	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/0/F_N	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T423	30.66 bar	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/0/Pc_bar	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T424	15 °C	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/0/T_tank_C	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T425	0.6	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/0/Cd	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T426	665.8 N	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/1/F_N	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T427	32.87 bar	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/1/Pc_bar	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T428	15 °C	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/1/T_tank_C	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T429	0.7	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/1/Cd	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)

ID	Value	Class	Source	Key	Source tag / formula
T430	704.3 N	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/2/F_N	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T431	34.7 bar	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/2/Pc_bar	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T432	15 °C	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/2/T_tank_C	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T433	0.8	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/2/Cd	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T434	666.4 N	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/3/F_N	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T435	32.9 bar	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/3/Pc_bar	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T436	20 °C	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/3/T_tank_C	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T437	0.6	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/3/Cd	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T438	719.2 N	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/4/F_N	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T439	35.41 bar	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/4/Pc_bar	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T440	20 °C	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/4/T_tank_C	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T441	0.7	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/4/Cd	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T442	763.8 N	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/5/F_N	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T443	37.52 bar	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/5/Pc_bar	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T444	20 °C	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/5/T_tank_C	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T445	0.8	SIM	CMB v2.4	/operating_points/pressure_limits/full_stroke_failure_S230/v/5/Cd	SIM grain_q1d v2 steady, initial port (full_stroke_failure_case_v2.json)
T446	375 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/1/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T447	1909896 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/1/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T448	8.994	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/1/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T449	192.2 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/1/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T450	1.551	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/1/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T451	250 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/2/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T452	1320214 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/2/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T453	7.336	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/2/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T454	188.4 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/2/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,

ID	Value	Class	Source	Key	Source tag / formula
T455	2.744	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/2/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T456	200 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/3/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T457	1084840 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/3/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T458	6.583	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/3/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T459	184.6 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/3/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T460	3.575	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/3/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T461	75 N	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/6/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T462	498407 Pa	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/6/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T463	4.342	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/6/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T464	149 s	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/6/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T465	9.018	SIM	CMB v2.4	/operating_points/throttle_points_20C_initial_port/6/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92,
T466	2500000 Pa	LEAD	CMB v2.4	/decisions/normal_software_Pc_cap_Pa	LEAD decision 2026-10-12
T467	4500000 Pa	LEAD	CMB v2.4	/decisions/chamber_MEOP_Pa	LEAD decision 2026-10-12 (CHANGELOG): basis = pintle full-stroke failure, 30 C tank, Cd 0.8 (data/v2/full_stroke_failure_case_v2.json)
T468	0.25	DERIVED	CMB v2.4	/stability/criteria [regex dP/Pc >= ([\d.]+)]	(class set by generator: DERIVED)
T469	44.33 bar	SIM	CMB v2.4	/meop_basis_v2p4/v/basis_bar	SIM meop_basis_v2p4.json
T470	13.72 mm	DESIGN	CMB v2.4	/gas_path/nozzle/throat_diameter_m	v1 geometry (data/v2/nozzle_contour_v2.csv) [display = raw x 1000 (m to mm)]
T471	19.41 mm	DESIGN	CMB v2.4	/gas_path/nozzle/exit_diameter_m	v1 geometry [display = raw x 1000 (m to mm)]
T472	2	DERIVED	CMB v2.4	/gas_path/nozzle/expansion_ratio	DERIVED v2 re-check with S230 (data/v2/nozzle_perf_v2.csv): attached to 15 % thrust, incipient lip separation at 12.5 % (Schmucker eps_sep 1.94; Summerfield attached), side-load bound < 0.01 N; eps 2.
T473	147.9 mm²	DERIVED	CMB v2.4	/gas_path/nozzle/throat_area_m2	DERIVED [display = raw x 1e+06 (m² to mm²)]
T474	3.199	SIM	CMB v2.4	/gas_path/nozzle/pe_over_pa_at_max	CEA shifting, S230 max point (nozzle_perf_v2.csv)
T475	1401 m/s	SIM	CMB v2.4	/operating_points/max_thrust/cstar_delivered_m_s	CEA x eta_c* 0.92 (chamber_port_state_v2.json)
T476	1.147	SIM	CMB v2.4	/operating_points/max_thrust/gamma_chamber	CEA
T477	28.72 kg/kmol	SIM	CMB v2.4	/operating_points/max_thrust/MW_chamber	CEA

ID	Value	Class	Source	Key	Source tag / formula
T478	338.6 mm	DERIVED	CMB v2.4	/gas_path/overall/gas_path_length_injector_face_to_exit_m	DERIVED [display = raw x 1000 (m to mm)]
T479	284.6 cm³	DERIVED	CMB v2.4	/gas_path/overall/free_chamber_volume_initial_m3	DERIVED (pre + post + convergent + initial port) [display = raw x 1e+06 (m³ to cm³)]
T480	1.924 m	DERIVED	CMB v2.4	/gas_path/overall/L_star_initial_m	DERIVED V_free / A_t
T481	53.9 mm	DESIGN	CMB v2.4	/gas_path/chamber_gas_diameter_m	v1 (pre/post-chamber gas-side diameter); changes if the post-chamber insulator thickens (OI-4) [display = raw x 1000 (m to mm)]
T482	63.5 mm	DESIGN	CMB v2.4	/gas_path/case/OD_m	mechanical.json v1 [display = raw x 1000 (m to mm)]
T483	3.175 mm	DESIGN	CMB v2.4	/gas_path/case/wall_m	mechanical.json v1 [display = raw x 1000 (m to mm)]
T484	3257 K	SIM	CMB v2.4	/operating_points/max_thrust/Tc_ideal_K	CEA
T485	2757 K	SIM	CMB v2.4	/operating_points/max_thrust/T0_delivered_K	CEA x eta_c*^2
T486	isomolded/extruded fine-grain graphite, bulk density >= 1.80-1.85 g/cm3, ash <= 0.1 %; g...	DERIVED	CMB v2.4	/gas_path/nozzle/graphite	DERIVED (erosion model sensitivity to density/porosity)
T487	0.4 s	LEAD	CMB v2.4	/light_windows/v/flight_first_start	LEAD; REPORT v2.4 §7.3
T488	0.65 s	LEAD	CMB v2.4	/light_windows/v/flight_relight	LEAD; REPORT v2.4 §7.3
T489	0.35 s	LEAD	CMB v2.4	/light_windows/v/stand	LEAD; REPORT v2.4 §7.3
T490	>= 3; limited by propane (0.055 g/start) and N2O torch flow (0.30 g/start)	DERIVED	CMB v2.4	/igniter/starts_per_flight	DERIVED
T491	Restartable: ≥ 3 starts per flight. Command to 90 % thrust ≤ 0.5 s (target 0.3 s), repea...	DERIVED	REQ rev 2	§2. Top-level row "ENG-04" col 1	DERIVED / LEAD
T492	0.81 s	REQ	GNC v3	/requirements/ignition/relight_planning_value_s	REQ lead decision 2026-10-13 (the GNC rule adopted)
T493	1.41 s	REQ	GNC v3	/requirements/ignition/relight_planning_value_s/option_c_unprimed_s	REQ lead decision 2026-10-13, option (c) PROVISIONAL
T494	7.366 s	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/nominal_fsw_v2.6_planning_1.4s/burn_time_s	(class set by generator: SIM)
T495	76 g	SIM	CMB v2.4	/relight_purge/v/N2O_vented_g/plan_max	SIM/DERIVED on ASSUMED Cd and slip brackets; LEAD lines 160, 161, 164; Mechanical line 162
T496	Accumulated burn per flight sized by the duty cycles in gnc.json v3. Grain replaced each...	ASSUMED	REQ rev 2	§2. Top-level row "ENG-06" col 1	ASSUMED / SIM
T497	8	DERIVED	RV2	line 441 [regex O/F (\d+)]	RV2 line 441 (independent review text)
T498	3000000 Pa	LEAD	CMB v2.4	/decisions/ecu_overpressure_abort_Pa	LEAD decision 2026-10-12 (~30 bar, Electronics implements; value approximate until Electronics fixes filter/latency)
T499	32.48 bar	DERIVED	ELE v2	/sensors/pressure_transducers/hw_ovp [regex =\s*([\d.]+) bar abs]	DERIVED pc_transducer_v2 C [nominal trip of the hardware comparator]
T500	40.6 bar	SIM	CMB v2.4	/late_light_and_meop/v/decomposition_peak_bar/pad_15C/rounded	SIM late_light_tables_v2p4.json, late_light_spike_v2p4.json on ASSUMED burn time and fuel fraction
T501	43.7 bar	SIM	CMB v2.4	/late_light_and_meop/v/decomposition_peak_bar/pad_15C/sharp	SIM late_light_tables_v2p4.json, late_light_spike_v2p4.json on ASSUMED burn time and fuel fraction
T502	0.67 bar	SIM	CMB v2.4	/meop_basis_v2p4/v/margin_to_MEOP_bar	SIM meop_basis_v2p4.json
T503	48.9–75.5 bar [OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs]	SIM	CMB v2.4	/late_light_and_meop/v/fuel_assisted_relight_bar/burn_1ms	SIM late_light_tables_v2p4.json, late_light_spike_v2p4.json on ASSUMED burn time and fuel fraction
T504	MEOP NOT shown → FMEA FA-1 (single-control hazard)	SIM	CMB v2.4	/late_light_and_meop/v/fuel_assisted_status	SIM late_light_tables_v2p4.json, late_light_spike_v2p4.json on ASSUMED burn time and fuel fraction
T505	6308000 Pa	DERIVED	CMB v2.4	/operating_points/pressure_limits/blocked_nozzle_bound_Pa	CoolProp Psat at 30 C (full_stroke_failure_case_v2.json)
T506	6750000 Pa	LEAD	CMB v2.4	/decisions/proof_pressure_Pa	LEAD decision 2026-10-12: 1.5 x MEOP; also covers the blocked-nozzle bound (tank pressure, 63.1 bar at 30 C) below yield
T507	21.9 bar	SIM	CMB v2.4	/meop_basis_v2p4/v/proof_margin_min_bar	SIM meop_basis_v2p4.json
T508	7000000 Pa	LEAD	CMB v2.4	/operating_points/tank_MEOP_Pa	LEAD v0
T509	45–56.5 bar	DERIVED	CMB v2.4	/operating_points/tank_pressure_range_Pa	CoolProp Psat 15-25 C (n2o_validation_v2.json: CoolProp vs ESDU <= 0.98 % to 300 K) [display = raw x 1e-05 (Pa to bar)]
T510	24 °C	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-09:" [regex T_tank <= (\d+) C]	REQ bullet

ID	Value	Class	Source	Key	Source tag / formula
T511	55.28 bar	LEAD	CL 176 lines	L172(4) [regex (\d+) C / ([\d.]+) bar abs]	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T512	TO-DEFINE	TO-DEFINE	—		[Tank interlock guard band = total channel uncertainty (Electronics states it; CHANGELOG 172 (4))]
T513	90 bar	LEAD	CL 176 lines	L172(2) [regex quasi-static (\d+) bar abs]	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T514	45 bar	DERIVED	generator	formula: lower end of combustion.json tank_pressure_range_Pa	DERIVED by build_datasheet.py: lower end of combustion.json tank_pressure_range_Pa ← T509
T515	DECISION: FA-1 structural load = 90 bar abs quasi-static at head and aft for all parts (...	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T516	90 bar	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex structural load = (\d+) bar abs]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T517	92 bar	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex margins also at (\d+) bar abs]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T518	102 bar	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex and at (\d+) bar abs \(sensitivity]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T519	0.25 ms	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex natural period of ([\d.]+)-([\d.]+) ms]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T520	1 ms	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex natural period of ([\d.]+)-([\d.]+) ms]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T521	1.8 kN	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex thrust spike ([\d.]+) kN x (\d+) ms]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T522	4 ms	LEAD	CL 176 lines	L173[sentence containing "DECISION: FA-1 structural load"] [regex thrust spike ([\d.]+) kN x (\d+) ms]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T523	On Combustion's FA-1 envelope (02-combustion/data/v2p4a/fa1_envelope_v2p4a.json 64b32815...	LEAD	CL 176 lines	L173[sentence containing "FA-1 envelope (" | "head peak"]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T524	O/F bounding within the construct (closed-volume max at O/F 3.5); fuel-basis sensitivity...	LEAD	CL 176 lines	L173[sentence containing "O/F bounding"]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T525	Pressure DLF: <= 1.007 for periods <= 0.1 ms (shell: quasi-static valid); 1.07-1.20 for ...	LEAD	CL 176 lines	L173[sentence containing "Pressure DLF"]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T526	Missing datum (fuel inventory at light) remains TO-VERIFY;	LEAD	CL 176 lines	L173[sentence containing "Missing datum"]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T527	IGN-1 with the dynamic Pc channel is the physical evidence	LEAD	CL 176 lines	L173[sentence containing "physical evidence"]	lead (decision) 2026-10-13 (line sha 828198ce9fb339e9)
T528	64b328150025fa1e: matches the citation	DERIVED	generator	FA1 sha256/16 against the citation in CHANGELOG L173	generator check on this build
T529	83.04 bar	SIM	FA1 v2.4a	/envelope/vapour_once|grid_OF4-12|rounded_Cd1.0/peak_head_bar/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T530	84.76 bar	SIM	FA1 v2.4a	/envelope/vapour_once|grid_OF4-12|sharp_Cd0.65/peak_head_bar/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T531	88.02 bar	SIM	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/peak_head_bar/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T532	0.9513–0.9986	SIM	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/aft_over_head_peak	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T533	3.65 ms	SIM	FA1 v2.4a	/envelope/vapour_once|incl_OF3-3.5|all/above45_head_total_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T534	4.45 ms	SIM	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/above45_head_total_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T535	1 ms	SIM	FA1 v2.4a	/envelope/vapour_once|incl_OF3-3.5|all/above67p5_head_total_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T536	1.3 ms	SIM	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/above67p5_head_total_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence

ID	Value	Class	Source	Key	Source tag / formula
T537	0 ms	SIM	FA1 v2.4a	/envelope/vapour_once|incl_OF3-3.5|all/above90_head_total_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T538	0 ms	SIM	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/above90_head_total_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T539	725.9 bar·ms	SIM	FA1 v2.4a	/envelope/vapour_once|incl_OF3-3.5|all/impulse_above_P0_25ms_head_bar_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T540	756.9 bar·ms	SIM	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/impulse_above_P0_25ms_head_bar_ms/max	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T541	integral of max(P - P0, 0) over 0-25 ms after the light, bar.ms; P0 = cold-flow pressure...	SIM	FA1 v2.4a	/metric_definitions/impulse	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T542	1694 N	ASSUMED	FA1 v2.4a	/envelope/vapour_once|incl_OF3-3.5|all/F_peak_QSx0.975_N/max	Cf band ASSUMED; quasi-steady Cf DERIVED
T543	1768 N	ASSUMED	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/F_peak_QSx0.975_N/max	Cf band ASSUMED; quasi-steady Cf DERIVED
T544	1839, 1891, 2879, 3053, 3027, 2946 N	DERIVED	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/F_sdof_QSx0.975_N_max_by_period	DERIVED (load-history comparison only)
T545	3053 N	DERIVED	generator	formula: max over the SDOF periods of F_sdof_QSx0.975_N_max_by_period (upper bracket)	DERIVED by build_datasheet.py: max over the SDOF periods of F_sdof_QSx0.975_N_max_by_period (upper bracket) ← T544
T546	3484, 3571, 3593, 3597, 3598, 3599 N	DERIVED	FA1 v2.4a	/interim_load_comparison/thrust_rect_sdof_N	DERIVED (load-history comparison only)
T547	3484–3599 N	DERIVED	generator	formula: range over the SDOF periods of the interim rectangular thrust pulse	DERIVED by build_datasheet.py: range over the SDOF periods of the interim rectangular thrust pulse ← T546
T548	[{"P_equiv_static_abs_bar": 82.67792665875984, "DLF": 0.9956380742055589, "case": "vapou...	DERIVED	FA1 v2.4a	/envelope/vapour_once|grid_OF4-12|rounded_Cd1.0/P_head_equiv_static_by_period	DERIVED (load-history comparison only)
T549	88.98 bar	DERIVED	generator	formula: max over the SDOF periods of P_head_equiv_static_by_period (vapour_once|grid_OF4-12|rounded_Cd1.0)	DERIVED by build_datasheet.py: max over the SDOF periods of P_head_equiv_static_by_period (vapour_once|grid_OF4-12|rounded_Cd1.0) ← T548
T550	[{"P_equiv_static_abs_bar": 87.58098825326037, "DLF": 0.9949665913585832, "case": "as_wr...	DERIVED	FA1 v2.4a	/envelope/both|incl_OF3-3.5|all/P_head_equiv_static_by_period	DERIVED (load-history comparison only)
T551	102 bar	DERIVED	generator	formula: max over the SDOF periods of P_head_equiv_static_by_period (both|incl_OF3-3.5|all)	DERIVED by build_datasheet.py: max over the SDOF periods of P_head_equiv_static_by_period (both|incl_OF3-3.5|all) ← T550
T552	OF_grid: [4.0, 5.0, 6.0, 8.0, 10.0, 12.0]; OF_extension: [3.0, 3.5]; tf_ms: [1.0, 2.0, 3...	SIM	FA1 v2.4a	/grid	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T553	896	SIM	FA1 v2.4a	/grid/n_cases	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T554	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): a...	SIM	FA1 v2.4a	/tags/model	(class set by generator: SIM)
T555	acoustic/wave peaks along the 230 mm port (lumped nodes; transit ~0.3 ms hot, ~1 ms cold...	SIM	FA1 v2.4a	/not_modelled	SIM three-node vented network on ASSUMED inputs (fast part, t_res, tf, f_ret, q, O/F): an ESTIMATE, not acceptance evidence
T556	MISSING: the fuel-vapour (and aerosol/film) inventory in the warm 32.2 mm port at the mo...	TO-VERIFY	FA1 v2.4a	/fuel_vapour_inventory/status	(class set by generator: TO-VERIFY)
T557	FA-1 final load rule (lead 172 (2) + 173) closed in S-AX/joints: 90 bar abs quasi-static...	SIM	CL 176 lines	L176	mechanical (result) 2026-10-13 (line sha b8543f9910478072)
T558	TC-09 316L YIELD -0.40/-0.41/-0.43 (ult +0.80/+0.78/+0.66) → v3.5 TC-09 IN718 +2.81/+2....	SIM	CL 176 lines	L176[sentence containing "YIELD" | "J-FWD" | "bearings exceed" | "brinelling"]	mechanical (result) 2026-10-13 (line sha b8543f9910478072)
T559	PENDING	PENDING	—		[Independent review and lead decision on the Mechanical FA-1 margins (CHANGELOG 176) and on the CAD changes they propose]
T560	TO-VERIFY	TO-VERIFY	—		[FA-1 fuel-vapour inventory at light (data file: first-order bound only; lead decision: remains TO-VERIFY)]
T561	7 °	REQ	MEC v2	/gimbal/range_deg	REQ
T562	60.93 N	DERIVED	generator	formula: F_rated × sin(gimbal range 7°)	DERIVED by build_datasheet.py: F_rated × sin(gimbal range 7°) ← T034, T561
T563	MISSING	MISSING	—		[Physical worst-case thrust at the v2.4 MEOP basis: combustion.json v2.4 publishes the basis pressure only]

ID	Value	Class	Source	Key	Source tag / formula
T564	MISSING	MISSING	—		[Lateral thrust at the gimbal range for the physical worst case (needs the worst-case thrust)]
T565	0.8	LEAD	CL 176 lines	L172(4) [regex full-stroke Cd <= ([\d.]+)]	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T566	LEAD decision 2026-10-12 (CHANGELOG): basis = pintle full-stroke failure, 30 C tank, Cd ...	LEAD	CMB v2.4	/decisions/chamber_MEOP_Pa/tag	(class set by generator: LEAD)
T567	876 N at Pc [withheld: superseded value, SUP-04] (pintle stuck at full stroke, 30 °C tan...	SIM	ICD draft 0.2	§3. Loads row "Physical worst-case thrust" col 1	SIM, Combustion v2.3
T568	Pressure parts: SF ≥ 1.5 on yield and ≥ 2.0 on ultimate, at MEOP and at temperature. Cha...	LEAD	REQ rev 2	§2. Top-level row "ENG-09" col 1	LEAD
T569	Throttle floor ≤ 20 % of max (5:1, 100–500 N) with stable combustion. Stretch: 12.5 % (8...	DERIVED	REQ rev 2	§2. Top-level row "ENG-02" col 1	DERIVED (GNC MC) / LEAD
T570	20 %	DERIVED	GNC v3	/requirements/throttle/floor_frac_required_max	DERIVED 6-DOF MC (landing closure by floor) + hover rule [display = raw x 100 (fraction to %)]
T571	pintle (needle-in-seat), unchanged from v1	LEAD	CMB v2.4	/gas_path/injector/type	LEAD 2026-10-11
T572	5.5 mm	DESIGN	CMB v2.4	/gas_path/injector/seat_diameter_m	v1 [display = raw x 1000 (m to mm)]
T573	4 mm	DESIGN	CMB v2.4	/gas_path/injector/stroke_m	v1 [display = raw x 1000 (m to mm)]
T574	0.7	ASSUMED	CMB v2.4	/gas_path/injector/Cd	ASSUMED pintle Cd (0.6-0.8)
T575	Cd_eff (vs Dyer) >= 0.41 measured in N2O cold flow keeps the 15 C ceiling >= 500 N (HEM-...	DERIVED	CMB v2.4	/gas_path/injector/Cd_acceptance	DERIVED
T576	MKS HV69 (MKS HV69, MKS no. S0022012 (Lindinger item 9788343))	DATASHEET	ELE v2	/actuators/pintle_throttle/model	DATASHEET via retailer copies of MKS data (TO-CONFIRM with MKS); see pintle_mini_servo_v1.json
T577	21 g	DATASHEET	ELE v2	/actuators/pintle_throttle/datasheet_class/mass_g	DATASHEET via retailer copies of MKS data (TO-CONFIRM with MKS); see pintle_mini_servo_v1.json
T578	PRAIL 7.49 V typ (TPS552882 + LTC4365, see pintle_rail)	DESIGN	ELE v2	/actuators/pintle_throttle/rail	(class set by generator: DESIGN)
T579	10 µm	DESIGN	CMB v2.4	/valves_and_actuators/pintle_actuator/position_resolution_m	v1; v2 check: at the 12.5 % floor dF/dh = 369 N/mm (plant fit), so 2 % of 500 N needs 27 um and 2 % of the floor thrust 3.4 um (the closed Pc loop then rests on Pc resolution); gnc.json v2: resolution
T580	Throttle response: t10–90 ≤ max(0.15 s, 0.8·ΔF / 1000 N/s). Ramp (amended 2026-10-13, le...	DERIVED	REQ rev 2	§2. Top-level row "ENG-03" col 1	DERIVED (GNC)
T581	2500 N/s	LEAD	ELE v2	/eng03_pintle_rate/start_ramp	COMB/GNC
T582	1347 N/s	SIM	ELE v2	/eng03_pintle_rate/rate_map/summary/nominal/open_min_N_per_s	latency ASSUMED; tau_p, tau_v FIT to the nonlinear drive simulation (3.5 MPa runs, grid tau_p [0.004, 0.006, 0.0085, 0.01] s x tau_v [0.002, 0.004, 0.006, 0.008, 0.01] s, minimum of the maximum shape
T583	1188 N/s	SIM	ELE v2	/eng03_pintle_rate/rate_map/summary/corner/open_min_N_per_s	(class set by generator: SIM)
T584	2635 N/s	SIM	ELE v2	/eng03_pintle_rate/rate_map/summary/corner/close_min_N_per_s	(class set by generator: SIM)
T585	57/95	SIM	ELE v2	/eng03_pintle_rate/rate_map/summary/nominal/states_open_ge_2500	(class set by generator: SIM)
T586	29/95	SIM	ELE v2	/eng03_pintle_rate/rate_map/summary/corner/states_open_ge_2500	(class set by generator: SIM)
T587	3.4 ms	DERIVED	CMB v2.4	/operating_points/chamber_fill_time_constant_s	DERIVED L*/(c*_del Gamma^2); step-response fit 0.004 s (1 ms sampling) [display = raw x 1000 (s to ms)]
T588	Shutdown: command to < 10 % thrust ≤ 0.10 s. Met with a mandatory pre-MECO throttle-down...	DERIVED	REQ rev 2	§2. Top-level row "ENG-05" col 1	DERIVED / LEAD
T589	No combustion basis for 0.08 s. Start transient: Pc overshoot <= 0.34 % for full-stroke ...	SIM	CMB v2.4	/valves_and_actuators/pintle_actuator/speed_checks/conclusion	(class set by generator: SIM)
T590	two-axis Cardan gimbal of the whole thrust chamber; split engine collar (GM-01/02) clamp...	DERIVED	MEC v2	/gimbal/architecture	DERIVED
T591	8 °	DERIVED	MEC v2	/gimbal/mechanical_stop_deg	DERIVED
T592	at the mid-burn CG of the gimballed group (founder decision FINAL 2026-10-12); re-centre...	DERIVED	MEC v2	/gimbal/pivot_location	DERIVED
T593	105.6 mm	DERIVED	MEC v2	/gimbal/pivot_from_injector_face_m	DERIVED+ESTIMATE [display = raw x 1000 (m to mm)]

ID	Value	Class	Source	Key	Source tag / formula
T594	233 mm	DERIVED	MEC v2	/gimbal/pivot_from_nozzle_exit_m	DERIVED+ESTIMATE [display = raw x 1000 (m to mm)]
T595	4 × 608-2RS (8×22×7); pins 8 h6 17-4PH H1150	TO-VERIFY	MEC v2	/gimbal/bearings	DATASHEET-PENDING
T596	Savox SB-2292SG	DATASHEET	ELE v2	/actuators/tvc_A/model	(class set by generator: DATASHEET)
T597	0.081 kg	DATASHEET	ELE v2	/actuators/tvc_A/mass_kg	DATASHEET
T598	3.04 N·m	DATASHEET	ELE v2	/actuators/tvc_A/stall_torque_Nm/v/7.4V	DATASHEET (22/31/45 kg-cm; lower of two conflicting vendor sets)
T599	857 °/s	DATASHEET	ELE v2	/actuators/tvc_A/noload_speed_dps/v/7.4V	DATASHEET (0.085/0.07/0.055 s/60deg)
T600	333 Hz	DATASHEET	ELE v2	/actuators/tvc_A/update_rate_Hz	DATASHEET
T601	4.8	LEAD	MEC v2	/tvc_actuation/gear_ratio_n	LEAD-DECISION
T602	0.0208 °	DERIVED	MEC v2	/tvc_actuation/resolution_deg_engine	DERIVED
T603	0.0417 °	DERIVED	MEC v2	/tvc_actuation/backlash_deg_engine	DERIVED
T604	100 °/s	REQ	GNC v3	/requirements/gimbal/rate_min_dps	REQ lead decision 2026-10-12 (ENG-07 raised from 60 deg/s), from SIM m2_mc6 v2 (command slew limit 100 deg/s in every baseline run) + rate comparators
T605	34.9 rad/s²	REQ	GNC v3	/requirements/gimbal/angular_accel_min_rad_s2	REQ-derived from SIM m2_ctrlfix: flight software v2.6 commands at most 2000 deg/s2
T606	as built: worst-direction slew at 7 deg 111.2 deg/s at 7.4 V, 118.3 at 7.6 V (>= 100); 9...	DERIVED	MEC v2	/tvc_actuation/ENG-07_rate_status	(class set by generator: DERIVED)
T607	Structure (gimbal, mounts, fasteners): SF ≥ 1.5 on yield and ≥ 2.0 on ultimate under des...	LEAD	REQ rev 2	§2. Top-level row "ENG-10" col 1	LEAD / DERIVED
T608	37.95 N·m	DERIVED	MEC v2	/tvc_actuation/design_loads/v/M_engine_GS_Nm	DERIVED
T609	26.5 N·m	REQ	GNC v3	/design_limit_loads/hinge_moment_design_Nm	REQ lead decision 2026-10-12 (actuator stall load, mechanical.json v1 8.4 V)
T610	2442–1608 N·m/rad	SIM	MEC v2	/tvc_actuation/axis_stiffness_table/v/centre/mech_T	FEA+ASSUMED
T611	Gimbal axis stiffness (structure plus rods) ≥ 3,100 N·m/rad per axis at every pose (hard...	DERIVED	REQ rev 2	§2. Top-level row "ENG-11" col 1	DERIVED (GNC csi_v3)
T612	16–27.8 Hz	SIM	MEC v2	/tvc_actuation/rocking_mode_Hz	FEA+ASSUMED
T613	first rocking mode 16.0-27.8 Hz (excl. J_int) vs the 25 Hz minimum: below it whenever th...	DERIVED	MEC v2	/tvc_actuation/ENG-11_status	(class set by generator: DERIVED)
T614	3.19 N·m	DERIVED	SLEW v2	/loads_Nm/L_v1_3.19	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T615	95.24 °/s	DERIVED	SLEW v2	/cases/6.6V|L_v1_3.19/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T616	111.1 °/s	DERIVED	SLEW v2	/cases/7.4V|L_v1_3.19/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T617	150 °/s	DERIVED	SLEW v2	/cases/8.4V|L_v1_3.19/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T618	4.867 N·m	DERIVED	SLEW v2	/loads_Nm/L_v2_terms	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T619	83.12 °/s	DERIVED	SLEW v2	/cases/6.6V|L_v2_terms/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T620	99.73 °/s	DERIVED	SLEW v2	/cases/7.4V|L_v2_terms/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T621	140 °/s	DERIVED	SLEW v2	/cases/8.4V|L_v2_terms/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136

ID	Value	Class	Source	Key	Source tag / formula
T622	5.175 N·m	DERIVED	SLEW v2	/loads_Nm/L_v2_hose_at_20Nm_rad	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T623	80.9 °/s	DERIVED	SLEW v2	/cases/6.6V|L_v2_hose_at_20Nm_rad/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T624	97.65 °/s	DERIVED	SLEW v2	/cases/7.4V|L_v2_hose_at_20Nm_rad/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T625	138.2 °/s	DERIVED	SLEW v2	/cases/8.4V|L_v2_hose_at_20Nm_rad/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T626	3.683 N·m	DERIVED	SLEW v2	/loads_Nm/L_v34_O3b_coil_EI0.5	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T627	91.68 °/s	DERIVED	SLEW v2	/cases/6.6V|L_v34_O3b_coil_EI0.5/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T628	107.8 °/s	DERIVED	SLEW v2	/cases/7.4V|L_v34_O3b_coil_EI0.5/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T629	147.1 °/s	DERIVED	SLEW v2	/cases/8.4V|L_v34_O3b_coil_EI0.5/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T630	0 N·m	DERIVED	SLEW v2	/loads_Nm/no_load	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T631	118.3 °/s	DERIVED	SLEW v2	/cases/6.6V|no_load/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T632	132.7 °/s	DERIVED	SLEW v2	/cases/7.4V|no_load/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T633	168.9 °/s	DERIVED	SLEW v2	/cases/8.4V|no_load/worst_any_motion_at_7deg/dps	Asked (independent GNC review REV-GNC-1 / coordinator 2026-10-13; gnc.json requests): recompute the engine slew rate with the v2 non-inertial hinge terms (sum 4.867 N m incl. the hose upper term 2.136
T634	−69.75, −69.75, −238.1, 69.75, 69.75, 355.2 mm	DERIVED	PROP v3.4.1	/results/assembly/neutral/bounds_mm	(class set by generator: DERIVED)
T635	z0_mm: 105.0; z1_mm: 110.0; r_engine_swept_mm: 60.08; by_engine: ENG-FD-03#0; r_ring_swe...	TO-VERIFY	CHK v3.4.1	/swept_worst_band	(class set by generator: TO-VERIFY)
T636	≥ 140 mm over the whole engine zone	REQ	ICD draft 0.2	§2. Engine bay row "Engine bay inner diameter" col 1	REQ, founder decision 2026-10-12
T637	z = −0.2377 to +0.252 m about the pivot (aft closure at the 8° stop to the top of the pi...	DERIVED	ICD draft 0.2	§2. Engine bay row "Engine zone, engine frame" col 1	DERIVED+ESTIMATE, mechanical.json v2; closing sweep in CAD v3.4 (v2.1)
T638	2.761 kg	DERIVED	MP v3.4.1	/states/wet/engine/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T639	2.678 kg	DERIVED	MP v3.4.1	/states/mid/engine/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T640	2.595 kg	DERIVED	MP v3.4.1	/states/burnout/engine/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T641	2.337 kg	DERIVED	MP v3.4.1	/states/nograin/engine/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T642	0.228 kg	DERIVED	MP v3.4.1	/states/wet/ring/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]

ID	Value	Class	Source	Key	Source tag / formula
T643	0.228 kg	DERIVED	MP v3.4.1	/states/mid/ring/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T644	0.228 kg	DERIVED	MP v3.4.1	/states/burnout/ring/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T645	0.228 kg	DERIVED	MP v3.4.1	/states/nograin/ring/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T646	0.01816 kg	DERIVED	MP v3.4.1	/states/wet/horn/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T647	0.01816 kg	DERIVED	MP v3.4.1	/states/mid/horn/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T648	0.01816 kg	DERIVED	MP v3.4.1	/states/burnout/horn/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T649	0.01816 kg	DERIVED	MP v3.4.1	/states/nograin/horn/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T650	0.01841 kg	DERIVED	MP v3.4.1	/states/wet/pushrod/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T651	0.01841 kg	DERIVED	MP v3.4.1	/states/mid/pushrod/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T652	0.01841 kg	DERIVED	MP v3.4.1	/states/burnout/pushrod/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T653	0.01841 kg	DERIVED	MP v3.4.1	/states/nograin/pushrod/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T654	1.364 kg	DERIVED	MP v3.4.1	/states/wet/fixed/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T655	1.364 kg	DERIVED	MP v3.4.1	/states/mid/fixed/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T656	1.364 kg	DERIVED	MP v3.4.1	/states/burnout/fixed/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T657	1.364 kg	DERIVED	MP v3.4.1	/states/nograin/fixed/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T658	4.389 kg	DERIVED	MP v3.4.1	/states/wet/all/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T659	4.306 kg	DERIVED	MP v3.4.1	/states/mid/all/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T660	4.223 kg	DERIVED	MP v3.4.1	/states/burnout/all/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T661	3.965 kg	DERIVED	MP v3.4.1	/states/nograin/all/m_kg	(class set by generator: DERIVED) [CAD volume × density and COTS masses (massprops)]
T662	0.233 m; exit protrudes about 0.03 m below the aft datum	DERIVED	ICD draft 0.2	§2. Engine bay row "Nozzle exit below the pivot" col 1	DERIVED+ESTIMATE
T663	±7° on the vector norm; 8° conical mechanical stop	REQ	ICD draft 0.2	§2. Engine bay row "Gimbal range" col 1	REQ ENG-07
T664	Thrust rings GM-09 (upper, thrust path) and GM-10 (lower: stop ring, TVC servo base)	DERIVED	ICD draft 0.2	§2. Engine bay row "Airframe attachment" col 1	DERIVED, CAD v3.3
T665	Ring outer radius 69.75 mm in the 140 mm bore; 0.25 mm radial gap	DERIVED	ICD draft 0.2	§2. Engine bay row "Ring fit" col 1	DERIVED, CAD v3.3
T666	16 × M5 radial screws. Bolt pattern, edge distance and wall bearing check: (v2.1)	DERIVED	ICD draft 0.2	§2. Engine bay row "Ring fasteners" col 1	ESTIMATE
T667	Fixed module above the gimbal zone: occupies the bay from +0.252 m to about +0.355 m (en...	LEAD	ICD draft 0.2	§2. Engine bay row "Main-valve module (MV-P, balanced poppet)" col 1	LEAD 2026-10-13 / Mechanical
T668	Option A/O3b: coil on the engine case; fixed 3/8 in wall line at az 219.5°, r 64.2 mm, f...	REQ	ICD draft 0.2	§2. Engine bay row "Feed across the gimbal" col 1	REQ ENG-08, amended 2026-10-13
T669	500 N (Pc 25 bar, software cap)	REQ	ICD draft 0.2	§3. Loads row "Rated thrust" col 1	REQ ENG-01
T670	650 N held for 3 frames on a 20 ms filter. Valid airborne, lit and with no drogue load	LEAD	ICD draft 0.2	§3. Loads row "Over-thrust abort" col 1	LEAD 2026-10-13, provisional (REV-GNC-1 RG-16 to RG-19 open)
T671	45 bar / 67.5 bar	LEAD	ICD draft 0.2	§3. Loads row "Chamber MEOP / proof" col 1	LEAD ENG-09
T672	61 N at 500 N; 107 N at 876 N	DERIVED	ICD draft 0.2	§3. Loads row "Lateral thrust at 7°" col 1	DERIVED (F sin 7°)
T673	35.3 N·m engine moment. Rod forces 441 / 552 / 609 N react into the TVC brackets on GM-10	LEAD	ICD draft 0.2	§3. Loads row "TVC design moment" col 1	LEAD 2026-10-12
T674	35.8 m/s² (B1 dispersed max), 33.7 m/s² (B1 p99), 27.7 m/s² (A1 p99)	SIM	ICD draft 0.2	§3. Loads row "Peak axial specific force in flight" col 1	SIM, duty cycles v3
T675	50.3 m/s² (B1 p99), 40.3 m/s² (A1 p99). The 530–673 m/s² values in the "disp_max" files ...	SIM	ICD draft 0.2	§3. Loads row "Touchdown, leg contact" col 1	SIM, duty cycles v3
T676	≥ 3,100 N·m/rad at every pose, goal 5,200. The airframe side of the ring mounts is part ...	LEAD	ICD draft 0.2	§3. Loads row "Axis stiffness, structure plus rods" col 1	LEAD 2026-10-12

ID	Value	Class	Source	Key	Source tag / formula
T677	16.07 kg for A1 (drogue + fins); 15.42 kg for B hops	SIM	ICD draft 0.2	§4. Mass row "Liftoff mass" col 1	SIM, ledger v3d
T678	13.67 kg / 2.00 kg / 0.404 kg	DERIVED	ICD draft 0.2	§4. Mass row "Dry mass / N2O / grain" col 1	DERIVED ledger
T679	(v2.1): CAD v3.4 re-issues the gimballed and fixed masses. The main-valve module and tan...	MISSING	ICD draft 0.2	§4. Mass row "Engine hardware mass and CG" col 1	OPEN
T680	Thrust-ring bolt pattern, wall bearing and pull-out at 876 N plus lateral and moment loads	TO-DEFINE	ICD draft 0.2	§12 row "ICD-1" col 1	ICD open item
T681	Mechanical (v2.1) / Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-1" col 2	ICD owner
T682	Main-valve module envelope and mass at the zone top	TO-DEFINE	ICD draft 0.2	§12 row "ICD-2" col 1	ICD open item
T683	Mechanical (v2.1)	DESIGN	ICD draft 0.2	§12 row "ICD-2" col 2	ICD owner
T684	Upper-bay vent area and station; baro-port placement	TO-DEFINE	ICD draft 0.2	§12 row "ICD-3" col 1	ICD open item
T685	Lauty with Mechanical	DESIGN	ICD draft 0.2	§12 row "ICD-3" col 2	ICD owner
T686	Relief and vent port routing (tank relief, 9.5 MPa disc line, fill/vent)	TO-DEFINE	ICD draft 0.2	§12 row "ICD-4" col 1	ICD open item
T687	Mechanical / Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-4" col 2	ICD owner
T688	Tank station and mounting; heater and insulation interface	TO-DEFINE	ICD draft 0.2	§12 row "ICD-5" col 1	ICD open item
T689	Mechanical / Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-5" col 2	ICD owner
T690	Launch-site bay temperature limit	TO-DEFINE	ICD draft 0.2	§12 row "ICD-6" col 1	ICD open item
T691	Founders	DESIGN	ICD draft 0.2	§12 row "ICD-6" col 2	ICD owner
T692	Pad slope specification and leg design	TO-DEFINE	ICD draft 0.2	§12 row "ICD-7" col 1	ICD open item
T693	Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-7" col 2	ICD owner
T694	Drogue opening load, bay volume, ejection method, swivel and release fitting	TO-DEFINE	ICD draft 0.2	§12 row "ICD-8" col 1	ICD open item
T695	Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-8" col 2	ICD owner
T696	Ground plume impingement on the feet	TO-DEFINE	ICD draft 0.2	§12 row "ICD-9" col 1	ICD open item
T697	Combustion	DESIGN	ICD draft 0.2	§12 row "ICD-9" col 2	ICD owner
T698	Airframe first bending mode ≥ 10 Hz on the real tube	TO-DEFINE	ICD draft 0.2	§12 row "ICD-10" col 1	ICD open item
T699	Lauty, then GNC re-check	DESIGN	ICD draft 0.2	§12 row "ICD-10" col 2	ICD owner
T700	Engine hardware mass and CG from CAD v3.4	TO-DEFINE	ICD draft 0.2	§12 row "ICD-11" col 1	ICD open item
T701	Mechanical (v2.1)	DESIGN	ICD draft 0.2	§12 row "ICD-11" col 2	ICD owner
T702	Wall-line P-clips (3, at z 110 / 170 / 230 mm engine frame) and the coil fixed-end clamp...	TO-DEFINE	ICD draft 0.2	§12 row "ICD-12" col 1	ICD open item
T703	Lauty with Mechanical	DESIGN	ICD draft 0.2	§12 row "ICD-12" col 2	ICD owner
T704	Harness routing across the gimbal	TO-DEFINE	ICD draft 0.2	§12 row "ICD-13" col 1	ICD open item
T705	Electronics / Mechanical, TO-DEFINE	DESIGN	ICD draft 0.2	§12 row "ICD-13" col 2	ICD owner
T706	Head-end bleed overboard port (A1 provision): size, station, routing; balanced or axial ...	TO-DEFINE	ICD draft 0.2	§12 row "ICD-14" col 1	ICD open item
T707	Mechanical (v2.1) / Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-14" col 2	ICD owner
T708	Drogue canopy type and oscillation data at the relight condition (≤ 2° design attitude)	TO-DEFINE	ICD draft 0.2	§12 row "ICD-15" col 1	ICD open item
T709	Lauty	DESIGN	ICD draft 0.2	§12 row "ICD-15" col 2	ICD owner
T710	0	DERIVED	CHK v3.4.1	/poses/0/interference_not_whitelisted	(class set by generator: DERIVED)
T711	41	DERIVED	CHK v3.4.1	/poses/0/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T712	0.5 mm	DERIVED	CHK v3.4.1	/poses/0/min_unexpected_mm	(class set by generator: DERIVED)
T713	0 mm	DERIVED	CHK v3.4.1	/poses/0/wall_min_mm	(class set by generator: DERIVED)
T714	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/0/wall_min_by	(class set by generator: DERIVED)
T715	0	DERIVED	CHK v3.4.1	/poses/1/interference_not_whitelisted	(class set by generator: DERIVED)
T716	46	DERIVED	CHK v3.4.1	/poses/1/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T717	0.5 mm	DERIVED	CHK v3.4.1	/poses/1/min_unexpected_mm	(class set by generator: DERIVED)
T718	0 mm	DERIVED	CHK v3.4.1	/poses/1/wall_min_mm	(class set by generator: DERIVED)
T719	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/1/wall_min_by	(class set by generator: DERIVED)
T720	0	DERIVED	CHK v3.4.1	/poses/2/interference_not_whitelisted	(class set by generator: DERIVED)
T721	45	DERIVED	CHK v3.4.1	/poses/2/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T722	0.5 mm	DERIVED	CHK v3.4.1	/poses/2/min_unexpected_mm	(class set by generator: DERIVED)
T723	0 mm	DERIVED	CHK v3.4.1	/poses/2/wall_min_mm	(class set by generator: DERIVED)
T724	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/2/wall_min_by	(class set by generator: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T725	0	DERIVED	CHK v3.4.1	/poses/3/interference_not_whitelisted	(class set by generator: DERIVED)
T726	50	DERIVED	CHK v3.4.1	/poses/3/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T727	0.399 mm	DERIVED	CHK v3.4.1	/poses/3/min_unexpected_mm	(class set by generator: DERIVED)
T728	0 mm	DERIVED	CHK v3.4.1	/poses/3/wall_min_mm	(class set by generator: DERIVED)
T729	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/3/wall_min_by	(class set by generator: DERIVED)
T730	0	DERIVED	CHK v3.4.1	/poses/4/interference_not_whitelisted	(class set by generator: DERIVED)
T731	50	DERIVED	CHK v3.4.1	/poses/4/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T732	0.345 mm	DERIVED	CHK v3.4.1	/poses/4/min_unexpected_mm	(class set by generator: DERIVED)
T733	0 mm	DERIVED	CHK v3.4.1	/poses/4/wall_min_mm	(class set by generator: DERIVED)
T734	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/4/wall_min_by	(class set by generator: DERIVED)
T735	0	DERIVED	CHK v3.4.1	/poses/5/interference_not_whitelisted	(class set by generator: DERIVED)
T736	53	DERIVED	CHK v3.4.1	/poses/5/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T737	0.5 mm	DERIVED	CHK v3.4.1	/poses/5/min_unexpected_mm	(class set by generator: DERIVED)
T738	0 mm	DERIVED	CHK v3.4.1	/poses/5/wall_min_mm	(class set by generator: DERIVED)
T739	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/5/wall_min_by	(class set by generator: DERIVED)
T740	0	DERIVED	CHK v3.4.1	/poses/6/interference_not_whitelisted	(class set by generator: DERIVED)
T741	53	DERIVED	CHK v3.4.1	/poses/6/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T742	0.487 mm	DERIVED	CHK v3.4.1	/poses/6/min_unexpected_mm	(class set by generator: DERIVED)
T743	0 mm	DERIVED	CHK v3.4.1	/poses/6/wall_min_mm	(class set by generator: DERIVED)
T744	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/6/wall_min_by	(class set by generator: DERIVED)
T745	0	DERIVED	CHK v3.4.1	/poses/7/interference_not_whitelisted	(class set by generator: DERIVED)
T746	53	DERIVED	CHK v3.4.1	/poses/7/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T747	0.487 mm	DERIVED	CHK v3.4.1	/poses/7/min_unexpected_mm	(class set by generator: DERIVED)
T748	0 mm	DERIVED	CHK v3.4.1	/poses/7/wall_min_mm	(class set by generator: DERIVED)
T749	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/7/wall_min_by	(class set by generator: DERIVED)
T750	0	DERIVED	CHK v3.4.1	/poses/8/interference_not_whitelisted	(class set by generator: DERIVED)
T751	55	DERIVED	CHK v3.4.1	/poses/8/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T752	0.5 mm	DERIVED	CHK v3.4.1	/poses/8/min_unexpected_mm	(class set by generator: DERIVED)
T753	0 mm	DERIVED	CHK v3.4.1	/poses/8/wall_min_mm	(class set by generator: DERIVED)
T754	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/8/wall_min_by	(class set by generator: DERIVED)
T755	0	DERIVED	CHK v3.4.1	/poses/9/interference_not_whitelisted	(class set by generator: DERIVED)
T756	50	DERIVED	CHK v3.4.1	/poses/9/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T757	0.294 mm	DERIVED	CHK v3.4.1	/poses/9/min_unexpected_mm	(class set by generator: DERIVED)
T758	0 mm	DERIVED	CHK v3.4.1	/poses/9/wall_min_mm	(class set by generator: DERIVED)
T759	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/9/wall_min_by	(class set by generator: DERIVED)
T760	0	DERIVED	CHK v3.4.1	/poses/10/interference_not_whitelisted	(class set by generator: DERIVED)
T761	53	DERIVED	CHK v3.4.1	/poses/10/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T762	0.421 mm	DERIVED	CHK v3.4.1	/poses/10/min_unexpected_mm	(class set by generator: DERIVED)
T763	0 mm	DERIVED	CHK v3.4.1	/poses/10/wall_min_mm	(class set by generator: DERIVED)
T764	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/10/wall_min_by	(class set by generator: DERIVED)
T765	0	DERIVED	CHK v3.4.1	/poses/11/interference_not_whitelisted	(class set by generator: DERIVED)
T766	47	DERIVED	CHK v3.4.1	/poses/11/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T767	0.5 mm	DERIVED	CHK v3.4.1	/poses/11/min_unexpected_mm	(class set by generator: DERIVED)
T768	0 mm	DERIVED	CHK v3.4.1	/poses/11/wall_min_mm	(class set by generator: DERIVED)
T769	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/11/wall_min_by	(class set by generator: DERIVED)
T770	0	DERIVED	CHK v3.4.1	/poses/12/interference_not_whitelisted	(class set by generator: DERIVED)
T771	53	DERIVED	CHK v3.4.1	/poses/12/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T772	0.378 mm	DERIVED	CHK v3.4.1	/poses/12/min_unexpected_mm	(class set by generator: DERIVED)
T773	0 mm	DERIVED	CHK v3.4.1	/poses/12/wall_min_mm	(class set by generator: DERIVED)
T774	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/12/wall_min_by	(class set by generator: DERIVED)
T775	0	DERIVED	CHK v3.4.1	/poses/13/interference_not_whitelisted	(class set by generator: DERIVED)
T776	50	DERIVED	CHK v3.4.1	/poses/13/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T777	0.238 mm	DERIVED	CHK v3.4.1	/poses/13/min_unexpected_mm	(class set by generator: DERIVED)
T778	0 mm	DERIVED	CHK v3.4.1	/poses/13/wall_min_mm	(class set by generator: DERIVED)
T779	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/13/wall_min_by	(class set by generator: DERIVED)
T780	0	DERIVED	CHK v3.4.1	/poses/14/interference_not_whitelisted	(class set by generator: DERIVED)
T781	53	DERIVED	CHK v3.4.1	/poses/14/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T782	0.378 mm	DERIVED	CHK v3.4.1	/poses/14/min_unexpected_mm	(class set by generator: DERIVED)
T783	0 mm	DERIVED	CHK v3.4.1	/poses/14/wall_min_mm	(class set by generator: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T784	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/14/wall_min_by	(class set by generator: DERIVED)
T785	0	DERIVED	CHK v3.4.1	/poses/15/interference_not_whitelisted	(class set by generator: DERIVED)
T786	49	DERIVED	CHK v3.4.1	/poses/15/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T787	0.5 mm	DERIVED	CHK v3.4.1	/poses/15/min_unexpected_mm	(class set by generator: DERIVED)
T788	0 mm	DERIVED	CHK v3.4.1	/poses/15/wall_min_mm	(class set by generator: DERIVED)
T789	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/15/wall_min_by	(class set by generator: DERIVED)
T790	0	DERIVED	CHK v3.4.1	/poses/16/interference_not_whitelisted	(class set by generator: DERIVED)
T791	55	DERIVED	CHK v3.4.1	/poses/16/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T792	0.421 mm	DERIVED	CHK v3.4.1	/poses/16/min_unexpected_mm	(class set by generator: DERIVED)
T793	0 mm	DERIVED	CHK v3.4.1	/poses/16/wall_min_mm	(class set by generator: DERIVED)
T794	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/16/wall_min_by	(class set by generator: DERIVED)
T795	0	DERIVED	CHK v3.4.1	/poses/17/interference_not_whitelisted	(class set by generator: DERIVED)
T796	56	DERIVED	CHK v3.4.1	/poses/17/unexpected_close_lt_3mm	(class set by generator: DERIVED)
T797	0.5 mm	DERIVED	CHK v3.4.1	/poses/17/min_unexpected_mm	(class set by generator: DERIVED)
T798	0 mm	DERIVED	CHK v3.4.1	/poses/17/wall_min_mm	(class set by generator: DERIVED)
T799	ENG-FD-11#0	DERIVED	CHK v3.4.1	/poses/17/wall_min_by	(class set by generator: DERIVED)
T800	Sonderborg, Denmark, DK-6400	REQ	BOM v3.4.1	/delivery	(class set by generator: REQ)
T801	prices not given: every line TO-QUOTE unless dated	DESIGN	BOM v3.4.1	/currency	(class set by generator: DESIGN)
T802	95	DERIVED	BOM v3.4.1	/n_lines	(class set by generator: DERIVED)
T803	129	DERIVED	BOM v3.4.1	/n_instances	(class set by generator: DERIVED)
T804	0.058–0.097 kg	ASSUMED	BOM v3.4.1	/provision_mass_estimate_kg/estimate_range_kg	A1 head-end bleed provision (lead 2026-10-13), NOT modelled in CAD v3.4.1; range = ESTIMATE of taps, tube+tee, crossing hose and bulkhead (excl. valves)
T805	6061-T6 (case, head, closures, collar, ring, frame, tank); 316L (aft section, flight bas...	DERIVED	MEC v2	/materials	DERIVED
T806	21.92 MW/m²	SIM	CMB v2.4	/thermal_and_loads/peak_convective_heat_flux_W_m2	SIM Bartz at the throat, wall 600 K, max thrust, S230 (thermal_bc_summary_v2.json) [display = raw x 1e-06 (W/m² to MW/m²)]
T807	10259 W/(m²·K)	SIM	CMB v2.4	/thermal_and_loads/peak_h_g_W_m2K	SIM
T808	2737 K	SIM	CMB v2.4	/thermal_and_loads/T_aw_throat_K	SIM
T809	27.12–31.55 MW/m²	SIM	CMB v2.4	/thermal_and_loads/peak_incident_flux_cold_wall_transient_W_m2	SIM coupled insert model at start (wall near ambient), nominal/conservative (erosion_S230/erosion_dc_v2.json) [display = raw x 1e-06 (W/m² to MW/m²)]
T810	448.1 K	LEAD	ELE v2	/abort_matrix/causes/17/thresholds/kCaseOverTemp_K	LEAD 2026-10-13 correction, MECH pre36 final
T811	523.1 K	ASSUMED	ELE v2	/abort_matrix/causes/17/thresholds/kNozzleHolderOverTemp_K	ASSUMED
T812	1.6 mm	DESIGN	CMB v2.4	/gas_path/liner_grain_zone/thickness_m	v1 (grain zone only: the residual web >= 1 mm shields the case) [display = raw x 1000 (m to mm)]
T813	Structure/thermal: pre-chamber liner TC-06 3.6 mm (D_pre 49.9 mm); case over-temperature...	LEAD	REQ rev 2	§Amendments after rev 2 bullet "Structure/thermal"	REQ bullet
T814	phenolic, thickness set by thermal FEA; 1-D bound needs >= 5 mm and still exceeds the 19...	DERIVED	CMB v2.4	/gas_path/post_chamber_insulator	DERIVED data/v2/soakback_wall_v2.json
T815	replace the insert when dA_t/A_t reaches 5 % (measured pin gauge / bore scan after each ...	DERIVED	CMB v2.4	/thermal_and_loads/throat_replacement_rule	DERIVED
T816	mica/vermiculite 900 C rating vs 1595 C FE: fails by 695 K	DERIVED	CMB v2.4	/nozzle_joint/v/NZ03_gasket	DERIVED nozzle_joint_v2p4.json, nozzle_lip_melt_v2p4.json; FEA Mechanical; materials as recalled TO-VERIFY
T817	2.0 mm carbon-phenolic flange + 0.5 mm flexible-graphite seal at the lip face	DERIVED	CMB v2.4	/nozzle_joint/v/proposal_J1	DERIVED nozzle_joint_v2p4.json, nozzle_lip_melt_v2p4.json; FEA Mechanical; materials as recalled TO-VERIFY
T818	N2 pressure-decay leak check at 67.5 bar before and after every firing (criterion TO-DEF...	DERIVED	CMB v2.4	/nozzle_joint/v/H19_control	DERIVED nozzle_joint_v2p4.json, nozzle_lip_melt_v2p4.json; FEA Mechanical; materials as recalled TO-VERIFY
T819	thermal FEA v1.1 (B1/A1/A2, TS-1 relight, TS-2 soak, servo mount, TC-06 sizing, Al vs 31...	PENDING	MEC v2	/open_items/3	(class set by generator: PENDING)
T820	ABS (FDM printed, 100 % infill); PETG backup	REQ	CMB v2.4	/decisions/fuel	REQ founder/LEAD; ABS card c* within +/-0.3 % across compositions/hf sources (cea_cards.py)
T821	3.17 L	LEAD	CMB v2.4	/decisions/N2O_load_and_tank/v/V_tank_m3	LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill) [display = raw x 1000 (m³ to L)]

ID	Value	Class	Source	Key	Source tag / formula
T822	Self-pressurised N2O conditioned 15-25 C (heater blanket for cold days, removed at lifto...	DERIVED	CMB v2.4	/decisions/tank_pressurisation	DERIVED data/v2/supercharge_trade_v2.json: N2 pad 60 bar gives +3.4 bar at MECO vs heating to 25 C +4.3 bar; regulated N2 (0.21 L at 300 bar) adds a pressure system for +13 bar
T823	0.92	ASSUMED	CMB v2.4	/operating_points/basis [regex eta_c* ([\d.]+)]	(class set by generator: ASSUMED)
T824	0.975	ASSUMED	CMB v2.4	/operating_points/basis [regex eta_cf ([\d.]+)]	(class set by generator: ASSUMED)
T825	491 kg/(m²·s)	SIM	CMB v2.4	/operating_points/max_thrust/G_ox_initial_kg_m2s	SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975)
T826	r = a * G_ox^n (oxidiser-flux law, local G_ox along the port)	ASSUMED	CMB v2.4	/gas_path/grain/regression_law/form	(class set by generator: ASSUMED)
T827	5.77e-05 m/s·(kg/m²s)^-n	ASSUMED	CMB v2.4	/gas_path/grain/regression_law/a	ASSUMED band x0.6-1.5 [m/s (kg/m2/s)^-n]
T828	0.5	ASSUMED	CMB v2.4	/gas_path/grain/regression_law/n	ASSUMED band 0.35-0.68
T829	No verified N2O/ABS law in accessible literature; every result is checked at x0.6 and x1...	ASSUMED	CMB v2.4	/gas_path/grain/regression_law/note	(class set by generator: ASSUMED)
T830	02-combustion/T1-T4-REGRESSION-SPEC.md, v2.4	DESIGN	T14 v2.4	header (version line)	document header
T831	116.8 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent of ENG-GR-01 over assembly-manifest instance bounds (instance prefix ENG-GR-01#)
T832	115 mm	DERIVED	MAN v3.4.1	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent of ENG-GR-02 over assembly-manifest instance bounds (instance prefix ENG-GR-02#)
T833	not stated in the BOM name or notes of ENG-GR-01/02: rounded entrance TO-VERIFY in CAD	TO-VERIFY	BOM v3.4.1	/lines/*[ENG-GR-01, ENG-GR-02]/name+notes (generator text search)	generator check of BOM text
T834	53.5 mm	DESIGN	CMB v2.4	/gas_path/grain/outer_diameter_m	combustion.json v1 [m] [display = raw x 1000 (m to mm)]
T835	230 mm	DERIVED	CMB v2.4	/gas_path/grain/length_m	DERIVED v2 baseline (grain trade 2026-10-12; v1 0.200) [m] [display = raw x 1000 (m to mm)]
T836	single centred straight round port	LEAD	CMB v2.4	/gas_path/grain/port/type	LEAD decision 2026-10-12
T837	25 mm	DESIGN	CMB v2.4	/gas_path/grain/port/initial_diameter_m	combustion.json v1 [m] [display = raw x 1000 (m to mm)]
T838	14.25 mm	DERIVED	CMB v2.4	/gas_path/grain/port/initial_web_m	DERIVED (OD - port)/2 [display = raw x 1000 (m to mm)]
T839	Rounded port entrance r >= 4 mm is now REQUIRED on every grain, flight and stand (decomp...	LEAD	CL 176 lines	L160(4)	lead (decision) 2026-10-13 (line sha c3c13e6adb771e7a)
T840	1000 kg/m³	ASSUMED	CMB v2.4	/gas_path/grain/density_kg_m3	ASSUMED printed ABS [kg/m3]; filament 1040
T841	0.404 kg	DERIVED	CMB v2.4	/gas_path/grain/mass_kg	DERIVED geometry x 1000 kg/m3
T842	0.4244 kg	DERIVED	MP v3.4.1	/grain_mass_wet_kg	(class set by generator: DERIVED)
T843	web >= 1 mm at the x1.5 regression bound for every flight duty cycle	DERIVED	CMB v2.4	/gas_path/grain/residual_web_rule	DERIVED: ABS thermal layer kappa/r 0.1-0.3 mm (soakback_v2.json grain_zone); >= 1 mm shields the case
T844	forward face on the pre-chamber insulator shoulder, aft face on the post-chamber insulat...	DESIGN	CMB v2.4	/gas_path/grain/retention	(class set by generator: DESIGN)
T845	tank → NC main isolation valve (off-gimbal) → -6 PTFE-lined braided hose across the gi...	DESIGN	CMB v2.4	/valves_and_actuators/architecture	(class set by generator: DESIGN)
T846	Dyer et al. AIAA 2007-5702 (SPI/HEM blend); HEM and SPI bound it (G ratio 0.55-0.68 and ...	LITERATURE	CMB v2.4	/gas_path/injector/flow_model	(class set by generator: LITERATURE)
T847	sintered <= 15 um, Cv >= 2.2	DERIVED	CMB v2.4	/gas_path/injector/filter	DERIVED: seat gap at the 12.5 % floor = lift 0.207 mm x sin 15 deg = 54 um; filter <= 1/3 of the smallest gap; Cv from the line budget (feed_line_checks_v2.json); v1 said 40 um
T848	3	DERIVED	CMB v2.4	/valves_and_actuators/isolation_valve/Cv_min	DERIVED feed budget (feed_line_checks_v2.json): valve Cv 3 + filter Cv 2.2 give 1.17 bar total at 0.26 kg/s vs the 2.70 bar budget; Cv 1.5 + Cv 1.0 give 2.85 bar
T849	7000000 Pa	LEAD	CMB v2.4	/valves_and_actuators/isolation_valve/MEOP_Pa	LEAD
T850	0.15 s	DERIVED	CMB v2.4	/valves_and_actuators/isolation_valve/close_time_max_s	DERIVED
T851	required (<= 7.5 MPa) between the isolation valve and the pintle	DESIGN	CMB v2.4	/valves_and_actuators/isolation_valve/trapped_liquid_relief	v1 / mechanical.json v1
T852	**Coast leak ceiling:** 0.0055 g/s (§3.5).	DERIVED	RCMB 2.4	line 1904	RCMB line 1904
T853	-6 accepted: 6.3 m/s at 0.241 kg/s nominal, 6.5-7.1 m/s at 0.26 kg/s (15-25 C); hose fri...	DERIVED	CMB v2.4	/valves_and_actuators/hose	DERIVED feed_line_checks_v2.json (answers Mechanical REQUEST 2026-10-12)

ID	Value	Class	Source	Key	Source tag / formula
T854	first start of a flight/test (P_feed < 0.5 P_tank): pintle to a bleed lift (0.10 mm) BEF...	DERIVED	CMB v2.4	/valves_and_actuators/priming_rule/v/0	DERIVED: isentropic compression bound N2 847-913 K, N2O vapour 614-646 K at 45-60 bar (feed_line_checks_v2.json)
T855	head top (Mechanical 2026-10-12) via sense line to the fixed side	DESIGN	CMB v2.4	/gas_path/pc_tap/location	mechanical CHANGELOG
T856	1/8 in OD (1.6 mm ID) SS, length <= 0.3 m, transducer dead volume <= 0.3 cm3	DERIVED	CMB v2.4	/gas_path/pc_tap/sense_line	DERIVED data/v2/feed_line_checks_v2.json: lowest f_n per length (dead volume 0.1-0.5 cm3) {"L0.20m": [164.0, 325.0], "L0.30m": [125.0, 230.0], "L0.45m": [94.0, 160.0], "L0.60m": [75.0, 123.0]} Hz; pha
T857	adopted: rolling-guide carriage, spring 0.25 N m at closed + 0.12 N m/rad (+/-10 %); C2 ...	LEAD	CMB v2.4	/valves_and_actuators/pintle_actuator/lead_decision_v2p3	LEAD decision 2026-10-12 (CHANGELOG), on the Combustion v2.3 recommendation
T858	0.2–0.3 kg	ASSUMED	CMB v2.4	/igniter/mass_kg	ASSUMED: body 50 g, plug 10 g, two micro solenoids 60-120 g, propane cylinder 50 g, lines/fittings 50 g
T859	spark-ignited N2O/propane torch, radial into the pre-chamber	DESIGN	CMB v2.4	/igniter/type	ignition_v2.json
T860	3 bar	DERIVED	CMB v2.4	/igniter/torch_sizing/v/Pc_torch_bar	DERIVED/CEA ignition_v2.json
T861	5.5	DERIVED	CMB v2.4	/igniter/torch_sizing/v/OF	DERIVED/CEA ignition_v2.json
T862	2 mm	DERIVED	CMB v2.4	/igniter/torch_sizing/v/throat_d_m	DERIVED/CEA ignition_v2.json [display = raw x 1000 (m to mm)]
T863	0.503 g/s	DERIVED	CMB v2.4	/igniter/torch_sizing/v/mdot_n2o_kg_s	DERIVED/CEA ignition_v2.json [display = raw x 1000 (kg/s to g/s)]
T864	0.091 g/s	DERIVED	CMB v2.4	/igniter/torch_sizing/v/mdot_propane_kg_s	DERIVED/CEA ignition_v2.json [display = raw x 1000 (kg/s to g/s)]
T865	0.131 mm	DERIVED	CMB v2.4	/igniter/N2O_orifice_d_m	DERIVED Dyer at 20 C, Cd 0.65 ASSUMED (0.128-0.137 mm for 10-25 C) [display = raw x 1000 (m to mm)]
T866	0.254 mm	DERIVED	CMB v2.4	/igniter/propane_orifice_d_m	DERIVED at 20 C, Cd 0.80 ASSUMED; calibrate both by weighing [display = raw x 1000 (m to mm)]
T867	12 V	DERIVED	CMB v2.4	/igniter/electrical/v/spark_supply_V	DERIVED; electronics.json v1 IGN_HV coil mode + SOL1/SOL2
T868	0.05 J	DERIVED	CMB v2.4	/igniter/electrical/v/spark_energy_J	DERIVED; electronics.json v1 IGN_HV coil mode + SOL1/SOL2
T869	50 Hz	DERIVED	CMB v2.4	/igniter/electrical/v/rate_Hz	DERIVED; electronics.json v1 IGN_HV coil mode + SOL1/SOL2
T870	radial boss on the head section at x_face 0.015 m, axis inclined 30 deg inward/aft so th...	DERIVED	CMB v2.4	/igniter/envelope/v/mount	DERIVED (answers Mechanical REQUEST 2026-10-12)
T871	PURGE (A1 relight only)	LEAD	CMB v2.4	/igniter/sequence/v/start/0/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T872	−2.9 s	LEAD	CMB v2.4	/igniter/sequence/v/start/0/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T873	MV latch and opener ramp (0.35 s) with the pintle seated; V1 and V2 open; both taps flow...	LEAD	CMB v2.4	/igniter/sequence/v/start/0/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T874	E0	LEAD	CMB v2.4	/igniter/sequence/v/start/1/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T875	0 s	LEAD	CMB v2.4	/igniter/sequence/v/start/1/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T876	START or RELIGHT at T_fire: MV opens with the pintle seated (already open after PURGE); ...	LEAD	CMB v2.4	/igniter/sequence/v/start/1/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T877	torch gate	LEAD	CMB v2.4	/igniter/sequence/v/start/2/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T878	0.16 s	LEAD	CMB v2.4	/igniter/sequence/v/start/2/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T879	no E1 by E0 + 0.16 s → TORCH_FAULT (A1); no main N2O admitted	LEAD	CMB v2.4	/igniter/sequence/v/start/2/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T880	E1	LEAD	CMB v2.4	/igniter/sequence/v/start/3/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T881	0.09 s	LEAD	CMB v2.4	/igniter/sequence/v/start/3/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T882	torch flame confirmed on P_torch AND photodiode (2oo2, FA-1 (a)), 10 ms persistence; pin...	LEAD	CMB v2.4	/igniter/sequence/v/start/3/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4

ID	Value	Class	Source	Key	Source tag / formula
T883	W	LEAD	CMB v2.4	/igniter/sequence/v/start/4/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T884	0.81 s	LEAD	CMB v2.4	/igniter/sequence/v/start/4/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T885	abort deadline W from the lift command: 0.40 s flight first start, 0.65 s flight relight...	LEAD	CMB v2.4	/igniter/sequence/v/start/4/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T886	E2	LEAD	CMB v2.4	/igniter/sequence/v/start/5/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T887	0.442 s	LEAD	CMB v2.4	/igniter/sequence/v/start/5/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T888	light: Pc >= 0.8 bar above the pre-lift baseline on filtered AND raw Pc for 10 ms, above...	LEAD	CMB v2.4	/igniter/sequence/v/start/5/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T889	E3	LEAD	CMB v2.4	/igniter/sequence/v/start/6/epoch	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T890	MISSING	LEAD	CMB v2.4	/igniter/sequence/v/start/6/t_s	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T891	thrust >= 90 % of the command (ENG-04 latency, §7.7)	LEAD	CMB v2.4	/igniter/sequence/v/start/6/action	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T892	−2.9 s	SIM	CMB v2.4	/relight_purge/v/timing_s/PURGE_command_rel_E0	SIM/DERIVED on ASSUMED Cd and slip brackets; LEAD lines 160, 161, 164; Mechanical line 162
T893	0.35 s	SIM	CMB v2.4	/relight_purge/v/timing_s/MV_ramp	SIM/DERIVED on ASSUMED Cd and slip brackets; LEAD lines 160, 161, 164; Mechanical line 162
T894	2.5 s	SIM	CMB v2.4	/relight_purge/v/timing_s/flow	SIM/DERIVED on ASSUMED Cd and slip brackets; LEAD lines 160, 161, 164; Mechanical line 162
T895	−0.05 s	SIM	CMB v2.4	/relight_purge/v/timing_s/flow_end_rel_E0	SIM/DERIVED on ASSUMED Cd and slip brackets; LEAD lines 160, 161, 164; Mechanical line 162
T896	tank 15-25 C (ENG-12); T_tank <= 24 C at ARM and at ignition (liquid-line interlock, §2....	LEAD	CMB v2.4	/igniter/sequence/v/arm_preconditions/0	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T897	pintle seated, encoder healthy; MV-P closed; bleed valves V1 and V2 closed, drivers inhi...	LEAD	CMB v2.4	/igniter/sequence/v/arm_preconditions/1	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T898	P_torch auto-zero; ARM refused if P_torch > 1.05 bar abs	LEAD	CMB v2.4	/igniter/sequence/v/arm_preconditions/2	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T899	Pc and P_torch transducers 0-70 bar class (ENG-09 >= 70 bar); P_feed, P_tank healthy; Pc...	LEAD	CMB v2.4	/igniter/sequence/v/arm_preconditions/3	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T900	igniter continuity (spark coil, SOL1, SOL2); photodiode healthy (needed for the 2oo2 E1)	LEAD	CMB v2.4	/igniter/sequence/v/arm_preconditions/4	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T901	relight only: chamber vented; RELIGHT refused if a case thermocouple reads > 433.15 K	LEAD	CMB v2.4	/igniter/sequence/v/arm_preconditions/5	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T902	MV closed during coasts (LEAD); the v2.3 text "the isolation valve may stay open between...	LEAD	CMB v2.4	/igniter/sequence/v/relight/0	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T903	A1 relight: PURGE at E0 - 2.90 s (2.5 s of flow, theta <= 2 deg, balanced or axial vent)...	LEAD	CMB v2.4	/igniter/sequence/v/relight/1	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T904	settling: liquid at the tank outlet from E0 - 2.90 s - t_settle (§2.4 note); RELIGHT ref...	LEAD	CMB v2.4	/igniter/sequence/v/relight/2	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T905	light window W 0.65 s; E0 → E2 max 0.626 s on the worst purged line state (rg01_line_st...	LEAD	CMB v2.4	/igniter/sequence/v/relight/3	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T906	pre-MECO throttle-down to <= 75 % (ENG-05)	LEAD	CMB v2.4	/igniter/sequence/v/shutdown/0	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T907	pintle closes at full drive speed	LEAD	CMB v2.4	/igniter/sequence/v/shutdown/1	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T908	MV de-energised on seat confirmation (encoder at seat and Pc < 2 bar abs) or 0.30 s afte...	LEAD	CMB v2.4	/igniter/sequence/v/shutdown/2	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4

ID	Value	Class	Source	Key	Source tag / formula
T909	trapped liquid relieves to the tank through S1 (§2.7); ground: 30 s hold then safing	LEAD	CMB v2.4	/igniter/sequence/v/shutdown/3	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T910	pintle closed; torch solenoids closed (SOL1 not after SOL2); MV closed (abort is MV-first)	LEAD	CMB v2.4	/igniter/sequence/v/abort_if_no_ignition/0	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T911	no retry into a pooled chamber; ground: wait >= 5 s, N2 purge, safing (§10.2)	LEAD	CMB v2.4	/igniter/sequence/v/abort_if_no_ignition/1	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T912	flight: no retry; vehicle-level response per GNC	LEAD	CMB v2.4	/igniter/sequence/v/abort_if_no_ignition/2	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T913	A5 (Pc > 8 bar for 5 ms) is DELETED (LEAD line 160): it tripped on 98 % of the late-ligh...	LEAD	CMB v2.4	/igniter/sequence/v/abort_if_no_ignition/3	LEAD (CHANGELOG 153-164) + DERIVED (REPORT v2.4 §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T914	REQUIRED of Electronics; FA-1 accepted for design continuation only; flight acceptance n...	LEAD	CMB v2.4	/igniter/sequence/v/fa1_supervision/v/status	LEAD line 164
T915	FA-1 stays accepted for design continuation only; flight acceptance needs IGN-1 and Mech...	LEAD	CL 176 lines	L170(2)	lead (decision) 2026-10-13 (line sha 0e5d7d63a5e8dd26)
T916	LEAD lines 160, 164; REPORT v2.4 §7.4	LEAD	CMB v2.4	/abort_matrix_v2p4/tag	(class set by generator: LEAD)
T917	no E2 within W of the lift command (0.40 / 0.65 / 0.35 s)	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A2	LEAD lines 160, 164; REPORT v2.4 §7.4
T918	v2.9: no light by the relative rule (rise >= kIgnRise_Pa over P0, floor, kIgnConfirmRelN...	DESIGN	ELE v2	/abort_matrix/causes/14/trigger	(class set by generator: DESIGN)
T919	80000	LEAD	ELE v2	/abort_matrix/causes/14/thresholds/kIgnRise_Pa	LEAD/COMB
T920	10	LEAD	ELE v2	/abort_matrix/causes/14/thresholds/kIgnConfirmRelN	LEAD/COMB
T921	0.4	LEAD	ELE v2	/abort_matrix/causes/14/thresholds/kIgnWindowFirst_s	LEAD 2026-10-13
T922	0.65	LEAD	ELE v2	/abort_matrix/causes/14/thresholds/kIgnWindowRelightLong_s	LEAD 2026-10-13
T923	0.35	LEAD	ELE v2	/abort_matrix/causes/14/thresholds/kIgnWindowFirstStand_s	LEAD 2026-10-13 fallback (b)
T924	30 bar abort raw persistence 3 → 5 ms; comparator stays the fast path for a blocked thr...	LEAD	CL 176 lines	L160(2)	lead (decision) 2026-10-13 (line sha c3c13e6adb771e7a)
T925	Pc > 30 bar raw for 5 ms	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A4	LEAD lines 160, 164; REPORT v2.4 §7.4
T926	firmware Pc above the over-pressure limit outside IGNITION	DESIGN	ELE v2	/abort_matrix/causes/0/trigger	(class set by generator: DESIGN)
T927	3000000	LEAD	ELE v2	/abort_matrix/causes/0/thresholds/kPcOverPressureAbs_Pa	LEAD via COMB
T928	0.003	DESIGN	ELE v2	/abort_matrix/causes/0/thresholds/kPcOverPressureTime_s	DESIGN
T929	A5 DELETED as an abort, flight and stand; replaced by a LIGHT_SPIKE log event (raw and f...	LEAD	CL 176 lines	L160(1)	lead (decision) 2026-10-13 (line sha c3c13e6adb771e7a)
T930	DELETED → LIGHT_SPIKE log	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A5	LEAD lines 160, 164; REPORT v2.4 §7.4
T931	8.0 bar abs for 5 ms from E1 to the ramp start (unchanged); with the 10 ms light confirm...	LEAD	ELE v2	/start_monitors_v2p9/A5	LEAD/COMB
T932	A7 START high side in flight must neither switch the Pc loop to open loop nor set teleme...	LEAD	CL 176 lines	L160(3)	lead (decision) 2026-10-13 (line sha c3c13e6adb771e7a)
T933	RE-12 pre-decided: if the SIL pool replay gives a stand A7 trip rate > 1 %, blank the st...	LEAD	CL 176 lines	L166(7)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T934	Pc loss or implausibility, bands and the flight START high-side rule (Electronics sets v...	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A7	LEAD lines 160, 164; REPORT v2.4 §7.4
T935	HOLD band from E1 (lo P0 - 0.3 bar), START band E2→E3 (lo = A3 threshold on the map, hi...	LEAD	ELE v2	/start_monitors_v2p9/A7	LEAD/COMB + DESIGN
T936	Bleed closure confirmed from P_mid is a pad go/no-go (launch-commit item) on every fligh...	LEAD	CL 176 lines	L168(3)	lead (decision) 2026-10-13 (line sha 3ad186c5fa9490ff)
T937	bleed valve fault or closure not confirmed by E0 - 0.05 s → BLEED_FAULT to the FC (FC a...	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A14	LEAD lines 160, 164; REPORT v2.4 §7.4
T938	v2.9 head-end bleed: pad BIT step failed, V1 closure not confirmed by E0 + kHbleedCloseC...	DESIGN	ELE v2	/abort_matrix/causes/11/trigger	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T939	0.15	DESIGN	ELE v2	/abort_matrix/causes/11/thresholds/kHbleedCloseConfirm_s	DESIGN
T940	0.2	DESIGN	ELE v2	/abort_matrix/causes/11/thresholds/kHbleedLeakTime_s	DESIGN
T941	0.1	DESIGN	ELE v2	/abort_matrix/causes/11/thresholds/kHbleedPathBy_s	DESIGN
T942	pintle or chamber anomaly during PURGE: inhibit torch, close MV	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A15	LEAD lines 160, 164; REPORT v2.4 §7.4
T943	no matching cause in firmware v2.9	MISSING	ELE v2	/abort_matrix/causes (generator search)	generator search of the cause list
T944	FA-1: (a) E1 declared only on P_torch AND photodiode (2oo2) and (b) torch supervision E1...	LEAD	CL 176 lines	L164(8)	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T945	torch loss on BOTH sensors >= 5 ms between E1 and E2: close pintle and MV	LEAD	CMB v2.4	/abort_matrix_v2p4/v/A16	LEAD lines 160, 164; REPORT v2.4 §7.4
T946	RE-05: E1 = torch confirmed (2oo2, FA-1 (a)) AND [closure confirmed OR closure deadline ...	LEAD	CL 176 lines	L166(3)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T947	no torch flame (P_torch) by the gate: abort before the pintle step	DESIGN	ELE v2	/abort_matrix/causes/5/trigger	(class set by generator: DESIGN)
T948	0.16	LEAD	ELE v2	/abort_matrix/causes/5/thresholds/kTorchGate_s	[DESIGN; COMB v2.4 CONFIRMED 2026-10-13: E0→E1 max 0.1494 s over 1e6 samples,
T949	200000	DESIGN	ELE v2	/abort_matrix/causes/5/thresholds/kTorchLitAbs_Pa	DESIGN, ESTIMATE
T950	3	DESIGN	ELE v2	/abort_matrix/causes/5/thresholds/kTorchConfirmN	DESIGN
T951	RE-02 light-rule baseline fallback: P0 > +check → keep the measured P0 (never substitut...	LEAD	CL 176 lines	L166(1)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T952	P0 = mean of filtered/raw gauge Pc over [E1 - 20 ms, E1]; |P0| > residual + 0.1 bar → a...	LEAD	ELE v2	/light_rule_v2p9/rule	LEAD/COMB
T953	RE-08: an invalid case TC is treated as hot, like bay and tank: both case TCs valid requ...	LEAD	CL 176 lines	L166(6)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T954	case TC (az 90/270, armed only while the MV is latched or OPEN) or nozzle holder above l...	DESIGN	ELE v2	/abort_matrix/causes/17/trigger	(class set by generator: DESIGN)
T955	448.1	LEAD	ELE v2	/abort_matrix/causes/17/thresholds/kCaseOverTemp_K	LEAD 2026-10-13 correction, MECH pre36 final
T956	523.1	ASSUMED	ELE v2	/abort_matrix/causes/17/thresholds/kNozzleHolderOverTemp_K	ASSUMED
T957	333.1	DERIVED	ELE v2	/abort_matrix/causes/17/thresholds/kServoBracketOverTemp_K	DS-derived
T958	0.2	DESIGN	ELE v2	/abort_matrix/causes/17/thresholds/kOverTempTime_s	MECH 2026-10-13: 0.2 s hold
T959	RE-28: no in-flight recovery path after an ECU reset; reset → SAFE, MV and bleed closed...	LEAD	CL 176 lines	L166(8)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T960	hardware watchdog reset: reboot lands in SAFE; a lit-session marker becomes a relight lo...	DESIGN	ELE v2	/abort_matrix/causes/27/trigger	(class set by generator: DESIGN)
T961	0.05	DESIGN	ELE v2	/abort_matrix/causes/27/thresholds/kHwWatchdog_s	DESIGN
T962	Drogue release predicate restated, interim term limited to the invalid case: RELEASE iff...	LEAD	CL 176 lines	L168(1)	lead (decision) 2026-10-13 (line sha 3ad186c5fa9490ff)
T963	FC (LEAD decision 2026-10-13 CHANGELOG): release at E2 AND [measured Pc >= 3.0 bar abs O...	DESIGN	ELE v2	/drogue_release_support/rule_owner	(no source tag; class set by generator: DESIGN)
T964	ground procedure, not a firmware cause	DESIGN	generator	abort map SAFE: firmware column	generator classification of the decided item
T965	every abort de-energises the NC valve at once and commands the pintle closed in parallel...	LEAD	ELE v2	/abort_matrix/actions	(class set by generator: LEAD)
T966	0.93	SIM	CMB v2.4	/stability/points_initial_port/v/1.0/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T967	6.5	SIM	CMB v2.4	/stability/points_initial_port/v/1.0/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T968	4.7	SIM	CMB v2.4	/stability/points_initial_port/v/1.0/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T969	1460 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/1.0/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T970	1125 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/1.0/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T971	2.17 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/1.0/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)

ID	Value	Class	Source	Key	Source tag / formula
T972	1.58	SIM	CMB v2.4	/stability/points_initial_port/v/0.75/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T973	10.2	SIM	CMB v2.4	/stability/points_initial_port/v/0.75/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T974	7.5	SIM	CMB v2.4	/stability/points_initial_port/v/0.75/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T975	1477 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.75/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T976	1138 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.75/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T977	1.61 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.75/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T978	2.78	SIM	CMB v2.4	/stability/points_initial_port/v/0.5/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T979	17	SIM	CMB v2.4	/stability/points_initial_port/v/0.5/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T980	12.5	SIM	CMB v2.4	/stability/points_initial_port/v/0.5/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T981	1499 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.5/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T982	1155 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.5/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T983	1.07 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.5/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T984	5.86	SIM	CMB v2.4	/stability/points_initial_port/v/0.25/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T985	32.1	SIM	CMB v2.4	/stability/points_initial_port/v/0.25/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T986	23.7	SIM	CMB v2.4	/stability/points_initial_port/v/0.25/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T987	1532 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.25/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T988	1180 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.25/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T989	0.56 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.25/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T990	7.18	SIM	CMB v2.4	/stability/points_initial_port/v/0.2/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T991	36.8	SIM	CMB v2.4	/stability/points_initial_port/v/0.2/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T992	27.2	SIM	CMB v2.4	/stability/points_initial_port/v/0.2/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T993	1539 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.2/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T994	1186 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.2/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T995	0.47 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.2/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T996	10.54	SIM	CMB v2.4	/stability/points_initial_port/v/0.125/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T997	52.2	SIM	CMB v2.4	/stability/points_initial_port/v/0.125/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T998	38.6	SIM	CMB v2.4	/stability/points_initial_port/v/0.125/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T999	1544 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.125/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1000	1190 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.125/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1001	0.33 Hz	SIM	CMB v2.4	/stability/points_initial_port/v/0.125/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1002	injector dP/Pc >= 0.25 everywhere (ceiling rule); chug gain margin from the lumped feed/...	DERIVED	CMB v2.4	/stability/criteria	(class set by generator: DERIVED)
T1003	classical hybrid low-frequency (thermal-lag/boundary-layer) instability 0.3-2 Hz band is...	TO-VERIFY	CMB v2.4	/stability/hybrid_LF_note	(class set by generator: TO-VERIFY)
T1004	0.01941 m	DESIGN	CMB v2.4	/gas_path/nozzle/exit_diameter_m	v1 geometry
T1005	Version v2.1a for validation and v2.2 for the time-matched afterburn, 2026-10-12. Owner:...	SIM	RPL v2.2b	line 3	RPL line 3
T1006	2.676 De	SIM	PVAL v2.1a	/correlations/own_f_2s_rans/x_md_cfd	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1007	2.663 De	DERIVED	PVAL v2.1a	/correlations/own_f_2s_rans/x_md_lewis_carlson	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]

ID	Value	Class	Source	Key	Source tag / formula
T1008	3.959 De	SIM	PVAL v2.1α	/correlations/own_f_2s_rans/spacing_first_mach	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1009	4.101 De	SIM	PVAL v2.1α	/correlations/own_f_2s_rans/spacing_mean3_mach	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1010	4.816 De	DERIVED	PVAL v2.1α	/correlations/own_f_2s_rans/spacing_pp_tam	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1011	14.9 De	SIM	PVAL v2.1α	/gci/core_length_Z095_De/fine	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1012	2222 K	SIM	PVAL v2.1α	/gci/T_axis_10De_K/fine	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1013	2058 K	SIM	PVAL v2.1α	/gci/T_axis_20De_K/fine	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1014	500.8 N	SIM	PVAL v2.1α	/thrust_vs_1d/own_f_2s_rans/F_cfd	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1015	505.8 N	DERIVED	PVAL v2.1α	/one_d/F_ideal_lambda_N	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1016	−0.9829 %	SIM	PVAL v2.1α	/thrust_vs_1d/own_f_2s_rans/dev_vs_ideal_lambda_pct	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1017	0.2542 kg/s	SIM	PVAL v2.1α	/thrust_vs_1d/own_f_2s_rans/mdot_cfd	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1018	0.2558 kg/s	DERIVED	PVAL v2.1α	/one_d/mdot	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1019	−0.6526 %	SIM	PVAL v2.1α	/thrust_vs_1d/own_f_2s_rans/mdot_dev_pct	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1020	v1 (combustion.json v1): Pc 24.99 bar, O/F 7.76, T0 = eta_c*^2 Tc = 2844.9 K. All grid l...	SIM	RPL v2.2b	§0. Checkpoint bullet "v1 (combustion.json v1)"	RPL bullet
T1021	2.012	DERIVED	PVAL v2.1α	/one_d/Me	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1022	3.236	DERIVED	PVAL v2.1α	/one_d/pe_over_pa	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1023	2.623	DERIVED	PVAL v2.1α	/one_d/Mj	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1024	51.94 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1006, T1004
T1025	grid differences within 2 sigma of the temporal scatter: GCI not meaningful; report the ...	SIM	PVAL v2.1α	/gci_resolvability/x_mach_disk_De/verdict	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1026	76.85 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1008, T1004
T1027	79.6 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1009, T1004
T1028	6.542 %	SIM	PVAL v2.1α	/gci/spacing_mean_3_De/gci_fine_pct	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1029	oscillatory: 3 x max-deviation bound	SIM	PVAL v2.1α	/gci/spacing_mean_3_De/note	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1030	3.688	SIM	PVAL v2.1α	/gci/M_max_before_disk/fine	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1031	289.2 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1011, T1004
T1032	0.1522 %	SIM	PVAL v2.1α	/gci/thrust_exit_N/gci_fine_pct	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1033	monotone without observed convergence (p < 0.5): 3 x max-deviation bound	SIM	PVAL v2.1α	/gci/thrust_exit_N/note	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]

ID	Value	Class	Source	Key	Source tag / formula
T1034	44.9 De	SIM	PVAL v2.1a	/jet_front_x_De/own_f_2s_rans	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1035	2.709 De	SIM	PVAL v2.1a	/v2op/x_md_cfd	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1036	2.666 De	DERIVED	PVAL v2.1a	/v2op/x_md_lewis_carlson	(class set by generator: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1037	−1.087 %	SIM	PVAL v2.1a	/v2op/thrust_dev_vs_ideal_lambda_pct	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1038	same coarse grid and models as own_c_2s_rans (v1 point O/F 7.76); v2 - v1 differences in...	SIM	PVAL v2.1a	/v2op/note	(class set by generator: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1039	adds 172-182 K at O/F 7.76 (temporal band 172-178 K; time-matched fine maximum 179.2 K; ...	SIM	RPL v2.2b	§0. Checkpoint bullet "adds 172"	RPL bullet
T1040	adds nothing at the v2 point for an equilibrium exhaust (O/F 10.4, oxidiser-rich; 0.44 K...	SIM	RPL v2.2b	§0. Checkpoint bullet "adds nothing"	RPL bullet
T1041	adds up to 1476 K at the fuel-rich 12.5 % floor.	SIM	RPL v2.2b	§0. Checkpoint bullet "adds up to"	RPL bullet
T1042	**Soot and radiation.** There is no soot model and no radiative transfer. The synthetic ...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Soot and radiation"	RPL bullet
T1043	**Finite-rate chemistry and turbulence-chemistry interaction.** Afterburning is a fast-c...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Finite-rate chemistry"	RPL bullet
T1044	**Exhaust gas model.** The exhaust is a calorically perfect gas with frozen calibrated g...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Exhaust gas model"	RPL bullet
T1045	**Geometry and setting.** The model is axisymmetric only (no gimbal angle, no 3-D struct...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Geometry and setting"	RPL bullet
T1046	**Duration.** Simulated time is 3-4 ms. The far field (> 15 De) is not statistically sta...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Duration"	RPL bullet
T1047	**No acoustics or plume loads on the vehicle.**	SIM	RPL v2.2b	§10. What the model cannot show bullet "**No acoustics"	RPL bullet
T1048	0 unchanged, 1 changed since review (05-plume-cfd/REPORT.md), 0 not found	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register
T1049	REV-PLUME-1	DERIVED	SUP 7 reviews	/reviews/1/id	(class set by generator: DERIVED)
T1050	ACCEPT WITH COMMENTS	DERIVED	SUP 7 reviews	/reviews/1/verdict	(class set by generator: DERIVED)
T1051	RC-10 thrust GCI R=1.09 diverges: reword; RC-11 correlation claim wording; RC-19 present...	DERIVED	SUP 7 reviews	/reviews/1/comments	(class set by generator: DERIVED)
T1052	1 unchanged, 0 changed since review (none), 0 not found	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register
T1053	REV-PLUME-2	DERIVED	SUP 7 reviews	/reviews/6/id	(class set by generator: DERIVED)
T1054	ACCEPT	DERIVED	SUP 7 reviews	/reviews/6/verdict	(class set by generator: DERIVED)
T1055	R2-12 summary wording (lead file) corrected: HANDBACK-SUMMARY d12c8862bf5524a4	DERIVED	SUP 7 reviews	/reviews/6/comments	(class set by generator: DERIVED)
T1056	v2	DESIGN	MEC v2	/version	(class set by generator: DESIGN)
T1057	PENDING	PENDING	—		[no published margin table found by the generator]
T1058	LC1	DESIGN	FEALC v1	/cases/0/id	(class set by generator: DESIGN)
T1059	Steady max thrust, gimbal neutral	DESIGN	FEALC v1	/cases/0/title	(class set by generator: DESIGN)
T1060	2499311 Pa	DESIGN	FEALC v1	/cases/0/chamber_Pa	(class set by generator: DESIGN)
T1061	500 N	DESIGN	FEALC v1	/cases/0/thrust_N	(class set by generator: DESIGN)
T1062	1.5	REQ	FEALC v1	/cases/0/criterion/SF_y	(class set by generator: REQ)
T1063	2	REQ	FEALC v1	/cases/0/criterion/SF_u	(class set by generator: REQ)
T1064	LC2	DESIGN	FEALC v1	/cases/1/id	(class set by generator: DESIGN)
T1065	Chamber MEOP + thermal state at the critical time	DESIGN	FEALC v1	/cases/1/title	(class set by generator: DESIGN)
T1066	4500000 Pa	DESIGN	FEALC v1	/cases/1/chamber_Pa	(class set by generator: DESIGN)
T1067	1.5	REQ	FEALC v1	/cases/1/criterion/SF_y	(class set by generator: REQ)
T1068	2	REQ	FEALC v1	/cases/1/criterion/SF_u	(class set by generator: REQ)
T1069	LC2F	DESIGN	FEALC v1	/cases/2/id	(class set by generator: DESIGN)
T1070	Pintle full-stroke failure thrust (structure)	DESIGN	FEALC v1	/cases/2/title	(class set by generator: DESIGN)
T1071	4500000 Pa	DESIGN	FEALC v1	/cases/2/chamber_Pa	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T1072	907.7 N	DESIGN	FEALC v1	/cases/2/thrust_N	(class set by generator: DESIGN)
T1073	1	REQ	FEALC v1	/cases/2/criterion/SF_y	(class set by generator: REQ)
T1074	1.5	REQ	FEALC v1	/cases/2/criterion/SF_u	(class set by generator: REQ)
T1075	LC3	DESIGN	FEALC v1	/cases/3/id	(class set by generator: DESIGN)
T1076	Max thrust + max gimbal (both axes) + lateral and angular acceleration limit loads	DESIGN	FEALC v1	/cases/3/title	(class set by generator: DESIGN)
T1077	2499311 Pa	DESIGN	FEALC v1	/cases/3/chamber_Pa	(class set by generator: DESIGN)
T1078	500 N	DESIGN	FEALC v1	/cases/3/thrust_N	(class set by generator: DESIGN)
T1079	1.5	REQ	FEALC v1	/cases/3/criterion/SF_y	(class set by generator: REQ)
T1080	2	REQ	FEALC v1	/cases/3/criterion/SF_u	(class set by generator: REQ)
T1081	LC4	DESIGN	FEALC v1	/cases/4/id	(class set by generator: DESIGN)
T1082	Hydrostatic proof 1.5 x MEOP, room temperature (chamber and tank)	DESIGN	FEALC v1	/cases/4/title	(class set by generator: DESIGN)
T1083	6750000 Pa	DESIGN	FEALC v1	/cases/4/chamber_Pa	(class set by generator: DESIGN)
T1084	1	REQ	FEALC v1	/cases/4/criterion/SF_y	(class set by generator: REQ)
T1085	BN	DESIGN	FEALC v1	/cases/5/id	(class set by generator: DESIGN)
T1086	Blocked-nozzle bound: chamber at tank pressure (30 C), hot	DESIGN	FEALC v1	/cases/5/title	(class set by generator: DESIGN)
T1087	6308000 Pa	DESIGN	FEALC v1	/cases/5/chamber_Pa	(class set by generator: DESIGN)
T1088	0 N	DESIGN	FEALC v1	/cases/5/thrust_N	(class set by generator: DESIGN)
T1089	1	REQ	FEALC v1	/cases/5/criterion/SF_y	(class set by generator: REQ)
T1090	LC5	DESIGN	FEALC v1	/cases/6/id	(class set by generator: DESIGN)
T1091	Transient thermal over the flight duty cycle, structural at the critical instants	DESIGN	FEALC v1	/cases/6/title	(class set by generator: DESIGN)
T1092	1.5	REQ	FEALC v1	/cases/6/criterion/SF_y	(class set by generator: REQ)
T1093	2	REQ	FEALC v1	/cases/6/criterion/SF_u	(class set by generator: REQ)
T1094	LC6	DESIGN	FEALC v1	/cases/7/id	(class set by generator: DESIGN)
T1095	Landing shock and handling/transport	DESIGN	FEALC v1	/cases/7/title	(class set by generator: DESIGN)
T1096	1.5	REQ	FEALC v1	/cases/7/criterion/SF_y	(class set by generator: REQ)
T1097	2	REQ	FEALC v1	/cases/7/criterion/SF_u	(class set by generator: REQ)
T1098	MODAL	DESIGN	FEALC v1	/cases/8/id	(class set by generator: DESIGN)
T1099	Gimballed engine on its actuators: first modes vs ENG-11	DESIGN	FEALC v1	/cases/8/title	(class set by generator: DESIGN)
T1100	LC-TB70 (53.4 bar abs, 34 ms) to Mechanical for the assembled FEA.	LEAD	CL 176 lines	L166(10)[sentence containing "LC-TB70"]	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T1101	id: OI-TB-MEOP; severity: ACCEPTED; issue: S6 throat block Pc 53.4 bar abs > MEOP for 72...	SIM	ELE v2	/open_issues/3	(class set by generator: SIM)
T1102	Mechanical adds a <= 3 ms, 75.5 bar abs head-pulse check of the case and closures (dynam... [OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs]	LEAD	CL 176 lines	L164(8)[sentence containing "Mechanical adds"]	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T1103	PENDING	PENDING	—		[FEA margins for load case LC1]
T1104	PENDING	PENDING	—		[FEA margins for load case LC2]
T1105	PENDING	PENDING	—		[FEA margins for load case LC2F]
T1106	PENDING	PENDING	—		[FEA margins for load case LC3]
T1107	PENDING	PENDING	—		[FEA margins for load case LC4]
T1108	PENDING	PENDING	—		[FEA margins for load case BN]
T1109	PENDING	PENDING	—		[FEA margins for load case LC5]
T1110	PENDING	PENDING	—		[FEA margins for load case LC6]
T1111	PENDING	PENDING	—		[FEA margins for load case MODAL]
T1112	REVISE	DERIVED	SUP 7 reviews	/reviews/3/verdict	(class set by generator: DERIVED)
T1113	RE-01 A5 still an abort; spike source not in SIL; RE-02 BASE_AMB fallback gives false E2...	DERIVED	SUP 7 reviews	/reviews/3/blocking	(class set by generator: DERIVED)
T1114	Electronics v2.10 publishes to the hub after archiving v2.9	LEAD	CL 176 lines	L166(10)[sentence containing "v2.10"]	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T1115	PJRC Teensy 4.1 (i.MX RT1062 600 MHz, native SDIO microSD, 3 CAN incl. 1 CAN-FD)	DESIGN	ELE v2	/ecu/mcu	(class set by generator: DESIGN)
T1116	120, 60, 1.6 mm	DESIGN	ELE v2	/ecu/pcb_outline_mm	DESIGN v1 (4-layer, hand-solderable)
T1117	128, 68, 32 mm	DESIGN	ELE v2	/ecu/enclosure_envelope_mm	DESIGN v1 reservation (PETG or 1 mm Al); +25 mm connector/cable clearance on both long sides

ID	Value	Class	Source	Key	Source tag / formula
T1118	v2.4 (KiCad 7.0.11, generated from hardware/kicad_gen/ecu_design.py; v2.4 = head-end ble...	DESIGN	ELE v2	/ecu/schematic/version	(class set by generator: DESIGN)
T1119	0 errors, 0 warnings, 9 sheets	DERIVED	ELE v2	/ecu/schematic/erc	KiCad eeschema ERC (GUI on Xvfb)
T1120	components 449, nets 277, pins 1256 of 1244 intended, errors 0, warnings 0	DERIVED	ELE v2	/ecu/schematic/netlist_crosscheck	check_netlist.py v1
T1121	TI_RPM0026A_PLACEHOLDER (NOT FOR FABRICATION); Pololu_U3V70_WireAttach (wire pads only);...	TO-VERIFY	ELE v2	/ecu/schematic/placeholders	(class set by generator: TO-VERIFY)
T1122	v2.9	DESIGN	ELE v2	/firmware/version	(class set by generator: DESIGN)
T1123	PJRC Teensy 4.1 (i.MX RT1062); HAL for the hardware NOT written (core runs in SIL and te...	TO-VERIFY	ELE v2	/firmware/target	(class set by generator: TO-VERIFY)
T1124	287 passed, 0 failed	SIM	ELE v2	/firmware/unit_tests/v2.9	(class set by generator: SIM)
T1125	src/core/engine_params.h: [["combustion.json", "v2.3", "d9ec54fbf450e243", "current"], [...	DESIGN	ELE v2	/inputs/generated_headers	(class set by generator: DESIGN)
T1126	ISO 11898-2 CAN, classic 2.0B 1 Mbit/s, 11-bit IDs; CAN-FD-capable transceivers so FD is...	DESIGN	ELE v2	/can_interface/physical	DESIGN
T1127	PC: M3234-000005-070BG; P_TORCH: M3234-000005-070BG; P_FEED: M3234-000005-170BG; P_TANK:...	DESIGN	ELE v2	/sensors/pressure_transducers/selection	DESIGN pc_transducer_v2; lead line 73 (3)
T1128	Pc J601, P_feed J602 (FS-01 MV-outlet block, Mechanical), P_tank J603, P_torch J901 (pc_...	DESIGN	ELE v2	/sensors/pressure_transducers/connectors	(class set by generator: DESIGN)
T1129	31.47 bar g = 32.48 bar abs nominal on the 070BG; band 30.90-34.06 bar abs; firmware abo...	DERIVED	ELE v2	/sensors/pressure_transducers/hw_ovp	DERIVED pc_transducer_v2 C
T1130	0.25 V	DESIGN	ELE v2	/sensors/pressure_transducers/fault_window/min	DESIGN
T1131	4.75 V	DESIGN	ELE v2	/sensors/pressure_transducers/fault_window/max	4.75 V = 74.4 bar g with the 0-70 bar g (070BG) transducer
T1132	TC1 type K case az 90 (MAX31856 U604); TC2 type K case az 270 (U605); TC3 type K nozzle ...	DESIGN	ELE v2	/sensors/thermocouples/channels	(class set by generator: DESIGN)
T1133	448.1 K	LEAD	ELE v2	/sensors/thermocouples/case_abort	LEAD 2026-10-13 correction, MECH pre36 final
T1134	433.1 K	LEAD	ELE v2	/sensors/thermocouples/relight_refusal	LEAD 2026-10-13
T1135	DS18B20 ×2 tank wall + #3 bay (TVC bracket) on 1-Wire (J608/J614); NTC 10k pintle-servo ...	DESIGN	ELE v2	/sensors/temperature_other	(class set by generator: DESIGN)
T1136	AS5048A x3 (gimbal A/B J610/J611, pintle crank J612), SPI1 1 MHz over cable	DESIGN	ELE v2	/sensors/encoders	(class set by generator: DESIGN)
T1137	1 kN S-type or compression load cell, 2 mV/V (TO-QUOTE)	TO-VERIFY	ELE v2	/sensors/load_cell/model	(class set by generator: TO-VERIFY)
T1138	FC side (ADIS16470 per the GNC/FC record): filter delay UNKNOWN (over_thrust)	TO-VERIFY	ELE v2	/sensors/imu	(class set by generator: TO-VERIFY)
T1139	1000 Hz	DESIGN	ELE v2	/loop_rates_and_latency/pc_control_Hz	DESIGN
T1140	4000 Hz	DESIGN	ELE v2	/loop_rates_and_latency/pc_adc_Hz	DESIGN
T1141	1000 Hz	DESIGN	ELE v2	/loop_rates_and_latency/tvc_shaping_Hz	DESIGN
T1142	333 Hz	DATASHEET	ELE v2	/loop_rates_and_latency/servo_pwm_Hz	DATASHEET servo
T1143	500 Hz	DESIGN	ELE v2	/loop_rates_and_latency/fc_command_Hz	DESIGN (>= 500 Hz brief)
T1144	500 Hz	DESIGN	ELE v2	/loop_rates_and_latency/telemetry_fast_Hz	DESIGN
T1145	50 Hz	DESIGN	ELE v2	/loop_rates_and_latency/telemetry_slow_Hz	DESIGN
T1146	1000 Hz	DESIGN	ELE v2	/loop_rates_and_latency/sd_log_Hz	DESIGN (+4 kHz raw Pc block)
T1147	1000 Hz	DESIGN	ELE v2	/loop_rates_and_latency/safety_monitor_Hz	DESIGN
T1148	50 Hz	DESIGN	ELE v2	/loop_rates_and_latency/tlm_relight_Hz	DESIGN protocol.h
T1149	100 Hz	LEAD	ELE v2	/loop_rates_and_latency/fc_thrust_Hz	GNC m2_fc_thrust v1.1 / LEAD

ID	Value	Class	Source	Key	Source tag / formula
T1150	0.05 s	DESIGN	ELE v2	/loop_rates_and_latency/hw_watchdog	DESIGN
T1151	0.02 s	DESIGN	ELE v2	/link_loss/degraded_after	DESIGN
T1152	0.1 s	DESIGN	ELE v2	/link_loss/lost_after	DESIGN
T1153	fc_cmd_to_ecu_tick_max: 0.0012; ecu_to_servo_pwm_max: 0.005; servo_internal_assumed: 0.0...	DERIVED	ELE v2	/loop_rates_and_latency/latency_s	DERIVED budget (CAN 1 Mbit/s 8-byte frame <= 0.135 ms; PWM frame 3 ms; transducer tau 0.9 ms; sinc3+decimation ~1.4 ms)
T1154	80000 Pa	LEAD	ELE v2	/light_rule_v2p9/kIgnRise_Pa	LEAD/COMB
T1155	10	LEAD	ELE v2	/light_rule_v2p9/kIgnConfirmRelN	LEAD/COMB
T1156	20	LEAD	ELE v2	/light_rule_v2p9/kIgnPreN	LEAD/COMB
T1157	1.8e5 Pa abs until B-PC-1 shows residual <= 0.3 bar, then 2.0e5 (kBpc1Shown)	LEAD	ELE v2	/light_rule_v2p9/kIgnFloorAbs_Pa	LEAD/COMB
T1158	residual + 0.1 bar (0.7 bar now, 0.4 bar after B-PC-1)	LEAD	ELE v2	/light_rule_v2p9/kIgnBaseCheck_Pa	LEAD
T1159	300	SIM	ELE v2	/light_rule_v2p9/sil_a1_relight_mc/runs_lit	(class set by generator: SIM)
T1160	0	SIM	ELE v2	/light_rule_v2p9/sil_a1_relight_mc/light_rule_aborts	(class set by generator: SIM)
T1161	0	SIM	ELE v2	/light_rule_v2p9/sil_a1_relight_mc/unlit_false_lights	(class set by generator: SIM)
T1162	17 ms	SIM	ELE v2	/light_rule_v2p9/sil_a1_relight_mc/E2_minus_light_ms/p50	(class set by generator: SIM)
T1163	18 ms	SIM	ELE v2	/light_rule_v2p9/sil_a1_relight_mc/E2_minus_light_ms/max	(class set by generator: SIM)
T1164	SIL (ASSUMED/SIM plant); target E2 <= ~15 ms after light NOT met: 16-18 ms = plant rise ...	SIM	ELE v2	/light_rule_v2p9/sil_a1_relight_mc/tag	(class set by generator: SIM)
T1165	(Pc - P0)/(Pc_map_lo - P0) < 0.40 for 20 ms in RUN; Pc_map_lo = min(40 ms-lagged map, in...	LEAD	ELE v2	/start_monitors_v2p9/A3	LEAD/COMB + DESIGN
T1166	first RUN tick with the ramp done and Pc >= P0 + 0.9 (setpoint - P0), or E2 + 0.6 s	DESIGN	ELE v2	/start_monitors_v2p9/E3	DESIGN
T1167	17	DESIGN	ELE v2	/abort_matrix/causes/0/code	(class set by generator: DESIGN)
T1168	PC_OVERPRESSURE	DESIGN	ELE v2	/abort_matrix/causes/0/cause	(class set by generator: DESIGN)
T1169	18	DESIGN	ELE v2	/abort_matrix/causes/1/code	(class set by generator: DESIGN)
T1170	HW_OVERPRESSURE	DESIGN	ELE v2	/abort_matrix/causes/1/cause	(class set by generator: DESIGN)
T1171	hardware comparator latch (31.47 bar g on the 070BG): ARM bus already off in hardware	DESIGN	ELE v2	/abort_matrix/causes/1/trigger	(class set by generator: DESIGN)
T1172	0.002	DESIGN	ELE v2	/abort_matrix/causes/1/thresholds/kHwOvpTime_s	DESIGN
T1173	19	DESIGN	ELE v2	/abort_matrix/causes/2/code	(class set by generator: DESIGN)
T1174	TVC_SUPPLY	DESIGN	ELE v2	/abort_matrix/causes/2/cause	(class set by generator: DESIGN)
T1175	pad only: predicted both-stall TVC plug voltage below the minimum	DESIGN	ELE v2	/abort_matrix/causes/2/trigger	(class set by generator: DESIGN)
T1176	6.6	LEAD	ELE v2	/abort_matrix/causes/2/thresholds/kTvcSupplyMin_V	LEAD 2026-10-12, V_val
T1177	0.1	DESIGN	ELE v2	/abort_matrix/causes/2/thresholds/kTvcSupplyTime_s	DESIGN
T1178	20	DESIGN	ELE v2	/abort_matrix/causes/3/code	(class set by generator: DESIGN)
T1179	OVERCURRENT	DESIGN	ELE v2	/abort_matrix/causes/3/cause	(class set by generator: DESIGN)
T1180	firmware e-fuse before the first start	DESIGN	ELE v2	/abort_matrix/causes/3/trigger	(class set by generator: DESIGN)
T1181	22	DESIGN	ELE v2	/abort_matrix/causes/3/thresholds/kEfuse_A	DESIGN
T1182	1	DESIGN	ELE v2	/abort_matrix/causes/3/thresholds/kEfuseTime_s	DESIGN
T1183	21	DESIGN	ELE v2	/abort_matrix/causes/4/code	(class set by generator: DESIGN)
T1184	MAINT_JAM	DESIGN	ELE v2	/abort_matrix/causes/4/cause	(class set by generator: DESIGN)
T1185	ARM refused: a pintle jam recorded in NV memory and not cleared (bearing A rule)	DESIGN	ELE v2	/abort_matrix/causes/4/trigger	(class set by generator: DESIGN)
T1186	22	DESIGN	ELE v2	/abort_matrix/causes/5/code	(class set by generator: DESIGN)
T1187	TORCH_FAULT	DESIGN	ELE v2	/abort_matrix/causes/5/cause	(class set by generator: DESIGN)
T1188	23	DESIGN	ELE v2	/abort_matrix/causes/6/code	(class set by generator: DESIGN)
T1189	RELIGHT_LOCKOUT	DESIGN	ELE v2	/abort_matrix/causes/6/cause	(class set by generator: DESIGN)
T1190	ARM/IGNITE/RELIGHT refused after an abort following a lit burn or on a relight, until CL...	DESIGN	ELE v2	/abort_matrix/causes/6/trigger	(class set by generator: DESIGN)
T1191	24	DESIGN	ELE v2	/abort_matrix/causes/7/code	(class set by generator: DESIGN)
T1192	OVER_THRUST	DESIGN	ELE v2	/abort_matrix/causes/7/cause	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T1193	flight RUN: FC_THRUST above the threshold on consecutive fresh valid frames (with or wit...	DESIGN	ELE v2	/abort_matrix/causes/7/trigger	(class set by generator: DESIGN)
T1194	650	LEAD	ELE v2	/abort_matrix/causes/7/thresholds/kOverThrust_N	LEAD 2026-10-13 ADOPTED
T1195	3	LEAD	ELE v2	/abort_matrix/causes/7/thresholds/kOverThrustFrames	LEAD 2026-10-13 ADOPTED
T1196	100	LEAD	ELE v2	/abort_matrix/causes/7/thresholds/kFcThrustRate_Hz	GNC m2_fc_thrust v1.1 / LEAD
T1197	0.05	DESIGN	ELE v2	/abort_matrix/causes/7/thresholds/kFcThrustStale_s	GNC decision 2026-10-13
T1198	25	DESIGN	ELE v2	/abort_matrix/causes/8/code	(class set by generator: DESIGN)
T1199	MODE_MISMATCH	DESIGN	ELE v2	/abort_matrix/causes/8/cause	(class set by generator: DESIGN)
T1200	ARM refused: strap invalid, or strap and load-cell reading disagree	DESIGN	ELE v2	/abort_matrix/causes/8/trigger	(class set by generator: DESIGN)
T1201	26	DESIGN	ELE v2	/abort_matrix/causes/9/code	(class set by generator: DESIGN)
T1202	BAY_HOT	DESIGN	ELE v2	/abort_matrix/causes/9/cause	(class set by generator: DESIGN)
T1203	ARM refused: bay temperature above the limit or no valid reading	DESIGN	ELE v2	/abort_matrix/causes/9/trigger	(class set by generator: DESIGN)
T1204	318.1	LEAD	ELE v2	/abort_matrix/causes/9/thresholds/kBayArmMax_K	LEAD approx 45 C, TO-DEFINE
T1205	27	DESIGN	ELE v2	/abort_matrix/causes/10/code	(class set by generator: DESIGN)
T1206	TANK_HOT	DESIGN	ELE v2	/abort_matrix/causes/10/cause	(class set by generator: DESIGN)
T1207	ARM refused: tank temperature (max of wall DS18B20 and Tsat(P_tank)) above the limit or ...	DESIGN	ELE v2	/abort_matrix/causes/10/trigger	(class set by generator: DESIGN)
T1208	297.1	LEAD	ELE v2	/abort_matrix/causes/10/thresholds/kTankIgnMax_K	LEAD 2026-10-13
T1209	28	DESIGN	ELE v2	/abort_matrix/causes/11/code	(class set by generator: DESIGN)
T1210	BLEED_FAULT	DESIGN	ELE v2	/abort_matrix/causes/11/cause	(class set by generator: DESIGN)
T1211	1	DESIGN	ELE v2	/abort_matrix/causes/12/code	(class set by generator: DESIGN)
T1212	CMD	DESIGN	ELE v2	/abort_matrix/causes/12/cause	(class set by generator: DESIGN)
T1213	FC ABORT frame 0x010 or CMD_STATE ABORT	DESIGN	ELE v2	/abort_matrix/causes/12/trigger	(class set by generator: DESIGN)
T1214	2	DESIGN	ELE v2	/abort_matrix/causes/13/code	(class set by generator: DESIGN)
T1215	HARD_START	DESIGN	ELE v2	/abort_matrix/causes/13/cause	(class set by generator: DESIGN)
T1216	Pc above the over-pressure limit during IGNITION	DESIGN	ELE v2	/abort_matrix/causes/13/trigger	(class set by generator: DESIGN)
T1217	3000000	LEAD	ELE v2	/abort_matrix/causes/13/thresholds/kPcOverPressureAbs_Pa	LEAD via COMB
T1218	0.003	DESIGN	ELE v2	/abort_matrix/causes/13/thresholds/kPcOverPressureTime_s	DESIGN
T1219	3	DESIGN	ELE v2	/abort_matrix/causes/14/code	(class set by generator: DESIGN)
T1220	NO_IGNITION	DESIGN	ELE v2	/abort_matrix/causes/14/cause	(class set by generator: DESIGN)
T1221	4	DESIGN	ELE v2	/abort_matrix/causes/15/code	(class set by generator: DESIGN)
T1222	FLAMEOUT	DESIGN	ELE v2	/abort_matrix/causes/15/cause	(class set by generator: DESIGN)
T1223	RUN: gauge Pc below the floor or below a fraction of the expected Pc	DESIGN	ELE v2	/abort_matrix/causes/15/trigger	(class set by generator: DESIGN)
T1224	150000	DESIGN	ELE v2	/abort_matrix/causes/15/thresholds/kFlameoutPc_Pa	DESIGN v2.2-v2.8.3, legacy rule only
T1225	0.35	DESIGN	ELE v2	/abort_matrix/causes/15/thresholds/kFlameoutFrac	DESIGN v2.2-v2.8.3, legacy rule only
T1226	0.02	LEAD	ELE v2	/abort_matrix/causes/15/thresholds/kFlameoutTime_s	LEAD/COMB 2026-10-13
T1227	5	DESIGN	ELE v2	/abort_matrix/causes/16/code	(class set by generator: DESIGN)
T1228	PC_BAND	DESIGN	ELE v2	/abort_matrix/causes/16/cause	(class set by generator: DESIGN)
T1229	RUN after settling: |Pc - Pc_ref| above both band limits	DESIGN	ELE v2	/abort_matrix/causes/16/trigger	(class set by generator: DESIGN)
T1230	0.3	DESIGN	ELE v2	/abort_matrix/causes/16/thresholds/kPcBandFrac	DESIGN
T1231	200000	DESIGN	ELE v2	/abort_matrix/causes/16/thresholds/kPcBandMin_Pa	DESIGN
T1232	0.15	DESIGN	ELE v2	/abort_matrix/causes/16/thresholds/kPcBandTime_s	DESIGN
T1233	0.3	DESIGN	ELE v2	/abort_matrix/causes/16/thresholds/kRunSettle_s	DESIGN
T1234	6	DESIGN	ELE v2	/abort_matrix/causes/17/code	(class set by generator: DESIGN)
T1235	OVER_TEMP	DESIGN	ELE v2	/abort_matrix/causes/17/cause	(class set by generator: DESIGN)
T1236	7	DESIGN	ELE v2	/abort_matrix/causes/18/code	(class set by generator: DESIGN)
T1237	TANK_OVERPRESSURE	DESIGN	ELE v2	/abort_matrix/causes/18/cause	(class set by generator: DESIGN)
T1238	P_tank above limit in ARMED/PREPRESS (never aborts a burn)	DESIGN	ELE v2	/abort_matrix/causes/18/trigger	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T1239	6500000	DESIGN	ELE v2	/abort_matrix/causes/18/thresholds/kTankOverPressure_Pa	DESIGN
T1240	0.2	DESIGN	ELE v2	/abort_matrix/causes/18/thresholds/kTankOverPressureTime_s	DESIGN
T1241	8	DESIGN	ELE v2	/abort_matrix/causes/19/code	(class set by generator: DESIGN)
T1242	GIMBAL_FAULT	DESIGN	ELE v2	/abort_matrix/causes/19/cause	(class set by generator: DESIGN)
T1243	command-model vs encoder mismatch while burning (allowance grows with the model rate)	DESIGN	ELE v2	/abort_matrix/causes/19/trigger	(class set by generator: DESIGN)
T1244	0.02618	DESIGN	ELE v2	/abort_matrix/causes/19/thresholds/kGimbalFaultErr_rad	DESIGN
T1245	0.05	DESIGN	ELE v2	/abort_matrix/causes/19/thresholds/kGimbalFaultTime_s	DESIGN
T1246	0.04	DESIGN	ELE v2	/abort_matrix/causes/19/thresholds/kGimbalFaultRateWin_s	DESIGN v2.2
T1247	9	DESIGN	ELE v2	/abort_matrix/causes/20/code	(class set by generator: DESIGN)
T1248	PINTLE_FAULT	DESIGN	ELE v2	/abort_matrix/causes/20/cause	(class set by generator: DESIGN)
T1249	pintle rail lost (LTC4365 trip / regulator failure), jam, encoder invalid at ARM	DESIGN	ELE v2	/abort_matrix/causes/20/trigger	(class set by generator: DESIGN)
T1250	5.5	DESIGN	ELE v2	/abort_matrix/causes/20/thresholds/kPrailLost_V	DESIGN
T1251	0.002	DESIGN	ELE v2	/abort_matrix/causes/20/thresholds/kPrailLostTime_s	DESIGN
T1252	0.1	DESIGN	ELE v2	/abort_matrix/causes/20/thresholds/kPintleStallTime_s	DESIGN, LEAD approved
T1253	0.05	DESIGN	ELE v2	/abort_matrix/causes/20/thresholds/kPintleStallTimeI_s	DESIGN, LEAD approved
T1254	10	DESIGN	ELE v2	/abort_matrix/causes/21/code	(class set by generator: DESIGN)
T1255	LINK_LOST	DESIGN	ELE v2	/abort_matrix/causes/21/cause	(class set by generator: DESIGN)
T1256	no valid CMD_FAST: SAFE/ARMED/PREPRESS → SAFE; RUN → SHUTDOWN by default	DESIGN	ELE v2	/abort_matrix/causes/21/trigger	(class set by generator: DESIGN)
T1257	0.02	DESIGN	ELE v2	/abort_matrix/causes/21/thresholds/kLinkDegraded_s	DESIGN
T1258	0.1	DESIGN	ELE v2	/abort_matrix/causes/21/thresholds/kLinkLost_s	DESIGN
T1259	11	DESIGN	ELE v2	/abort_matrix/causes/22/code	(class set by generator: DESIGN)
T1260	PREPRESS_FAIL	DESIGN	ELE v2	/abort_matrix/causes/22/cause	(class set by generator: DESIGN)
T1261	feed not primed (P_feed > fraction of P_tank) in time, bleed lift not confirmed, MV not ...	DESIGN	ELE v2	/abort_matrix/causes/22/trigger	(class set by generator: DESIGN)
T1262	0.6	DESIGN	ELE v2	/abort_matrix/causes/22/thresholds/kPrepressTimeout_s	DESIGN
T1263	0.8	DESIGN	ELE v2	/abort_matrix/causes/22/thresholds/kFeedPrimedFrac	DESIGN
T1264	12	DESIGN	ELE v2	/abort_matrix/causes/23/code	(class set by generator: DESIGN)
T1265	PC_SENSOR	DESIGN	ELE v2	/abort_matrix/causes/23/cause	(class set by generator: DESIGN)
T1266	Pc transducer output outside the window: IGNITION and stand abort; flight RUN continues ...	DESIGN	ELE v2	/abort_matrix/causes/23/trigger	(class set by generator: DESIGN)
T1267	0.25	DESIGN	ELE v2	/abort_matrix/causes/23/thresholds/kPcSensorMin_V	DESIGN
T1268	4.75	DERIVED	ELE v2	/abort_matrix/causes/23/thresholds/kPcSensorMax_V	4.75 V = 74.4 bar g with the 0-70 bar g (070BG) transducer
T1269	0.005	DESIGN	ELE v2	/abort_matrix/causes/23/thresholds/kPcSensorLossStand_s	DESIGN
T1270	13	DESIGN	ELE v2	/abort_matrix/causes/24/code	(class set by generator: DESIGN)
T1271	BATT_LOW	DESIGN	ELE v2	/abort_matrix/causes/24/cause	(class set by generator: DESIGN)
T1272	VBAT low in pad states before the first start (flag in flight); ARM refused below the re...	DESIGN	ELE v2	/abort_matrix/causes/24/trigger	(class set by generator: DESIGN)
T1273	6.6	DESIGN	ELE v2	/abort_matrix/causes/24/thresholds/kBattLow_V	DESIGN
T1274	0.2	DESIGN	ELE v2	/abort_matrix/causes/24/thresholds/kBattLowTime_s	DESIGN
T1275	8.1	LEAD	ELE v2	/abort_matrix/causes/24/thresholds/kBattArmMin_V	LEAD 2026-10-12 ARM rule, rest voltage
T1276	14	DESIGN	ELE v2	/abort_matrix/causes/25/code	(class set by generator: DESIGN)
T1277	MV_FAULT	DESIGN	ELE v2	/abort_matrix/causes/25/cause	(class set by generator: DESIGN)
T1278	MV CLOSED switch missing at ARM; direct relight OPEN switch missing by the deadline: sta...	DESIGN	ELE v2	/abort_matrix/causes/25/trigger	(class set by generator: DESIGN)
T1279	0.95	DESIGN	ELE v2	/abort_matrix/causes/25/thresholds/kMvOpenBy_s	DESIGN

ID	Value	Class	Source	Key	Source tag / formula
T1280	0.5	DESIGN	ELE v2	/abort_matrix/causes/25/thresholds/kMvCloseConfirm_s	DESIGN
T1281	15	DESIGN	ELE v2	/abort_matrix/causes/26/code	(class set by generator: DESIGN)
T1282	ARM_LOST	DESIGN	ELE v2	/abort_matrix/causes/26/cause	(class set by generator: DESIGN)
T1283	arming plug removed or ARM bus not on 50 ms after the request	DESIGN	ELE v2	/abort_matrix/causes/26/trigger	(class set by generator: DESIGN)
T1284	16	DESIGN	ELE v2	/abort_matrix/causes/27/code	(class set by generator: DESIGN)
T1285	WATCHDOG	DESIGN	ELE v2	/abort_matrix/causes/27/cause	(class set by generator: DESIGN)
T1286	LiPo 2S 2.2 Ah (16.3 Wh), usable 13 Wh	DESIGN	ELE v2	/power_budget/battery	(class set by generator: DESIGN)
T1287	typ: 1.25; peak: 2.6	DERIVED	ELE v2	/power_budget/logic_W	DERIVED
T1288	1.73 Wh	DERIVED	ELE v2	/power_budget/flight_day_Wh	DERIVED (30 min pad hold + 60 s flight + 3 starts)
T1289	2.79 Wh	DERIVED	ELE v2	/power_budget/static_fire_session_Wh	DERIVED
T1290	35.7 A	DERIVED	ELE v2	/power_budget/peak_current_A	DERIVED worst case: 3 servos stalled
T1291	6.11 V	DERIVED	ELE v2	/power_budget/min_voltage_at_ECU_V	DERIVED at 3.5 V/cell OCV, ASSUMED 10 mOhm pack
T1292	F101: 25 A ATM main; F102: 2 A 1206 logic; F301: 15 A ATM ARM bus; F501: 3 A 1206 12 V b...	DESIGN	ELE v2	/power_budget/fusing	(class set by generator: DESIGN)
T1293	TO-CHECK: MV opener stall 9.2 A at 7.4 V (DATASHEET) / 10.4 A at 8.4 V (ASSUMED) + pintl...	MISSING	ELE v2	/power_budget/arm_bus_peak_vs_F301	OPEN
T1294	2 NC valves in series (V1 at the engine-side tee of the T4 crown + FD-01 radial taps via...	DESIGN	ELE v2	/head_end_bleed_provision/architecture	DESIGN / MECH v3.4.1 relay
T1295	PREPRESS steps S0-S4 (valve pattern vs P_mid); any failure → BLEED_FAULT	DESIGN	ELE v2	/head_end_bleed_provision/pad_bit	(class set by generator: DESIGN)
T1296	PURGE state: MV open, pintle seated, V1+V2 open, t_purge 2.5 s; pre-scheduled (Cmd PURGE...	LEAD	ELE v2	/head_end_bleed_provision/purge	LEAD + DESIGN
T1297	invisible on P_feed (0.016 bar) and Pc (SIL -0.02..-0.05 bar); seen only by the P_mid le...	SIM	ELE v2	/head_end_bleed_provision/in_burn_stuck_open	(class set by generator: SIM)
T1298	valve: Burkert 6027 HP (stainless, G1/4, PEEK/FKM seat, NC, orifice 1.0 mm: 0-250 bar DC...	TO-VERIFY	ELE v2	/head_end_bleed_provision/coil	(class set by generator: TO-VERIFY)
T1299	MV latch drops → MV-P spring + pressure bias close it (dashpot ~30 ms ESTIMATE); the pi...	DESIGN	ELE v2	/fail_safe_states/ECU power loss	(class set by generator: DESIGN)
T1300	hardware watchdog (50 ms) resets MCU and drops ARM bus -> same as above; reboot lands in...	DESIGN	ELE v2	/fail_safe_states/MCU hang	(class set by generator: DESIGN)
T1301	see link_loss	DESIGN	ELE v2	/fail_safe_states/link loss	(class set by generator: DESIGN)
T1302	RUN: open-loop feed-forward map on P_feed + fault flag; IGNITION: abort	DESIGN	ELE v2	/fail_safe_states/Pc sensor fault	(class set by generator: DESIGN)
T1303	flag; RUN: continue on servo commands (default) unless mismatch abort criterion met	DESIGN	ELE v2	/fail_safe_states/gimbal encoder fault	(class set by generator: DESIGN)
T1304	ARM bus dead: igniter, pintle and main valve power removed regardless of firmware	DESIGN	ELE v2	/fail_safe_states/arming plug removed	(class set by generator: DESIGN)
T1305	LTC4365 OV disconnect 8.06-8.34 V: servo unpowered, spring closes; PRAIL_OK low drops th...	DESIGN	ELE v2	/fail_safe_states/pintle rail fail-high	(class set by generator: DESIGN)
T1306	hardware comparator latch 31.47 bar g removes the ARM bus independent of firmware; firmw...	DESIGN	ELE v2	/fail_safe_states/Pc over-pressure	(class set by generator: DESIGN)
T1307	OI-IGN-DET	DESIGN	ELE v2	/open_issues/0/id	(class set by generator: DESIGN)
T1308	HIGH (cross-discipline)	DESIGN	ELE v2	/open_issues/0/severity	(class set by generator: DESIGN)
T1309	light-detection margin at the v2.8 ignition lift: S5 dispersed pad-start MC 31/300 NO_IG...	TO-VERIFY	ELE v2	/open_issues/0/issue	(class set by generator: TO-VERIFY)
T1310	Combustion (threshold and lift) + Electronics (detection logic)	DESIGN	ELE v2	/open_issues/0/owner	(class set by generator: DESIGN)
T1311	OI-RELIGHT-A1	DESIGN	ELE v2	/open_issues/1/id	(class set by generator: DESIGN)
T1312	HIGH	DESIGN	ELE v2	/open_issues/1/severity	(class set by generator: DESIGN)
T1313	A1 relight not frozen (RG-01): re-routed segment volume and the GNC re-fly pending; drai...	TO-VERIFY	ELE v2	/open_issues/1/issue	(class set by generator: TO-VERIFY)
T1314	Mechanical / Electronics / GNC	DESIGN	ELE v2	/open_issues/1/owner	(class set by generator: DESIGN)
T1315	OI-GIMBAL-JAM	DESIGN	ELE v2	/open_issues/2/id	(class set by generator: DESIGN)
T1316	MEDIUM	DESIGN	ELE v2	/open_issues/2/severity	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T1317	S6 gimbal_jam_moving_cmd FAIL (small-amplitude jam undetected)	TO-VERIFY	ELE v2	/open_issues/2/issue	(class set by generator: TO-VERIFY)
T1318	Electronics + GNC	DESIGN	ELE v2	/open_issues/2/owner	(class set by generator: DESIGN)
T1319	OI-TB-MEOP	DESIGN	ELE v2	/open_issues/3/id	(class set by generator: DESIGN)
T1320	ACCEPTED	DESIGN	ELE v2	/open_issues/3/severity	(class set by generator: DESIGN)
T1321	S6 throat block Pc 53.4 bar abs > MEOP for 72 ms	TO-VERIFY	ELE v2	/open_issues/3/issue	(class set by generator: TO-VERIFY)
T1322	Mechanical FEA	DESIGN	ELE v2	/open_issues/3/owner	(class set by generator: DESIGN)
T1323	OI-HAL	DESIGN	ELE v2	/open_issues/4/id	(class set by generator: DESIGN)
T1324	HIGH	DESIGN	ELE v2	/open_issues/4/severity	(class set by generator: DESIGN)
T1325	hardware HAL not written; TLM_SLOW 0x110 undefined; 53 ARM double-promotion warnings	TO-VERIFY	ELE v2	/open_issues/4/issue	(class set by generator: TO-VERIFY)
T1326	Electronics	DESIGN	ELE v2	/open_issues/4/owner	(class set by generator: DESIGN)
T1327	OI-FC	DESIGN	ELE v2	/open_issues/5/id	(class set by generator: DESIGN)
T1328	HIGH	DESIGN	ELE v2	/open_issues/5/severity	(class set by generator: DESIGN)
T1329	GNC tracker / FSW not yet in the flight computer; IMU filter delay unknown; RG-16 mass b...	TO-VERIFY	ELE v2	/open_issues/5/issue	(class set by generator: TO-VERIFY)
T1330	FC owner + GNC	DESIGN	ELE v2	/open_issues/5/owner	(class set by generator: DESIGN)
T1331	OI-TIMEBASE	DESIGN	ELE v2	/open_issues/6/id	(class set by generator: DESIGN)
T1332	MEDIUM	DESIGN	ELE v2	/open_issues/6/severity	(class set by generator: DESIGN)
T1333	no common time base yet for DAQ, CAN analyser and ECU	TO-VERIFY	ELE v2	/open_issues/6/issue	(class set by generator: TO-VERIFY)
T1334	Electronics	DESIGN	ELE v2	/open_issues/6/owner	(class set by generator: DESIGN)
T1335	OI-PFEED-SPOF	DESIGN	ELE v2	/open_issues/7/id	(class set by generator: DESIGN)
T1336	MEDIUM	DESIGN	ELE v2	/open_issues/7/severity	(class set by generator: DESIGN)
T1337	P_feed is a single point for the relight path choice (invalid → UNKNOWN → prep path) a...	TO-VERIFY	ELE v2	/open_issues/7/issue	(class set by generator: TO-VERIFY)
T1338	Electronics	DESIGN	ELE v2	/open_issues/7/owner	(class set by generator: DESIGN)
T1339	OI-LATE-FLIP	DESIGN	ELE v2	/open_issues/8/id	(class set by generator: DESIGN)
T1340	MEDIUM	DESIGN	ELE v2	/open_issues/8/severity	(class set by generator: DESIGN)
T1341	a coast encoder loss after the last TLM_RELIGHT frame flips the path late (S10 evidence ...	TO-VERIFY	ELE v2	/open_issues/8/issue	(class set by generator: TO-VERIFY)
T1342	GNC planning	DESIGN	ELE v2	/open_issues/8/owner	(class set by generator: DESIGN)
T1343	OI-B1-WIND	DESIGN	ELE v2	/open_issues/9/id	(class set by generator: DESIGN)
T1344	MEDIUM	DESIGN	ELE v2	/open_issues/9/severity	(class set by generator: DESIGN)
T1345	B1 launch commit mean wind < 6 m/s until B-PNT-1 shows the drive lag inside the MC tau	TO-VERIFY	ELE v2	/open_issues/9/issue	(class set by generator: TO-VERIFY)
T1346	Electronics bench	DESIGN	ELE v2	/open_issues/9/owner	(class set by generator: DESIGN)
T1347	OI-RAIL	DESIGN	ELE v2	/open_issues/10/id	(class set by generator: DESIGN)
T1348	MEDIUM	DESIGN	ELE v2	/open_issues/10/severity	(class set by generator: DESIGN)
T1349	LTC4365 UV ASSUMED; Q803 enhancement; RPM0026A footprint placeholder; ARM-bus peak vs F301	TO-VERIFY	ELE v2	/open_issues/10/issue	(class set by generator: TO-VERIFY)
T1350	Electronics	DESIGN	ELE v2	/open_issues/10/owner	(class set by generator: DESIGN)
T1351	OI-MV-CAM	DESIGN	ELE v2	/open_issues/11/id	(class set by generator: DESIGN)
T1352	LOW	DESIGN	ELE v2	/open_issues/11/severity	(class set by generator: DESIGN)
T1353	MV-P opener cam-limited 61 ms opening (bench B-MV-1)	TO-VERIFY	ELE v2	/open_issues/11/issue	(class set by generator: TO-VERIFY)
T1354	Mechanical + Electronics	DESIGN	ELE v2	/open_issues/11/owner	(class set by generator: DESIGN)
T1355	OI-TRIM	DESIGN	ELE v2	/open_issues/12/id	(class set by generator: DESIGN)
T1356	LOW	DESIGN	ELE v2	/open_issues/12/severity	(class set by generator: DESIGN)
T1357	map-trim plateau (Pc loop trim authority +/-40 %) not yet characterised on hardware	TO-VERIFY	ELE v2	/open_issues/12/issue	(class set by generator: TO-VERIFY)
T1358	Electronics + Combustion	DESIGN	ELE v2	/open_issues/12/owner	(class set by generator: DESIGN)
T1359	OI-DOC	DESIGN	ELE v2	/open_issues/13/id	(class set by generator: DESIGN)
T1360	LOW	DESIGN	ELE v2	/open_issues/13/severity	(class set by generator: DESIGN)
T1361	pc_transducer_v2 labels P_torch J610 (is J901); protocol.h FC_THRUST valid-window commen...	TO-VERIFY	ELE v2	/open_issues/13/issue	(class set by generator: TO-VERIFY)
T1362	Electronics	DESIGN	ELE v2	/open_issues/13/owner	(class set by generator: DESIGN)
T1363	OI-A5-LATE	DESIGN	ELE v2	/open_issues/14/id	(class set by generator: DESIGN)
T1364	HIGH	DESIGN	ELE v2	/open_issues/14/severity	(class set by generator: DESIGN)
T1365	v2.9 light rule (10 ms) + unchanged A5 (8.0 bar abs, 5 ms): S6 late lights 0.20/0.25 s i...	TO-VERIFY	ELE v2	/open_issues/14/issue	(class set by generator: TO-VERIFY)
T1366	LEAD / Combustion	DESIGN	ELE v2	/open_issues/14/owner	(class set by generator: DESIGN)

ID	Value	Class	Source	Key	Source tag / formula
T1367	OI-TEB-FALSE-LIGHT	DESIGN	ELE v2	/open_issues/15/id	(class set by generator: DESIGN)
T1368	MEDIUM	DESIGN	ELE v2	/open_issues/15/severity	(class set by generator: DESIGN)
T1369	S9 +1.034 bar post-zero bias: unlit filtered 1.87 bar abs > floor 1.8 → false E2 → RUN...	TO-VERIFY	ELE v2	/open_issues/15/issue	(class set by generator: TO-VERIFY)
T1370	LEAD / Electronics (B-PC-1)	DESIGN	ELE v2	/open_issues/15/owner	(class set by generator: DESIGN)
T1371	OI-E2-LATENCY	DESIGN	ELE v2	/open_issues/16/id	(class set by generator: DESIGN)
T1372	LOW	DESIGN	ELE v2	/open_issues/16/severity	(class set by generator: DESIGN)
T1373	E2 - light 16-18 ms in SIL (target ~15 ms): plant rise + 50 Hz filter + 10 ms persistenc...	TO-VERIFY	ELE v2	/open_issues/16/issue	(class set by generator: TO-VERIFY)
T1374	Combustion / Electronics	DESIGN	ELE v2	/open_issues/16/owner	(class set by generator: DESIGN)
T1375	OI-A3-MARGIN	DESIGN	ELE v2	/open_issues/17/id	(class set by generator: DESIGN)
T1376	MEDIUM	DESIGN	ELE v2	/open_issues/17/severity	(class set by generator: DESIGN)
T1377	A3 ratio minimum 0.454 at the A1 relight MC (threshold 0.40) during the start ramp; afte...	TO-VERIFY	ELE v2	/open_issues/17/issue	(class set by generator: TO-VERIFY)
T1378	Combustion	DESIGN	ELE v2	/open_issues/17/owner	(class set by generator: DESIGN)
T1379	OI-BLEED-COIL	DESIGN	ELE v2	/open_issues/18/id	(class set by generator: DESIGN)
T1380	HIGH	DESIGN	ELE v2	/open_issues/18/severity	(class set by generator: DESIGN)
T1381	Burkert 6027 HP: 24 V DC listed, ECU valve rail +12V_ARM (U3V70F12); 2 × 16 W → +2.7 A ...	TO-VERIFY	ELE v2	/open_issues/18/issue	(class set by generator: TO-VERIFY)
T1382	Electronics / Mechanical	DESIGN	ELE v2	/open_issues/18/owner	(class set by generator: DESIGN)
T1383	OI-BLEED-VOL	DESIGN	ELE v2	/open_issues/19/id	(class set by generator: DESIGN)
T1384	MEDIUM	DESIGN	ELE v2	/open_issues/19/severity	(class set by generator: DESIGN)
T1385	unreached vapour with taps (2 deg 4.84 cm3, 10 deg 15.91 cm3) from an UNPUBLISHED Mechan...	TO-VERIFY	ELE v2	/open_issues/19/issue	(class set by generator: TO-VERIFY)
T1386	Mechanical	DESIGN	ELE v2	/open_issues/19/owner	(class set by generator: DESIGN)
T1387	OI-GNC-V3	DESIGN	ELE v2	/open_issues/20/id	(class set by generator: DESIGN)
T1388	MEDIUM	DESIGN	ELE v2	/open_issues/20/severity	(class set by generator: DESIGN)
T1389	gnc.json v3 requirements.ignition no longer carries option_c_unprimed_s / stale_or_missi...	TO-VERIFY	ELE v2	/open_issues/20/issue	(class set by generator: TO-VERIFY)
T1390	GNC	DESIGN	ELE v2	/open_issues/20/owner	(class set by generator: DESIGN)
T1391	RV-1c (GNC reference vehicle on mechanical.json v2, ledger v3d, parametric; real airfram...	ASSUMED	GNC v3	/reference_vehicle/name	(class set by generator: ASSUMED)
T1392	2.03 m	ASSUMED	GNC v3	/reference_vehicle/length_overall_m	ASSUMED REQUIREMENTS ref. vehicle ~2.0 m + 0.03 m nozzle protrusion
T1393	0.146 m	DERIVED	GNC v3	/reference_vehicle/body_diameter_m	DERIVED engine-bay ID 0.140 m (mechanical.json v1 required_engine_bay_ID_m, founder decision 2026-10-12, final) + 2 × 3 mm PETG wall ASSUMED
T1394	13.67 kg	DERIVED	GNC v3	/reference_vehicle/dry_mass_kg	DERIVED mass ledger v3d (mechanical.json v2 engine + fixed TVC module; ASSUMED airframe, legs, avionics, recovery)
T1395	0.404 kg	DESIGN	GNC v3	/reference_vehicle/fuel_grain_mass_kg	combustion.json v2.3 grain
T1396	2 kg	REQ	GNC v3	/reference_vehicle/n2o_load_kg	REQ lead decision 2026-10-12 (tank 2.00 kg / 3.17 L; OI-1 option (a))
T1397	3.167 L	REQ	GNC v3	/reference_vehicle/tank_volume_L	REQ lead decision 2026-10-12
T1398	16.07 kg	SIM	GNC v3	/reference_vehicle/liftoff_mass_kg	SIM nominal A1 flight (ledger v3d + 2.00 kg, roll RCS provision only)
T1399	14.77 kg	SIM	GNC v3	/reference_vehicle/landing_mass_kg	SIM nominal A1 flight
T1400	3.156	DERIVED	GNC v3	/lift_capability/rv1c_liftoff_T_W	DERIVED 500 N / (m_liftoff g0) at the MC load
T1401	n: 4; foot_radius_m: 0.4; stroke_m: 0.1; k_per_leg_N_m: 2500.0; c_per_leg_N_s_m: 120.0; ...	ASSUMED	GNC v3	/reference_vehicle/landing_legs	ASSUMED (Lauty to design; MC disperses k, c x U(0.8, 1.2), mu U(0.3, 0.7))
T1402	4 aft fins, span 0.12 m, root/tip chord 0.20/0.10 m, 3 mm GF/PETG, 0.30 kg	DERIVED	GNC v3	/reference_vehicle/fins	DERIVED descent/coast trade (REPORT B)
T1403	Tail-first aero (CNa, CP, base/plume interaction) is UNKNOWN to better than ~x2: only CF...	TO-VERIFY	GNC v3	/reference_vehicle/aero/unknowns	(class set by generator: TO-VERIFY)
T1404	300 m	REQ	GNC v3	/flight_profile/mission_plan_v3/v/A1/apogee_target_m	(class set by generator: REQ)
T1405	A1 apogee target 285 m ADOPTED under line 168 (4) (0 exhausted runs at the 73 g purge; c...	LEAD	CL 176 lines	L175(1)	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)

ID	Value	Class	Source	Key	Source tag / formula
T1406	285 m	LEAD	CL 176 lines	`L175(1) [regex A1 apogee target (\d+) m]`	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T1407	A1 tail draws: the re-fly flies the purge at the ECU cap (<= 73.0 g) plus the +0.60 kg v...	LEAD	CL 176 lines	`L168(4)`	lead (decision) 2026-10-13 (line sha 3ad186c5fa9490ff)
T1408	Pc VALID is now a launch-commit condition.	LEAD	CL 176 lines	`L175(4)[sentence containing "launch-commit"]`	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T1409	BASELINE flight	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/A1/role`	(class set by generator: REQ)
T1410	A-D: TVC ascent, pre-MECO throttle-down, nose-first coast with aft fins, nose drogue at ...	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/A1/profile`	(class set by generator: REQ)
T1411	303.5 m	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/nominal_relight_planning_1.81s/h_max`	(class set by generator: SIM)
T1412	64.73 m	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/nominal_relight_planning_1.81s/h_ign`	(class set by generator: SIM)
T1413	7.397 s	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/nominal_relight_planning_1.81s/burn_time_s`	(class set by generator: SIM)
T1414	1.177 kg	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/nominal_relight_planning_1.81s/n2o_burned`	(class set by generator: SIM)
T1415	0.3975	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/nominal_relight_planning_1.81s/reserve_frac`	(class set by generator: SIM)
T1416	development hop (continuous burn, 20 % floor)	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/B1/role`	(class set by generator: REQ)
T1417	62 m	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/B1/h_target_m`	REQ lead decision 2026-10-13 (cap PROVISIONAL until T1-T4)
T1418	62 m PROVISIONAL (lead 2026-10-13: the GNC 62.9 m rounded down) until the T1-T4 regressi...	LEAD	GNC v3	`/flight_profile/mission_plan_v3/v/B1/h_target_status`	(class set by generator: LEAD)
T1419	85 m	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/B1/h_sizing_m`	REQ lead decision 2026-10-13
T1420	Cap rule ADOPTED as the method to set the B1 cap after T1-T4; 62 m stays provisional and...	LEAD	CL 176 lines	`L161(7)`	lead (decision) 2026-10-13 (line sha 175f8cd442f74c80)
T1421	development hop (unchanged)	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/B2/role`	(class set by generator: REQ)
T1422	50 m	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/B2/h_target_m`	(class set by generator: REQ)
T1423	30-100 m (charter): B2 then B1, before A1	REQ	GNC v3	`/flight_profile/mission_plan_v3/v/first_flights`	(class set by generator: REQ)
T1424	Launch commit PROVISIONAL for A1 and B1: W10 < 6 m/s AND sigma_u(3 m) <= 1.0 m/s, 10-min...	LEAD	CL 176 lines	`L161(6)`	lead (decision) 2026-10-13 (line sha 175f8cd442f74c80)
T1425	Mission B1: hop cap 62 m PROVISIONAL (was 85 m) until T1-T4 regression data; B1 flies on...	LEAD	REQ rev 2	`§Amendments after rev 2 bullet "Mission B1"`	REQ bullet
T1426	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 50...	SIM	GNC v3	`/relight_planning_v28/tag`	(class set by generator: SIM)
T1427	500 runs	SIM	GNC v3	`/relight_planning_v28/tag [regex (\d+) runs]`	(class set by generator: SIM) [runs per case]
T1428	0.992	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_none/P_by_state/liq`	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1429	0.9796–0.9969	DERIVED	generator	`formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)`	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1428, T1427
T1430	0.992	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_none/P_by_state/vapU`	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1431	0.9796–0.9969	DERIVED	generator	`formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)`	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1430, T1427
T1432	0.994	SIM	GNC v3	`/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_none/P_by_state/vapF`	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN

ID	Value	Class	Source	Key	Source tag / formula
T1433	0.9825–0.998	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1432, T1427
T1434	0.27	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_none/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1435	0.2329–0.3106	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1434, T1427
T1436	0.992	SIM	GNC v3	/relight_planning_v28/v/sweep_and_full_by_base/A1_none/1.41/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1437	0.9796–0.9969	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1436, T1427
T1438	0.988	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_pc/P_by_state/liq	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1439	0.9741–0.9945	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1438, T1427
T1440	0.994	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_pc/P_by_state/vapU	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1441	0.9825–0.998	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1440, T1427
T1442	0.992	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_pc/P_by_state/vapF	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1443	0.9796–0.9969	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1442, T1427
T1444	0.36	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1_pc/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1445	0.3191–0.403	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1444, T1427
T1446	0.992	SIM	GNC v3	/relight_planning_v28/v/sweep_and_full_by_base/A1_pc/1.41/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1447	0.9796–0.9969	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1446, T1427
T1448	0.988	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_none/P_by_state/liq	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1449	0.9741–0.9945	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1448, T1427
T1450	0.992	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_none/P_by_state/vapU	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1451	0.9796–0.9969	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1450, T1427
T1452	0.99	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_none/P_by_state/vapF	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN

ID	Value	Class	Source	Key	Source tag / formula
T1453	0.9768–0.9957	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1452, T1427
T1454	0.272	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_none/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1455	0.2348–0.3126	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1454, T1427
T1456	0.992	SIM	GNC v3	/relight_planning_v28/v/sweep_and_full_by_base/A1J_none/1.41/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1457	0.9796–0.9969	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1456, T1427
T1458	0.98	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_pc/P_by_state/liq	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1459	0.9636–0.9891	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1458, T1427
T1460	0.988	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_pc/P_by_state/vapU	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1461	0.9741–0.9945	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1460, T1427
T1462	0.99	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_pc/P_by_state/vapF	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1463	0.9768–0.9957	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1462, T1427
T1464	0.356	SIM	GNC v3	/flight_profile/mission_plan_v3/v/A1/relight_v2.8/P_by_line_state_at_planning_value/A1J_pc/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1465	0.3153–0.3989	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1464, T1427
T1466	0.994	SIM	GNC v3	/relight_planning_v28/v/sweep_and_full_by_base/A1J_pc/1.41/P_fault_1p37	SIM m2_relight_v28 v1.2 (sim6/run6 v2.7.1-dev start-delay quantile option, iface_v3c, 500 runs, seed 101 + start-delay seed 20261018, FSW v2.7 k_est_asc 2) on SIM Combustion RC-03 E0→E2 + SIL/DESIGN
T1467	0.9825–0.998	DERIVED	generator	formula: Wilson 95 % interval, n = 500 runs (from the GNC tag)	DERIVED by build_datasheet.py: Wilson 95 % interval, n = 500 runs (from the GNC tag) ← T1466, T1427
T1468	0.81 s	REQ	GNC v3	/relight_planning_v28/v/planning_value_s	REQ lead decision 2026-10-13 (the GNC rule adopted)
T1469	0.996	SIM	GNC v3	/b1_regression_band/v/lead_cap/finding [regex P ([\d.]+) \[([\d.]+), ([\d.]+)\]]	(class set by generator: SIM)
T1470	0.986	SIM	GNC v3	/b1_regression_band/v/lead_cap/finding [regex P ([\d.]+) \[([\d.]+), ([\d.]+)\]]	(class set by generator: SIM)
T1471	0.999	SIM	GNC v3	/b1_regression_band/v/lead_cap/finding [regex P ([\d.]+) \[([\d.]+), ([\d.]+)\]]	(class set by generator: SIM)
T1472	B1 at the lead's 62 m cap: the nominal flight closes the +25 % rule at the worst band co...	SIM	GNC v3	/b1_regression_band/v/lead_cap/finding	(class set by generator: SIM)
T1473	planning value 0.81 s ADOPTED by the lead (2026-10-13, the GNC rule; the 0.6 s + sequenc...	LEAD	GNC v3	/relight_planning_v28/v/status	(class set by generator: LEAD)
T1474	simulation and analysis only; no item here is a bench or flight validation	SIM	GNC v3	/publication_note/not_physical_acceptance	(class set by generator: SIM)
T1475	CONDICIONADO	DERIVED	SUP 7 reviews	/reviews/4/verdict	(class set by generator: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T1476	RG-01 relight on the current design basis not re-flown	DERIVED	SUP 7 reviews	/reviews/4/blocking	(class set by generator: DERIVED)
T1477	GNC RG-01 re-fly started on these tables	LEAD	CL 176 lines	L164(9)	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T1478	REV-COMB-1	DERIVED	SUP 7 reviews	/reviews/0/id	(class set by generator: DERIVED)
T1479	2026-10-12/13	DERIVED	SUP 7 reviews	/reviews/0/date	(class set by generator: DERIVED)
T1480	REVISE	DERIVED	SUP 7 reviews	/reviews/0/verdict	(class set by generator: DERIVED)
T1481	02-combustion (REPORT.md v2.3, combustion.json v2.3, HANDBACK-SUMMARY.md)	DERIVED	SUP 7 reviews	/reviews/0/deliverable	(class set by generator: DERIVED)
T1482	RC-01 ignition/abort logic; RC-02 hard-start bound	DERIVED	SUP 7 reviews	/reviews/0/blocking	(class set by generator: DERIVED)
T1483	0 of 3 reviewed files unchanged; changed since review: 02-combustion/REPORT.md, interfac...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1484	2026-10-12/13	DERIVED	SUP 7 reviews	/reviews/1/date	(class set by generator: DERIVED)
T1485	05-plume-cfd (REPORT.md)	DERIVED	SUP 7 reviews	/reviews/1/deliverable	(class set by generator: DERIVED)
T1486	0 of 1 reviewed files unchanged; changed since review: 05-plume-cfd/REPORT.md	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1487	REV-GNC-1	DERIVED	SUP 7 reviews	/reviews/2/id	(class set by generator: DERIVED)
T1488	2026-10-13	DERIVED	SUP 7 reviews	/reviews/2/date	(class set by generator: DERIVED)
T1489	REVISE	DERIVED	SUP 7 reviews	/reviews/2/verdict	(class set by generator: DERIVED)
T1490	01-systems-gnc v3 (REPORT-v3.md, gnc.json v3) with addendum on v3.1 staging (FC_THRUST)	DERIVED	SUP 7 reviews	/reviews/2/deliverable	(class set by generator: DERIVED)
T1491	RG-01 relight evidence on superseded 21.4 cm3 segment; flag blind to vapour	DERIVED	SUP 7 reviews	/reviews/2/blocking	(class set by generator: DERIVED)
T1492	5 of 5 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1493	REV-ELEC-1	DERIVED	SUP 7 reviews	/reviews/3/id	(class set by generator: DERIVED)
T1494	2026-10-13	DERIVED	SUP 7 reviews	/reviews/3/date	(class set by generator: DERIVED)
T1495	06-electronics v2.9 (firmware v2.9, SIL sil_v2p9, KiCad ECU v2.4, BOM v2.1, bench_plan_v...	DERIVED	SUP 7 reviews	/reviews/3/deliverable	(class set by generator: DERIVED)
T1496	10 of 10 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1497	CHANGELOG line 166	LEAD	SUP 7 reviews	/reviews/3/lead_disposition	(class set by generator: LEAD)
T1498	REV-GNC-2	DERIVED	SUP 7 reviews	/reviews/4/id	(class set by generator: DERIVED)
T1499	2026-10-13	DERIVED	SUP 7 reviews	/reviews/4/date	(class set by generator: DERIVED)
T1500	01-systems-gnc/staging_v3.1/rev1 (REPORT-GNC-rev1.md, rev1_items.json, FSW record v1.4, ...	DERIVED	SUP 7 reviews	/reviews/4/deliverable	(class set by generator: DERIVED)
T1501	6 of 6 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1502	CHANGELOG line 168	LEAD	SUP 7 reviews	/reviews/4/lead_disposition	(class set by generator: LEAD)
T1503	REV-COMB-2	DERIVED	SUP 7 reviews	/reviews/5/id	(class set by generator: DERIVED)
T1504	2026-10-13	DERIVED	SUP 7 reviews	/reviews/5/date	(class set by generator: DERIVED)
T1505	REVISE (targeted)	DERIVED	SUP 7 reviews	/reviews/5/verdict	(class set by generator: DERIVED)
T1506	02-combustion v2.4 (REPORT.md, REVISION-RESPONSE.md, CF-4-SPEC.md, T1-T4-REGRESSION-SPEC...	DERIVED	SUP 7 reviews	/reviews/5/deliverable	(class set by generator: DERIVED)
T1507	5 of 5 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1508	CHANGELOG line 172	LEAD	SUP 7 reviews	/reviews/5/lead_disposition	(class set by generator: LEAD)
T1509	2026-10-13	DERIVED	SUP 7 reviews	/reviews/6/date	(class set by generator: DERIVED)
T1510	05-plume-cfd v2.2b (REPORT.md)	DERIVED	SUP 7 reviews	/reviews/6/deliverable	(class set by generator: DERIVED)
T1511	1 of 1 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T1512	CHANGELOG line 172	LEAD	SUP 7 reviews	/reviews/6/lead_disposition	(class set by generator: LEAD)
T1513	R2-01 line 166 (3)/(4) not implemented; R2-02 two stand safing rules; R2-03 FA-1 pulse s...	DERIVED	SUP 7 reviews	/reviews/5/major	(class set by generator: DERIVED)
T1514	REV-GNC-1 reviewer resumed for the rev1 response; RG-01 stays BLOCKING until the re-fly ...	LEAD	CL 176 lines	L161(12)	lead (decision) 2026-10-13 (line sha 175f8cd442f74c80)

ID	Value	Class	Source	Key	Source tag / formula
T1515	GNC rev2 (response to REV-GNC-2 and lead lines 164-172) STAGED in 01-systems-gnc/staging...	SIM	CL 176 lines	L174	gnc (result) 2026-10-13 (line sha abaa78ccd482cd5a)
T1516	Independent review REV-GNC-3 of rev2 and amendments G, K, V launched (max). gnc.json v3....	LEAD	CL 176 lines	L175(8)	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T1517	fuel-assisted fast part at an A1 relight, 48.9-75.5 bar at 1 ms [OVR-07: single point at ASSUMED O/F 8; superseded by the FA-1 envelope (head peak 83.0 to 88.0 bar abs) and the structural load 90 bar abs]	DERIVED	CMB v2.4	/fmea_input/v/FA-1/hazard	DERIVED; LEAD lines 160, 164; REPORT v2.4 §9.1
T1518	A1 relight lights within W 0.65 s with P_abort <= 1e-3, venting <= 76 g, theta <= 2 deg	LEAD	CMB v2.4	/cf4_gate/v/claim	LEAD line 164 (accepted as the gate); physical execution by the founders
T1519	CF-4 spec ACCEPTED as the gate for the relight claim (founders execute); Combustion publ...	LEAD	CL 176 lines	L164(5)	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T1520	**Record at 1 kHz:** raw Pc and filtered Pc; P0 (the log v10 `ign_ref_hPa`); the rise (`...	DESIGN	BENCH v2	§IGN-1 / T1 (whole section)	BENCH §IGN-1 / T1
T1521	A5 DELETED as an abort, flight and stand; replaced by a LIGHT_SPIKE log event (raw and f...	LEAD	CL 176 lines	L160(1)[sentence containing "IGN-1"]	lead (decision) 2026-10-13 (line sha c3c13e6adb771e7a)
T1522	post-zero residual: sets `kBpc1Shown` (floor 1.8 → 2.0 bar abs, baseline check 0.7 → 0...	DESIGN	BENCH v2	§1. Order and gates row "3" col 2	
T1523	B-PC-1 measures zero TC and 20 s drift (Mechanical supplies the transducer temperature M...	LEAD	CL 176 lines	L166(5)[sentence containing "B-PC-1"]	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T1524	RE-06 HAL is an Electronics open item, owner Electronics, required before B-PC-1.	LEAD	CL 176 lines	L166(9)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T1525	+0.60 kg bleed-valve mass: no trim below 275 m now; the trigger is the selected valve pa...	LEAD	CL 176 lines	L175(2)	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T1526	PENDING	PENDING	—		[Structural FEA margins (see §9)]
T1527	BLOCKING	MISSING	generator	open-item status RG-01	status summarised by the generator from the evidence in the row
T1528	DESIGN CONTINUATION ONLY; margins reported, not reviewed	MISSING	generator	open-item status FA-1	status summarised by the generator from the evidence in the row
T1529	NOT RUN (founders)	MISSING	generator	open-item status CF-4	status summarised by the generator from the evidence in the row
T1530	NOT RUN (founders)	MISSING	generator	open-item status IGN-1	status summarised by the generator from the evidence in the row
T1531	NOT RUN (founders)	MISSING	generator	open-item status B-PC-1	status summarised by the generator from the evidence in the row
T1532	TO-VERIFY	TO-VERIFY	generator	open-item status BLEED-MASS	status summarised by the generator from the evidence in the row
T1533	NOT MET (structure only)	MISSING	generator	open-item status ENG-11	status summarised by the generator from the evidence in the row
T1534	MARGINAL	TO-VERIFY	generator	open-item status ENG-07	status summarised by the generator from the evidence in the row
T1535	PENDING	PENDING	generator	open-item status FEA	status summarised by the generator from the evidence in the row
T1536	TO-VERIFY	TO-VERIFY	generator	open-item status ENVELOPE	status summarised by the generator from the evidence in the row
T1537	TO-VERIFY	TO-VERIFY	generator	open-item status GRAIN-R4	status summarised by the generator from the evidence in the row
T1538	REVISE (targeted)	PENDING	generator	open-item status REV-COMB-2	status summarised by the generator from the evidence in the row
T1539	IN PROGRESS	PENDING	generator	open-item status COMB-v2.4a	status summarised by the generator from the evidence in the row
T1540	TO-DEFINE	TO-DEFINE	generator	open-item status INTERLOCK-GB	status summarised by the generator from the evidence in the row
T1541	IN PROGRESS	PENDING	generator	open-item status ELEC-v2.10	status summarised by the generator from the evidence in the row
T1542	Analysis, then static fire	REQ	REQ rev 2	§2. Top-level row "ENG-01" col 3	
T1543	ENG-01 at the cap (CHANGELOG 170): the 25 bar software Pc cap stays; if T1 measures the ...	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-01 at the cap (CHANGELOG 170): the 25 bar so" [sentence containing "ENG-01"]	REQ amendment bullet
T1544	Analysis, cold flow, static fire T3 with 1 kHz Pc	REQ	REQ rev 2	§2. Top-level row "ENG-02" col 3	
T1545	SIL, B-PNT-1, static fire	REQ	REQ rev 2	§2. Top-level row "ENG-03" col 3	

ID	Value	Class	Source	Key	Source tag / formula
T1546	Analysis, static fire	REQ	REQ rev 2	§2. Top-level row "ENG-04" col 3	
T1547	ENG-04 / relight: deterministic relight (RELIGHT at planned T_fire;	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-04 / relight: deterministic relight (RELIGHT" [sentence containing "ENG-04"]	REQ amendment bullet
T1548	Analysis, B-PNT-1 closing time at 25 bar, static fire	REQ	REQ rev 2	§2. Top-level row "ENG-05" col 3	
T1549	ENG-05 / hard start (stand): unignited-inventory allowable set by the two-volume relief ...	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-05 / hard start (stand): unignited-inventory" [sentence containing "ENG-05"]	REQ amendment bullet
T1550	Analysis, static fire bore measurement	REQ	REQ rev 2	§2. Top-level row "ENG-06" col 3	
T1551	SIL, B-TVC-1	REQ	REQ rev 2	§2. Top-level row "ENG-07" col 3	
T1552	No actuator saturation episode longer than 0.1 s during the landing burn. Logged in flig...	DERIVED	REQ rev 2	§2. Top-level row "ENG-07b" col 1	DERIVED (GNC)
T1553	MC, flight logs	REQ	REQ rev 2	§2. Top-level row "ENG-07b" col 3	
T1554	Reflected servo rotor inertia J_int ≤ 1e-3 kg m² at the output (provisional), unless a s...	DERIVED	REQ rev 2	§2. Top-level row "ENG-07c" col 1	DERIVED (GNC) / ASSUMED
T1555	B-TVC-1	REQ	REQ rev 2	§2. Top-level row "ENG-07c" col 3	
T1556	Flex gain margin ≥ 6 dB at the gimbal rocking mode. ECU notch required, with separate al...	DERIVED	REQ rev 2	§2. Top-level row "ENG-07d" col 1	DERIVED (GNC csi_v3)
T1557	CSI analysis, B-TVC-1	REQ	REQ rev 2	§2. Top-level row "ENG-07d" col 3	
T1558	Gimbal sweep fits the airframe ICD: bay ID ≥ 140 mm, zone [−0.238, +0.252] m. Feed acros...	REQ	REQ rev 2	§2. Top-level row "ENG-08" col 1	REQ / DERIVED
T1559	CAD, B-FD-1	REQ	REQ rev 2	§2. Top-level row "ENG-08" col 3	
T1560	FEA, hydro proof	REQ	REQ rev 2	§2. Top-level row "ENG-09" col 3	
T1561	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition.	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-09: MEOP 45 bar kept; ignition interlock T_t" [sentence containing "ENG-09"]	REQ amendment bullet
T1562	ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa, pro...	REQ	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172): " [sentence containing "ENG-09"]	REQ amendment bullet
T1563	FEA, hand calculations	REQ	REQ rev 2	§2. Top-level row "ENG-10" col 3	
T1564	Modal and stiffness FEA, B-TVC-1	REQ	REQ rev 2	§2. Top-level row "ENG-11" col 3	
T1565	Trade studies, static fire	REQ	REQ rev 2	§2. Top-level row "ENG-12" col 3	
T1566	ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM.	REQ	REQ rev 2	§Amendments after rev 2 bullet "ENG-12: tank temperature range at ARM 15-24 C (w" [sentence containing "ENG-12"]	REQ amendment bullet
T1567	Engine controller (ECU) works with a Teensy-class flight computer. It provides closed-lo...	DERIVED	REQ rev 2	§2. Top-level row "ENG-13" col 1	DERIVED / LEAD
T1568	Bench, SIL, HIL	REQ	REQ rev 2	§2. Top-level row "ENG-13" col 3	
T1569	Cost: cheapest viable. Every BOM line has a source and price, or is marked TO-QUOTE with...	REQ	REQ rev 2	§2. Top-level row "ENG-14" col 1	REQ
T1570	BOM review	REQ	REQ rev 2	§2. Top-level row "ENG-14" col 3	
T1571	Manufacturability: printed parts on a hobby FDM printer; metal parts from stock tube and...	REQ	REQ rev 2	§2. Top-level row "ENG-15" col 1	REQ
T1572	CAD review	REQ	REQ rev 2	§2. Top-level row "ENG-15" col 3	
T1573	Safety: N2O-compatible, oxygen-clean wetted paths (PFPE lubricant only); passive relief ...	REQ	REQ rev 2	§2. Top-level row "ENG-16" col 1	REQ / LEAD
T1574	Review, bench	REQ	REQ rev 2	§2. Top-level row "ENG-16" col 3	
T1575	B-PWR-1, B-PWR-2	DESIGN	BENCH v2	§1. Order and gates row "1" col 1	
T1576	rails, e-fuse, ARM interlock	DESIGN	BENCH v2	§1. Order and gates row "1" col 2	
T1577	bare ECU, bench supply	DESIGN	BENCH v2	§1. Order and gates row "1" col 3	
T1578	B-ECU-1	DESIGN	BENCH v2	§1. Order and gates row "2" col 1	
T1579	I/O, CAN, watchdog, log v10	DESIGN	BENCH v2	§1. Order and gates row "2" col 2	
T1580	step 1	DESIGN	BENCH v2	§1. Order and gates row "2" col 3	
T1581	B-PC-1	DESIGN	BENCH v2	§1. Order and gates row "3" col 1	
T1582	step 2, N2 0-70 bar g, reference gauge	DESIGN	BENCH v2	§1. Order and gates row "3" col 3	
T1583	B-MV-1, B-PNT-1, B-TVC-1	DESIGN	BENCH v2	§1. Order and gates row "4" col 1	
T1584	actuator timing, ENG-03, MV switch ladder	DESIGN	BENCH v2	§1. Order and gates row "4" col 2	
T1585	step 2	DESIGN	BENCH v2	§1. Order and gates row "4" col 3	

ID	Value	Class	Source	Key	Source tag / formula
T1586	B-BLEED-1 (new)	DESIGN	BENCH v2	§1. Order and gates row "5" col 1	
T1587	head-end bleed: pad BIT, P_mid comparator, V1 closure confirmation time	DESIGN	BENCH v2	§1. Order and gates row "5" col 2	
T1588	steps 2-4, valves + tee	DESIGN	BENCH v2	§1. Order and gates row "5" col 3	
T1589	B-IGN-1, B-FC-1 (+ drogue telemetry)	DESIGN	BENCH v2	§1. Order and gates row "6" col 1	
T1590	spark EMI on Pc, FC link, TLM_FAST d1 bits	DESIGN	BENCH v2	§1. Order and gates row "6" col 2	
T1591	step 2, FC	DESIGN	BENCH v2	§1. Order and gates row "6" col 3	
T1592	HIL-OT-1, HIL-TR-1	DESIGN	BENCH v2	§1. Order and gates row "7" col 1	
T1593	over-thrust and transient replay	DESIGN	BENCH v2	§1. Order and gates row "7" col 2	
T1594	steps 2, 6	DESIGN	BENCH v2	§1. Order and gates row "7" col 3	
T1595	LEAK-1, CF-3, CF-4	DESIGN	BENCH v2	§1. Order and gates row "8" col 1	
T1596	feed leak, ignition-lift table, segment void	DESIGN	BENCH v2	§1. Order and gates row "8" col 2	
T1597	Mechanical / Combustion rigs	DESIGN	BENCH v2	§1. Order and gates row "8" col 3	
T1598	IGN-1, T1-T4	DESIGN	BENCH v2	§1. Order and gates row "9" col 1	
T1599	relative light rule on hardware, start latency, F_acc	DESIGN	BENCH v2	§1. Order and gates row "9" col 2	
T1600	hot-fire stand	DESIGN	BENCH v2	§1. Order and gates row "9" col 3	
T1601	B-PNT-1	REQ	REQ rev 2	§4. Gating early bench tests row "B-PNT-1" col 0	
T1602	Commanded travel ≥ 103°; unpowered spring closing at the MEOP-equivalent load; guide μ ≤...	REQ	REQ rev 2	§4. Gating early bench tests row "B-PNT-1" col 2	
T1603	1× MKS HV69 (EUR 62.99 at Lindinger, catalogue price incl. AT VAT; TO-QUOTE to DK-6400)	REQ	REQ rev 2	§4. Gating early bench tests row "B-PNT-1" col 1	
T1604	B-TVC-1	REQ	REQ rev 2	§4. Gating early bench tests row "B-TVC-1" col 0	
T1605	Measure servo output stiffness K_s at 6.6 and 7.4 V, damping, J_int and the rotor/struct...	REQ	REQ rev 2	§4. Gating early bench tests row "B-TVC-1" col 2	
T1606	2× SB-2292SG (TO-QUOTE) with the n 4.8 linkage on a 0.0545 kg·m² inertia dummy	REQ	REQ rev 2	§4. Gating early bench tests row "B-TVC-1" col 1	
T1607	B-PWR-1/2	REQ	REQ rev 2	§4. Gating early bench tests row "B-PWR-1/2" col 0	
T1608	Source and branch resistance against the 15 / 20 mΩ assumptions.	REQ	REQ rev 2	§4. Gating early bench tests row "B-PWR-1/2" col 2	
T1609	Flight battery pack and harness	REQ	REQ rev 2	§4. Gating early bench tests row "B-PWR-1/2" col 1	
T1610	B-FD-1	REQ	REQ rev 2	§4. Gating early bench tests row "B-FD-1" col 0	
T1611	Hinge stiffness while pressurised ≤ 20 N·m/rad per axis.	REQ	REQ rev 2	§4. Gating early bench tests row "B-FD-1" col 2	
T1612	Gimbal hose	REQ	REQ rev 2	§4. Gating early bench tests row "B-FD-1" col 1	
T1613	P1 liquid at the pintle inlet <= 0.293 s after reaching the lift	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/0	LEAD line 164 (accepted as the gate); physical execution by the founders
T1614	P2 >= 0.35 s continuous liquid or inside the void budget	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/1	LEAD line 164 (accepted as the gate); physical execution by the founders
T1615	P3 cap clears FD-01 crown <= 2.3 s after flow start	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/2	LEAD line 164 (accepted as the gate); physical execution by the founders
T1616	P4 each tap >= 80 % of model flow	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/3	LEAD line 164 (accepted as the gate); physical execution by the founders
T1617	P5 <= 76 g vented (IS-1, IS-2)	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/4	LEAD line 164 (accepted as the gate); physical execution by the founders
T1618	P6 closure by E0 - 0.05 s and BLEED_FAULT on a held-open valve	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/5	LEAD line 164 (accepted as the gate); physical execution by the founders
T1619	P7 no unfiltered path to the seat	LEAD	CMB v2.4	/cf4_gate/v/pass_criteria/6	LEAD line 164 (accepted as the gate); physical execution by the founders
T1620	Grain mass before and after each firing (balance resolution <= 0.1 g), dried to constant...	DESIGN	T14 v2.4	§T1-T4 regression bullet "Grain mass before"	T14 bullet
T1621	Port diameter at six stations, x/L = 0.10, 0.30, 0.50, 0.70, 0.90 and 0.97 from the head...	DESIGN	T14 v2.4	§T1-T4 regression bullet "Port diameter at six"	T14 bullet
T1622	Time histories: Pc at 1 kHz, oxidiser mass flow from the tank load cell and P_feed, thru...	DESIGN	T14 v2.4	§T1-T4 regression bullet "Time histories"	T14 bullet

ID	Value	Class	Source	Key	Source tag / formula
T1623	At least two oxidiser fluxes: T1 at 100 % and the T2 throttle steps, so that n is fitted...	DESIGN	T14 v2.4	§T1-T4 regression bullet "At least two oxidiser"	T14 bullet
T1624	port per combustion grain v2 S230 (lead relay; combustion.json v1 was a helical port); r...	DESIGN	BOM v3.4.1	/lines/52/notes	BOM notes
T1625	0.003 s	DESIGN	ELE v2	/abort_matrix/causes/0/thresholds/kPcOverPressureTime_s	DESIGN
T1626	5 ms	LEAD	CL 176 lines	L166(4) [regex (\d+) ms persistence]	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T1627	NOT in gnc.json v3 requirements.ignition (key removed; FC rule source to be re-confirmed...	DESIGN	ELE v2	/relight_interface/fc_planning_from_gnc/option_c_unprimed_s	(no source tag; class set by generator: DESIGN)
T1628	OI-GNC-V3: lead verified gnc.json v3 (3e97c378cfebee95) carries requirements.ignition.re...	LEAD	CL 176 lines	L161(9)	lead (decision) 2026-10-13 (line sha 175f8cd442f74c80)
T1629	FALLBACK only (lead line 126); no design (coordinator)	DESIGN	ELE v2	/relight_interface/head_end_bleed/status	(no source tag; class set by generator: DESIGN)
T1630	RG-01 design basis = Combustion's minimal set (3): coil re-route + pre-scheduled 2-tap p...	LEAD	CL 176 lines	L164(1)	lead (decision) 2026-10-13 (line sha 7d6ed784b634742d)
T1631	Combustion data/v2/full_stroke_failure_case_v2.json max F_N 907.7 N (v1 helix, 30 C, Cd ...	DESIGN	FEALC v1	/cases/2/src/thrust	(class set by generator: DESIGN)
T1632	4. Coast nose-first, stabilised by 4 aft fins, to a **300 m** apogee.	REQ	REQ rev 2	line 19	REQ line 19
T1633	M2. Mass properties from the CAD v3.3 solids (engine + gimbal, CG pivot iterated in CAD)...	DESIGN	MEC v2	/status	(class set by generator: DESIGN)
T1634	v3.4.1	DESIGN	PROP v3.4.1	/cad_version	(class set by generator: DESIGN)
T1635	CAD v3.4: HV69 in-line pintle stack with rolling guide, feed parts (hose/elbow/lines), s...	PENDING	MEC v2	/open_items/0	(class set by generator: PENDING)
T1636	4.392 kg	DERIVED	BOM v3.4.1	/mass_sum_kg	(class set by generator: DERIVED)
T1637	Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts	DESIGN	BOM v3.4.1	/lines/92/material_name	(class set by generator: DESIGN)
T1638	v2.4	DESIGN	CMB v2.4	/version	(class set by generator: DESIGN)
T1639	combustion.json /abort_matrix_v2p4/v/A4: annotate, ACTIVE, shown 1×; CHANGELOG.md L160(2...	DERIVED	generator	override OVR-01 targets (this build)	generator target check on this build
T1640	ACTIVE	DERIVED	generator	override OVR-01 status (this build)	generator check on this build
T1641	combustion.json /igniter/sequence/v/start/0/action: annotate, ACTIVE, shown 1×; combusti...	DERIVED	generator	override OVR-02 targets (this build)	generator target check on this build
T1642	ACTIVE	DERIVED	generator	override OVR-02 status (this build)	generator check on this build
T1643	combustion.json /igniter/sequence/v/abort_if_no_ignition/1: annotate, ACTIVE, shown 1×	DERIVED	generator	override OVR-03 targets (this build)	generator target check on this build
T1644	ACTIVE	DERIVED	generator	override OVR-03 status (this build)	generator check on this build
T1645	combustion.json /decisions/tank_pressurisation: annotate, ACTIVE, shown 1×; combustion.j...	DERIVED	generator	override OVR-04 targets (this build)	generator target check on this build
T1646	ACTIVE	DERIVED	generator	override OVR-04 status (this build)	generator check on this build
T1647	combustion.json /valves_and_actuators/isolation_valve/trapped_liquid_relief: replace, AC...	DERIVED	generator	override OVR-05 targets (this build)	generator target check on this build
T1648	ACTIVE	DERIVED	generator	override OVR-05 status (this build)	generator check on this build
T1649	combustion.json /decisions/chamber_MEOP_Pa/tag: replace, ACTIVE, shown 1×; REQUIREMENTS....	DERIVED	generator	override OVR-06 targets (this build)	generator target check on this build
T1650	ACTIVE	DERIVED	generator	override OVR-06 status (this build)	generator check on this build
T1651	annotation of every displayed text quoting the single-point value: ACTIVE, shown 9×	DERIVED	generator	override OVR-07 targets (this build)	generator target check on this build
T1652	ACTIVE	DERIVED	generator	override OVR-07 status (this build)	generator check on this build
T1653	combustion.json /igniter/sequence/v/start/0/action: annotate, ACTIVE, shown 1×	DERIVED	generator	override OVR-08 targets (this build)	generator target check on this build

ID	Value	Class	Source	Key	Source tag / formula
T1654	ACTIVE	DERIVED	generator	override OVR-08 status (this build)	generator check on this build
T1655	CHANGELOG.md L168(1): annotate, ACTIVE, shown 1×	DERIVED	generator	override OVR-09 targets (this build)	generator target check on this build
T1656	ACTIVE	DERIVED	generator	override OVR-09 status (this build)	generator check on this build
T1657	combustion.json /operating_points/pressure_limits/full_stroke_failure_S230_max_bar: WITH...	DERIVED	generator	withheld SUP-01 keys (this build)	generator check on this build
T1658	1 displayed field masked	DERIVED	generator	withheld SUP-01 masks (this build)	generator count on this build
T1659	yes	DERIVED	generator	withheld SUP-01 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T1660	combustion.json /thermal_and_loads/flights_per_insert_A1: WITHHELD	DERIVED	generator	withheld SUP-02 keys (this build)	generator check on this build
T1661	0 displayed fields masked	DERIVED	generator	withheld SUP-02 masks (this build)	generator count on this build
T1662	yes	DERIVED	generator	withheld SUP-02 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T1663	combustion.json /valves_and_actuators/pintle_actuator/seat_and_seal_review_v2p3/v/b_seat...	DERIVED	generator	withheld SUP-03 keys (this build)	generator check on this build
T1664	0 displayed fields masked	DERIVED	generator	withheld SUP-03 masks (this build)	generator count on this build
T1665	yes	DERIVED	generator	withheld SUP-03 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T1666	no key listed	DERIVED	generator	withheld SUP-04 keys (this build)	generator check on this build
T1667	3 displayed fields masked	DERIVED	generator	withheld SUP-04 masks (this build)	generator count on this build
T1668	yes	DERIVED	generator	withheld SUP-04 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T1669	Combustion value (ENG-12). CAD material density differs from the Combustion and BOM-note...	DERIVED	generator	contradiction C-02	generator summary of the cited sides
T1670	CHANGELOG 166 (4): filtered channel, 5 ms. Electronics v2.10 to implement (REV-ELEC-1).	DERIVED	generator	contradiction C-03	generator summary of the cited sides
T1671	deleted (LIGHT_SPIKE log event). REV-ELEC-1 RE-01 blocking; Electronics v2.10.	DERIVED	generator	contradiction C-04	generator summary of the cited sides
T1672	present (lead verified, CHANGELOG 161 (9)). Electronics fixes the lookup (OI-GNC-V3).	DERIVED	generator	contradiction C-05	generator summary of the cited sides
T1673	CHANGELOG 164 (1): decided purge path. stale text in electronics.json v2; Electronics v2...	DERIVED	generator	contradiction C-06	generator summary of the cited sides
T1674	v2.4 MEOP basis (pressure); the thrust at that basis is MISSING. Combustion to publish t...	DERIVED	generator	contradiction C-07	generator summary of the cited sides
T1675	CHANGELOG 175 (1). gnc.json v3.1 (published after REV-GNC-3) and the REQUIREMENTS missio...	DERIVED	generator	contradiction C-08	generator summary of the cited sides
T1676	both shown (scopes differ). BOM line covers body and plug; solenoids and propane supply ...	DERIVED	generator	contradiction C-09	generator summary of the cited sides
T1677	CAD v3.4.1 for geometry, BOM and mass; mechanical.json v2 for stiffness and loads until ...	DERIVED	generator	contradiction C-10	generator summary of the cited sides
T1678	combustion.json v2.4. regenerate engine_params.h in Electronics v2.10.	DERIVED	generator	contradiction C-11	generator summary of the cited sides
T1679	the decisions, via overrides OVR-01, OVR-02, OVR-03, OVR-04, OVR-05, OVR-06, OVR-07, OVR...	DERIVED	generator	contradiction C-12	generator summary of the cited sides
T1680	generator sum with the transport lock excluded, as the BOM note states. the difference e...	DERIVED	generator	contradiction C-13	generator summary of the cited sides
T1681	CAD v3.4.1 as built (no lead decision on the result yet). lead decision on the Mechanica...	DERIVED	generator	contradiction C-14	generator summary of the cited sides